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# py-mathx-lab Documentation

*Release 2.0.0*

Walter Weinmann

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Small, reproducible math experiments implemented in Python.

- **Audience:** curious engineers, students, and researchers
- **Idea:** each experiment is a self-contained runnable module with a short write-up
- **Goal:** a growing, searchable “lab notebook” of experiments



## START HERE

- *Background* - mathematical background for experiments
- *Development* - Makefile workflow, CI, coding conventions
- *Experiment Status* - document the processing status
- *Experiments Gallery* - experiment gallery (IDs, tags, how to run)
- *Getting started* - install, setup, and run your first experiment
- *Mathematical experimentation* - what “experiments” in mathematics mean and how to read this repo
- *PDF download* - download the PDF version of these docs
- *References* - bibliography and reading list
- *Valid Tags* - central directory of valid tags for experiments



## RUN ONE EXPERIMENT

```
make uv-check  
make venv  
make install-dev  
make run EXP=e001 ARGS="--out out/e001 --seed 1"
```



- 
- – **E129** - *E129: Euler lucky constants for  $n^2 + n + b$*
- 

## 3.1 Mathematical experimentation

Mathematical experimentation is the practice of using examples, computation, and visualization to discover structure, to generate conjectures, find counterexamples, estimate quantities, and build intuition that later supports (or refutes) formal arguments.

It is *not* “proof by computer output”. Instead, experiments are a disciplined way to ask better questions and to stress-test ideas-especially now that modern computers make it easy to explore large search spaces, high-precision numerics, and rich visualizations.

This repository, **py-mathx-lab**, is a small “lab notebook” of such experiments: compact, reproducible, and readable.

### 3.1.1 What counts as an “experiment” in mathematics?

An experiment is a finite procedure that produces evidence about a mathematical claim or object. Typical outcomes:

- **Conjecture generation:** patterns suggest statements that might be true (or false).
- **Counterexample search:** systematic exploration tries to break a hypothesis early.
- **Quantitative exploration:** estimate constants, rates, limits, or distributions.
- **Model checking:** validate (or invalidate) approximations and heuristics.
- **Visualization:** reveal structure that is hard to see symbolically.

The modern viewpoint-where computation is a genuine part of mathematical discovery-is widely discussed under the name *experimental mathematics* ([Bailey and Borwein, 2005, Bailey *et al.*, 2004, Borwein, 2005]). Some authors emphasize “plausible reasoning” supported by computation, paired with careful verification and eventual proof ([Borwein, 2009, Borwein and Bailey, 2008]). Other texts take a more problem-driven, exploratory style aimed at students and engineers ([Li *et al.*, 2003]), or present computation as a tool to formulate concrete research problems ([Arnold, 2015]).

### 3.1.2 Why computers change the game

Computers do not replace mathematical thinking-but they expand what is feasible to *inspect*:

- **Scale:** enumerate millions of cases (to find the first failure or build confidence).
- **Precision:** compute with high-precision floats or exact rationals/integers to avoid roundoff illusions.
- **Multiple lenses:** combine numerics, exact arithmetic, symbolic manipulation, and plotting.
- **Search:** automate discovery (parameter sweeps, optimization, random sampling, heuristics).

- **Reproducibility:** re-run the same pipeline with fixed seeds and pinned dependencies.

Used well, computation turns “I wonder if...” into “here is evidence, here are edge cases, and here is what we should prove next”.

### 3.1.3 Experiments vs. proofs

A proof is the end of the story; an experiment is often the beginning.

Experiments are excellent for:

- **disproving** statements (one counterexample ends it),
- identifying **what is actually true** (after a naive conjecture fails),
- suggesting **lemmas** and **invariants** that make a proof possible.

But experiments can also mislead. Common failure modes:

- floating point error and catastrophic cancellation,
- plotting artifacts,
- “pattern matching” based on too few samples,
- unintentional selection bias (“I tried the cases that worked”).

The goal is to use experiments to *reduce uncertainty*, not to hide it.

### 3.1.4 Targets for py-mathx-lab

py-mathx-lab aims to be a practical, long-lived collection of experiments with shared conventions:

1. **Reproducible runs**

Each experiment is runnable as a module, writes results to a single output directory, and (when relevant) uses a fixed seed.

2. **Readable code**

The code should be short, well-typed, and structured so readers can modify it.

3. **Useful artifacts**

Each experiment should generate at least one of:

- a figure
- a table / summary statistics
- a counterexample / witness object
- a short narrative explaining what was learned

4. **Clear boundaries**

An experiment write-up should state:

- what is being tested,
- what counts as “success” or “failure”,
- what might invalidate the result (precision limits, domain constraints, runtime limits).

5. **Traceability**

Each page includes references to books/papers that motivated the work or explain the background.

### 3.1.5 Repository conventions for experiments

An experiment page should usually include:

- **Goal** (one paragraph)
- **How to run** (a command that works from the repo root)



- **Parameters** (including defaults and ranges, if swept)
- **Results** (figures/tables and a short interpretation)
- **Notes / pitfalls** (numerical caveats, surprising behavior)
- **References** (bibliography keys)

In code, prefer:

- deterministic outputs (fixed seeds, stable sorting),
- explicit configuration objects / CLI arguments,
- sanity checks (dimension checks, bounds checks, invariants),
- cross-checks (e.g., float vs. exact, two independent formulas).

### 3.1.6 Examples of good “experiment themes”

The experiment format supports many domains, for example:

- **Numerical analysis:** error landscapes, stability regions, conditioning, Monte Carlo integration.
- **Number theory:** continued fractions and convergents, integer sequences, modular patterns, primality heuristics.
- **Geometry/topology:** random point clouds, curvature approximations, combinatorial invariants.
- **Optimization/probability:** stochastic search behavior, concentration phenomena, empirical distributions.

A concrete example of a rich, experiment-friendly topic is continued fractions, where it is natural to compute and plot convergents, partial quotient statistics, and periodicity phenomena ([Borwein *et al.*, 2014]).

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**Next steps:** start with *Getting started*, browse the *Valid Tags* directory, and then browse the *Experiments Gallery* gallery.

## 3.2 Valid Tags

This page defines the allowed tags for experiments in **py-mathx-lab**. Tags are used in the *Experiments Gallery* and individual experiment pages to categorize content.

### 3.2.1 Primary Tags (Domains)

These represent the broad mathematical area of the experiment.

| Tag                             | Description                                                                           |
|---------------------------------|---------------------------------------------------------------------------------------|
| <b>analysis</b>                 | Calculus, real/complex analysis, limits, and approximation.                           |
| <b>aps</b>                      | Arithmetic Progressions and related theorems.                                         |
| <b>conjecture-generation</b>    | Patterns suggest statements that might be true (or false).                            |
| <b>counterexample-search</b>    | Systematic exploration tries to break a hypothesis early.                             |
| <b>dirichlet-characters</b>     | Dirichlet characters modulo $q$ , orthogonality, and primitive characters.            |
| <b><math>L</math>-functions</b> | Dirichlet $L$ -functions and related analytic objects controlling prime distribution. |
| <b>model-checking</b>           | Validate (or invalidate) approximations and heuristics.                               |
| <b>number-theory</b>            | Properties of integers, divisibility, and prime numbers.                              |
| <b>prime-races</b>              | Prime number races (e.g., Chebyshev bias) comparing counts in residue classes.        |
| <b>quantitative-exploratic</b>  | Estimate constants, rates, limits, or distributions.                                  |
| <b>visualization</b>            | Reveal structure that is hard to see symbolically.                                    |

### 3.2.2 Secondary Tags (Topics & Methods)

These provide more specific detail about the techniques or subtopics involved.

| Tag                   | Description                                                                                      |
|-----------------------|--------------------------------------------------------------------------------------------------|
| arithmetic-functions  | Classical functions on integers (e.g., $\phi$ , $\mu$ , $\sigma$ , $\tau$ ) and their relations. |
| bounds                | Explicit mathematical bounds (e.g., on prime-counting functions).                                |
| carmichael-lambda     | Carmichael's lambda function $\lambda(n)$ , the exponent of $(\mathbb{Z}/n\mathbb{Z})^*$ .       |
| carmichael            | Carmichael numbers (absolute Fermat pseudoprimes).                                               |
| classification        | Grouping objects into classes based on shared properties.                                        |
| critical-line         | Zeta/L-function values on $\text{Re}(s)=1/2$ and related numerics.                               |
| dirichlet-convolution | Dirichlet convolution of arithmetic functions and identity checks.                               |
| dirichlet-series      | Dirichlet generating functions and related analytic tools.                                       |
| divisor-function      | Divisor functions such as $\sigma(n)=d(n)$ and $\sigma(n)$ , including record behavior.          |
| explicit-formula      | Explicit formulas connecting primes and zeros ( $\chi(x)$ , $\psi(x)$ , etc.).                   |
| exploration           | Open-ended search for patterns or properties.                                                    |
| factorization         | Integer factorization methods and hardness.                                                      |
| fermat                | Fermat numbers and Fermat primes.                                                                |
| gaps                  | Studies of the distribution of gaps between primes.                                              |
| generating-functions  | Ordinary/exponential generating functions (esp. partitions).                                     |
| gram-points           | Gram points and Gram's law heuristics for zeta zeros.                                            |
| hardy-z               | Hardy's Z-function and Riemann–Siegel theta function.                                            |
| heuristics            | Probabilistic or empirical models (e.g., Cramér's model for primes).                             |
| li-x                  | Logarithmic integral $\text{li}(x)$ and its relation to $\pi(x)$ .                               |
| liouville             | Liouville function $\lambda(n)$ and related parity questions for $\Omega(n)$ .                   |
| lucas-lehmer          | Lucas–Lehmer primality test for Mersenne numbers.                                                |
| mangoldt              | von Mangoldt function $\Lambda(n)$ and related Chebyshev functions $\psi(x)$ , $\theta(x)$ .     |
| mersenne              | Mersenne numbers $M_p = 2^p - 1$ and Mersenne primes.                                            |
| miller-rabin          | Miller–Rabin primality test and its counterexamples.                                             |
| mobius                | Möbius function $\mu(n)$ , Möbius inversion, squarefree indicators.                              |
| multiplicative        | Multiplicative arithmetic functions; Dirichlet convolution viewpoints.                           |
| numerics              | Heavy use of floating-point or high-precision computation.                                       |
| omega                 | Prime-factor counting functions $\omega(n)$ and $\Omega(n)$ (distinct vs with multiplicity).     |
| open-problems         | Related to famous unproven conjectures.                                                          |
| optimization          | Finding maxima, minima, or best-fit parameters.                                                  |
| partition             | Partition function $p(n)$ , identities, and asymptotics.                                         |
| perfect               | Related specifically to perfect, abundant, or deficient numbers.                                 |
| pnt                   | Prime Number Theorem and related asymptotics.                                                    |
| primes                | Prime number distribution, density, and related theorems.                                        |
| primorial             | Primorials, Euclid numbers, primorial primes.                                                    |
| pseudoprime           | Pseudoprimes, primality-test failures.                                                           |
| pseudoprimes          | Collective study of various types of pseudoprimes.                                               |
| riemann-zeta          | Riemann zeta function $\zeta(s)$ : series, Euler product, analytic continuation, zeros.          |
| search                | Systematic search through a large state space.                                                   |
| semiprime             | Semiprimes, RSA-type composites.                                                                 |
| sieving               | Sieve methods (Eratosthenes, Sundaram, Atkin, etc.).                                             |
| sigma                 | Related to the sum-of-divisors function $\sigma(n)$ .                                            |
| summatory             | Summatory functions (e.g., Mertens $M(x)$ , summatory totient $\Phi(x)$ ).                       |
| taylor                | Related to Taylor series and their approximations.                                               |
| totient               | Euler's totient function $\phi(n)$ , totient equations, summatory behavior.                      |
| twin                  | Twin primes and the twin prime conjecture.                                                       |
| klauber               | Klauber triangle and prime patterns between consecutive squares.                                 |
| sacks                 | Sacks spiral ( $r = \sqrt{n}$ , $\theta = 2\pi\sqrt{n}$ ) and related prime patterns.            |
| hex-spiral            | Integer spirals on a hexagonal lattice (hex-grid prime maps).                                    |
| ulam                  | Ulam spiral and related prime patterns in 2D.                                                    |
| wieferich             | Wieferich primes and related congruences.                                                        |
| wilson                | Wilson's theorem and Wilson primes.                                                              |

continues on next page

Table 1 – continued from previous page

| Tag                     | Description                                               |
|-------------------------|-----------------------------------------------------------|
| <code>zeta-zeros</code> | Nontrivial zeros, zero-counting $N(T)$ , root bracketing. |

### 3.2.3 Usage

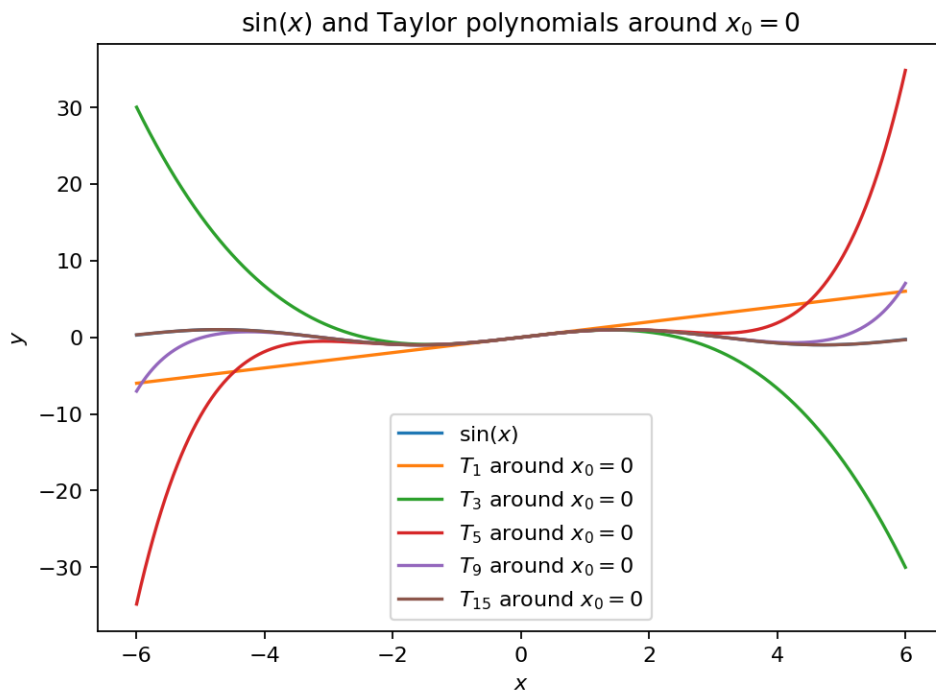
When adding a new experiment:

1. Choose at least one **Primary Tag** (Domain or Type).
2. Choose one or more **Secondary Tags** (Topics & Methods).
3. Add them to the `**Tags:**` line in your `.md` file.
4. Update the *Experiments Gallery* using the corresponding CSS classes (`tag-primary` for primary tags, `tag-secondary` for secondary tags).

## 3.3 Experiments Gallery

A compact, image-first overview of the experiments in **py-mathx-lab**.

### 3.3.1 E001 — Taylor Error Landscapes

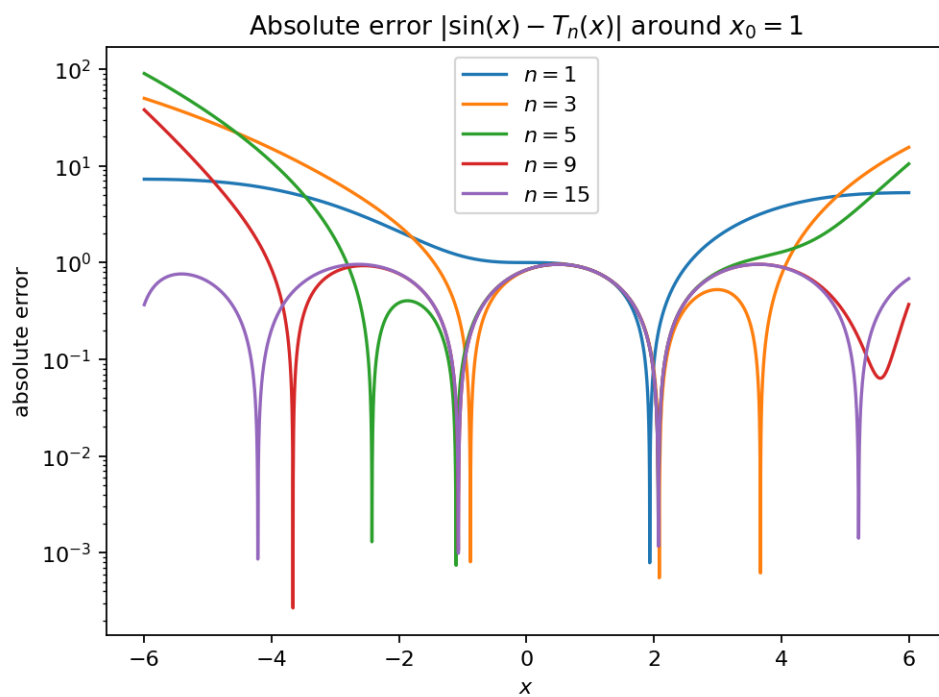
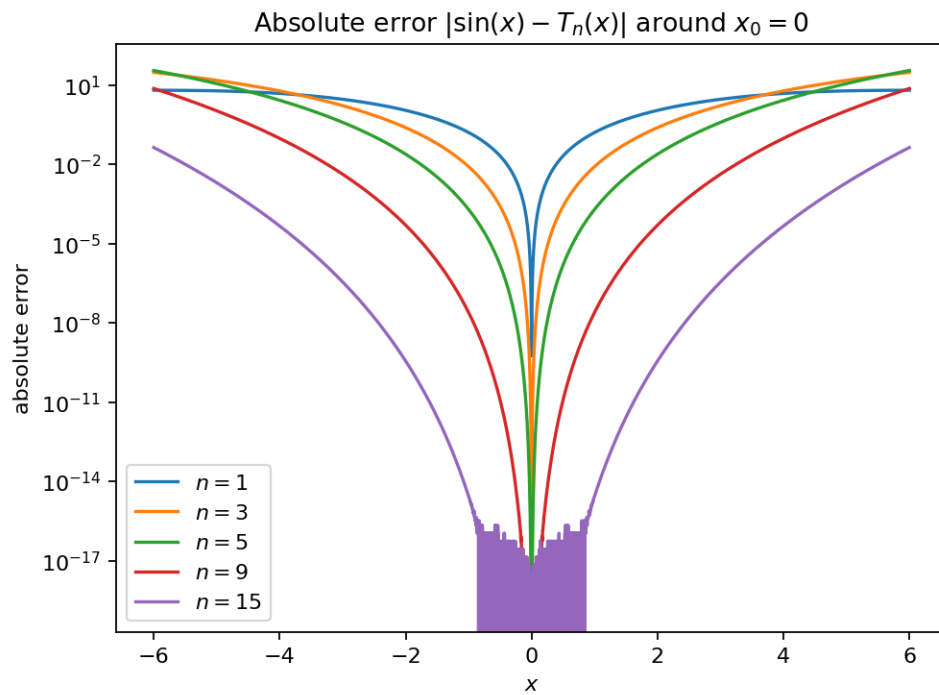


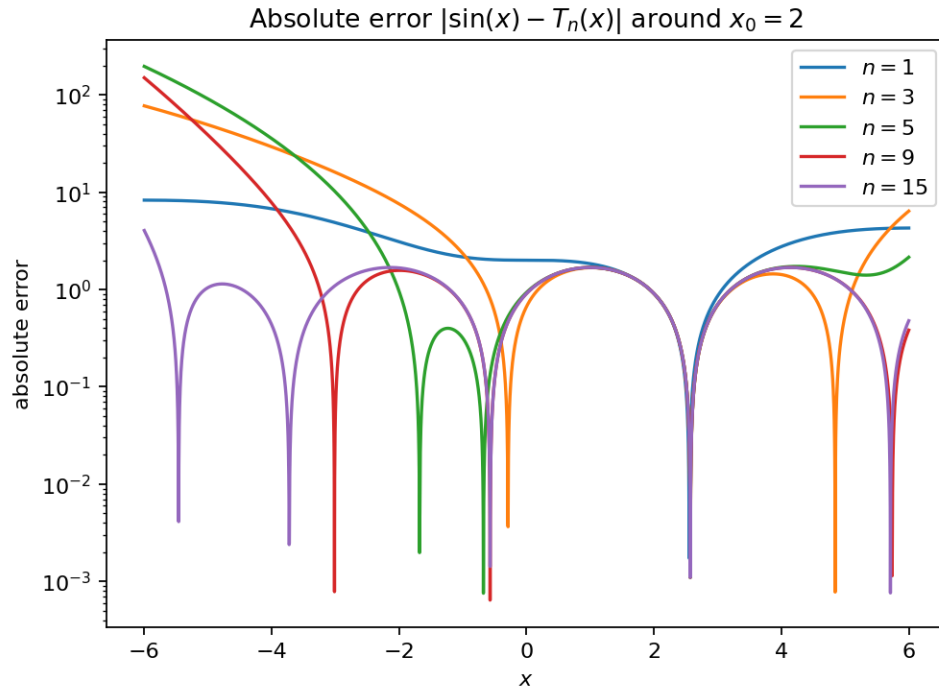
**Tags:** analysis, quantitative-exploration, visualization, numerics, taylor See: *Valid Tags*.

#### Goal

Build intuition for Taylor truncation error by visualizing the absolute error landscape  $|\sin(x) - T_n(x; x_0)|$  over a domain while varying:

- the polynomial degree  $n$ ,
- the expansion center  $x_0$ .





### Background (quick refresher)

If you want a short mathematical recap first, read: [Taylor series refresher](#).

### Research question

How does the approximation error of Taylor polynomials for  $\sin(x)$  depend on the polynomial degree  $n$  and the expansion center  $x_0$ , over a fixed domain of  $x$ ?

Concretely: for a chosen grid of  $(n, x_0)$ , what does the error landscape  $E_{n, x_0}(x) = |\sin(x) - T_n(x; x_0)|$  look like, and where do numerical artifacts start to dominate the truncation error?

### Why this qualifies as a mathematical experiment

This page is not just a worked example or a derivation — it is an *experiment* in the sense of **experimental mathematics**: a finite, reproducible procedure that produces **evidence** about how a mathematical object behaves.

For E001, the object is the family of Taylor polynomials  $T_n(x; x_0)$  for  $\sin(x)$ , and the observable is the error function  $E_{n, x_0}(x) = |\sin(x) - T_n(x; x_0)|$ . The experiment qualifies because it:

- **Explores a parameter space:** it varies degree  $n$  and center  $x_0$  and inspects how the entire error landscape changes.
- **Generates testable conjectures:** e.g. “there is a widening low-error region around  $x_0$  as  $n$  increases” and “improvement is not uniform across a fixed interval”.
- **Searches for failure modes / edge cases:** large  $|x - x_0|$ , high degrees, and wide domains can expose numerical artifacts that *look* like truncation error but are actually floating-point limitations.
- **Produces artifacts that can be checked independently:** plots and parameter snapshots make it easy to compare runs, reproduce the same conditions, and verify that an observed pattern is not accidental.

The goal is to turn “Taylor series should be good near  $x_0$ ” into *structured evidence* about **where** and **how** the approximation is good (or bad), which then informs what one would try to prove or bound formally.

- How does the truncation error  $|\sin(x) - T_n(x; x_0)|$  behave as we move away from the center  $x_0$ ?

- How many terms are needed to achieve a specific precision (e.g.,  $10^{-6}$ ) over a fixed interval?
- Does increasing the degree  $n$  always improve the result everywhere in the domain?

### Experiment design

- **Target function:**  $\sin(x)$
- **Evaluation:**  $x \in [-2\pi, 2\pi]$  (default)
- **Parameters:**
  - **degrees:**  $\{1, 3, 5, 7, 9\}$
  - **centers:**  $\{0, \pi/2, \pi\}$
- **Outputs:**
  - Plot of  $f(x)$  vs.  $T_n(x)$
  - Semi-log plot of  $|f(x) - T_n(x)|$

### How to run

```
make run EXP=e001 ARGS="--seed 1"
```

Artifacts are written under `out/e001/` (figures, parameters, and a short `report.md`).

### What to expect

Qualitatively (and this is what the plots should confirm):

- near  $x_0$ , the higher degree reduces the error quickly,
- away from  $x_0$ , the approximation can degrade even for higher degrees,
- changing  $x_0$  shifts the “low-error region”.

### Results

After running the experiment, include (or regenerate) the figures in the documentation. The canonical output location is `out/e001/`. For publishing, copy one representative “hero” image into `docs/_static/experiments/` (see “Gallery images” below).

### Notes / pitfalls

- Use **log-scale** for absolute error plots to see the full dynamic range.
- Be careful interpreting relative error near zeros of  $\sin(x)$ .
- Huge domains and high degrees can expose floating-point artifacts that are not truncation error.

### Extensions

- **Alternative functions:** Repeat the experiment for  $\exp(x)$  or  $1/(1-x)$ .
- **Relative error:** Plot  $|(f - T_n)/f|$  instead of absolute error.
- **Automatic degree selection:** Find the minimal  $n$  such that error  $< \epsilon$  on a given interval.

### Gallery images (recommended)

To keep the experiment gallery attractive and stable:

1. run the experiment locally,
2. pick one representative output figure,

3. copy it into the repo under:

```
docs/_static/experiments/e001_hero.png
```

This allows the docs to show thumbnails without depending on generated out/ artifacts.

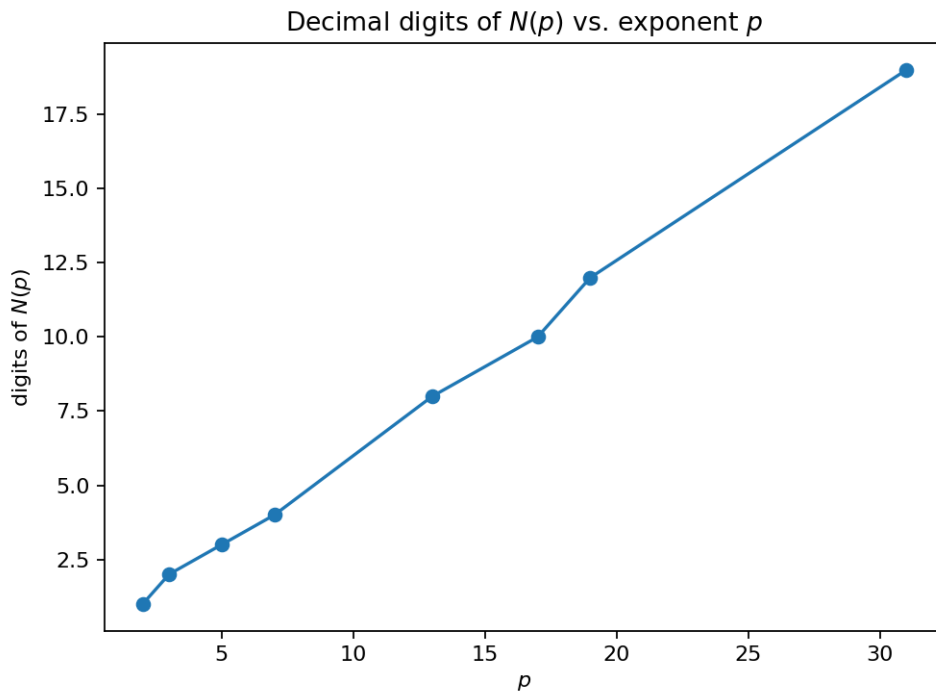
## References

See *References*. [Bailey and Borwein, 2005, Borwein, 2005, Borwein and Bailey, 2008]

## Related experiments

- *E082 — Zeta(s) series convergence* (E082 — Zeta(s) series convergence)
- *E083 — Series vs. Euler product ( )* (E083 — Series vs. Euler product ( ))
- *E084 —  $| (1/2+it) |$  growth snapshots* (E084 —  $| (1/2+it) |$  growth snapshots)
- *E085 — Dirichlet eta acceleration for (s)* (E085 — Dirichlet eta acceleration for (s))
- *E086 — Hardy Z(t) near zeros* (E086 — Hardy Z(t) near zeros)

### 3.3.2 E002: Even Perfect Numbers — Generator and Growth



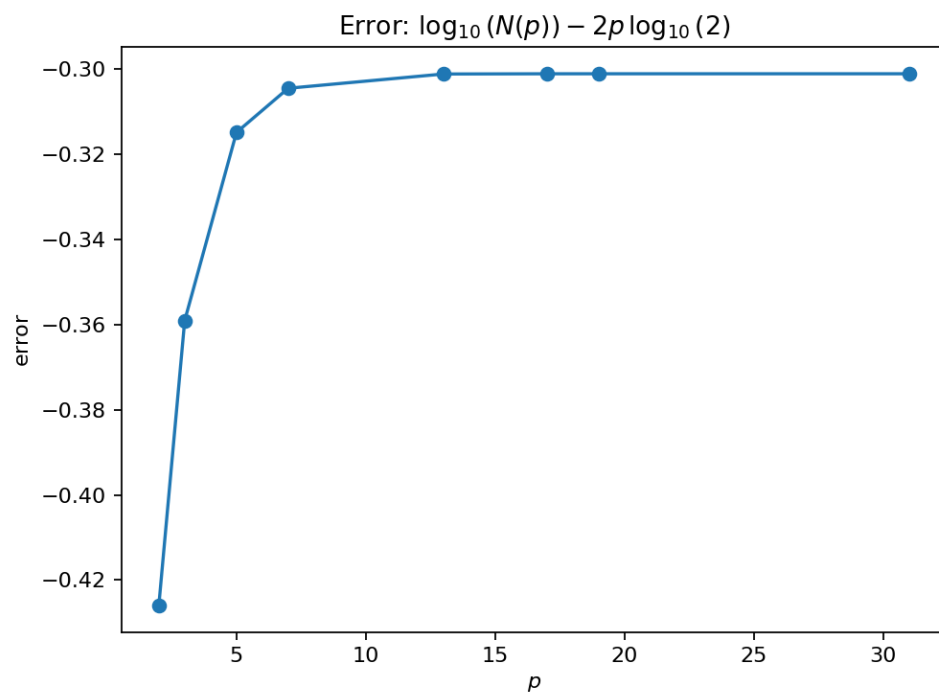
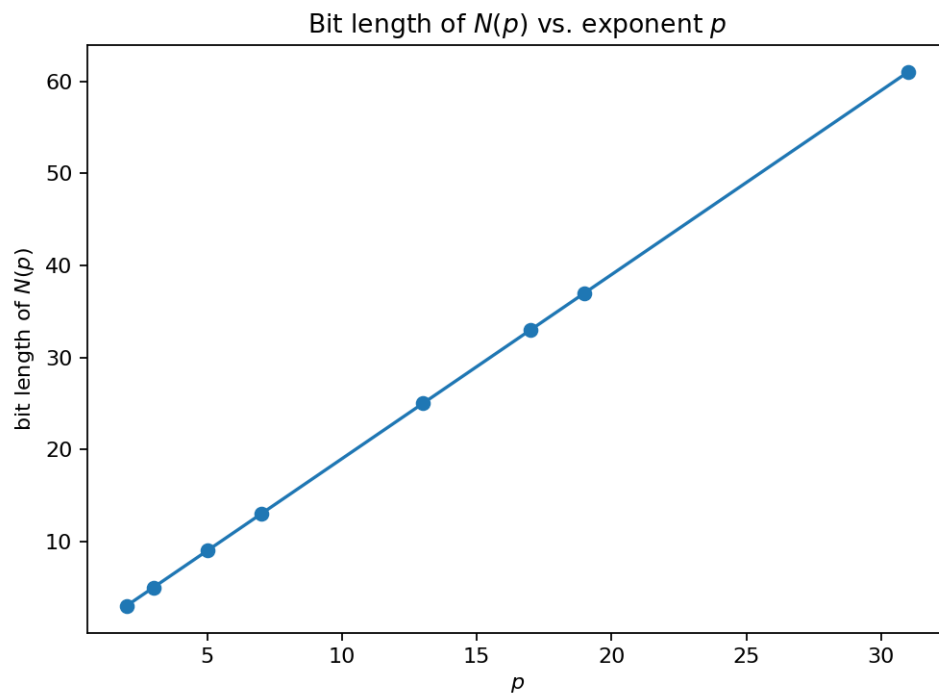
**Tags:** number-theory, quantitative-exploration, visualization, numerics See: *Valid Tags*.

## Highlights

- Generate even perfect numbers from known Mersenne exponents  $p$ .
- Plot digits and bit-length growth vs.  $p$ .
- Test the approximation  $\log_{10}(N(p)) \approx 2p \log_{10}(2)$ .

## Goal

Generate **even perfect numbers** from known Mersenne prime exponents  $p$  and visualize how fast the numbers grow. Measure growth via **digit count**, **bit length**, and simple logarithmic approximations.





## Research question

For

$$N(p) = 2^{p-1}(2^p - 1),$$

how do:

- `digits(N)`,
- `bit_length(N)`,
- `log10(N)`

scale with the exponent  $p$ ?

How accurate is the approximation

$$\log_{10}(N(p)) \approx 2p \log_{10}(2)?$$

## Why this qualifies as a mathematical experiment

The Euclid–Euler theorem tells us exactly what even perfect numbers look like, but it does not directly convey how quickly the objects become astronomically large. This experiment uses computation and visualization to build quantitative intuition and test simple asymptotic approximations.

## Experiment design

### Inputs

- A curated list of known Mersenne prime exponents, e.g.  $p \in \{2, 3, 5, 7, 13, 17, 19, 31, \dots\}$ .

### Observables

For each exponent  $p$ :

- $N(p)$  as a Python integer
- `digits(N(p))`
- `bit_length(N(p))`
- approximation error:

$$\Delta(p) = \log_{10}(N(p)) - 2p \log_{10}(2)$$

### Plots

- $p$  vs. `digits`
- $p$  vs. `bit length`
- $p$  vs.  $\Delta(p)$

## How to run

From the repo root:

```
make run EXP=e002
```

or:

```
uv run python -m mathxlab.experiments.e002
```

## Notes / pitfalls

- Avoid converting huge integers to decimal strings repeatedly in tight loops; compute digits using logs where possible.
- Use `int.bit_length()` for stable bit length (fast and exact).
- Keep the exponent list modest for fast docs builds; this is a growth experiment, not a search for new Mersenne primes.

## Extensions

- Compare growth to  $2^{2p}$  and quantify the relative gap.
- Add a “human scale” axis: compare  $\text{digits}(N(p))$  to common benchmarks (atoms in the observable universe, etc.).
- Pull the exponent list from a small data file so the experiment can be updated without code changes.

## References

See *References*.

[Caldwell, n.d., Voight, 1998, OEIS Foundation Inc., 2025]

## Related experiments

- *E005: Odd Perfect Numbers — Constraint Filter Pipeline* (Odd Perfect Numbers — Constraint Filter Pipeline)
- *E007: Mersenne growth (bits and digits)* (Mersenne growth (bits and digits))
- *E010: Even perfect numbers from Mersenne primes* (Even perfect numbers from Mersenne primes)
- *E084 —  $\lfloor (1/2+it) \rfloor$  growth snapshots* (E084 —  $\lfloor (1/2+it) \rfloor$  growth snapshots)
- *E097 —  $(n)/n$  landscape: deficient, perfect, abundant* (E097 —  $(n)/n$  landscape: deficient, perfect, abundant)

### 3.3.3 E003: Abundancy Index Landscape

**Tags:** number-theory, quantitative-exploration, visualization, numerics See: *Valid Tags*.

## Highlights

- Compute  $\sigma(1..N)$  via a divisor-sum sieve.
- Visualize the distribution of  $I(n) = \sigma(n)/n$ .
- Highlight perfect numbers as the razor-thin level set  $I(n) = 2$ .

## Goal

Visualize how **rare** perfect numbers are by plotting the distribution of the **abundancy index**

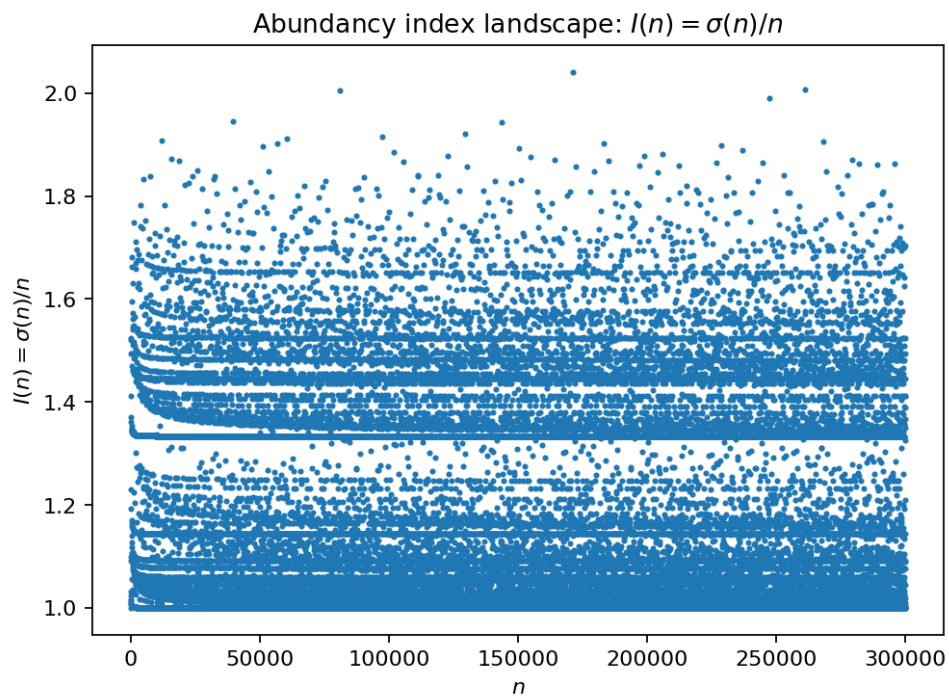
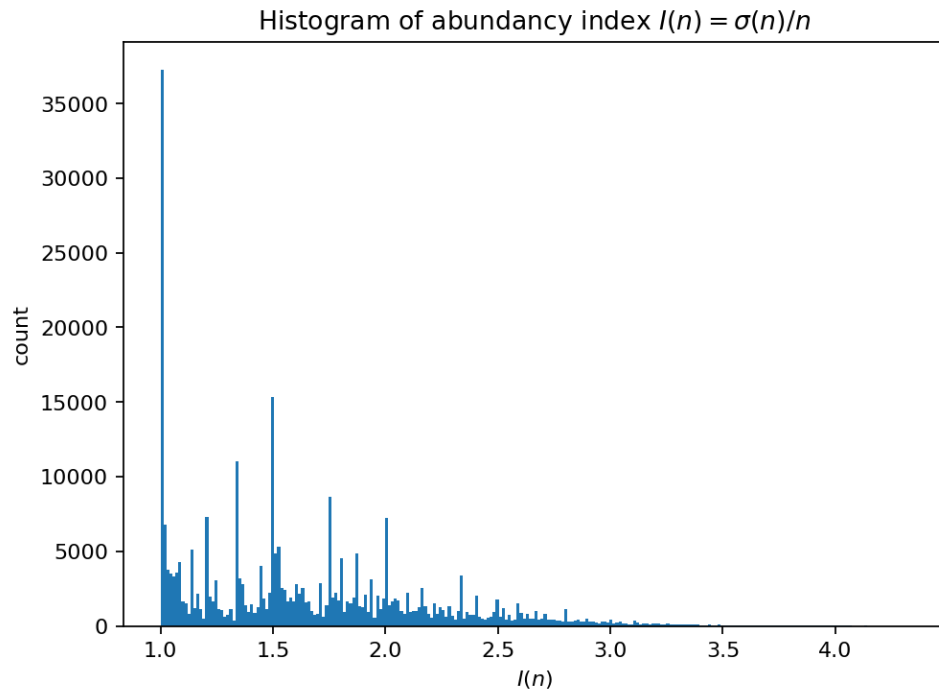
$$I(n) = \frac{\sigma(n)}{n}$$

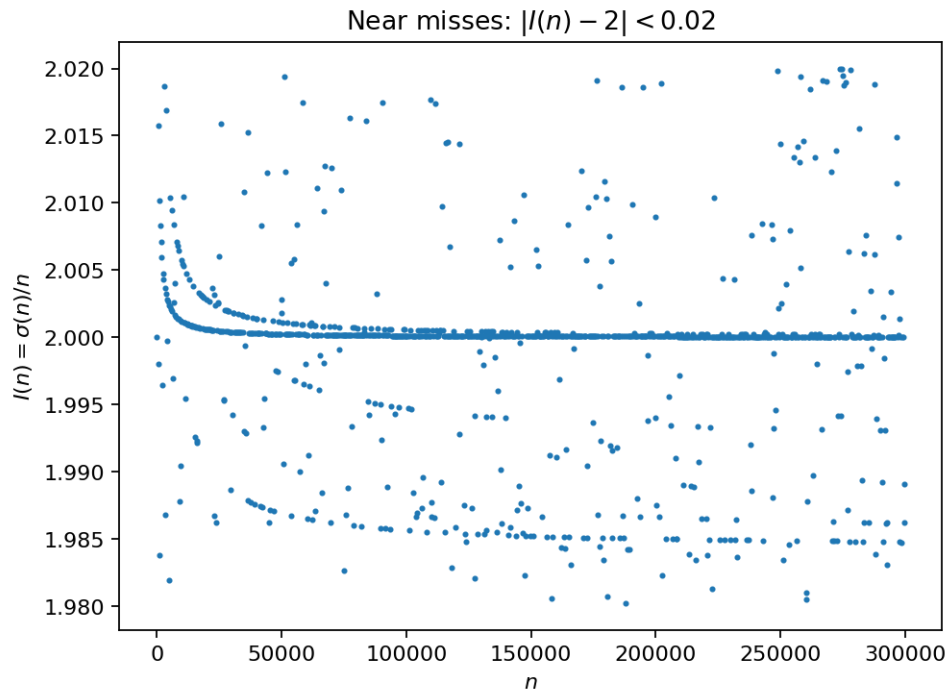
for integers  $n \leq N$ . Perfect numbers satisfy  $I(n) = 2$ .

## Research question

For a given bound  $N$ :

- What does the empirical distribution of  $I(n)$  look like?
- How frequently do we see values near 2?





- Where do the perfect numbers appear in the landscape?

### Why this qualifies as a mathematical experiment

The divisor-sum function  $\sigma(n)$  is highly structured but “spiky” and hard to intuit symbolically. Computational sweeps reveal qualitative structure (clusters, gaps, tails) and put perfect numbers into context.

### Experiment design

**Method:** compute  $\sigma(1), \dots, \sigma(N)$  via a divisor-sum sieve

Use the identity “each divisor contributes to its multiples”:

- initialize an array `sigma[0..N]` with zeros
- for each `d = 1..N`:
  - add `d` to `sigma[k]` for all multiples `k = d, 2d, 3d, ...`

This runs in about  $O(N \log N)$  time and avoids per-number factorization.

### Observables

- $I(n) = \sigma(n)/n$
- distance to perfection:  $|I(n) - 2|$
- classification:
  - deficient:  $I(n) < 2$
  - perfect:  $I(n) = 2$
  - abundant:  $I(n) > 2$

## Plots

- histogram of  $I(n)$  (or of  $I(n) - 1$ )
- scatter plot of  $n$  vs.  $I(n)$  (optionally with log-scale on  $n$ )
- highlight perfect numbers

## How to run

```
make run EXP=e003
```

or:

```
uv run python -m mathxlab.experiments.e003
```

## Notes / pitfalls

- $I(n)$  is rational. For classification, compare integers using  $\sigma(n)$  and  $2n$  rather than floats.
- For plots, floats are fine, but compute the “perfect” condition as `sigma[n] == 2*n`.
- Start with  $N \leq 1\,000\,000$  to keep runtime and memory reasonable.

## Extensions

- Repeat for different ranges and overlay histograms.
- Plot the top- $k$  values of  $I(n)$  and compare to known extremal families.
- Explore the “near misses” set (feeds directly into E006).

## References

See *References*.

[Voight, 1998, Weisstein, 2003, OEIS Foundation Inc., 2025]

## Related experiments

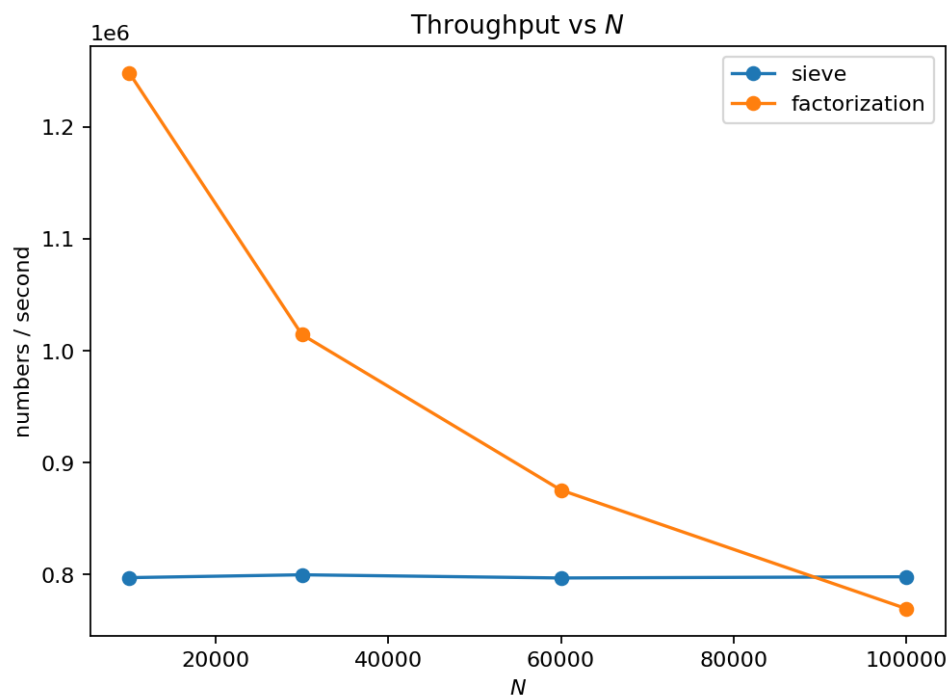
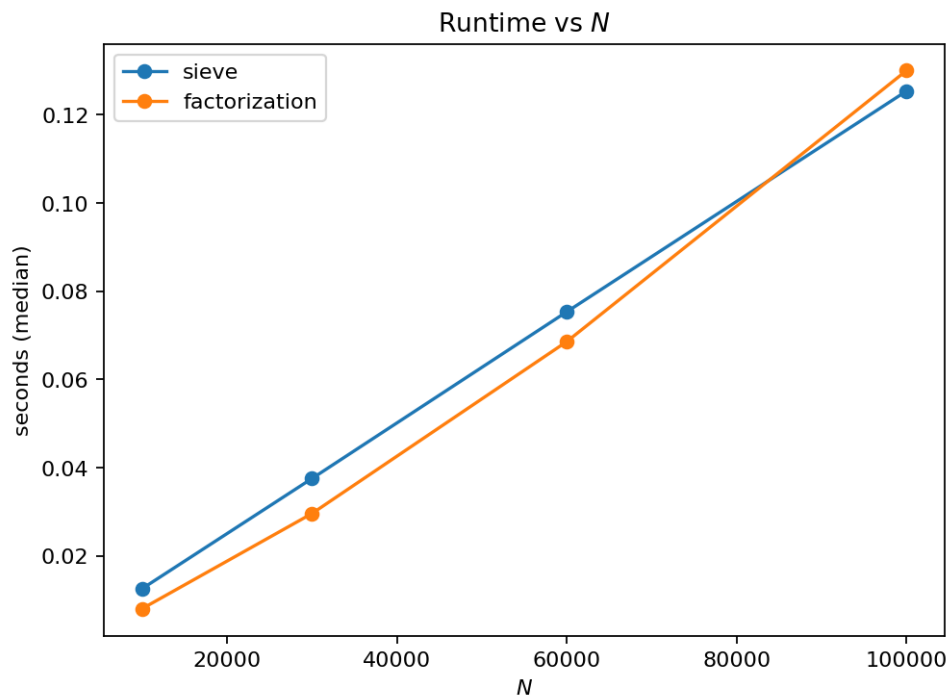
- *E002: Even Perfect Numbers — Generator and Growth* (Even Perfect Numbers — Generator and Growth)
- *E052: Totient ratio landscape* (Totient ratio landscape)
- *E059: Abundancy index landscape* (Abundancy index landscape)
- *E094 —  $(n)$  vs.  $\Omega(n)$ : Erdős–Kac normalization* (E094 —  $(n)$  vs.  $\Omega(n)$ : Erdős–Kac normalization)
- *E097 —  $(n)/n$  landscape: deficient, perfect, abundant* (E097 —  $(n)/n$  landscape: deficient, perfect, abundant)

### 3.3.4 E004: Computing $\sigma(n)$ at Scale — Sieve vs. Factorization

**Tags:** number-theory, quantitative-exploration, numerics, optimization See: *Valid Tags*.

## Highlights

- Benchmark bulk sieve vs. per-number factorization.
- Find runtime crossover points as  $N$  grows.
- Validate correctness on sampled inputs.



## Goal

Benchmark two practical ways to compute the sum-of-divisors function:

1. **Sieve method:** compute all  $\sigma(1..N)$  in one pass.
2. **Factorization method:** compute  $\sigma(n)$  per-number from the prime factorization.

## Research question

For a range of bounds  $N$ :

- which approach is faster?
- where is the crossover point?
- what are the memory tradeoffs?

## Why this qualifies as a mathematical experiment

Both methods are mathematically equivalent, but performance depends on constants, caching, and implementation details. This is a quantitative exploration of algorithmic behavior grounded in number-theory structure.

## Experiment design

### Method A: divisor-sum sieve (bulk computation)

Compute `sigma[1..N]` by adding each divisor to its multiples (as in E003).

### Method B: per-number factorization

Factor each  $n$  (e.g. using a precomputed prime list up to  $\sqrt{N}$ ) and compute:

If

$$n = \prod_{i=1}^k p_i^{a_i},$$

then

$$\sigma(n) = \prod_{i=1}^k \frac{p_i^{a_i+1} - 1}{p_i - 1}.$$

## Measurements

- wall-clock runtime vs.  $N$  (multiple trials, median)
- peak memory (rough estimate acceptable for v1)
- correctness cross-check: random sample where both methods agree

## Outputs

- plot: runtime vs.  $N$  for both methods
- short table:  $N$ , runtime(A), runtime(B), speedup

## How to run

```
make run EXP=e004
```

or:

```
uv run python -m mathxlab.experiments.e004
```

### Notes / pitfalls

- Factorization becomes expensive quickly; keep the factorization method limited to moderate  $N$ .
- Use integer arithmetic for correctness checks (`sigma[n] == 2*n` etc.).
- Report both runtime and *effective throughput* (numbers processed per second).

### Extensions

- Add a third method using smallest-prime-factor (SPF) sieve for fast factorization.
- Compare pure Python vs. `numpy` arrays for Method A.
- Turn the benchmark into a reusable utility for later experiments.

### References

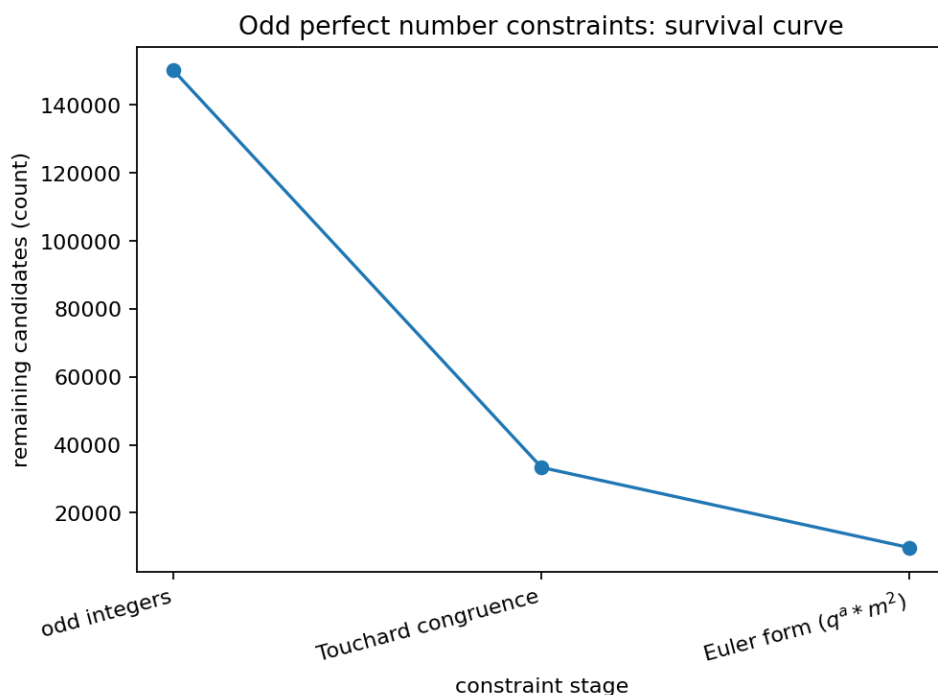
See *References*.

[Voight, 1998, Weisstein, 2003]

### Related experiments

- *E098 — Maximizers of  $(n)/n^{\wedge}$  across* (E098 — Maximizers of  $(n)/n^{\wedge}$  across )
- *E017: Sieve memory blow-up vs. segmented sieve* (Sieve memory blow-up vs. segmented sieve)
- *E051: Semiprimes: balanced vs. unbalanced factorization timing* (Semiprimes: balanced vs. unbalanced factorization timing)
- *E002: Even Perfect Numbers — Generator and Growth* (Even Perfect Numbers — Generator and Growth)
- *E003: Abundancy Index Landscape* (Abundancy Index Landscape)

## 3.3.5 E005: Odd Perfect Numbers — Constraint Filter Pipeline





**Tags:** number-theory, counterexample-search, visualization, search See: *Valid Tags*.

### Highlights

- Apply necessary constraints as staged filters on odd candidates.
- Plot survival curves (remaining candidates per constraint stage).
- Make the “open problem” constraints tangible at finite scales.

### Goal

Demonstrate *why brute-force search for odd perfect numbers is unrealistic* by applying known **necessary conditions** as a step-by-step filter pipeline and visualizing how the candidate pool collapses.

### Research question

Given odd integers up to a bound  $N$ :

- how many survive each constraint stage?
- which constraints are most “eliminating” in practice at small scales?
- what does a survival curve suggest about scalability?

### Why this qualifies as a mathematical experiment

Odd perfect numbers are an open problem. Theorems provide constraints rather than a classification. Computation makes these constraints tangible by measuring their elimination power on finite candidate sets.

### Experiment design

#### Candidate set

Start with odd integers  $1 < n \leq N$ .

#### Constraint stages (v1)

Use a conservative, well-known set of *necessary* conditions that can be checked mechanically:

1. **Euler form (shape constraint):** any odd perfect number must be of the form

$$n = q^\alpha m^2$$

where  $q$  is prime,  $\gcd(q, m) = 1$ , and

$$q \equiv \alpha \equiv 1 \pmod{4}.$$

2. **Congruence filter (Touchard-type):** odd perfect numbers satisfy strong congruence restrictions (implemented as one or two simple congruence checks supported by the references).
3. **Small-prime structure filters:** apply a few lightweight necessary conditions (e.g., reject candidates divisible by some specific small patterns if justified by the reference set).

For each stage, record “remaining candidates”.

### Output

- table: stage name, remaining count, elimination percentage
- plot: survival curve (stage index vs. remaining candidates)

## How to run

```
make run EXP=e005
```

or:

```
uv run python -m mathxlab.experiments.e005
```

## Notes / pitfalls

- Be explicit about what is *proved* vs. what is a heuristic. Only implement conditions that are truly necessary.
- At small  $N$ , some deep constraints won't show their full strength; the goal is the *pipeline idea*, not a record bound.
- Euler-form testing requires factorization; keep  $N$  small enough for trial division (v1).

## Extensions

- Add stronger bounds/constraints from the modern literature and compare elimination power.
- Replace trial division with an SPF sieve to scale the Euler-form test.
- Report not just counts but also the distribution of remaining prime-factor patterns.

## References

See *References*.

[Guy, 2004, Ochem and Rao, 2014, Stone, 2024, Voight, 1998]

## Related experiments

- *E053: Inverse totient multiplicities* (Inverse totient multiplicities)
- *E002: Even Perfect Numbers — Generator and Growth* (Even Perfect Numbers — Generator and Growth)
- *E010: Even perfect numbers from Mersenne primes* (Even perfect numbers from Mersenne primes)
- *E046: Prime-testing pipeline and tuning pitfalls* (Prime-testing pipeline and tuning pitfalls)
- *E095 — Squarefree filter:  $(n)=\Omega(n)$  when  $(n) \neq 0$*  (E095 — Squarefree filter:  $(n)=\Omega(n)$  when  $(n) \neq 0$ )

### 3.3.6 E006: Near Misses to Perfection

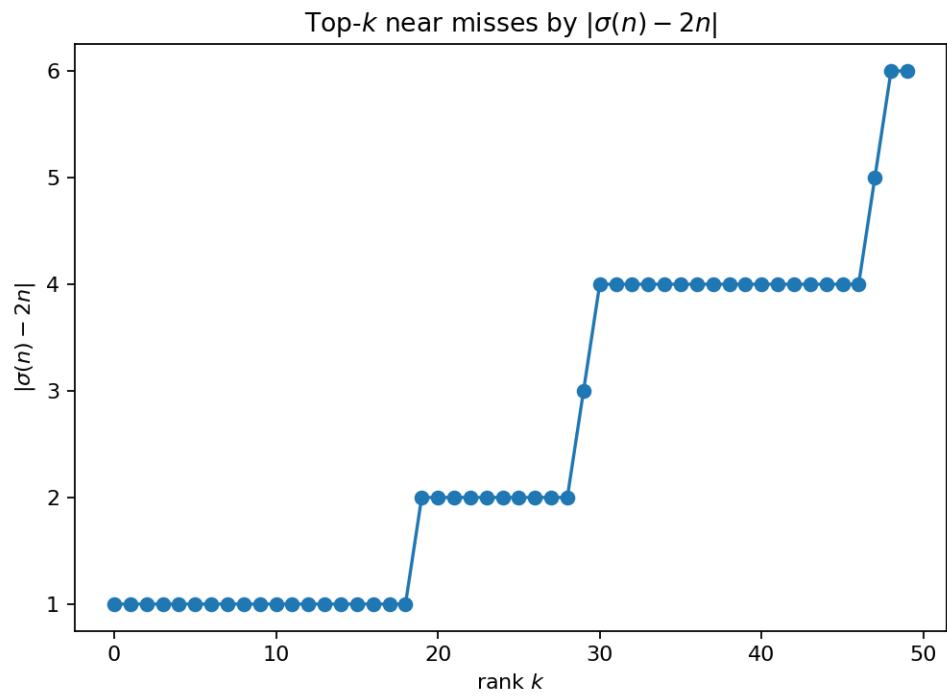
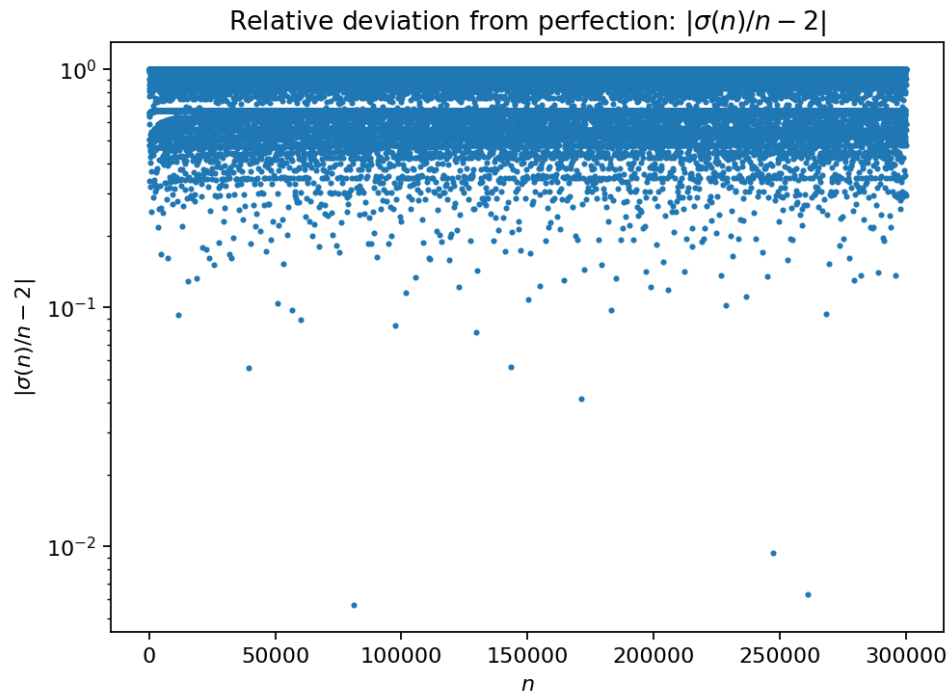
Tags: number-theory, conjecture-generation, visualization, numerics See: *Valid Tags*.

## Highlights

- Search for  $n$  with  $\sigma(n)$  unusually close to  $2n$ .
- Build leaderboards for absolute and relative deviation.
- Visualize how “near perfection” clusters by structure.

## Goal

Find and visualize integers whose divisor sum is **unusually close** to the perfect condition  $\sigma(n) = 2n$ , without being perfect.



### Research question

For integers  $n \leq N$ , which numbers minimize:

- absolute deviation:

$$D_1(n) = |\sigma(n) - 2n|$$

- relative deviation:

$$D_2(n) = \left| \frac{\sigma(n)}{n} - 2 \right|?$$

Do “near misses” cluster in recognizable families (highly composite, abundant, etc.)?

### Why this qualifies as a mathematical experiment

The perfect condition is a sharp equality. Studying the closest failures often reveals structure and suggests new questions (e.g., which multiplicative patterns drive  $\sigma(n)$  toward  $2n$ ).

### Experiment design

#### Computation

- Compute  $\sigma(1..N)$  via the divisor-sum sieve (as in E003).
- For each  $n$ , compute  $D_1(n)$  and  $D_2(n)$ .
- Keep the top- $k$  smallest deviations (excluding actual perfect numbers).

#### Outputs

- table: top- $k$  near misses (with  $n$ ,  $\sigma(n)$ ,  $D_1$ ,  $D_2$ )
- plot:  $n$  vs.  $D_2(n)$  (log-scale on  $D_2$  often helps)
- mark perfect numbers for reference

#### How to run

```
make run EXP=e006
```

or:

```
uv run python -m mathxlab.experiments.e006
```

### Notes / pitfalls

- Use integer comparisons to identify perfect numbers (`sigma[n] == 2*n`).
- For  $D_2$ , floats are fine for plotting, but store exact rational values for ranking when possible (e.g., compare  $|\sigma(n) - 2n|$  first, then normalize for reporting).
- Choose  $k$  small (e.g. 50 or 200) so the report stays readable.

### Extensions

- Repeat for different  $N$  and compare stability of the “near miss” leaderboard.
- Add a second leaderboard restricted to odd  $n$  only.
- Compare near misses to known abundant/deficient classifications and prime factorizations.

## References

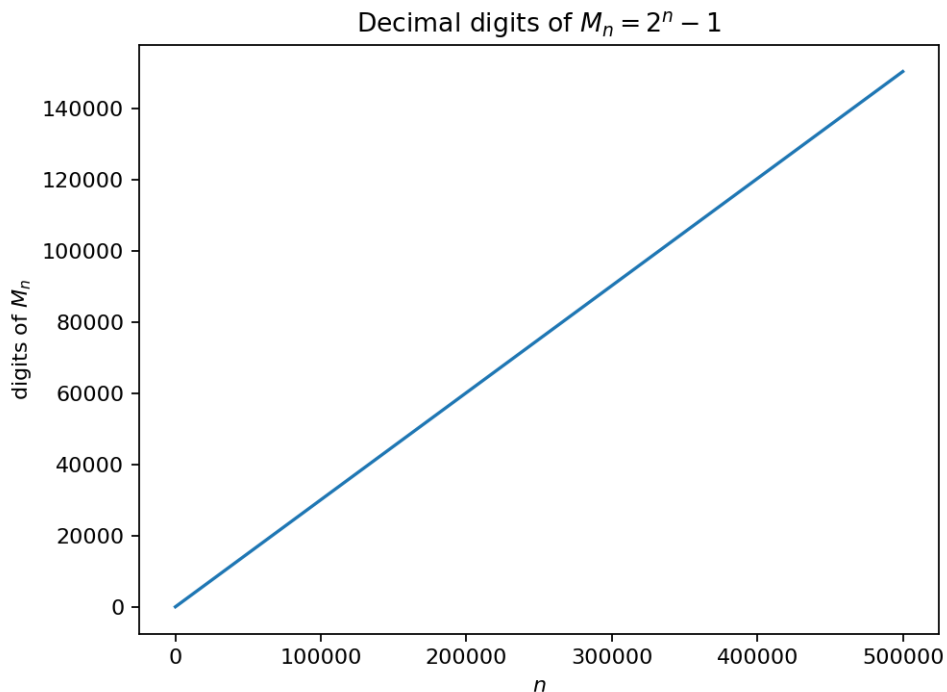
See *References*.

[Voight, 1998, Weisstein, 2003, OEIS Foundation Inc., 2025]

## Related experiments

- *E027: Record prime gaps vs.  $\log^2$  heuristic* (Record prime gaps vs.  $\log^2$  heuristic)
- *E029: Twin primes: observed vs. heuristic* (Twin primes: observed vs. heuristic)
- *E086 — Hardy  $Z(t)$  near zeros* (E086 — Hardy  $Z(t)$  near zeros)
- *E002: Even Perfect Numbers — Generator and Growth* (Even Perfect Numbers — Generator and Growth)
- *E003: Abundancy Index Landscape* (Abundancy Index Landscape)

### 3.3.7 E007: Mersenne growth (bits and digits)



**Tags:** number-theory, quantitative-exploration, visualization See: *Valid Tags*.

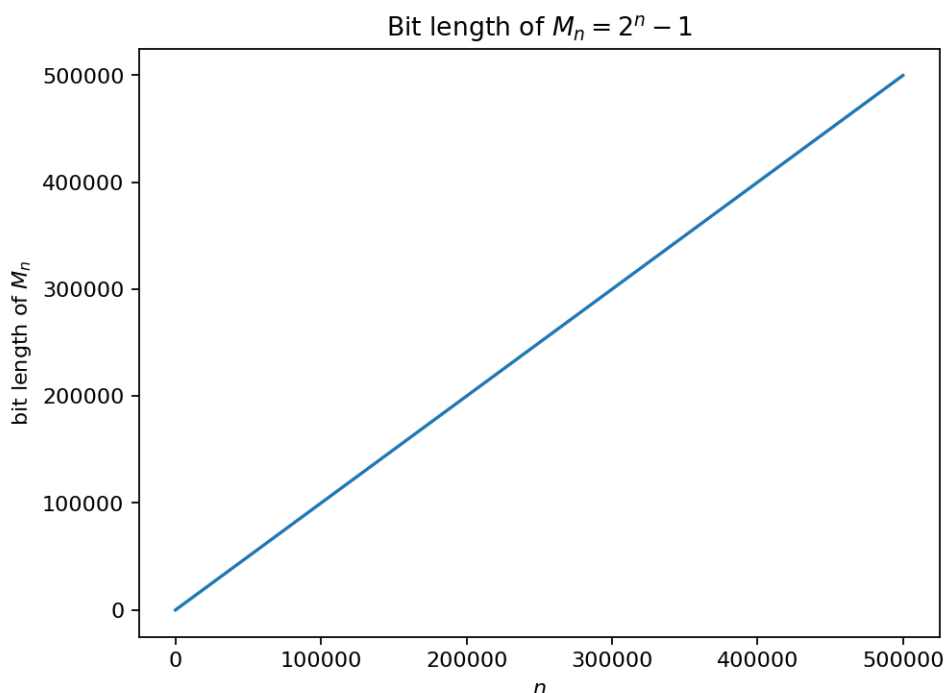
## Highlights

- Visualize how quickly  $M_n = 2^n - 1$  grows in **bits** and **decimal digits**.
- Compute size metrics *without* constructing huge integers.
- Provide simple rules-of-thumb for choosing feasible bounds in later experiments.

## Goal

Quantify and visualize the growth of Mersenne numbers  $M_n = 2^n - 1$  as  $n$  increases.

This experiment focuses on **size** (bit-length and decimal digits), because it determines what later algorithms (Lucas–Lehmer, sieving) can realistically handle on a laptop.



### Background (quick refresher)

- *Mersenne numbers and primes refresher*

### Research question

How fast does the size of  $M_n$  grow with  $n$ , and what simple formulas approximate:

- the number of bits of  $M_n$
- the number of decimal digits of  $M_n$

### Why this qualifies as a mathematical experiment

- **Finite procedure:** evaluate size formulas on a range  $n = 1..N$ .
- **Observable(s):** bit-length and digit count as functions of  $n$ .
- **Parameter space:** vary  $N$  (and optional sampling density).
- **Outcome:** visual evidence (curves) and tables usable as planning guidance.
- **Reproducibility:** parameters written to `params.json`; figures written to `figures/`.

### Experiment design

#### Computation

Key facts:

- $\text{bits}(2^n - 1) = n$  for  $n \geq 1$ .
- $\text{digits}(2^n - 1) = \lfloor n \log_{10}(2) \rfloor + 1$  for  $n \geq 1$ .

Compute both for a grid of  $n$  values.

## Outputs

- plot:  $n$  vs. bits
- plot:  $n$  vs. digits
- table: selected  $n$  with digits (e.g., 10, 100, 1k, 10k, ...)

## How to run

```
make run EXP=e007
```

or:

```
uv run python -m mathxlab.experiments.e007
```

## Notes / pitfalls

- Avoid converting huge  $M_n$  to decimal just to count digits. The logarithm formula is exact for the digit count.
- For very large  $n$ , use stable floating-point evaluation for  $n \log_{10}(2)$  (double precision is fine up to very large  $n$ ).
- This experiment is intentionally lightweight; it should run quickly.

## Extensions

- Overlay “feasibility bands” (e.g., digits thresholds) for what your other experiments can handle.
- Compare  $M_n$  to other fast-growing families (factorials, primorials) on a log scale.

## References

See *References*.

[contributors, 2025]

## Related experiments

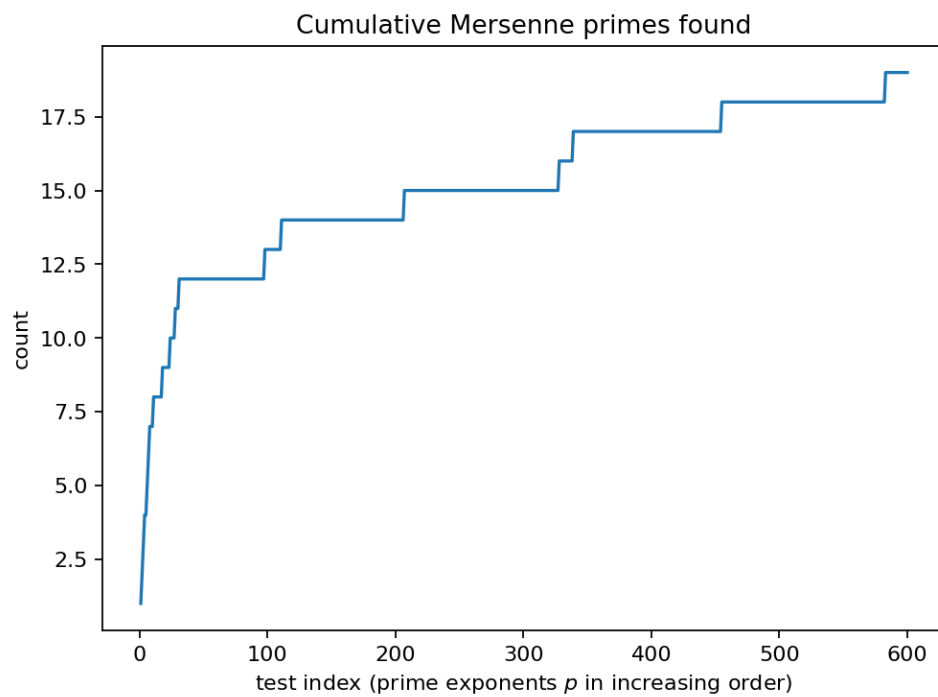
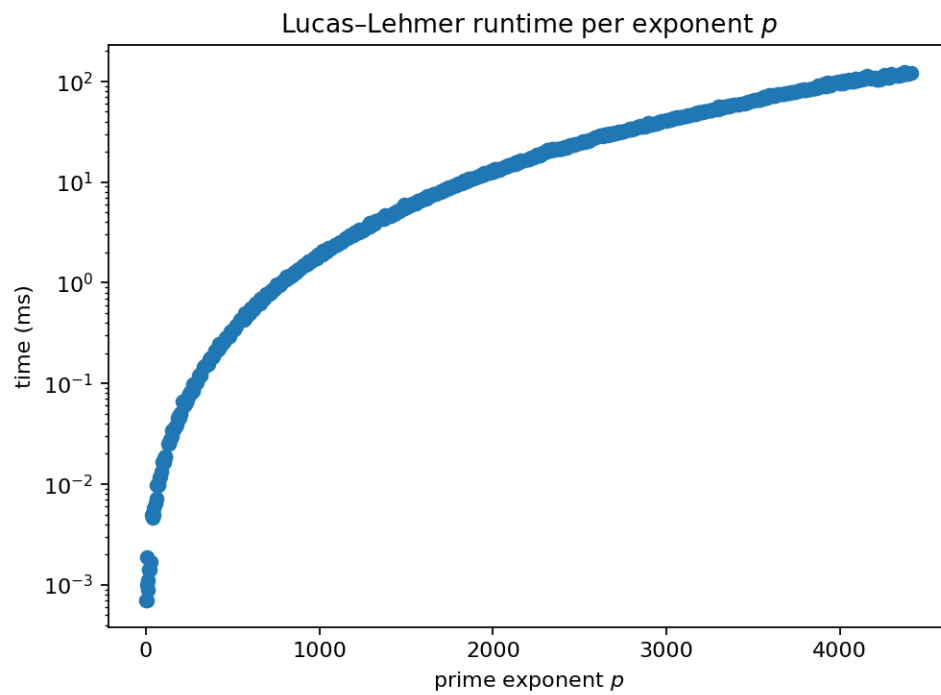
- *E002: Even Perfect Numbers — Generator and Growth* (Even Perfect Numbers — Generator and Growth)
- *E009: Small-factor scan for Mersenne numbers* (Small-factor scan for Mersenne numbers)
- *E010: Even perfect numbers from Mersenne primes* (Even perfect numbers from Mersenne primes)
- *E011: Heuristic rarity of Mersenne primes* (Heuristic rarity of Mersenne primes)
- *E084 —  $\lfloor (1/2+it) \rfloor$  growth snapshots* (E084 —  $\lfloor (1/2+it) \rfloor$  growth snapshots)

### 3.3.8 E008: Lucas–Lehmer scan (prime exponents)

Tags: number-theory, quantitative-exploration, visualization See: *Valid Tags*.

## Highlights

- Run the Lucas–Lehmer test for many prime exponents  $p$ .
- Record timing and residue behavior for composite vs. prime outcomes.
- Produce a reproducible “scan report” (which  $p$  were tested, which passed).





## Goal

Empirically explore Mersenne primality testing by scanning prime exponents  $p$  and applying the Lucas–Lehmer test to  $M_p = 2^p - 1$ .

## Background (quick refresher)

- *Mersenne numbers and primes refresher*

## Research question

For prime exponents  $p \leq P$ , how does:

- runtime of the Lucas–Lehmer test scale with  $p$ ?
- the distribution of final residues (mod  $M_p$ ) differ between prime and composite outcomes?

## Why this qualifies as a mathematical experiment

- **Finite procedure:** enumerate prime exponents up to a bound and test them.
- **Observable(s):** pass/fail, final residue, and runtime.
- **Parameter space:** vary  $P$  and optional “max tests” cap.
- **Outcome:** a dataset (tested  $p$  values) and plots/tables that suggest scaling behavior.
- **Failure modes:** naive implementations can be slow; parameters bound the search to keep runs feasible.

## Experiment design

### Computation

- Enumerate prime exponents  $p$  up to a bound  $P$ .
- For each  $p$ , compute  $M_p = 2^p - 1$  and run Lucas–Lehmer:
  - $s_0 = 4$
  - $s_{k+1} = s_k^2 - 2 \pmod{M_p}$
  - $M_p$  is prime iff  $s_{p-2} \equiv 0 \pmod{M_p}$ .
- Record (at minimum):  $p$ , pass/fail, final residue, and wall time.

### Outputs

- table: scanned exponents with result and time
- plot:  $p$  vs. runtime (often log-scale is helpful)
- plot: residue magnitude / patterns (optional)

### How to run

```
make run EXP=e008
```

or:

```
uv run python -m mathxlab.experiments.e008
```

### Notes / pitfalls

- Only run Lucas–Lehmer for **prime**  $p$ ; if  $p$  is composite then  $M_p$  is composite.
- The test is defined for odd prime  $p$ ; treat  $p = 2$  separately ( $M_2 = 3$  is prime).
- Keep bounds modest at first; the cost grows quickly with  $p$ .

### Extensions

- Add a “pre-sieve” that checks for small factors before running Lucas–Lehmer.
- Compare your timings to published implementations (as a sanity check, not a competition).

### References

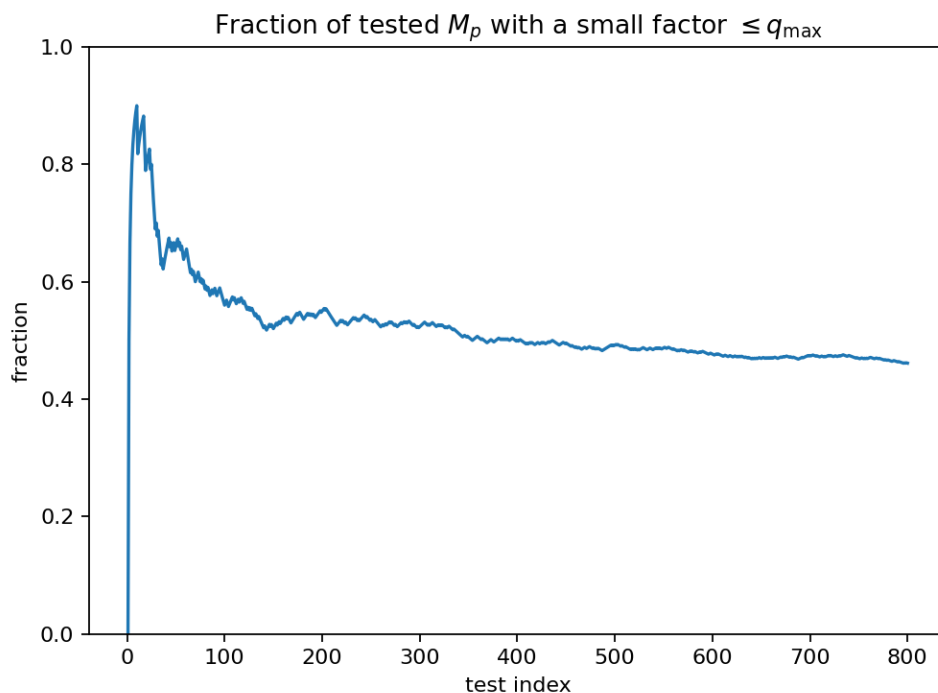
See [References](#).

[Caldwell, 2021, contributors, 2025, Mersenne Research, 2024]

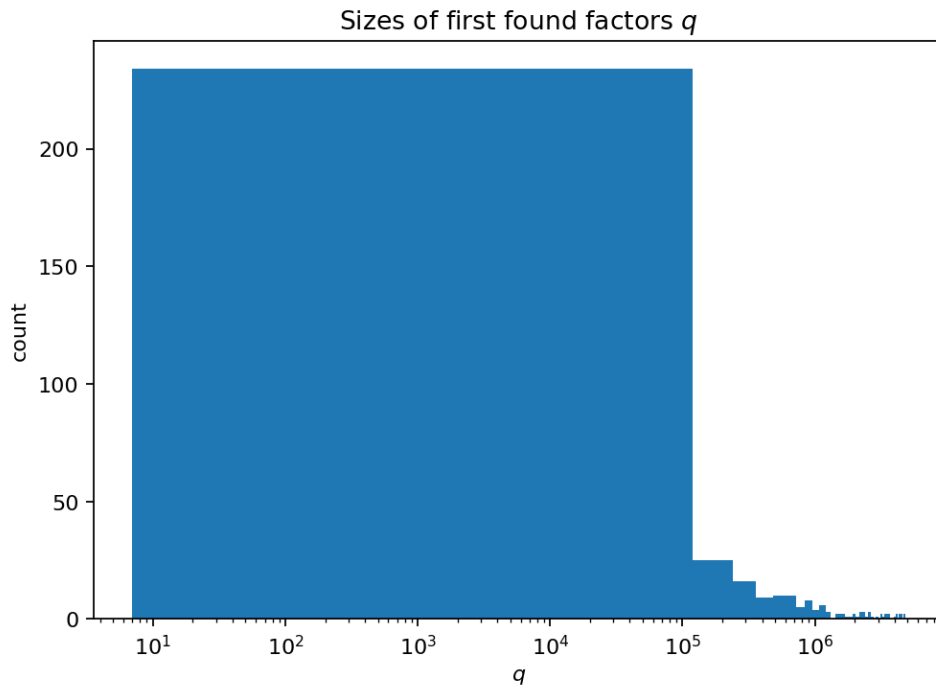
### Related experiments

- *E009: Small-factor scan for Mersenne numbers* (Small-factor scan for Mersenne numbers)
- *E042: Repunit primes (small  $k$  scan)* (Repunit primes (small  $k$  scan))
- *E048: Carmichael numbers: Korselt scan + Fermat counterexamples* (Carmichael numbers: Korselt scan + Fermat counterexamples)
- *E049: Wieferich primes (base 2): scan and quotient visualization* (Wieferich primes (base 2): scan and quotient visualization)
- *E002: Even Perfect Numbers — Generator and Growth* (Even Perfect Numbers — Generator and Growth)

## 3.3.9 E009: Small-factor scan for Mersenne numbers



Tags: number-theory, quantitative-exploration, visualization See: [Valid Tags](#).



### Highlights

- Quickly eliminate many  $M_p$  candidates by finding small prime factors.
- Use the structure of possible factors of  $M_p$  to reduce wasted work.
- Show how often “trivial” factors appear in a range of exponents.

### Goal

Explore how often Mersenne numbers  $M_p = 2^p - 1$  (with prime  $p$ ) have **small factors**, and how a lightweight sieve can filter candidates before running Lucas–Lehmer.

### Background (quick refresher)

- *Mersenne numbers and primes refresher*

### Research question

For prime exponents  $p \leq P$  and a factor bound  $q \leq Q$ :

- how many  $M_p$  have a factor below  $Q$ ?
- how much compute can be saved by sieving before Lucas–Lehmer?

### Why this qualifies as a mathematical experiment

- **Finite procedure:** scan a finite range of  $p$  and candidate factors.
- **Observable(s):** first small factor found (if any), counts by  $p$  and by factor size.
- **Parameter space:** vary  $P$  and  $Q$ .
- **Outcome:** tables/plots that show factor frequency and sieve effectiveness.
- **Failure modes:** incomplete sieve bounds can mislead; clearly report  $Q$  as a limitation.

## Experiment design

### Computation

For fixed prime exponent  $p$ , any prime factor  $q$  of  $M_p$  satisfies:

- $q \equiv 1 \pmod{2p}$  (a common necessary condition used to generate candidates)

A practical sieve approach:

- Generate candidate primes  $q$  of the form  $q = 2pk + 1$  up to  $Q$ .
- For each candidate prime  $q$ , check:
  - $2^p \equiv 1 \pmod{q}$

If true, then  $q$  divides  $M_p$ .

### Outputs

- table: per  $p$ , smallest factor found (or “none up to  $Q$ ”)
- plot:  $p$  vs. smallest factor (when present)
- summary: fraction of  $p$  eliminated by the sieve

### How to run

```
make run EXP=e009
```

or:

```
uv run python -m mathxlab.experiments.e009
```

### Notes / pitfalls

- This is a **bounded** sieve: “no factor found” only means “none found up to  $Q$ ”.
- Prime testing for  $q$  should be fast; for small  $Q$ , a simple deterministic method is fine.
- Report both  $P$  and  $Q$  prominently in the report.

### Extensions

- Combine with E008: run Lucas–Lehmer only on exponents not eliminated by the sieve.
- Track how much wall time is saved by the combined pipeline.

### References

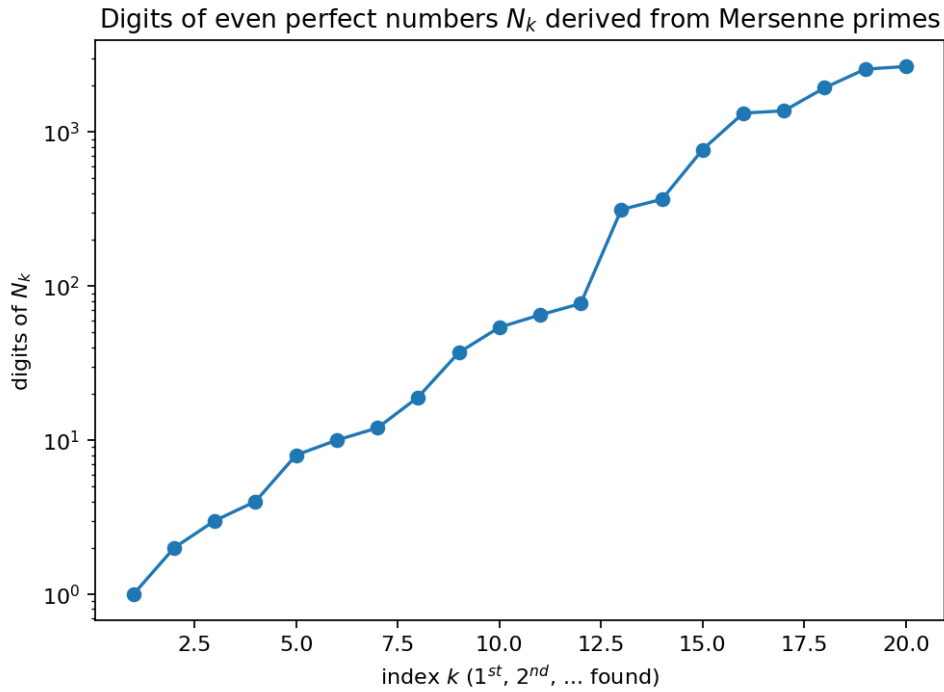
See *References*.

[Caldwell, 2021, contributors, 2025, Inc., 2025]

### Related experiments

- *E042: Repunit primes (small  $k$  scan)* (Repunit primes (small  $k$  scan))
- *E007: Mersenne growth (bits and digits)* (Mersenne growth (bits and digits))
- *E008: Lucas–Lehmer scan (prime exponents)* (Lucas–Lehmer scan (prime exponents))
- *E010: Even perfect numbers from Mersenne primes* (Even perfect numbers from Mersenne primes)
- *E011: Heuristic rarity of Mersenne primes* (Heuristic rarity of Mersenne primes)

### 3.3.10 E010: Even perfect numbers from Mersenne primes



**Tags:** number-theory, quantitative-exploration, visualization See: [Valid Tags](#).

#### Highlights

- Generate even perfect numbers via the Euclid–Euler theorem.
- Connect Mersenne prime exponents  $p$  to perfect numbers  $N = 2^{p-1}(2^p - 1)$ .
- Visualize growth and verify the defining property  $\sigma(N) = 2N$  for sampled cases.

#### Goal

Make the Mersenne  $\leftrightarrow$  perfect-number connection computationally explicit:

If  $M_p = 2^p - 1$  is prime, then:

$$N_p = 2^{p-1}(2^p - 1)$$

is an **even perfect number**.

#### Background (quick refresher)

- [Mersenne numbers and primes refresher](#)
- [Perfect numbers refresher](#)

#### Research question

Across the Mersenne prime exponents discovered within your scan bounds:

- how fast do the corresponding even perfect numbers grow?
- can we verify perfectness ( $\sigma(N) = 2N$ ) efficiently for these cases?

### Why this qualifies as a mathematical experiment

- **Finite procedure:** find a set of Mersenne primes within a finite bound and generate perfect numbers.
- **Observable(s):** size metrics (digits), and validation checks of  $\sigma(N) = 2N$ .
- **Parameter space:** vary the exponent bound and validation depth.
- **Outcome:** concrete examples + growth plots that support intuition.
- **Reproducibility:** exponents tested and successes recorded in artifacts.

### Experiment design

#### Computation

- Obtain a list of Mersenne prime exponents  $p$  from a scan (or a fixed list for small  $p$ ).
- For each  $p$ , compute  $N_p = 2^{p-1}(2^p - 1)$ .
- Verify perfectness for these cases:

$$\sigma(N_p) = 2N_p.$$

For modest  $p$ , exact computation is feasible; for larger  $p$ , report size metrics and skip expensive checks.

#### Outputs

- table:  $p$ ,  $M_p$  size,  $N_p$  size, and validation status
- plot:  $p$  vs. digits of  $N_p$
- optional: prime-factor structure display for small cases

#### How to run

```
make run EXP=e010
```

or:

```
uv run python -m mathxlab.experiments.e010
```

#### Notes / pitfalls

- Don't attempt divisor-sum sieves for huge  $N_p$ ; validation must be bounded and explicit.
- Clearly separate “constructed from theorem” (conditional on  $M_p$  being prime) from “validated by computation”.

#### Extensions

- Cross-link to E002 outputs for perfect numbers and compare growth on the same axes.
- Explore which parts of the perfectness check can be done symbolically using known factorization.

#### References

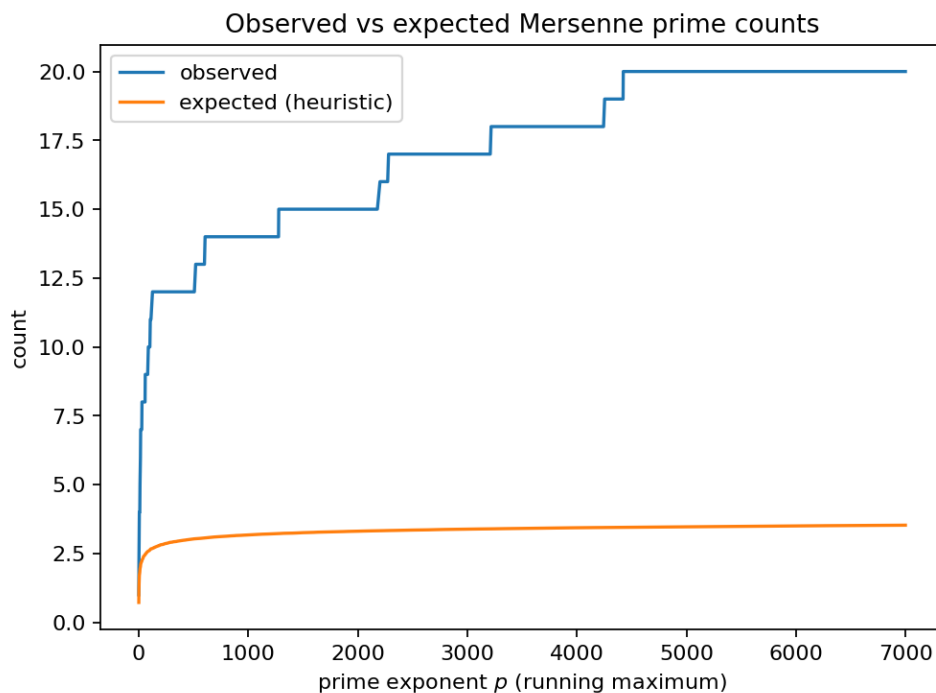
See *References*.

[Caldwell, n.d., contributors, 2025, Inc., 2025]

## Related experiments

- *E002: Even Perfect Numbers — Generator and Growth* (Even Perfect Numbers — Generator and Growth)
- *E007: Mersenne growth (bits and digits)* (Mersenne growth (bits and digits))
- *E009: Small-factor scan for Mersenne numbers* (Small-factor scan for Mersenne numbers)
- *E011: Heuristic rarity of Mersenne primes* (Heuristic rarity of Mersenne primes)
- *E097 —  $(n)/n$  landscape: deficient, perfect, abundant* (E097 —  $(n)/n$  landscape: deficient, perfect, abundant)

### 3.3.11 E011: Heuristic rarity of Mersenne primes



Tags: number-theory, quantitative-exploration, visualization See: [Valid Tags](#).

## Highlights

- Compare observed Mersenne-prime counts to a simple heuristic expectation curve.
- Visualize cumulative “found vs. expected” as exponent bounds increase.
- Keep the discussion explicitly empirical (evidence, not proof).

## Goal

Mersenne primes are extremely rare. This experiment compares:

- the observed count of Mersenne primes among prime exponents  $p \leq P$
- a lightweight heuristic curve that grows slowly with  $P$

The aim is intuition-building and trend visualization, not a rigorous model.

## Background (quick refresher)

- *Mersenne numbers and primes refresher*

## Research question

For increasing prime exponent bounds  $P$ :

- how does the cumulative count of “passes” from your scan grow?
- how does it compare to a simple expectation curve of the form:

$$E(P) \approx \sum_{p \leq P, p \text{ prime}} \frac{1}{p \ln 2}$$

## Why this qualifies as a mathematical experiment

- **Finite procedure:** run a finite scan and compute a finite expected sum.
- **Observable(s):** cumulative count of passes vs. cumulative expected value.
- **Parameter space:** vary  $P$  (and optionally cap runtime per test).
- **Outcome:** plots that show agreement/divergence and suggest questions for deeper study.
- **Failure modes:** small ranges are noisy; the experiment should clearly label bounds.

## Experiment design

### Computation

- From E008 (or a fixed list), get the set of exponents tested and which passed.
- For each bound  $P$ , compute:
  - $\text{Found}(P)$  = number of passing exponents  $p \leq P$
  - $E(P)$  = cumulative heuristic sum
- Plot both curves against  $P$ .

### Outputs

- plot:  $P$  vs.  $\text{Found}(P)$  and  $E(P)$
- table: selected checkpoints ( $P$ ,  $\text{Found}$ ,  $E$ )

### How to run

```
make run EXP=e011
```

or:

```
uv run python -m mathxlab.experiments.e011
```

### Notes / pitfalls

- The heuristic curve is a **comparison tool**, not a theorem.
- Results depend heavily on which exponents were actually tested; record the tested set in the report.
- A single large Mersenne prime can dominate impressions—keep axes and annotations clear.



## Extensions

- Compare multiple heuristics (still empirical) and see which best matches the observed range.
- Extend the scan bound and see whether the deviation grows or stabilizes.

## References

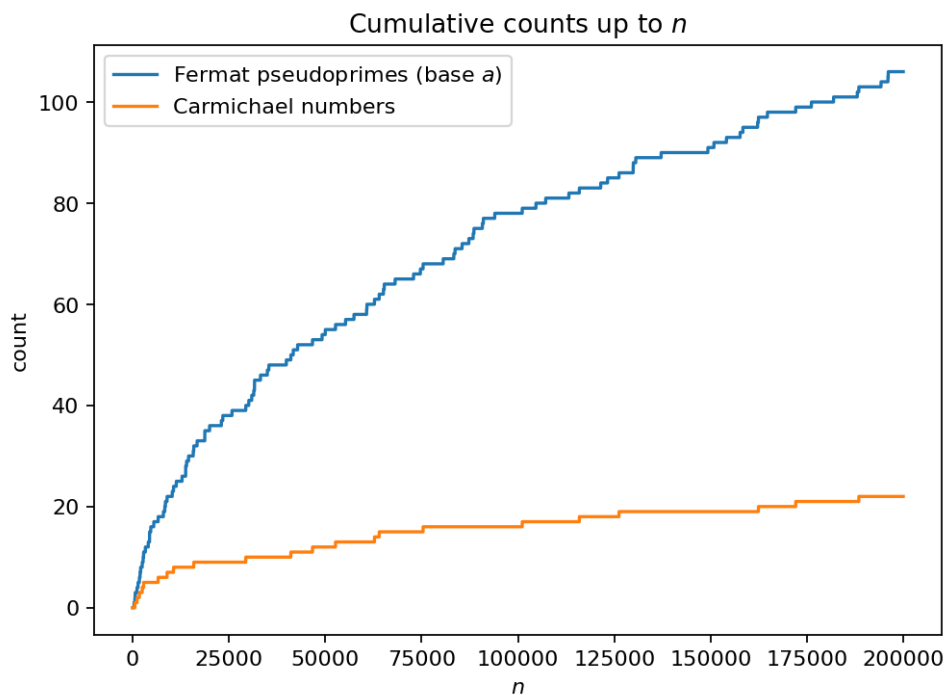
See *References*.

[Caldwell, 2021, Mersenne Research, 2025]

## Related experiments

- *E007: Mersenne growth (bits and digits)* (Mersenne growth (bits and digits))
- *E009: Small-factor scan for Mersenne numbers* (Small-factor scan for Mersenne numbers)
- *E010: Even perfect numbers from Mersenne primes* (Even perfect numbers from Mersenne primes)
- *E027: Record prime gaps vs.  $\log^2$  heuristic* (Record prime gaps vs.  $\log^2$  heuristic)
- *E029: Twin primes: observed vs. heuristic* (Twin primes: observed vs. heuristic)

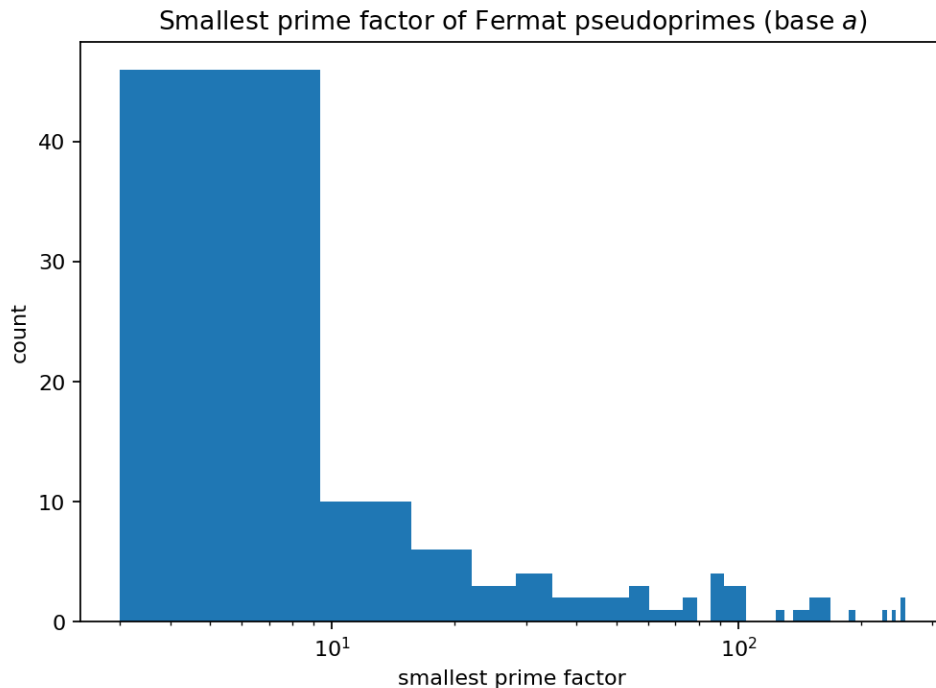
### 3.3.12 E012: Fermat pseudoprimes and Carmichael numbers (counterexamples)



Tags: number-theory, counterexample-search, visualization See: *Valid Tags*.

## Highlights

- Counterexamples to naive Fermat primality testing: base-a pseudoprimes and Carmichael numbers.
- Writes reproducible artifacts (`params.json`, `report.md`, and figures).
- Designed to surface patterns *and* “looks-true-until-it-breaks” behavior.



### Goal

Counterexamples to naive Fermat primality testing: base- $a$  pseudoprimes and Carmichael numbers.

### Background (quick refresher)

- *Prime numbers refresher*

### Research question

Which **prime-related claim, heuristic, or algorithm** breaks first under a clean, controlled computational sweep, and what does the smallest or clearest counterexample (or deviation) look like?

### Why this qualifies as a mathematical experiment

- **Finite procedure:** run a bounded search / sweep with recorded parameters.
- **Observable(s):** counts, gaps, residues, runtime scaling, or first counterexample witnesses.
- **Parameter space:** vary bounds (and sometimes algorithmic choices).
- **Outcome:** plots/tables + “witness objects” for failures.
- **Reproducibility:** outputs saved to `out/e012/` with a parameter snapshot.

### Experiment design

- **Computation:** bounded enumeration / sampling with explicit limits.
- **Outputs:** figures and a short `report.md` summarizing what was found.
- **Artifacts written:**
  - `figures/fig_01_cumulative_counts.png`
  - `figures/fig_02_smallest_factor_hist.png`
  - `params.json`
  - `report.md`

## How to run

```
make run EXP=e012
```

or:

```
uv run python -m mathxlab.experiments.e012
```

## Notes / pitfalls

- “No counterexample found” only means “none found within the configured bounds”.
- For probabilistic tests (when used), treat outcomes as *evidence*, not proof.

## Extensions

- Increase bounds and rerun (recording runtime and memory).
- Compare alternative heuristics or algorithms on the same parameter grid.
- Turn found deviations into new, tighter conjectures.

## References

See *References*.

## Related experiments

- *E048: Carmichael numbers: Korselt scan + Fermat counterexamples* (Carmichael numbers: Korselt scan + Fermat counterexamples)
- *E013: Prime-polynomial counterexamples (Euler’s  $n^2 + n + 41$ )* (Prime-polynomial counterexamples (Euler’s  $n^2 + n + 41$ ))
- *E014: Primorial  $\pm 1$  counterexamples* (Primorial  $\pm 1$  counterexamples)
- *E018: Miller–Rabin base choice counterexamples* (Miller–Rabin base choice counterexamples)
- *E047: Fermat numbers: Pépin test + factor witnesses* (Fermat numbers: Pépin test + factor witnesses)

### 3.3.13 E013: Prime-polynomial counterexamples (Euler’s $n^2 + n + 41$ )

Tags: number-theory, counterexample-search, visualization See: *Valid Tags*.

## Highlights

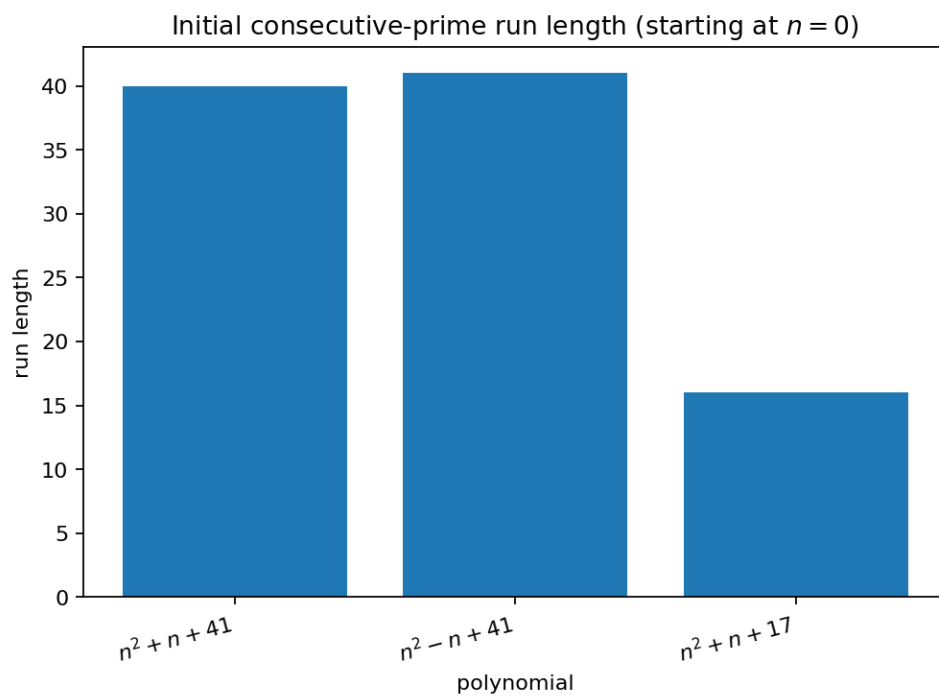
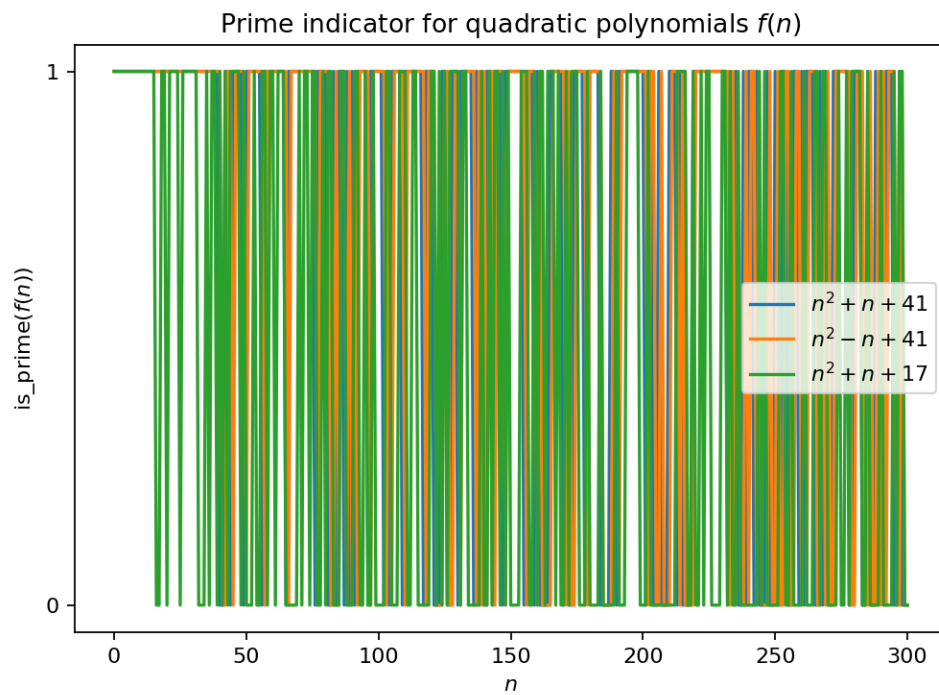
- Turn “prime-generating polynomials” folklore into a crisp counterexample (Euler’s  $n^2+n+41$ ).
- Writes reproducible artifacts (`params.json`, `report.md`, and figures).
- Designed to surface patterns *and* “looks-true-until-it-breaks” behavior.

## Goal

Turn “prime-generating polynomials” folklore into a crisp counterexample (Euler’s  $n^2+n+41$ ).

## Background (quick refresher)

- *Prime numbers refresher*



### Research question

Which **prime-related claim, heuristic, or algorithm** breaks first under a clean, controlled computational sweep, and what does the smallest or clearest counterexample (or deviation) look like?

### Why this qualifies as a mathematical experiment

- **Finite procedure:** run a bounded search / sweep with recorded parameters.
- **Observable(s):** counts, gaps, residues, runtime scaling, or first counterexample witnesses.
- **Parameter space:** vary bounds (and sometimes algorithmic choices).
- **Outcome:** plots/tables + “witness objects” for failures.
- **Reproducibility:** outputs saved to `out/e013/` with a parameter snapshot.

### Experiment design

- **Computation:** bounded enumeration / sampling with explicit limits.
- **Outputs:** figures and a short `report.md` summarizing what was found.
- **Artifacts written:**
  - `figures/fig_01_prime_indicator.png`
  - `figures/fig_02_initial_prime_run_lengths.png`
  - `params.json`
  - `report.md`

### How to run

```
make run EXP=e013
```

or:

```
uv run python -m mathxlab.experiments.e013
```

### Notes / pitfalls

- “No counterexample found” only means “none found within the configured bounds”.
- For probabilistic tests (when used), treat outcomes as *evidence*, not proof.

### Extensions

- Increase bounds and rerun (recording runtime and memory).
- Compare alternative heuristics or algorithms on the same parameter grid.
- Turn found deviations into new, tighter conjectures.

### References

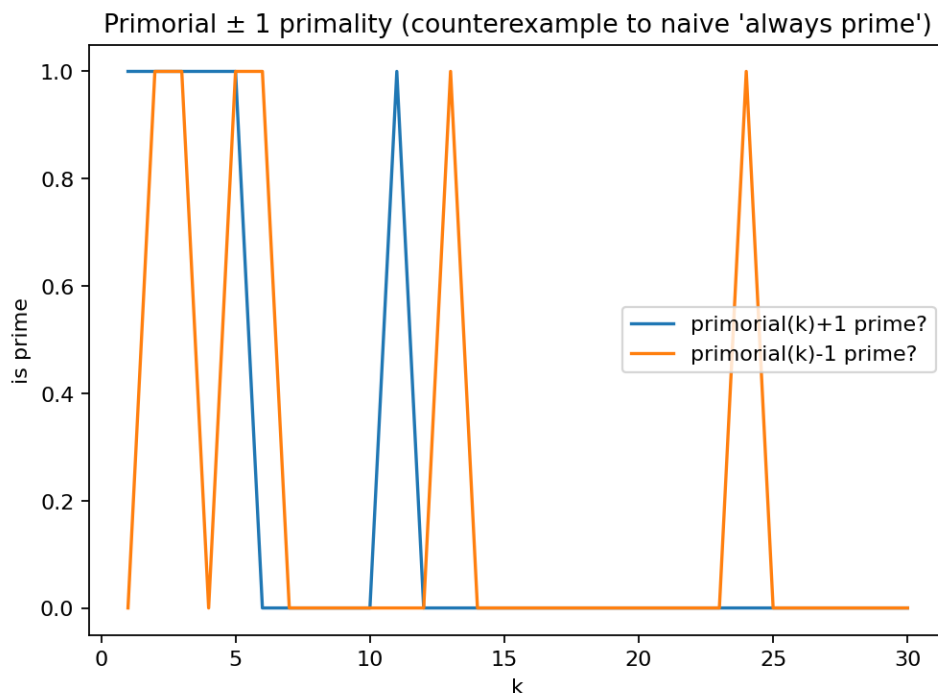
See *References*.

### Related experiments

- *E012: Fermat pseudoprimes and Carmichael numbers (counterexamples)* (Fermat pseudoprimes and Carmichael numbers (counterexamples))
- *E014: Primorial  $\pm 1$  counterexamples* (Primorial  $\pm 1$  counterexamples)
- *E018: Miller–Rabin base choice counterexamples* (Miller–Rabin base choice counterexamples)

- *E048: Carmichael numbers: Korselt scan + Fermat counterexamples* (Carmichael numbers: Korselt scan + Fermat counterexamples)
- *E068: Dirichlet  $L(s, \cdot)$ : series vs. Euler product (partial approximations)*. (Dirichlet  $L(s, \cdot)$ : series vs. Euler product (partial approximations).)
- *E127: Quadratic prime-run atlas ( $n^2 + a n + b$ )* (Quadratic prime-run atlas around (a,b))
- *E128: Quadratic modular obstructions (Euler-type)* (Modular obstructions that force failures)
- *E129: Euler lucky constants for  $n^2 + n + b$*  (Euler lucky constants for  $n^2 + n + b$ )

### 3.3.14 E014: Primorial $\pm 1$ counterexamples



Tags: number-theory, counterexample-search, visualization See: [Valid Tags](#).

#### Highlights

- This is a thin wrapper that follows the standard experiment template and delegates the actual computation to `:mod:mathxlab.experiments.prime_suite`.
- Writes reproducible artifacts (`params.json`, `report.md`, and figures).
- Designed to surface patterns *and* “looks-true-until-it-breaks” behavior.

#### Goal

This is a thin wrapper that follows the standard experiment template and delegates the actual computation to `:mod:mathxlab.experiments.prime_suite`.

#### Background (quick refresher)

- *Prime numbers refresher*

### Research question

Which **prime-related claim, heuristic, or algorithm** breaks first under a clean, controlled computational sweep, and what does the smallest or clearest counterexample (or deviation) look like?

### Why this qualifies as a mathematical experiment

- **Finite procedure:** run a bounded search / sweep with recorded parameters.
- **Observable(s):** counts, gaps, residues, runtime scaling, or first counterexample witnesses.
- **Parameter space:** vary bounds (and sometimes algorithmic choices).
- **Outcome:** plots/tables + “witness objects” for failures.
- **Reproducibility:** outputs saved to `out/e014/` with a parameter snapshot.

### Experiment design

- **Computation:** bounded enumeration / sampling with explicit limits.
- **Outputs:** figures and a short `report.md` summarizing what was found.
- **Artifacts written:**
  - `figures/fig_*.png`
  - `params.json`
  - `report.md`

### How to run

```
make run EXP=e014
```

or:

```
uv run python -m mathxlab.experiments.e014
```

### Notes / pitfalls

- “No counterexample found” only means “none found within the configured bounds”.
- For probabilistic tests (when used), treat outcomes as *evidence*, not proof.

### Extensions

- Increase bounds and rerun (recording runtime and memory).
- Compare alternative heuristics or algorithms on the same parameter grid.
- Turn found deviations into new, tighter conjectures.

### References

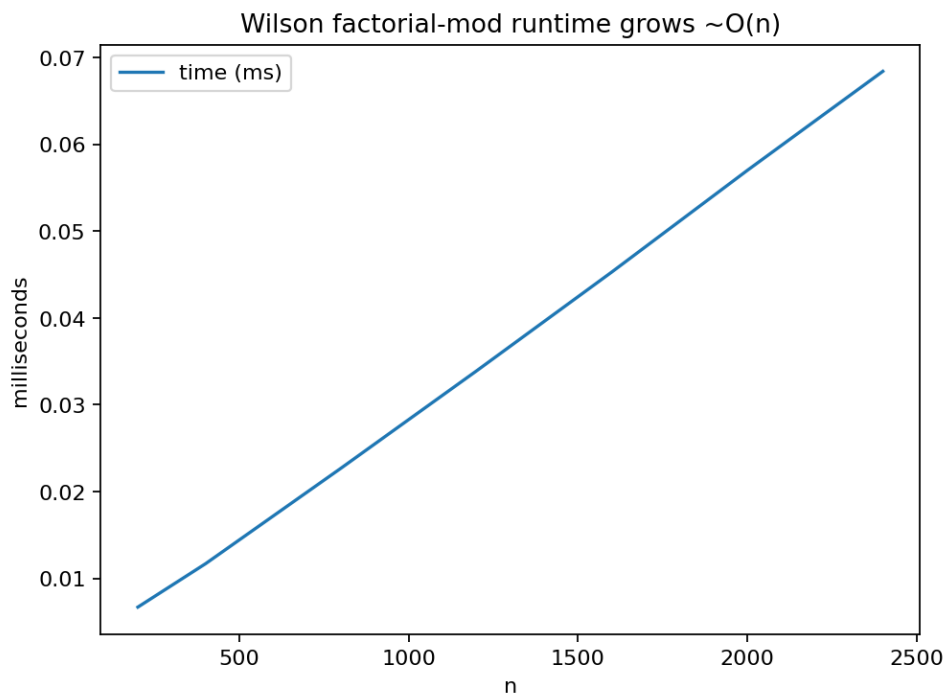
See *References*.

### Related experiments

- *E012: Fermat pseudoprimes and Carmichael numbers (counterexamples)* (Fermat pseudoprimes and Carmichael numbers (counterexamples))
- *E013: Prime-polynomial counterexamples (Euler’s  $n^2 + n + 41$ )* (Prime-polynomial counterexamples (Euler’s  $n^2 + n + 41$ ))
- *E018: Miller–Rabin base choice counterexamples* (Miller–Rabin base choice counterexamples)

- *E048: Carmichael numbers: Korselt scan + Fermat counterexamples* (Carmichael numbers: Korselt scan + Fermat counterexamples)
- *E005: Odd Perfect Numbers — Constraint Filter Pipeline* (Odd Perfect Numbers — Constraint Filter Pipeline)

### 3.3.15 E015: Wilson test infeasibility



Tags: number-theory, quantitative-exploration, visualization See: [Valid Tags](#).

#### Highlights

- This is a thin wrapper that follows the standard experiment template and delegates the actual computation to `:mod:mathxlab.experiments.prime_suite`.
- Writes reproducible artifacts (`params.json`, `report.md`, and figures).
- Designed to surface patterns *and* “looks-true-until-it-breaks” behavior.

#### Goal

This is a thin wrapper that follows the standard experiment template and delegates the actual computation to `:mod:mathxlab.experiments.prime_suite`.

#### Background (quick refresher)

- *Prime numbers refresher*

#### Research question

Which **prime-related claim, heuristic, or algorithm** breaks first under a clean, controlled computational sweep, and what does the smallest or clearest counterexample (or deviation) look like?



### Why this qualifies as a mathematical experiment

- **Finite procedure:** run a bounded search / sweep with recorded parameters.
- **Observable(s):** counts, gaps, residues, runtime scaling, or first counterexample witnesses.
- **Parameter space:** vary bounds (and sometimes algorithmic choices).
- **Outcome:** plots/tables + “witness objects” for failures.
- **Reproducibility:** outputs saved to `out/e015/` with a parameter snapshot.

### Experiment design

- **Computation:** bounded enumeration / sampling with explicit limits.
- **Outputs:** figures and a short `report.md` summarizing what was found.
- **Artifacts written:**
  - `figures/fig_*.png`
  - `params.json`
  - `report.md`

### How to run

```
make run EXP=e015
```

or:

```
uv run python -m mathxlab.experiments.e015
```

### Notes / pitfalls

- “No counterexample found” only means “none found within the configured bounds”.
- For probabilistic tests (when used), treat outcomes as *evidence*, not proof.

### Extensions

- Increase bounds and rerun (recording runtime and memory).
- Compare alternative heuristics or algorithms on the same parameter grid.
- Turn found deviations into new, tighter conjectures.

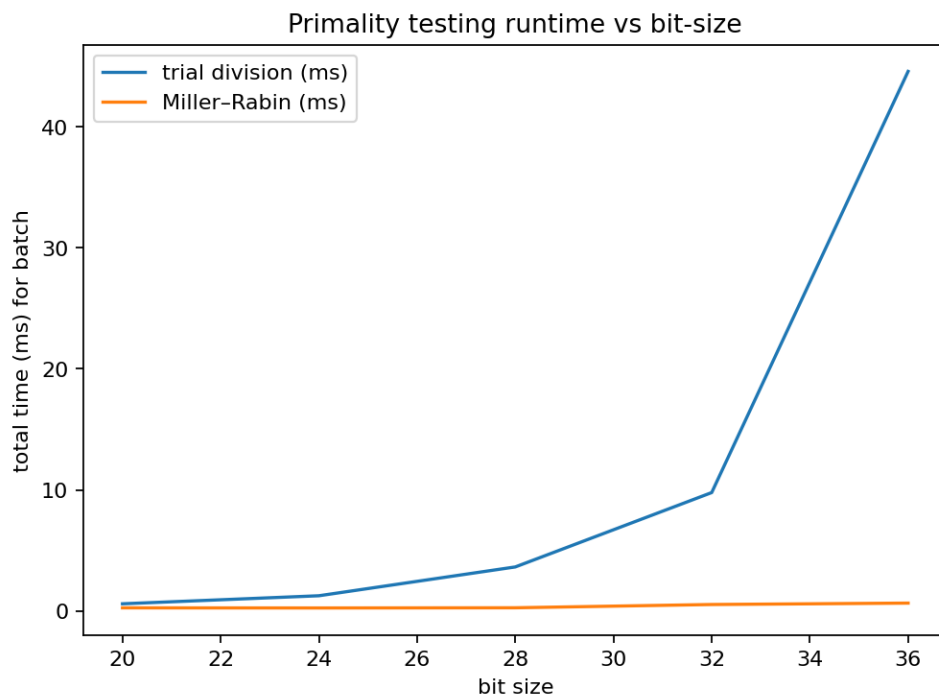
### References

See [References](#).

### Related experiments

- *E047: Fermat numbers: Pépin test + factor witnesses* (Fermat numbers: Pépin test + factor witnesses)
- *E002: Even Perfect Numbers — Generator and Growth* (Even Perfect Numbers — Generator and Growth)
- *E003: Abundancy Index Landscape* (Abundancy Index Landscape)
- *E007: Mersenne growth (bits and digits)* (Mersenne growth (bits and digits))
- *E008: Lucas–Lehmer scan (prime exponents)* (Lucas–Lehmer scan (prime exponents))

### 3.3.16 E016: Trial division vs. Miller–Rabin scaling



**Tags:** number-theory, quantitative-exploration, visualization See: [Valid Tags](#).

#### Highlights

- This is a thin wrapper that follows the standard experiment template and delegates the actual computation to `:mod:mathxlab.experiments.prime_suite`.
- Writes reproducible artifacts (`params.json`, `report.md`, and figures).
- Designed to surface patterns *and* “looks-true-until-it-breaks” behavior.

#### Goal

This is a thin wrapper that follows the standard experiment template and delegates the actual computation to `:mod:mathxlab.experiments.prime_suite`.

#### Background (quick refresher)

- *Prime numbers refresher*

#### Research question

Which **prime-related claim, heuristic, or algorithm** breaks first under a clean, controlled computational sweep, and what does the smallest or clearest counterexample (or deviation) look like?

#### Why this qualifies as a mathematical experiment

- **Finite procedure:** run a bounded search / sweep with recorded parameters.
- **Observable(s):** counts, gaps, residues, runtime scaling, or first counterexample witnesses.
- **Parameter space:** vary bounds (and sometimes algorithmic choices).
- **Outcome:** plots/tables + “witness objects” for failures.
- **Reproducibility:** outputs saved to `out/e016/` with a parameter snapshot.

### Experiment design

- **Computation:** bounded enumeration / sampling with explicit limits.
- **Outputs:** figures and a short `report.md` summarizing what was found.
- **Artifacts written:**
  - `figures/fig_*.png`
  - `params.json`
  - `report.md`

### How to run

```
make run EXP=e016
```

or:

```
uv run python -m mathxlab.experiments.e016
```

### Notes / pitfalls

- “No counterexample found” only means “none found within the configured bounds”.
- For probabilistic tests (when used), treat outcomes as *evidence*, not proof.

### Extensions

- Increase bounds and rerun (recording runtime and memory).
- Compare alternative heuristics or algorithms on the same parameter grid.
- Turn found deviations into new, tighter conjectures.

### References

See [References](#).

### Related experiments

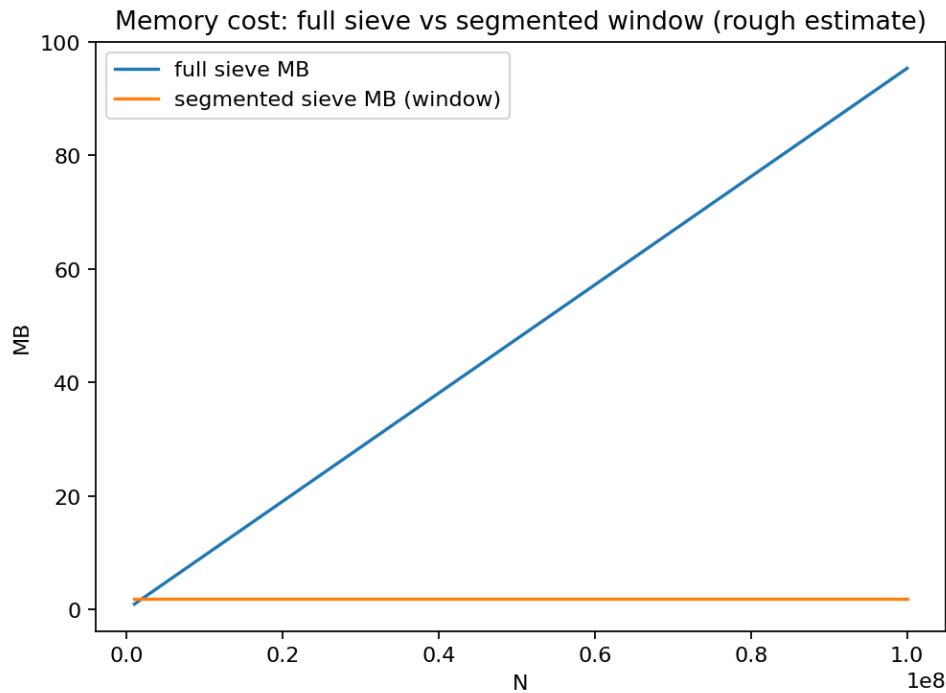
- *E044: Solovay–Strassen vs. Miller–Rabin (liars)* (Solovay–Strassen vs. Miller–Rabin (liars))
- *E105 — Mertens  $M(x)$ : scaling views* (E105 — Mertens  $M(x)$ : scaling views)
- *E119 — Summatory totient  $\Phi(x)$  scaling check* (E119 — Summatory totient  $\Phi(x)$  scaling check)
- *E018: Miller–Rabin base choice counterexamples* (Miller–Rabin base choice counterexamples)
- *E002: Even Perfect Numbers — Generator and Growth* (Even Perfect Numbers — Generator and Growth)

### 3.3.17 E017: Sieve memory blow-up vs. segmented sieve

**Tags:** number-theory, quantitative-exploration, visualization See: [Valid Tags](#).

#### Highlights

- This is a thin wrapper that follows the standard experiment template and delegates the actual computation to `:mod:mathxlab.experiments.prime_suite`.
- Writes reproducible artifacts (`params.json`, `report.md`, and figures).
- Designed to surface patterns *and* “looks-true-until-it-breaks” behavior.



### Goal

This is a thin wrapper that follows the standard experiment template and delegates the actual computation to `:mod:mathxlab.experiments.prime_suite`.

### Background (quick refresher)

- *Prime numbers refresher*

### Research question

Which **prime-related claim, heuristic, or algorithm** breaks first under a clean, controlled computational sweep, and what does the smallest or clearest counterexample (or deviation) look like?

### Why this qualifies as a mathematical experiment

- **Finite procedure:** run a bounded search / sweep with recorded parameters.
- **Observable(s):** counts, gaps, residues, runtime scaling, or first counterexample witnesses.
- **Parameter space:** vary bounds (and sometimes algorithmic choices).
- **Outcome:** plots/tables + “witness objects” for failures.
- **Reproducibility:** outputs saved to `out/e017/` with a parameter snapshot.

### Experiment design

- **Computation:** bounded enumeration / sampling with explicit limits.
- **Outputs:** figures and a short `report.md` summarizing what was found.
- **Artifacts written:**
  - `figures/fig_*.png`
  - `params.json`
  - `report.md`

## How to run

```
make run EXP=e017
```

or:

```
uv run python -m mathxlab.experiments.e017
```

## Notes / pitfalls

- “No counterexample found” only means “none found within the configured bounds”.
- For probabilistic tests (when used), treat outcomes as *evidence*, not proof.

## Extensions

- Increase bounds and rerun (recording runtime and memory).
- Compare alternative heuristics or algorithms on the same parameter grid.
- Turn found deviations into new, tighter conjectures.

## References

See *References*.

## Related experiments

- *E004: Computing  $\sigma(n)$  at Scale — Sieve vs. Factorization* (Computing  $\sigma(n)$  at Scale — Sieve vs. Factorization)
- *E002: Even Perfect Numbers — Generator and Growth* (Even Perfect Numbers — Generator and Growth)
- *E003: Abundancy Index Landscape* (Abundancy Index Landscape)
- *E007: Mersenne growth (bits and digits)* (Mersenne growth (bits and digits))
- *E008: Lucas–Lehmer scan (prime exponents)* (Lucas–Lehmer scan (prime exponents))

### 3.3.18 E018: Miller–Rabin base choice counterexamples

**Tags:** number-theory, counterexample-search, visualization See: *Valid Tags*.

## Highlights

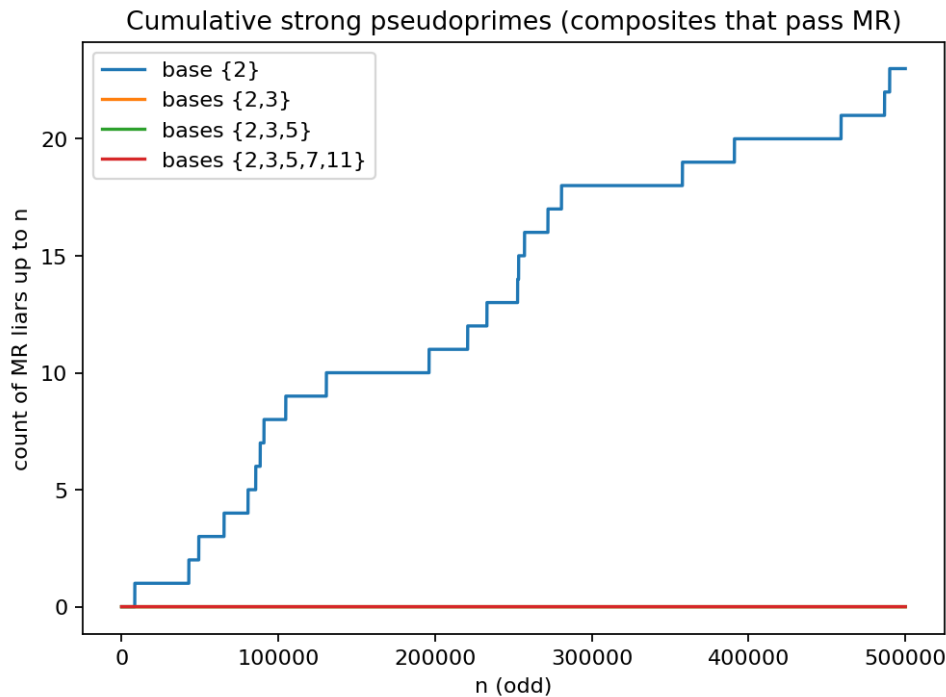
- This is a thin wrapper that follows the standard experiment template and delegates the actual computation to `:mod:mathxlab.experiments.prime_suite`.
- Writes reproducible artifacts (`params.json`, `report.md`, and figures).
- Designed to surface patterns *and* “looks-true-until-it-breaks” behavior.

## Goal

This is a thin wrapper that follows the standard experiment template and delegates the actual computation to `:mod:mathxlab.experiments.prime_suite`.

## Background (quick refresher)

- *Prime numbers refresher*



### Research question

Which **prime-related claim, heuristic, or algorithm** breaks first under a clean, controlled computational sweep, and what does the smallest or clearest counterexample (or deviation) look like?

### Why this qualifies as a mathematical experiment

- **Finite procedure:** run a bounded search / sweep with recorded parameters.
- **Observable(s):** counts, gaps, residues, runtime scaling, or first counterexample witnesses.
- **Parameter space:** vary bounds (and sometimes algorithmic choices).
- **Outcome:** plots/tables + “witness objects” for failures.
- **Reproducibility:** outputs saved to `out/e018/` with a parameter snapshot.

### Experiment design

- **Computation:** bounded enumeration / sampling with explicit limits.
- **Outputs:** figures and a short `report.md` summarizing what was found.
- **Artifacts written:**
  - `figures/fig_*.png`
  - `params.json`
  - `report.md`

### How to run

```
make run EXP=e018
```

or:

```
uv run python -m mathxlab.experiments.e018
```

### Notes / pitfalls

- “No counterexample found” only means “none found within the configured bounds”.
- For probabilistic tests (when used), treat outcomes as *evidence*, not proof.

### Extensions

- Increase bounds and rerun (recording runtime and memory).
- Compare alternative heuristics or algorithms on the same parameter grid.
- Turn found deviations into new, tighter conjectures.

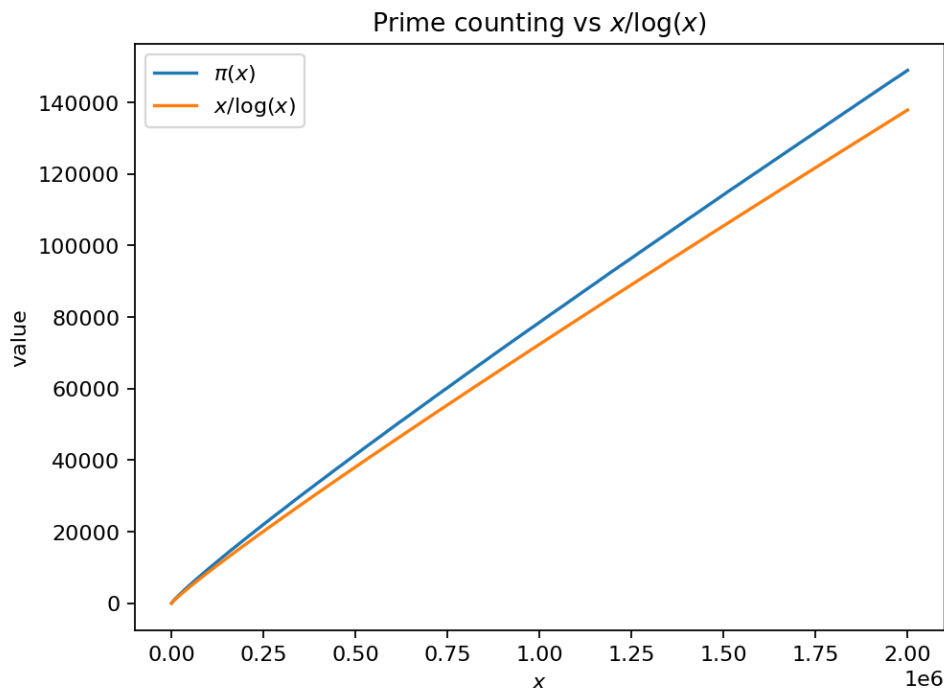
### References

See [References](#).

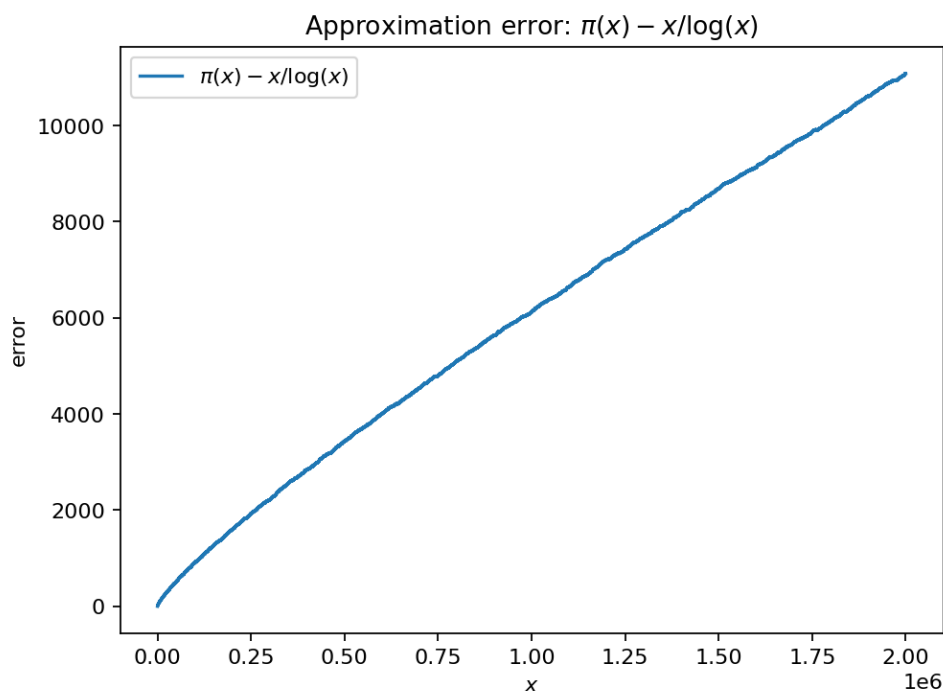
### Related experiments

- *E012: Fermat pseudoprimes and Carmichael numbers (counterexamples)* (Fermat pseudoprimes and Carmichael numbers (counterexamples))
- *E013: Prime-polynomial counterexamples (Euler’s  $n^2 + n + 41$ )* (Prime-polynomial counterexamples (Euler’s  $n^2 + n + 41$ ))
- *E014: Primorial  $\pm 1$  counterexamples* (Primorial  $\pm 1$  counterexamples)
- *E048: Carmichael numbers: Korselt scan + Fermat counterexamples* (Carmichael numbers: Korselt scan + Fermat counterexamples)
- *E049: Wieferich primes (base 2): scan and quotient visualization* (Wieferich primes (base 2): scan and quotient visualization)

### 3.3.19 E019: Prime density and PNT visualization



Tags: number-theory, quantitative-exploration, visualization See: [Valid Tags](#).



### Highlights

- This is a thin wrapper that follows the standard experiment template and delegates the actual computation to `:mod:mathxlab.experiments.prime_suite`.
- Writes reproducible artifacts (`params.json`, `report.md`, and figures).
- Designed to surface patterns *and* “looks-true-until-it-breaks” behavior.

### Goal

This is a thin wrapper that follows the standard experiment template and delegates the actual computation to `:mod:mathxlab.experiments.prime_suite`.

### Background (quick refresher)

- *Prime numbers refresher*

### Research question

Which **prime-related claim, heuristic, or algorithm** breaks first under a clean, controlled computational sweep, and what does the smallest or clearest counterexample (or deviation) look like?

### Why this qualifies as a mathematical experiment

- **Finite procedure:** run a bounded search / sweep with recorded parameters.
- **Observable(s):** counts, gaps, residues, runtime scaling, or first counterexample witnesses.
- **Parameter space:** vary bounds (and sometimes algorithmic choices).
- **Outcome:** plots/tables + “witness objects” for failures.
- **Reproducibility:** outputs saved to `out/e019/` with a parameter snapshot.



### Experiment design

- **Computation:** bounded enumeration / sampling with explicit limits.
- **Outputs:** figures and a short `report.md` summarizing what was found.
- **Artifacts written:**
  - `figures/fig_*.png`
  - `params.json`
  - `report.md`

### How to run

```
make run EXP=e019
```

or:

```
uv run python -m mathxlab.experiments.e019
```

### Notes / pitfalls

- “No counterexample found” only means “none found within the configured bounds”.
- For probabilistic tests (when used), treat outcomes as *evidence*, not proof.

### Extensions

- Increase bounds and rerun (recording runtime and memory).
- Compare alternative heuristics or algorithms on the same parameter grid.
- Turn found deviations into new, tighter conjectures.

### References

See [References](#).

### Related experiments

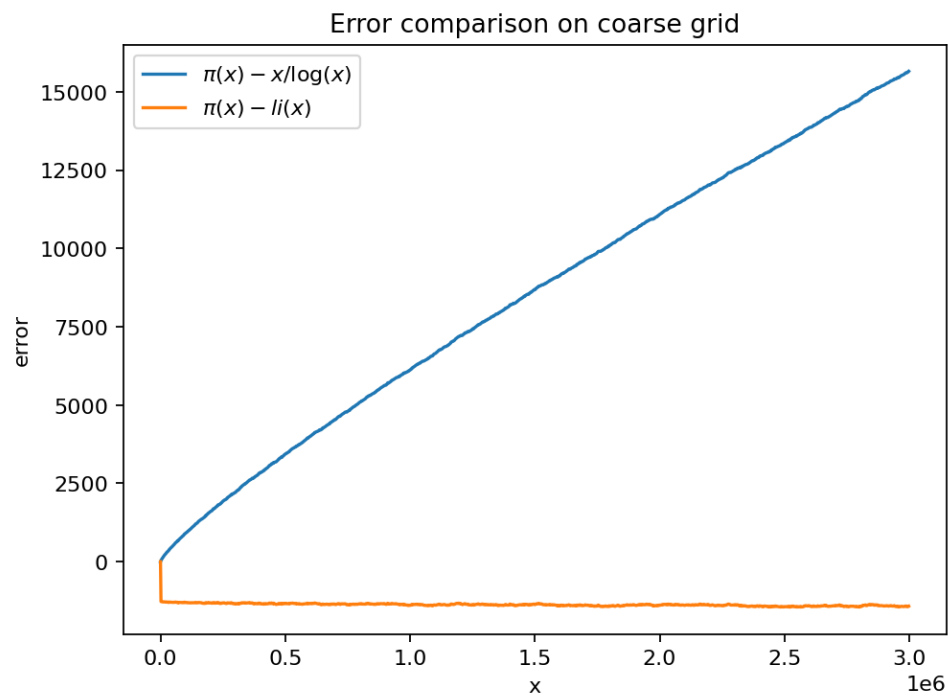
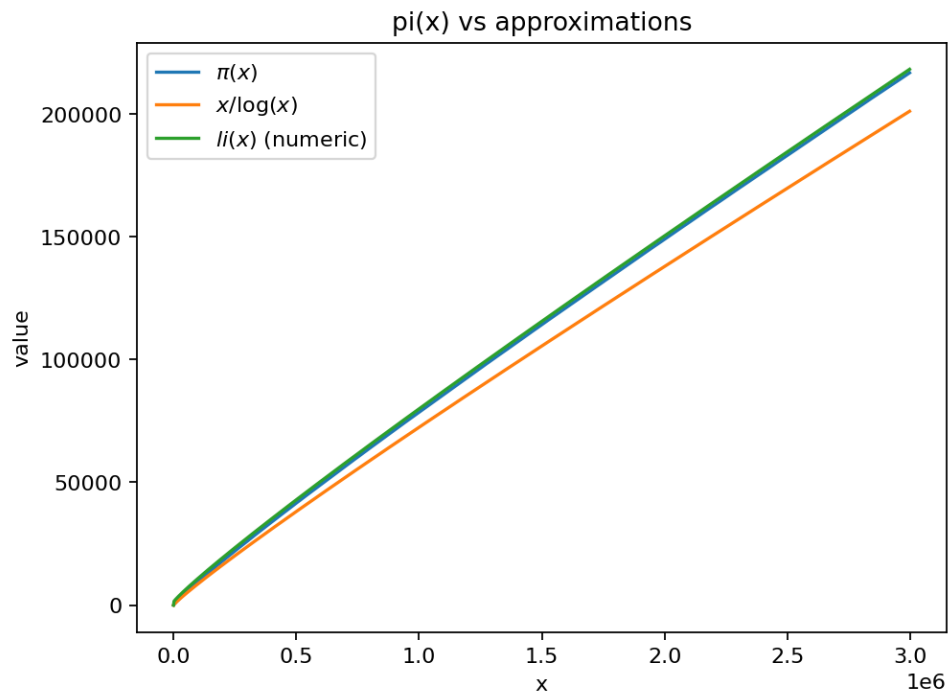
- *E054: Squarefree density via Möbius* (Squarefree density via Möbius)
- *E002: Even Perfect Numbers — Generator and Growth* (Even Perfect Numbers — Generator and Growth)
- *E003: Abundancy Index Landscape* (Abundancy Index Landscape)
- *E007: Mersenne growth (bits and digits)* (Mersenne growth (bits and digits))
- *E008: Lucas–Lehmer scan (prime exponents)* (Lucas–Lehmer scan (prime exponents))

### 3.3.20 E020: Compare $\pi(x)$ to $\text{li}(x)$ numerically

**Tags:** number-theory, quantitative-exploration, visualization See: [Valid Tags](#).

### Highlights

- This is a thin wrapper that follows the standard experiment template and delegates the actual computation to `:mod:mathxlab.experiments.prime_suite`.
- Writes reproducible artifacts (`params.json`, `report.md`, and figures).
- Designed to surface patterns *and* “looks-true-until-it-breaks” behavior.



## Goal

This is a thin wrapper that follows the standard experiment template and delegates the actual computation to `:mod:mathxlab.experiments.prime_suite`.

## Background (quick refresher)

- *Prime numbers refresher*

## Research question

Which **prime-related claim, heuristic, or algorithm** breaks first under a clean, controlled computational sweep, and what does the smallest or clearest counterexample (or deviation) look like?

## Why this qualifies as a mathematical experiment

- **Finite procedure:** run a bounded search / sweep with recorded parameters.
- **Observable(s):** counts, gaps, residues, runtime scaling, or first counterexample witnesses.
- **Parameter space:** vary bounds (and sometimes algorithmic choices).
- **Outcome:** plots/tables + “witness objects” for failures.
- **Reproducibility:** outputs saved to `out/e020/` with a parameter snapshot.

## Experiment design

- **Computation:** bounded enumeration / sampling with explicit limits.
- **Outputs:** figures and a short `report.md` summarizing what was found.
- **Artifacts written:**
  - `figures/fig_*.png`
  - `params.json`
  - `report.md`

## How to run

```
make run EXP=e020
```

or:

```
uv run python -m mathxlab.experiments.e020
```

## Notes / pitfalls

- “No counterexample found” only means “none found within the configured bounds”.
- For probabilistic tests (when used), treat outcomes as *evidence*, not proof.

## Extensions

- Increase bounds and rerun (recording runtime and memory).
- Compare alternative heuristics or algorithms on the same parameter grid.
- Turn found deviations into new, tighter conjectures.

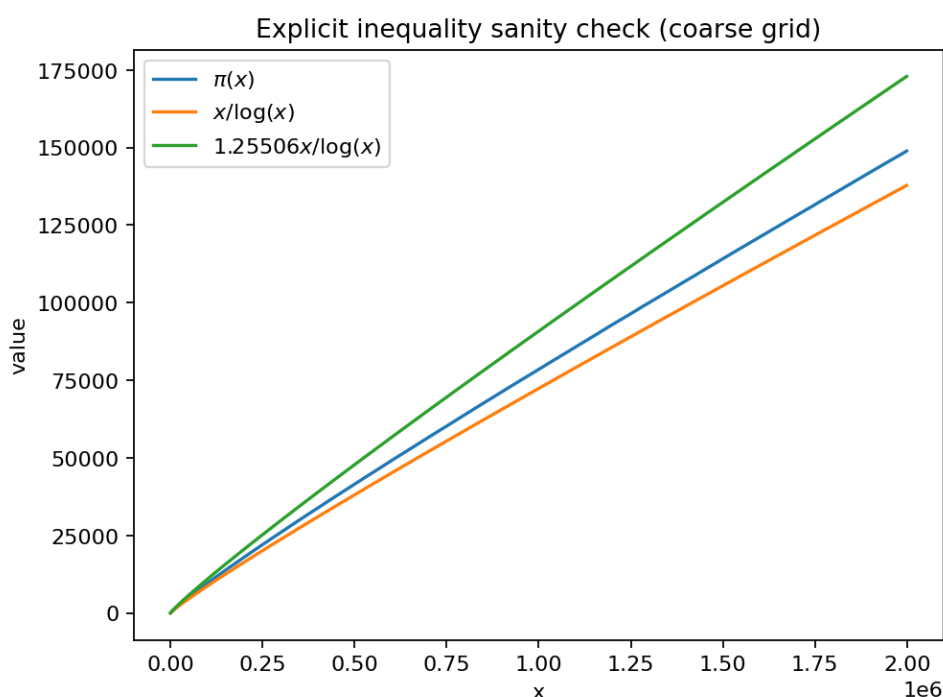
## References

See *References*.

## Related experiments

- *E002: Even Perfect Numbers — Generator and Growth* (Even Perfect Numbers — Generator and Growth)
- *E003: Abundancy Index Landscape* (Abundancy Index Landscape)
- *E007: Mersenne growth (bits and digits)* (Mersenne growth (bits and digits))
- *E008: Lucas–Lehmer scan (prime exponents)* (Lucas–Lehmer scan (prime exponents))
- *E009: Small-factor scan for Mersenne numbers* (Small-factor scan for Mersenne numbers)

### 3.3.21 E021: Explicit bounds sanity checks



Tags: number-theory, quantitative-exploration, visualization See: *Valid Tags*.

## Highlights

- This is a thin wrapper that follows the standard experiment template and delegates the actual computation to `:mod:mathxlab.experiments.prime_suite`.
- Writes reproducible artifacts (`params.json`, `report.md`, and figures).
- Designed to surface patterns *and* “looks-true-until-it-breaks” behavior.

## Goal

This is a thin wrapper that follows the standard experiment template and delegates the actual computation to `:mod:mathxlab.experiments.prime_suite`.

## Background (quick refresher)

- *Prime numbers refresher*

## Research question

Which **prime-related claim, heuristic, or algorithm** breaks first under a clean, controlled computational sweep, and what does the smallest or clearest counterexample (or deviation) look like?

## Why this qualifies as a mathematical experiment

- **Finite procedure:** run a bounded search / sweep with recorded parameters.
- **Observable(s):** counts, gaps, residues, runtime scaling, or first counterexample witnesses.
- **Parameter space:** vary bounds (and sometimes algorithmic choices).
- **Outcome:** plots/tables + “witness objects” for failures.
- **Reproducibility:** outputs saved to `out/e021/` with a parameter snapshot.

## Experiment design

- **Computation:** bounded enumeration / sampling with explicit limits.
- **Outputs:** figures and a short `report.md` summarizing what was found.
- **Artifacts written:**
  - `figures/fig_*.png`
  - `params.json`
  - `report.md`

## How to run

```
make run EXP=e021
```

or:

```
uv run python -m mathxlab.experiments.e021
```

## Notes / pitfalls

- “No counterexample found” only means “none found within the configured bounds”.
- For probabilistic tests (when used), treat outcomes as *evidence*, not proof.

## Extensions

- Increase bounds and rerun (recording runtime and memory).
- Compare alternative heuristics or algorithms on the same parameter grid.
- Turn found deviations into new, tighter conjectures.

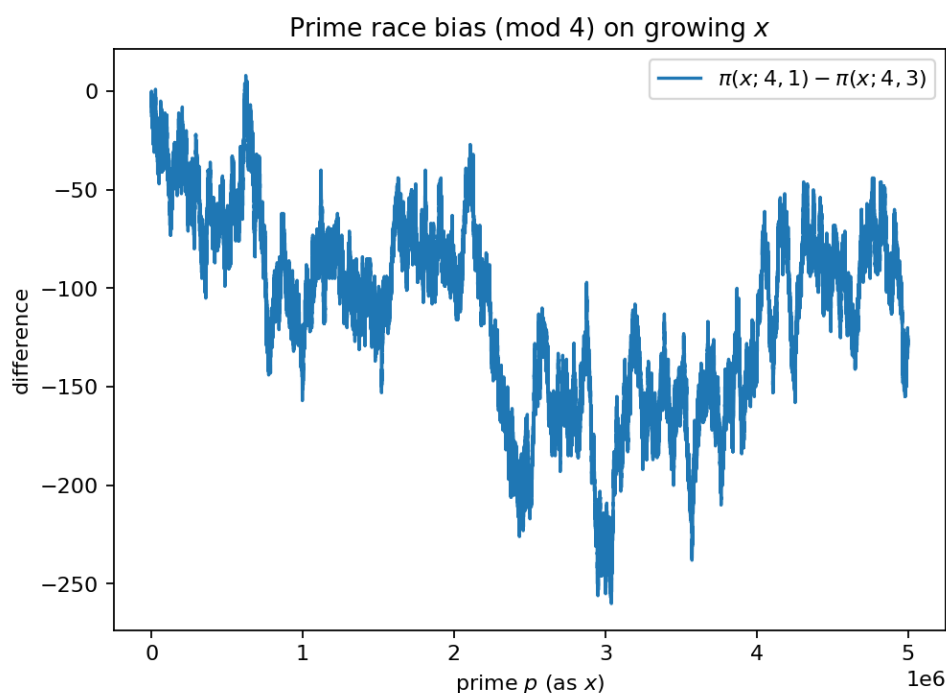
## References

See *References*.

### Related experiments

- *E077: Indicator via character orthogonality (sanity check)*. (Indicator via character orthogonality (sanity check).)
- *E002: Even Perfect Numbers — Generator and Growth* (Even Perfect Numbers — Generator and Growth)
- *E003: Abundancy Index Landscape* (Abundancy Index Landscape)
- *E007: Mersenne growth (bits and digits)* (Mersenne growth (bits and digits))
- *E008: Lucas–Lehmer scan (prime exponents)* (Lucas–Lehmer scan (prime exponents))

### 3.3.22 E022: Prime race modulo 4



Tags: number-theory, quantitative-exploration, visualization See: [Valid Tags](#).

### Highlights

- This is a thin wrapper that follows the standard experiment template and delegates the actual computation to `:mod:mathxlab.experiments.prime_suite`.
- Writes reproducible artifacts (`params.json`, `report.md`, and figures).
- Designed to surface patterns *and* “looks-true-until-it-breaks” behavior.

### Goal

This is a thin wrapper that follows the standard experiment template and delegates the actual computation to `:mod:mathxlab.experiments.prime_suite`.

### Background (quick refresher)

- *Prime numbers refresher*

## Research question

Which **prime-related claim, heuristic, or algorithm** breaks first under a clean, controlled computational sweep, and what does the smallest or clearest counterexample (or deviation) look like?

## Why this qualifies as a mathematical experiment

- **Finite procedure:** run a bounded search / sweep with recorded parameters.
- **Observable(s):** counts, gaps, residues, runtime scaling, or first counterexample witnesses.
- **Parameter space:** vary bounds (and sometimes algorithmic choices).
- **Outcome:** plots/tables + “witness objects” for failures.
- **Reproducibility:** outputs saved to `out/e022/` with a parameter snapshot.

## Experiment design

- **Computation:** bounded enumeration / sampling with explicit limits.
- **Outputs:** figures and a short `report.md` summarizing what was found.
- **Artifacts written:**
  - `figures/fig_*.png`
  - `params.json`
  - `report.md`

## How to run

```
make run EXP=e022
```

or:

```
uv run python -m mathxlab.experiments.e022
```

## Notes / pitfalls

- “No counterexample found” only means “none found within the configured bounds”.
- For probabilistic tests (when used), treat outcomes as *evidence*, not proof.

## Extensions

- Increase bounds and rerun (recording runtime and memory).
- Compare alternative heuristics or algorithms on the same parameter grid.
- Turn found deviations into new, tighter conjectures.

## References

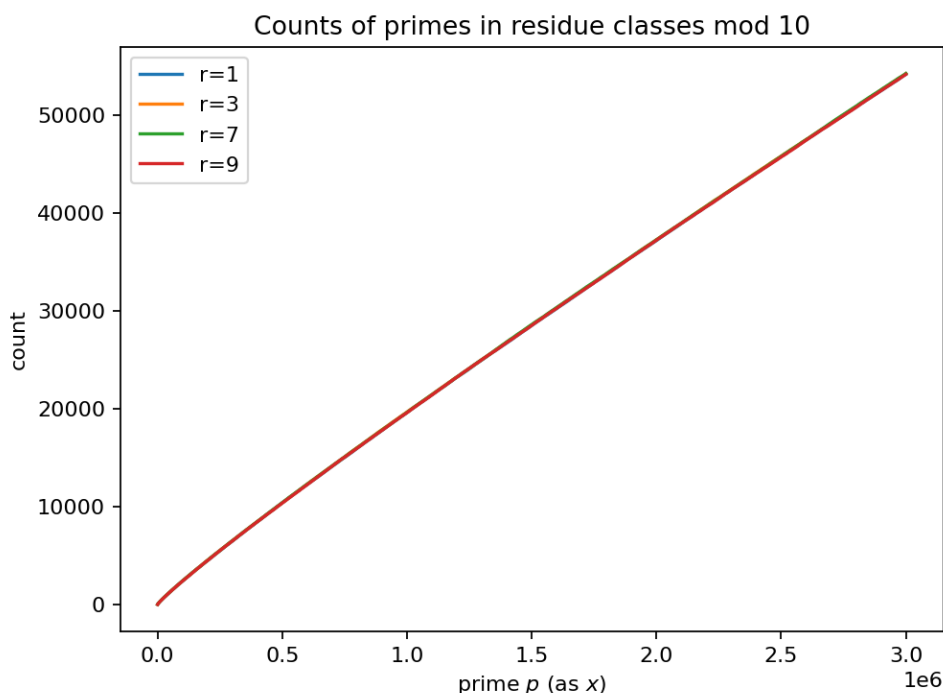
See *References*.

## Related experiments

- *E072: Prime race mod 4:  $\pi(x;4,3)$  vs.  $\pi(x;4,1)$ .* (Prime race mod 4:  $\pi(x;4,3)$  vs.  $\pi(x;4,1)$ .)
- *E073: Prime race mod 3:  $\pi(x;3,2)$  vs.  $\pi(x;3,1)$ .* (Prime race mod 3:  $\pi(x;3,2)$  vs.  $\pi(x;3,1)$ .)
- *E074: Prime race mod 8: leaderboard among 1,3,5,7.* (Prime race mod 8: leaderboard among 1,3,5,7.)
- *E075: Prime race statistic: distribution on a log-grid.* (Prime race statistic: distribution on a log-grid.)

- *E081: Prime race sign changes: first crossings table.* (Prime race sign changes: first crossings table.)

### 3.3.23 E023: Residue class distribution mod $q$



Tags: number-theory, quantitative-exploration, visualization See: [Valid Tags](#).

#### Highlights

- This is a thin wrapper that follows the standard experiment template and delegates the actual computation to `:mod:mathxlab.experiments.prime_suite`.
- Writes reproducible artifacts (`params.json`, `report.md`, and figures).
- Designed to surface patterns *and* “looks-true-until-it-breaks” behavior.

#### Goal

This is a thin wrapper that follows the standard experiment template and delegates the actual computation to `:mod:mathxlab.experiments.prime_suite`.

#### Background (quick refresher)

- *Prime numbers refresher*

#### Research question

Which **prime-related claim, heuristic, or algorithm** breaks first under a clean, controlled computational sweep, and what does the smallest or clearest counterexample (or deviation) look like?

#### Why this qualifies as a mathematical experiment

- **Finite procedure:** run a bounded search / sweep with recorded parameters.
- **Observable(s):** counts, gaps, residues, runtime scaling, or first counterexample witnesses.
- **Parameter space:** vary bounds (and sometimes algorithmic choices).
- **Outcome:** plots/tables + “witness objects” for failures.



- **Reproducibility:** outputs saved to `out/e023/` with a parameter snapshot.

### Experiment design

- **Computation:** bounded enumeration / sampling with explicit limits.
- **Outputs:** figures and a short `report.md` summarizing what was found.
- **Artifacts written:**
  - `figures/fig_*.png`
  - `params.json`
  - `report.md`

### How to run

```
make run EXP=e023
```

or:

```
uv run python -m mathxlab.experiments.e023
```

### Notes / pitfalls

- “No counterexample found” only means “none found within the configured bounds”.
- For probabilistic tests (when used), treat outcomes as *evidence*, not proof.

### Extensions

- Increase bounds and rerun (recording runtime and memory).
- Compare alternative heuristics or algorithms on the same parameter grid.
- Turn found deviations into new, tighter conjectures.

### References

See *References*.

### Related experiments

- *E113 — First prime in each residue class* (E113 — First prime in each residue class)
- *E070: Primes in residue classes:  $\pi(x; q, a)$* . (Primes in residue classes:  $\pi(x; q, a)$ .)
- *E075: Prime race statistic: distribution on a log-grid*. (Prime race statistic: distribution on a log-grid.)
- *E002: Even Perfect Numbers — Generator and Growth* (Even Perfect Numbers — Generator and Growth)
- *E003: Abundancy Index Landscape* (Abundancy Index Landscape)

### 3.3.24 E024: Ulam spiral structure

size 101: 10,201 numbers and 1,252 primes

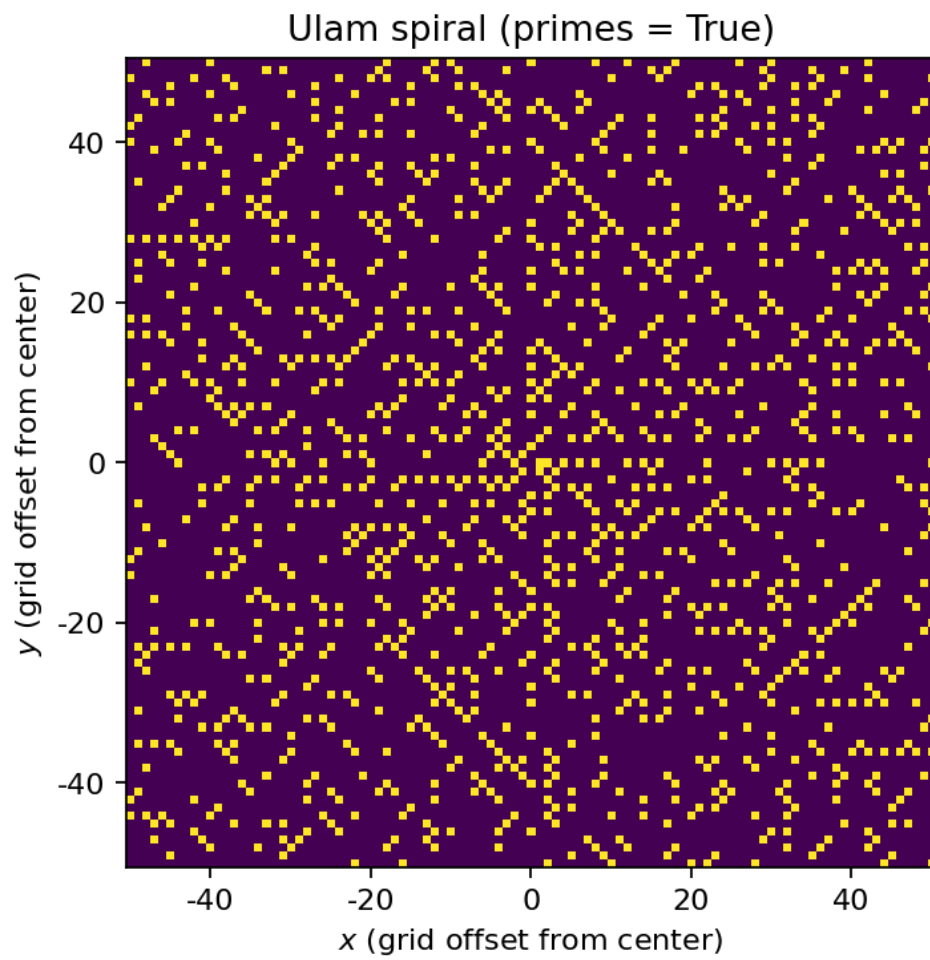
size 301: 90,601 numbers and 8,769 primes

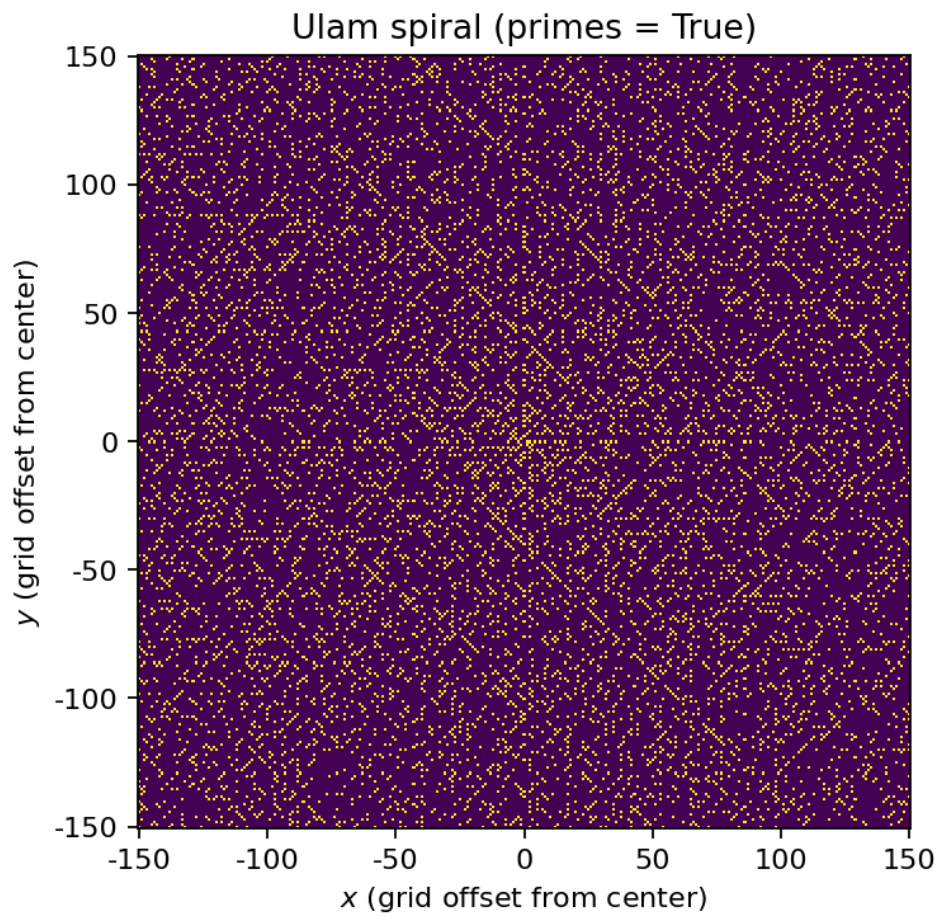
size 501: 251,001 numbers and 22,115 primes

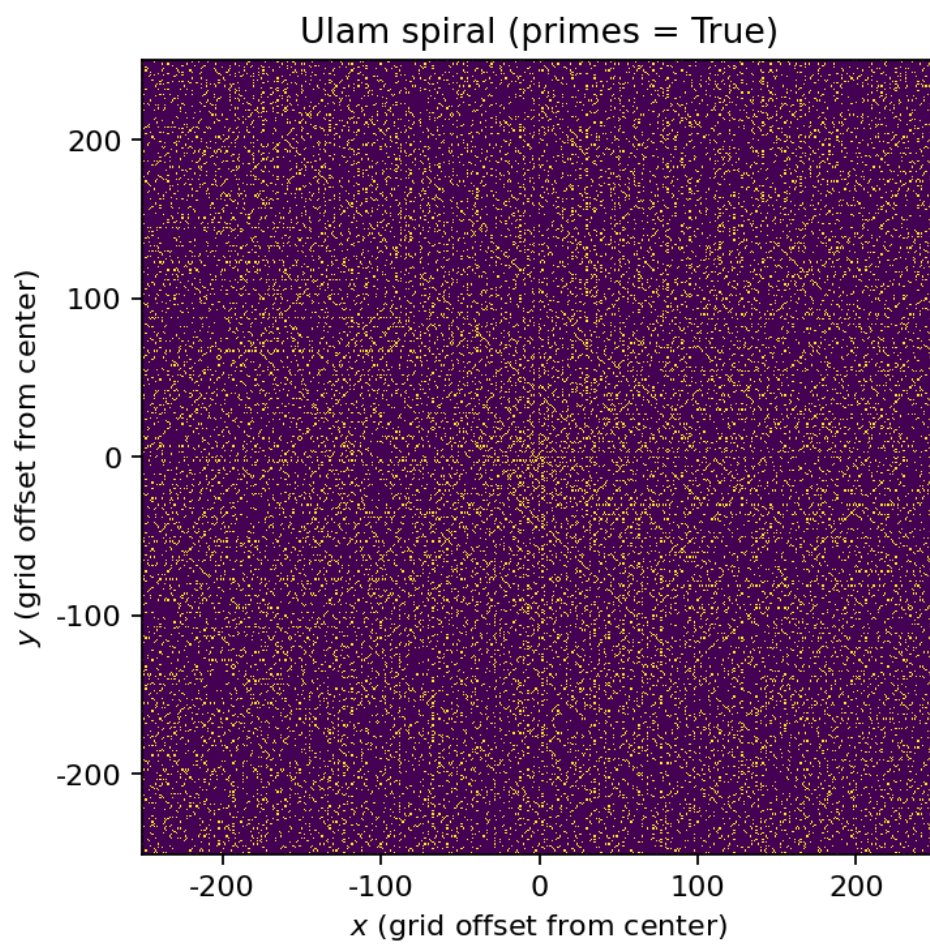
size 999: 998,001 numbers and 78,359 primes

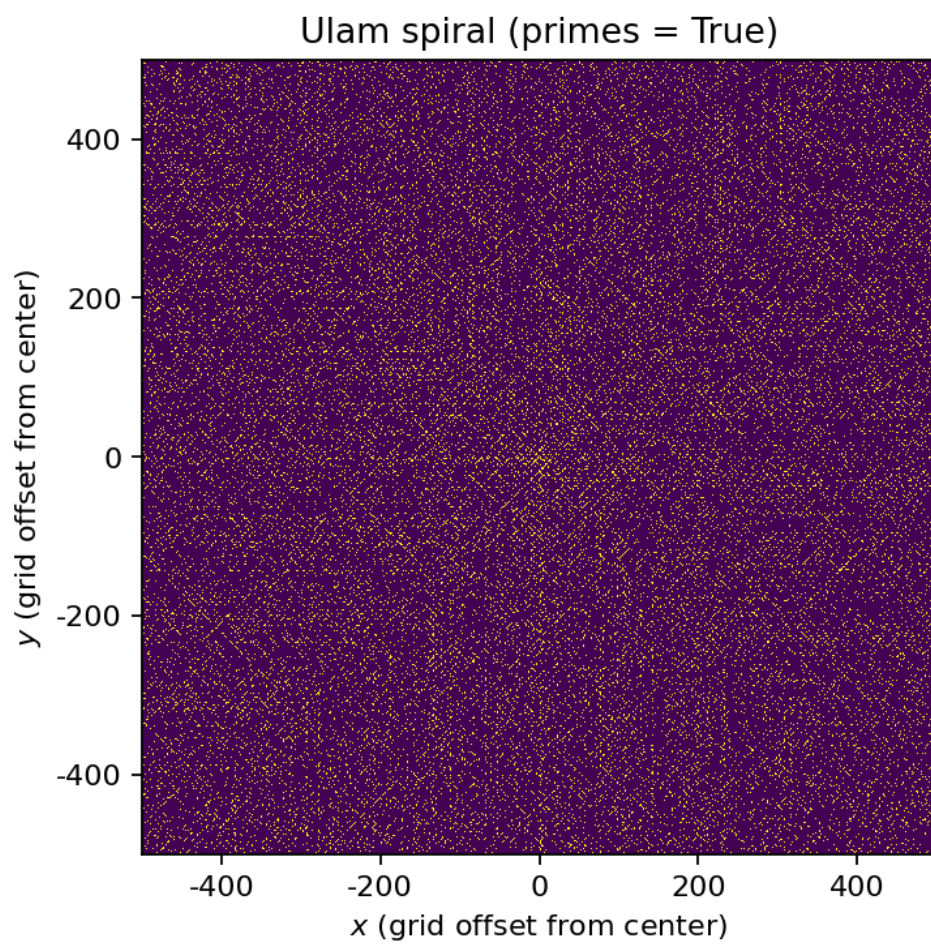
**Tags:** number-theory, quantitative-exploration, visualization

See: *Valid Tags*.









## Highlights

- Visual “pattern detector” for primes: highlights a diagonal structure that emerges in the Ulam spiral.
- Parameterizable run: vary `--size` from the command line (default: 301).
- Writes reproducible artifacts (`params.json`, `report.md`, and figures) into `out/e024/`.

## Goal

Render an Ulam spiral and highlight primes on the spiral grid to make diagonal structure (prime-rich diagonals) visible, then compare how that structure changes as the spiral size increases.

## Background (quick refresher)

### What is an Ulam spiral?

An **Ulam spiral** places the positive integers on a square spiral (starting at 1 in the center, then winding outward) and then marks which positions are prime. A striking visual phenomenon appears: **primes cluster along certain diagonals**. A common explanation is that many diagonals correspond to simple quadratic polynomials in the spiral index; some of these produce unusually many primes for small inputs. See [contributors, 2025].

### Optional background pages

- *Prime numbers refresher*
- *Quadratic polynomials (algebraic) refresher*
- *Euler’s prime-generating polynomial refresher*

## Research question

Which diagonals (or families of diagonals) in the Ulam spiral show unusually high prime density for a given window size, and how does that visual structure change as **size** increases?

## Why this qualifies as a mathematical experiment

- **Finite procedure:** build a finite spiral window and run a deterministic primality test up to `size^2`.
- **Observable(s):** diagonal structure, local “prime-rich” streaks, and how visibility changes with `size`.
- **Parameter space:** the window size `size` (and, if you extend it, rendering/annotation choices).
- **Outcome:** figures and a short report capturing the key observation and caveats.
- **Reproducibility:** outputs saved to `out/e024/` with a parameter snapshot.

## Experiment design

- **Computation:** map integers `1..size^2` to lattice coordinates on a square spiral; mark primes.
- **Outputs:** one or more figures showing prime positions on the spiral grid.
- **Artifacts written:**
  - `figures/fig_*.png`
  - `params.json`
  - `report.md`

## How to run

```
make run EXP=e024
```

Override the spiral size (must be **odd**):

```
make run EXP=e024 ARGS="--size 501"
```

Direct invocation (always works):

```
uv run --extra dev python -m mathxlab.experiments.e024 --out out/e024 --size 501
```

## Notes / pitfalls

- **size** must be **odd** so that the spiral has a single center cell.
- Larger **size** means more numbers ( $\text{size}^2$ ) and a longer run time.
- “Looks-true” trap: visual regularity suggests hypotheses but does not prove theorems about primes.

## Extensions

- Add a “diagonal density” quantitative overlay: estimate prime density per diagonal and plot it.
- Annotate diagonals with the corresponding quadratic polynomials (for small windows).
- Compare multiple sizes side-by-side (same styling and axis conventions).

## References

- Ulam spiral overview and basic properties: [contributors, 2025].
- Popular exposition and recreational-math context: [Gardner, 1983].
- Narrative-style discussion of mathematical discovery/pitfalls (useful for “looks-true” effects): [Hoffman, 1989].

See also *References*.

## Related experiments

- *E002: Even Perfect Numbers — Generator and Growth* (Even Perfect Numbers — Generator and Growth)
- *E003: Abundancy Index Landscape* (Abundancy Index Landscape)
- *E124: Klauber triangle structure* (Klauber triangle structure)
- *E125: Sacks spiral structure* (Sacks spiral structure)
- *E126: Hexagonal number spiral structure* (Hexagonal number spiral structure)

## Parameters (example)

- **size**: 301 (implied grid:  $301 \times 301$ , integers 1..90601)

## Recommended size range

The Ulam spiral is computed on a **size** × **size** grid, so runtime and memory scale roughly with  $\text{size}^2$  (because about  $\text{size}^2$  integers are classified as prime/not-prime).

A practical range that works well for most runs:

- **Minimum (still meaningful):** **size** = 101
- **Default / recommended:** **size** = 301

- **Comfortable upper range (typical laptop):** `size = 801` or `1001`
- **Above `~size = 1501`:** expect noticeably higher runtime (and potentially memory pressure), depending on the primality implementation.

**Rule of thumb:** start with `301`, then try `501`, `801`, `1001`. Increase further only if runtime remains acceptable.

### Summary

We render the classic Ulam spiral for the first 90,601 positive integers and highlight primes on the spiral grid. Even at this moderate scale, faint diagonal “streaks” are visible: some diagonals contain noticeably more primes than nearby diagonals.

### Key observations

- Several diagonals show higher prime density than the surrounding background.
- Many regions still look close to “noise-like” at this resolution; the strongest diagonal effects are easiest to see when zooming.

### Interpretation

The diagonal structure is a real phenomenon and is commonly explained by the fact that many diagonals correspond to simple quadratic polynomials in the spiral index; some quadratics produce unusually many primes for small inputs. This is therefore a useful “pattern detector” for prime-rich formulas — but it is not evidence that primes follow a simple global rule.

### Caveats

- Finite window: patterns can look stronger/weaker depending on the chosen size and zoom level.
- Visualization choices matter: marker size and interpolation can hide or exaggerate diagonal structure.
- Grid mapping details (exact spiral convention and indexing) affect which diagonals appear most prominent.

## 3.3.25 E025: Prime gaps are not monotone

**Tags:** number-theory, quantitative-exploration, visualization See: *Valid Tags*.

### Highlights

- This is a thin wrapper that follows the standard experiment template and delegates the actual computation to `:mod:mathxlab.experiments.prime_suite`.
- Writes reproducible artifacts (`params.json`, `report.md`, and figures).
- Designed to surface patterns *and* “looks-true-until-it-breaks” behavior.

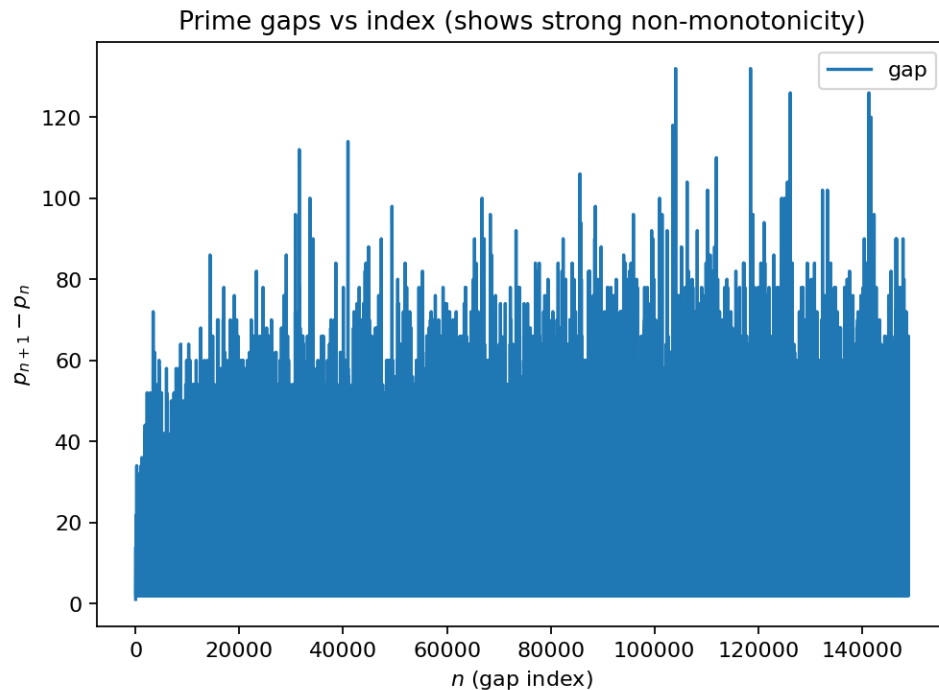
### Goal

This is a thin wrapper that follows the standard experiment template and delegates the actual computation to `:mod:mathxlab.experiments.prime_suite`.

### Background (quick refresher)

- *Prime numbers refresher*





### Research question

Which **prime-related claim, heuristic, or algorithm** breaks first under a clean, controlled computational sweep, and what does the smallest or clearest counterexample (or deviation) look like?

### Why this qualifies as a mathematical experiment

- **Finite procedure:** run a bounded search / sweep with recorded parameters.
- **Observable(s):** counts, gaps, residues, runtime scaling, or first counterexample witnesses.
- **Parameter space:** vary bounds (and sometimes algorithmic choices).
- **Outcome:** plots/tables + “witness objects” for failures.
- **Reproducibility:** outputs saved to `out/e025/` with a parameter snapshot.

### Experiment design

- **Computation:** bounded enumeration / sampling with explicit limits.
- **Outputs:** figures and a short `report.md` summarizing what was found.
- **Artifacts written:**
  - `figures/fig_*.png`
  - `params.json`
  - `report.md`

### How to run

```
make run EXP=e025
```

or:

```
uv run python -m mathxlab.experiments.e025
```

### Notes / pitfalls

- “No counterexample found” only means “none found within the configured bounds”.
- For probabilistic tests (when used), treat outcomes as *evidence*, not proof.

### Extensions

- Increase bounds and rerun (recording runtime and memory).
- Compare alternative heuristics or algorithms on the same parameter grid.
- Turn found deviations into new, tighter conjectures.

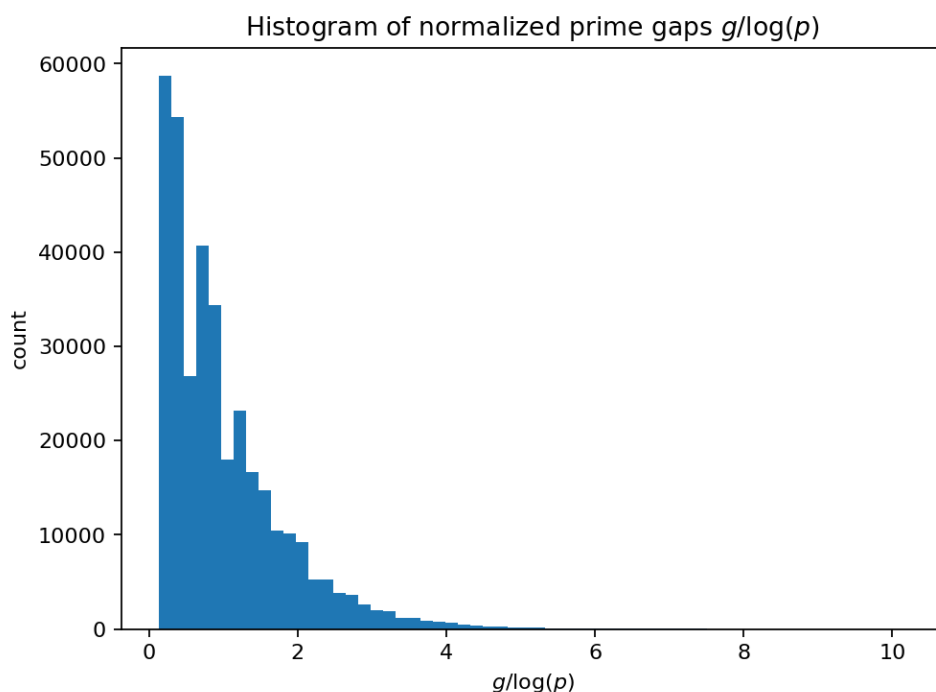
### References

See [References](#).

### Related experiments

- *E026: Normalized prime gaps* (Normalized prime gaps)
- *E028: Jumping champions (most frequent gaps)* (Jumping champions (most frequent gaps))
- *E033: Bounded gaps vs. twin primes (not the same)* (Bounded gaps vs. twin primes (not the same))
- *E027: Record prime gaps vs.  $\log^2$  heuristic* (Record prime gaps vs.  $\log^2$  heuristic)
- *E002: Even Perfect Numbers — Generator and Growth* (Even Perfect Numbers — Generator and Growth)

## 3.3.26 E026: Normalized prime gaps



Tags: number-theory, quantitative-exploration, visualization See: [Valid Tags](#).

## Highlights

- This is a thin wrapper that follows the standard experiment template and delegates the actual computation to `:mod:mathxlab.experiments.prime_suite`.
- Writes reproducible artifacts (`params.json`, `report.md`, and figures).
- Designed to surface patterns *and* “looks-true-until-it-breaks” behavior.

## Goal

This is a thin wrapper that follows the standard experiment template and delegates the actual computation to `:mod:mathxlab.experiments.prime_suite`.

## Background (quick refresher)

- *Prime numbers refresher*

## Research question

Which **prime-related claim, heuristic, or algorithm** breaks first under a clean, controlled computational sweep, and what does the smallest or clearest counterexample (or deviation) look like?

## Why this qualifies as a mathematical experiment

- **Finite procedure:** run a bounded search / sweep with recorded parameters.
- **Observable(s):** counts, gaps, residues, runtime scaling, or first counterexample witnesses.
- **Parameter space:** vary bounds (and sometimes algorithmic choices).
- **Outcome:** plots/tables + “witness objects” for failures.
- **Reproducibility:** outputs saved to `out/e026/` with a parameter snapshot.

## Experiment design

- **Computation:** bounded enumeration / sampling with explicit limits.
- **Outputs:** figures and a short `report.md` summarizing what was found.
- **Artifacts written:**
  - `figures/fig_*.png`
  - `params.json`
  - `report.md`

## How to run

```
make run EXP=e026
```

or:

```
uv run python -m mathxlab.experiments.e026
```

## Notes / pitfalls

- “No counterexample found” only means “none found within the configured bounds”.
- For probabilistic tests (when used), treat outcomes as *evidence*, not proof.

## Extensions

- Increase bounds and rerun (recording runtime and memory).
- Compare alternative heuristics or algorithms on the same parameter grid.
- Turn found deviations into new, tighter conjectures.

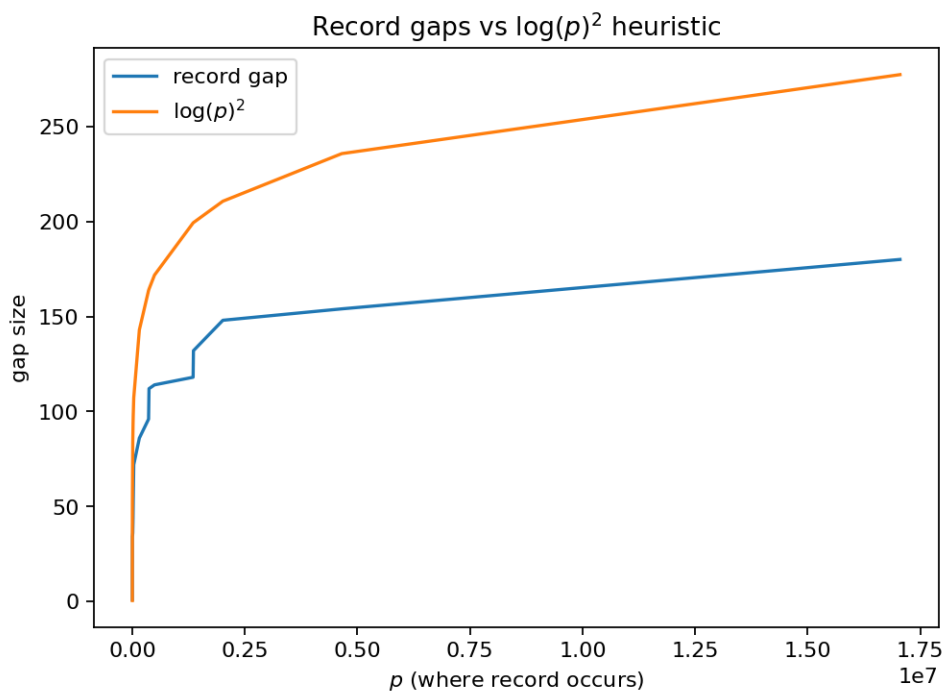
## References

See [References](#).

## Related experiments

- *E025: Prime gaps are not monotone* (Prime gaps are not monotone)
- *E028: Jumping champions (most frequent gaps)* (Jumping champions (most frequent gaps))
- *E033: Bounded gaps vs. twin primes (not the same)* (Bounded gaps vs. twin primes (not the same))
- *E027: Record prime gaps vs.  $\log^2$  heuristic* (Record prime gaps vs.  $\log^2$  heuristic)
- *E002: Even Perfect Numbers — Generator and Growth* (Even Perfect Numbers — Generator and Growth)

### 3.3.27 E027: Record prime gaps vs. $\log^2$ heuristic



Tags: number-theory, conjecture-generation, visualization See: [Valid Tags](#).

## Highlights

- This is a thin wrapper that follows the standard experiment template and delegates the actual computation to `mod:mathxlab.experiments.prime_suite`.
- Writes reproducible artifacts (`params.json`, `report.md`, and figures).
- Designed to surface patterns *and* “looks-true-until-it-breaks” behavior.

## Goal

This is a thin wrapper that follows the standard experiment template and delegates the actual computation to `:mod:mathxlab.experiments.prime_suite`.

## Background (quick refresher)

- *Prime numbers refresher*

## Research question

Which **prime-related claim, heuristic, or algorithm** breaks first under a clean, controlled computational sweep, and what does the smallest or clearest counterexample (or deviation) look like?

## Why this qualifies as a mathematical experiment

- **Finite procedure:** run a bounded search / sweep with recorded parameters.
- **Observable(s):** counts, gaps, residues, runtime scaling, or first counterexample witnesses.
- **Parameter space:** vary bounds (and sometimes algorithmic choices).
- **Outcome:** plots/tables + “witness objects” for failures.
- **Reproducibility:** outputs saved to `out/e027/` with a parameter snapshot.

## Experiment design

- **Computation:** bounded enumeration / sampling with explicit limits.
- **Outputs:** figures and a short `report.md` summarizing what was found.
- **Artifacts written:**
  - `figures/fig_*.png`
  - `params.json`
  - `report.md`

## How to run

```
make run EXP=e027
```

or:

```
uv run python -m mathxlab.experiments.e027
```

## Notes / pitfalls

- “No counterexample found” only means “none found within the configured bounds”.
- For probabilistic tests (when used), treat outcomes as *evidence*, not proof.

## Extensions

- Increase bounds and rerun (recording runtime and memory).
- Compare alternative heuristics or algorithms on the same parameter grid.
- Turn found deviations into new, tighter conjectures.

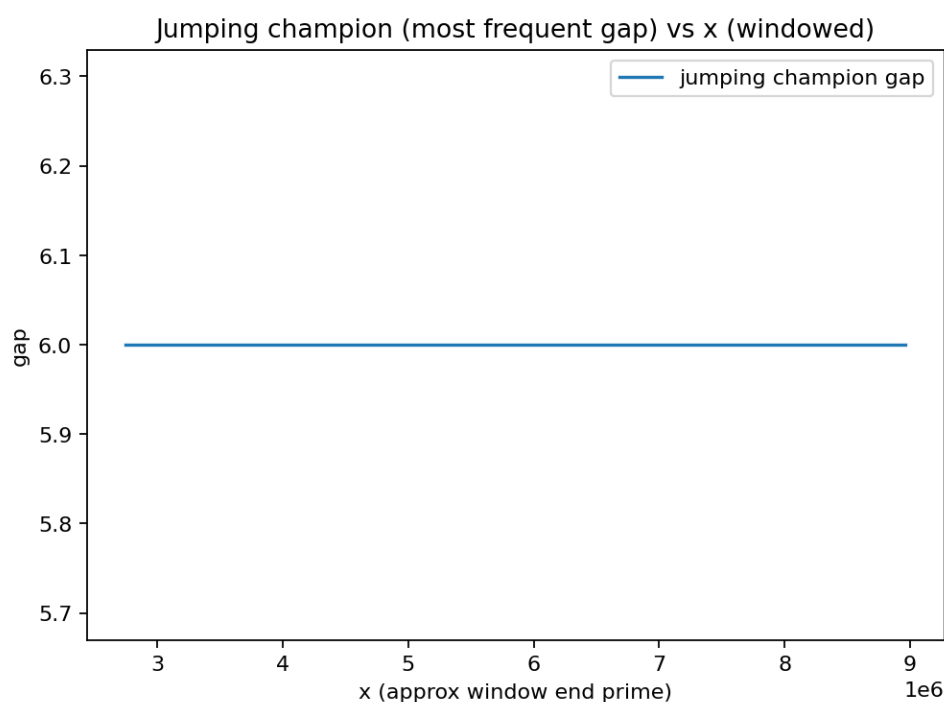
## References

See *References*.

## Related experiments

- *E029: Twin primes: observed vs. heuristic* (Twin primes: observed vs. heuristic)
- *E006: Near Misses to Perfection* (Near Misses to Perfection)
- *E011: Heuristic rarity of Mersenne primes* (Heuristic rarity of Mersenne primes)
- *E025: Prime gaps are not monotone* (Prime gaps are not monotone)
- *E026: Normalized prime gaps* (Normalized prime gaps)

### 3.3.28 E028: Jumping champions (most frequent gaps)



**Tags:** number-theory, quantitative-exploration, visualization See: *Valid Tags*.

## Highlights

- This is a thin wrapper that follows the standard experiment template and delegates the actual computation to `:mod:mathxlab.experiments.prime_suite`.
- Writes reproducible artifacts (`params.json`, `report.md`, and figures).
- Designed to surface patterns *and* “looks-true-until-it-breaks” behavior.

## Goal

This is a thin wrapper that follows the standard experiment template and delegates the actual computation to `:mod:mathxlab.experiments.prime_suite`.

## Background (quick refresher)

- *Prime numbers refresher*

## Research question

Which **prime-related claim, heuristic, or algorithm** breaks first under a clean, controlled computational sweep, and what does the smallest or clearest counterexample (or deviation) look like?

## Why this qualifies as a mathematical experiment

- **Finite procedure:** run a bounded search / sweep with recorded parameters.
- **Observable(s):** counts, gaps, residues, runtime scaling, or first counterexample witnesses.
- **Parameter space:** vary bounds (and sometimes algorithmic choices).
- **Outcome:** plots/tables + “witness objects” for failures.
- **Reproducibility:** outputs saved to `out/e028/` with a parameter snapshot.

## Experiment design

- **Computation:** bounded enumeration / sampling with explicit limits.
- **Outputs:** figures and a short `report.md` summarizing what was found.
- **Artifacts written:**
  - `figures/fig_*.png`
  - `params.json`
  - `report.md`

## How to run

```
make run EXP=e028
```

or:

```
uv run python -m mathxlab.experiments.e028
```

## Notes / pitfalls

- “No counterexample found” only means “none found within the configured bounds”.
- For probabilistic tests (when used), treat outcomes as *evidence*, not proof.

## Extensions

- Increase bounds and rerun (recording runtime and memory).
- Compare alternative heuristics or algorithms on the same parameter grid.
- Turn found deviations into new, tighter conjectures.

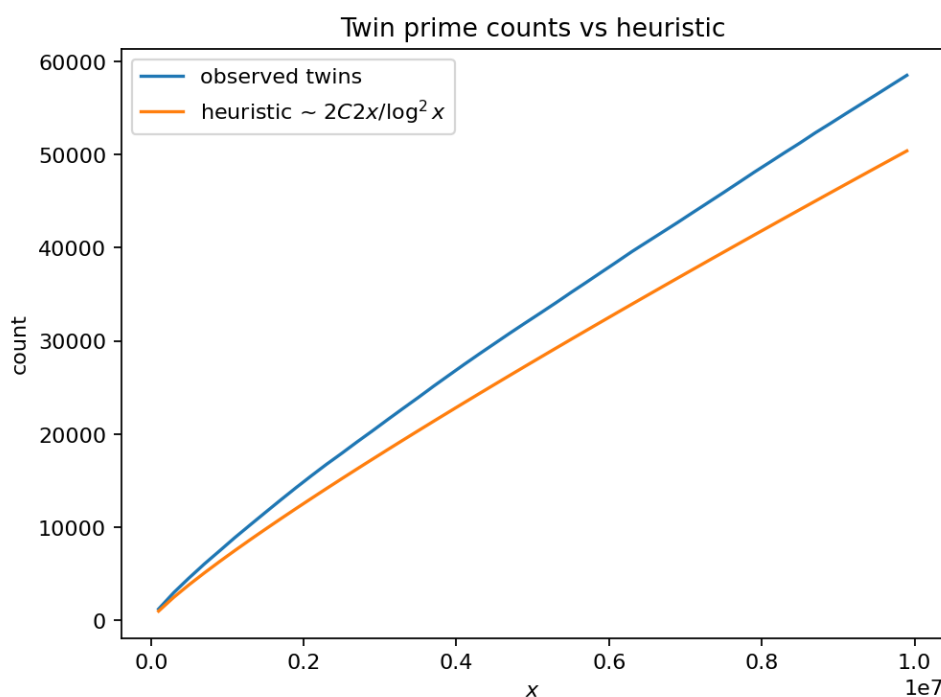
## References

See *References*.

### Related experiments

- *E025: Prime gaps are not monotone* (Prime gaps are not monotone)
- *E026: Normalized prime gaps* (Normalized prime gaps)
- *E033: Bounded gaps vs. twin primes (not the same)* (Bounded gaps vs. twin primes (not the same))
- *E027: Record prime gaps vs.  $\log^2$  heuristic* (Record prime gaps vs.  $\log^2$  heuristic)
- *E002: Even Perfect Numbers — Generator and Growth* (Even Perfect Numbers — Generator and Growth)

### 3.3.29 E029: Twin primes: observed vs. heuristic



Tags: number-theory, conjecture-generation, visualization See: [Valid Tags](#).

### Highlights

- This is a thin wrapper that follows the standard experiment template and delegates the actual computation to `:mod:mathxlab.experiments.prime_suite`.
- Writes reproducible artifacts (`params.json`, `report.md`, and figures).
- Designed to surface patterns *and* “looks-true-until-it-breaks” behavior.

### Goal

This is a thin wrapper that follows the standard experiment template and delegates the actual computation to `:mod:mathxlab.experiments.prime_suite`.

### Background (quick refresher)

- *Prime numbers refresher*



### Research question

Which **prime-related claim, heuristic, or algorithm** breaks first under a clean, controlled computational sweep, and what does the smallest or clearest counterexample (or deviation) look like?

### Why this qualifies as a mathematical experiment

- **Finite procedure:** run a bounded search / sweep with recorded parameters.
- **Observable(s):** counts, gaps, residues, runtime scaling, or first counterexample witnesses.
- **Parameter space:** vary bounds (and sometimes algorithmic choices).
- **Outcome:** plots/tables + “witness objects” for failures.
- **Reproducibility:** outputs saved to `out/e029/` with a parameter snapshot.

### Experiment design

- **Computation:** bounded enumeration / sampling with explicit limits.
- **Outputs:** figures and a short `report.md` summarizing what was found.
- **Artifacts written:**
  - `figures/fig_*.png`
  - `params.json`
  - `report.md`

### How to run

```
make run EXP=e029
```

or:

```
uv run python -m mathxlab.experiments.e029
```

### Notes / pitfalls

- “No counterexample found” only means “none found within the configured bounds”.
- For probabilistic tests (when used), treat outcomes as *evidence*, not proof.

### Extensions

- Increase bounds and rerun (recording runtime and memory).
- Compare alternative heuristics or algorithms on the same parameter grid.
- Turn found deviations into new, tighter conjectures.

### References

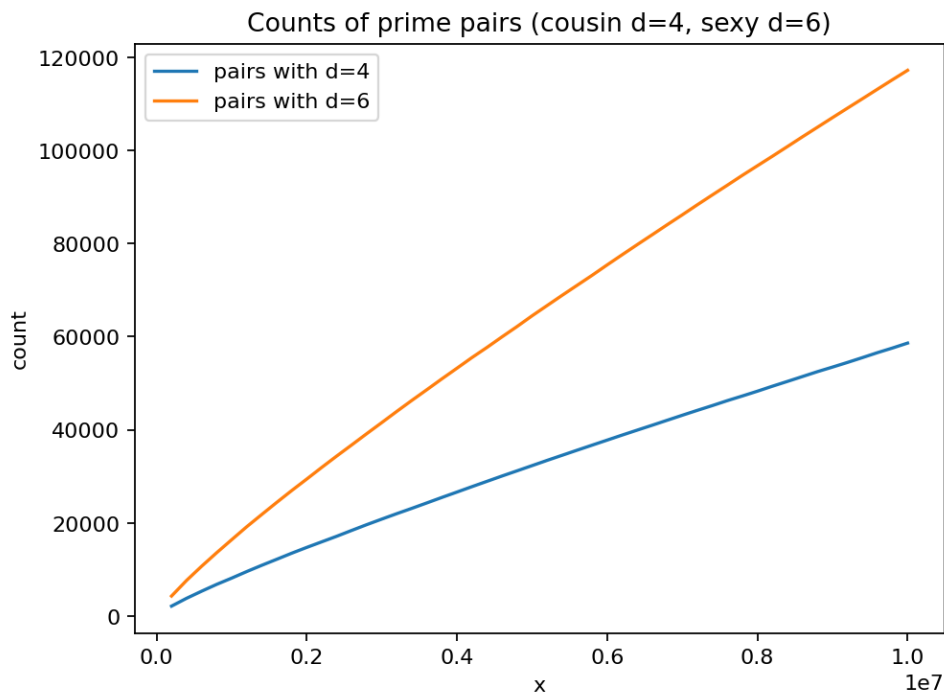
See *References*.

### Related experiments

- *E027: Record prime gaps vs.  $\log^2$  heuristic* (Record prime gaps vs.  $\log^2$  heuristic)
- *E006: Near Misses to Perfection* (Near Misses to Perfection)
- *E011: Heuristic rarity of Mersenne primes* (Heuristic rarity of Mersenne primes)
- *E033: Bounded gaps vs. twin primes (not the same)* (Bounded gaps vs. twin primes (not the same))

- *E034: Twin primes in sliding windows* (Twin primes in sliding windows)

### 3.3.30 E030: Cousin and sexy prime pairs



Tags: number-theory, quantitative-exploration, visualization See: [Valid Tags](#).

#### Highlights

- This is a thin wrapper that follows the standard experiment template and delegates the actual computation to `:mod:mathxlab.experiments.prime_suite`.
- Writes reproducible artifacts (`params.json`, `report.md`, and figures).
- Designed to surface patterns *and* “looks-true-until-it-breaks” behavior.

#### Goal

This is a thin wrapper that follows the standard experiment template and delegates the actual computation to `:mod:mathxlab.experiments.prime_suite`.

#### Background (quick refresher)

- *Prime numbers refresher*

#### Research question

Which **prime-related claim, heuristic, or algorithm** breaks first under a clean, controlled computational sweep, and what does the smallest or clearest counterexample (or deviation) look like?

#### Why this qualifies as a mathematical experiment

- **Finite procedure:** run a bounded search / sweep with recorded parameters.
- **Observable(s):** counts, gaps, residues, runtime scaling, or first counterexample witnesses.
- **Parameter space:** vary bounds (and sometimes algorithmic choices).
- **Outcome:** plots/tables + “witness objects” for failures.

- **Reproducibility:** outputs saved to `out/e030/` with a parameter snapshot.

### Experiment design

- **Computation:** bounded enumeration / sampling with explicit limits.
- **Outputs:** figures and a short `report.md` summarizing what was found.
- **Artifacts written:**
  - `figures/fig_*.png`
  - `params.json`
  - `report.md`

### How to run

```
make run EXP=e030
```

or:

```
uv run python -m mathxlab.experiments.e030
```

### Notes / pitfalls

- “No counterexample found” only means “none found within the configured bounds”.
- For probabilistic tests (when used), treat outcomes as *evidence*, not proof.

### Extensions

- Increase bounds and rerun (recording runtime and memory).
- Compare alternative heuristics or algorithms on the same parameter grid.
- Turn found deviations into new, tighter conjectures.

### References

See [References](#).

### Related experiments

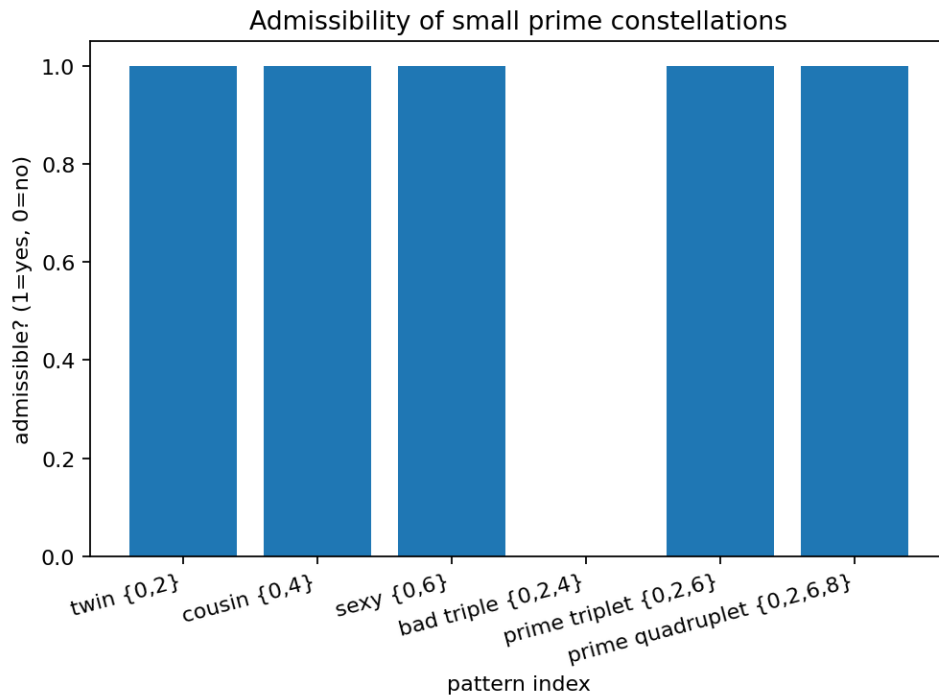
- *E002: Even Perfect Numbers — Generator and Growth* (Even Perfect Numbers — Generator and Growth)
- *E003: Abundancy Index Landscape* (Abundancy Index Landscape)
- *E007: Mersenne growth (bits and digits)* (Mersenne growth (bits and digits))
- *E008: Lucas–Lehmer scan (prime exponents)* (Lucas–Lehmer scan (prime exponents))
- *E009: Small-factor scan for Mersenne numbers* (Small-factor scan for Mersenne numbers)

### 3.3.31 E031: Admissibility and modular obstructions

**Tags:** number-theory, quantitative-exploration, visualization See: [Valid Tags](#).

### Highlights

- This is a thin wrapper that follows the standard experiment template and delegates the actual computation to `:mod:mathxlab.experiments.prime_suite`.
- Writes reproducible artifacts (`params.json`, `report.md`, and figures).
- Designed to surface patterns *and* “looks-true-until-it-breaks” behavior.



### Goal

This is a thin wrapper that follows the standard experiment template and delegates the actual computation to `:mod:mathxlab.experiments.prime_suite`.

### Background (quick refresher)

- *Prime numbers refresher*

### Research question

Which **prime-related claim, heuristic, or algorithm** breaks first under a clean, controlled computational sweep, and what does the smallest or clearest counterexample (or deviation) look like?

### Why this qualifies as a mathematical experiment

- **Finite procedure:** run a bounded search / sweep with recorded parameters.
- **Observable(s):** counts, gaps, residues, runtime scaling, or first counterexample witnesses.
- **Parameter space:** vary bounds (and sometimes algorithmic choices).
- **Outcome:** plots/tables + “witness objects” for failures.
- **Reproducibility:** outputs saved to `out/e031/` with a parameter snapshot.

### Experiment design

- **Computation:** bounded enumeration / sampling with explicit limits.
- **Outputs:** figures and a short `report.md` summarizing what was found.
- **Artifacts written:**
  - `figures/fig_*.png`
  - `params.json`
  - `report.md`

## How to run

```
make run EXP=e031
```

or:

```
uv run python -m mathxlab.experiments.e031
```

## Notes / pitfalls

- “No counterexample found” only means “none found within the configured bounds”.
- For probabilistic tests (when used), treat outcomes as *evidence*, not proof.

## Extensions

- Increase bounds and rerun (recording runtime and memory).
- Compare alternative heuristics or algorithms on the same parameter grid.
- Turn found deviations into new, tighter conjectures.

## References

See *References*.

## Related experiments

- *E002: Even Perfect Numbers — Generator and Growth* (Even Perfect Numbers — Generator and Growth)
- *E003: Abundancy Index Landscape* (Abundancy Index Landscape)
- *E007: Mersenne growth (bits and digits)* (Mersenne growth (bits and digits))
- *E008: Lucas–Lehmer scan (prime exponents)* (Lucas–Lehmer scan (prime exponents))
- *E009: Small-factor scan for Mersenne numbers* (Small-factor scan for Mersenne numbers)

### 3.3.32 E032: Prime triplets and quadruplets

**Tags:** number-theory, quantitative-exploration, visualization See: *Valid Tags*.

## Highlights

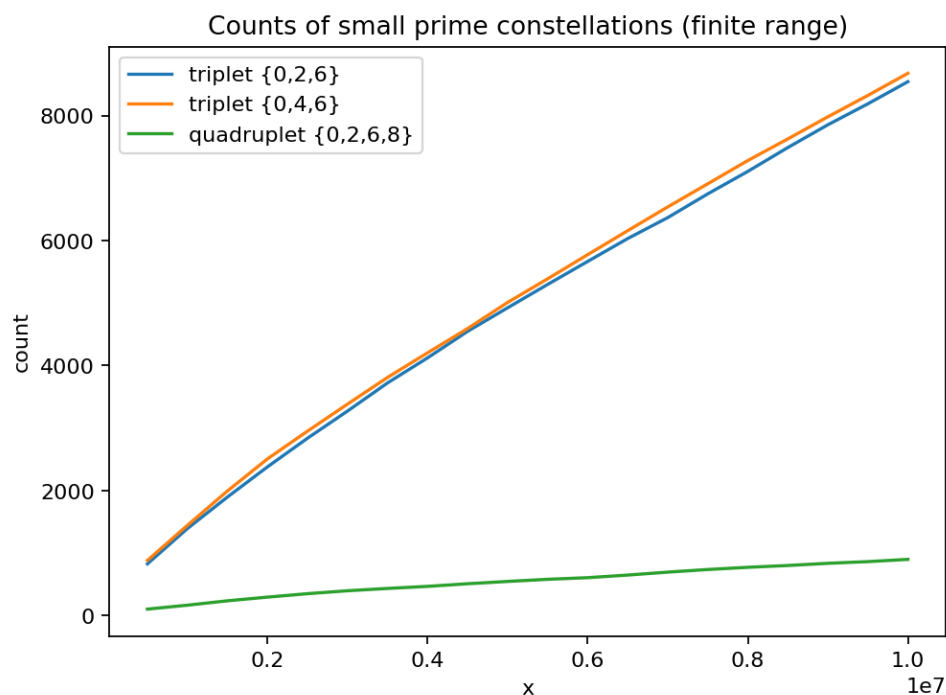
- This is a thin wrapper that follows the standard experiment template and delegates the actual computation to `:mod:mathxlab.experiments.prime_suite`.
- Writes reproducible artifacts (`params.json`, `report.md`, and figures).
- Designed to surface patterns *and* “looks-true-until-it-breaks” behavior.

## Goal

This is a thin wrapper that follows the standard experiment template and delegates the actual computation to `:mod:mathxlab.experiments.prime_suite`.

## Background (quick refresher)

- *Prime numbers refresher*



### Research question

Which **prime-related claim, heuristic, or algorithm** breaks first under a clean, controlled computational sweep, and what does the smallest or clearest counterexample (or deviation) look like?

### Why this qualifies as a mathematical experiment

- **Finite procedure:** run a bounded search / sweep with recorded parameters.
- **Observable(s):** counts, gaps, residues, runtime scaling, or first counterexample witnesses.
- **Parameter space:** vary bounds (and sometimes algorithmic choices).
- **Outcome:** plots/tables + “witness objects” for failures.
- **Reproducibility:** outputs saved to `out/e032/` with a parameter snapshot.

### Experiment design

- **Computation:** bounded enumeration / sampling with explicit limits.
- **Outputs:** figures and a short `report.md` summarizing what was found.
- **Artifacts written:**
  - `figures/fig_*.png`
  - `params.json`
  - `report.md`

### How to run

```
make run EXP=e032
```

or:

```
uv run python -m mathxlab.experiments.e032
```

### Notes / pitfalls

- “No counterexample found” only means “none found within the configured bounds”.
- For probabilistic tests (when used), treat outcomes as *evidence*, not proof.

### Extensions

- Increase bounds and rerun (recording runtime and memory).
- Compare alternative heuristics or algorithms on the same parameter grid.
- Turn found deviations into new, tighter conjectures.

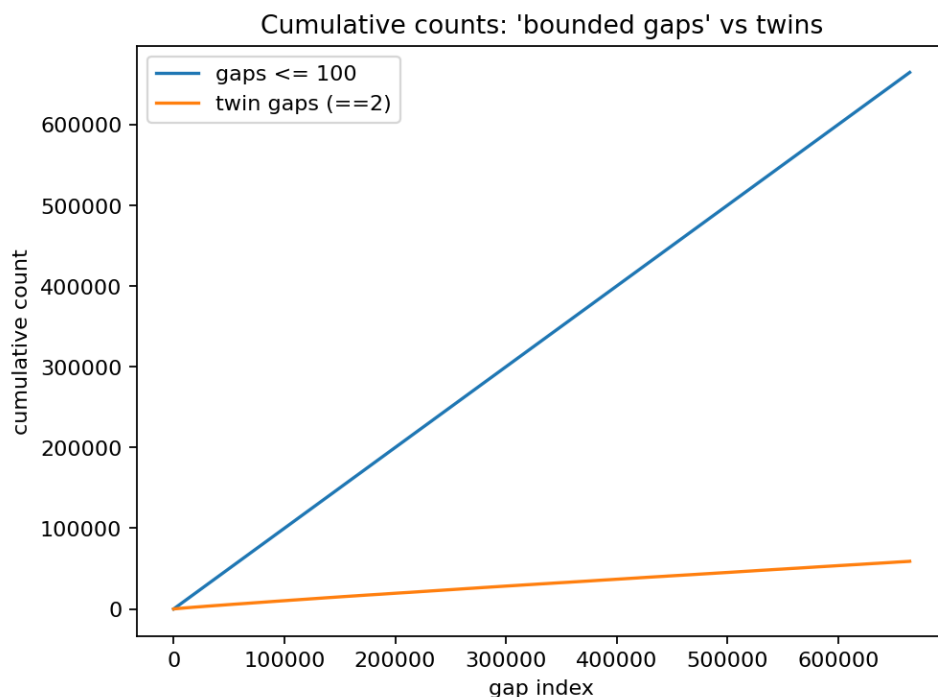
### References

See [References](#).

### Related experiments

- *E002: Even Perfect Numbers — Generator and Growth* (Even Perfect Numbers — Generator and Growth)
- *E003: Abundancy Index Landscape* (Abundancy Index Landscape)
- *E007: Mersenne growth (bits and digits)* (Mersenne growth (bits and digits))
- *E008: Lucas–Lehmer scan (prime exponents)* (Lucas–Lehmer scan (prime exponents))
- *E009: Small-factor scan for Mersenne numbers* (Small-factor scan for Mersenne numbers)

### 3.3.33 E033: Bounded gaps vs. twin primes (not the same)



Tags: number-theory, quantitative-exploration, visualization See: [Valid Tags](#).

## Highlights

- This is a thin wrapper that follows the standard experiment template and delegates the actual computation to `:mod:mathxlab.experiments.prime_suite`.
- Writes reproducible artifacts (`params.json`, `report.md`, and figures).
- Designed to surface patterns *and* “looks-true-until-it-breaks” behavior.

## Goal

This is a thin wrapper that follows the standard experiment template and delegates the actual computation to `:mod:mathxlab.experiments.prime_suite`.

## Background (quick refresher)

- *Prime numbers refresher*

## Research question

Which **prime-related claim, heuristic, or algorithm** breaks first under a clean, controlled computational sweep, and what does the smallest or clearest counterexample (or deviation) look like?

## Why this qualifies as a mathematical experiment

- **Finite procedure:** run a bounded search / sweep with recorded parameters.
- **Observable(s):** counts, gaps, residues, runtime scaling, or first counterexample witnesses.
- **Parameter space:** vary bounds (and sometimes algorithmic choices).
- **Outcome:** plots/tables + “witness objects” for failures.
- **Reproducibility:** outputs saved to `out/e033/` with a parameter snapshot.

## Experiment design

- **Computation:** bounded enumeration / sampling with explicit limits.
- **Outputs:** figures and a short `report.md` summarizing what was found.
- **Artifacts written:**
  - `figures/fig_*.png`
  - `params.json`
  - `report.md`

## How to run

```
make run EXP=e033
```

or:

```
uv run python -m mathxlab.experiments.e033
```

## Notes / pitfalls

- “No counterexample found” only means “none found within the configured bounds”.
- For probabilistic tests (when used), treat outcomes as *evidence*, not proof.



## Extensions

- Increase bounds and rerun (recording runtime and memory).
- Compare alternative heuristics or algorithms on the same parameter grid.
- Turn found deviations into new, tighter conjectures.

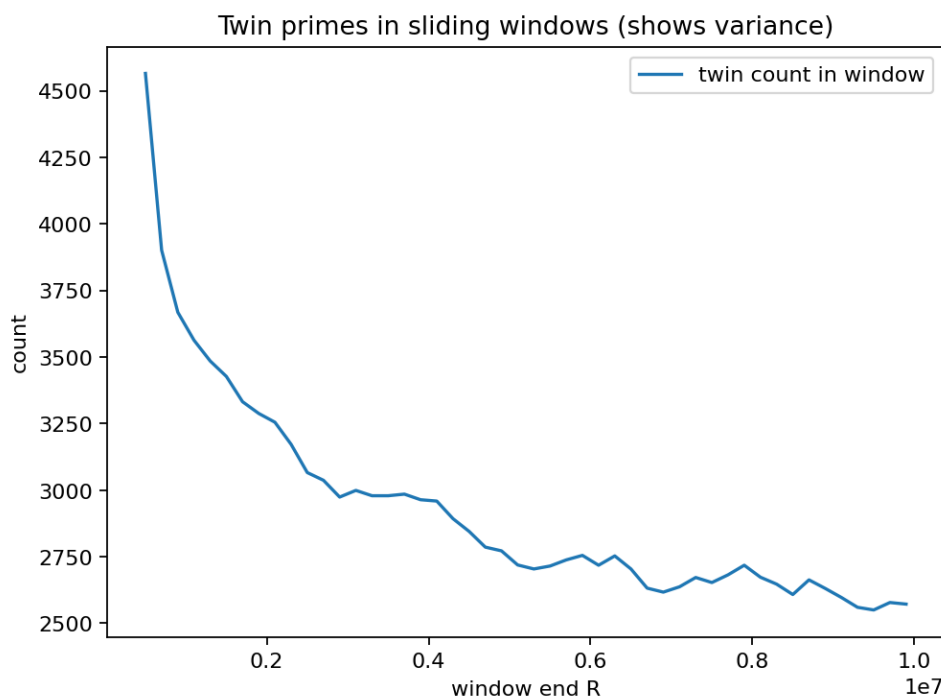
## References

See [References](#).

## Related experiments

- *E025: Prime gaps are not monotone* (Prime gaps are not monotone)
- *E026: Normalized prime gaps* (Normalized prime gaps)
- *E028: Jumping champions (most frequent gaps)* (Jumping champions (most frequent gaps))
- *E034: Twin primes in sliding windows* (Twin primes in sliding windows)
- *E027: Record prime gaps vs.  $\log^2$  heuristic* (Record prime gaps vs.  $\log^2$  heuristic)

### 3.3.34 E034: Twin primes in sliding windows



Tags: number-theory, quantitative-exploration, visualization See: [Valid Tags](#).

## Highlights

- This is a thin wrapper that follows the standard experiment template and delegates the actual computation to `mod:mathxlab.experiments.prime_suite`.
- Writes reproducible artifacts (`params.json`, `report.md`, and figures).
- Designed to surface patterns *and* “looks-true-until-it-breaks” behavior.

## Goal

This is a thin wrapper that follows the standard experiment template and delegates the actual computation to `:mod:mathxlab.experiments.prime_suite`.

## Background (quick refresher)

- *Prime numbers refresher*

## Research question

Which **prime-related claim, heuristic, or algorithm** breaks first under a clean, controlled computational sweep, and what does the smallest or clearest counterexample (or deviation) look like?

## Why this qualifies as a mathematical experiment

- **Finite procedure:** run a bounded search / sweep with recorded parameters.
- **Observable(s):** counts, gaps, residues, runtime scaling, or first counterexample witnesses.
- **Parameter space:** vary bounds (and sometimes algorithmic choices).
- **Outcome:** plots/tables + “witness objects” for failures.
- **Reproducibility:** outputs saved to `out/e034/` with a parameter snapshot.

## Experiment design

- **Computation:** bounded enumeration / sampling with explicit limits.
- **Outputs:** figures and a short `report.md` summarizing what was found.
- **Artifacts written:**
  - `figures/fig_*.png`
  - `params.json`
  - `report.md`

## How to run

```
make run EXP=e034
```

or:

```
uv run python -m mathxlab.experiments.e034
```

## Notes / pitfalls

- “No counterexample found” only means “none found within the configured bounds”.
- For probabilistic tests (when used), treat outcomes as *evidence*, not proof.

## Extensions

- Increase bounds and rerun (recording runtime and memory).
- Compare alternative heuristics or algorithms on the same parameter grid.
- Turn found deviations into new, tighter conjectures.

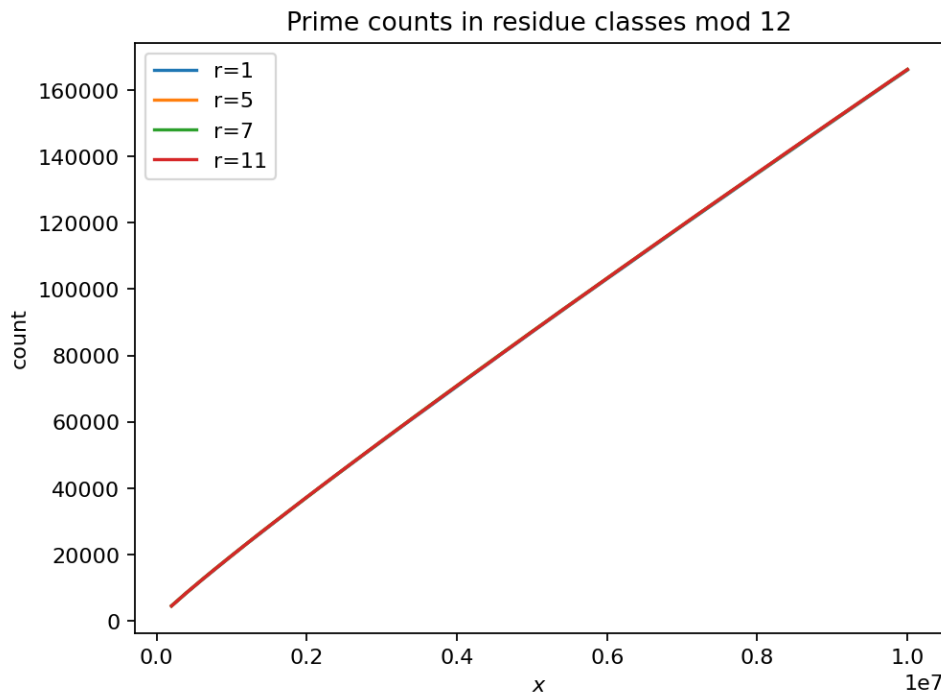
## References

See *References*.

## Related experiments

- *E033: Bounded gaps vs. twin primes (not the same)* (Bounded gaps vs. twin primes (not the same))
- *E029: Twin primes: observed vs. heuristic* (Twin primes: observed vs. heuristic)
- *E002: Even Perfect Numbers — Generator and Growth* (Even Perfect Numbers — Generator and Growth)
- *E003: Abundancy Index Landscape* (Abundancy Index Landscape)
- *E007: Mersenne growth (bits and digits)* (Mersenne growth (bits and digits))

### 3.3.35 E035: Primes in arithmetic progressions mod q



**Tags:** number-theory, quantitative-exploration, visualization See: *Valid Tags*.

## Highlights

- This is a thin wrapper that follows the standard experiment template and delegates the actual computation to `:mod:mathxlab.experiments.prime_suite`.
- Writes reproducible artifacts (`params.json`, `report.md`, and figures).
- Designed to surface patterns *and* “looks-true-until-it-breaks” behavior.

## Goal

This is a thin wrapper that follows the standard experiment template and delegates the actual computation to `:mod:mathxlab.experiments.prime_suite`.

## Background (quick refresher)

- *Prime numbers refresher*

## Research question

Which **prime-related claim, heuristic, or algorithm** breaks first under a clean, controlled computational sweep, and what does the smallest or clearest counterexample (or deviation) look like?

## Why this qualifies as a mathematical experiment

- **Finite procedure:** run a bounded search / sweep with recorded parameters.
- **Observable(s):** counts, gaps, residues, runtime scaling, or first counterexample witnesses.
- **Parameter space:** vary bounds (and sometimes algorithmic choices).
- **Outcome:** plots/tables + “witness objects” for failures.
- **Reproducibility:** outputs saved to `out/e035/` with a parameter snapshot.

## Experiment design

- **Computation:** bounded enumeration / sampling with explicit limits.
- **Outputs:** figures and a short `report.md` summarizing what was found.
- **Artifacts written:**
  - `figures/fig_*.png`
  - `params.json`
  - `report.md`

## How to run

```
make run EXP=e035
```

or:

```
uv run python -m mathxlab.experiments.e035
```

## Notes / pitfalls

- “No counterexample found” only means “none found within the configured bounds”.
- For probabilistic tests (when used), treat outcomes as *evidence*, not proof.

## Extensions

- Increase bounds and rerun (recording runtime and memory).
- Compare alternative heuristics or algorithms on the same parameter grid.
- Turn found deviations into new, tighter conjectures.

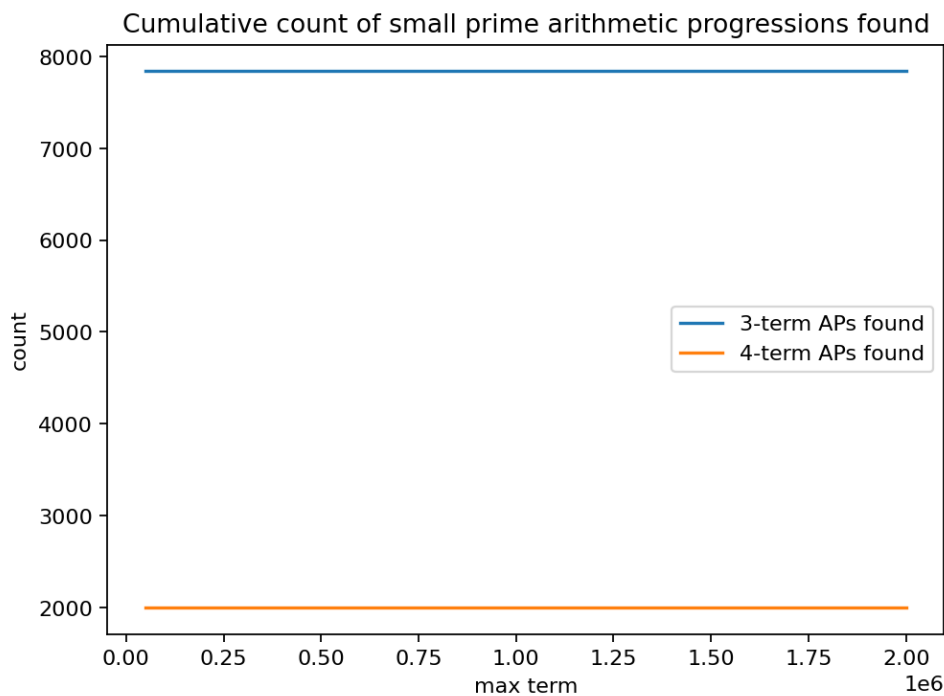
## References

See *References*.

## Related experiments

- *E036: Prime arithmetic progressions (small search)* (Prime arithmetic progressions (small search))
- *E076: Chebyshev  $(x;q,a)$ : weighted prime counts in progressions.* (Chebyshev  $(x;q,a)$ : weighted prime counts in progressions.)
- *E002: Even Perfect Numbers — Generator and Growth* (Even Perfect Numbers — Generator and Growth)
- *E003: Abundancy Index Landscape* (Abundancy Index Landscape)
- *E007: Mersenne growth (bits and digits)* (Mersenne growth (bits and digits))

### 3.3.36 E036: Prime arithmetic progressions (small search)



Tags: number-theory, quantitative-exploration, visualization See: [Valid Tags](#).

## Highlights

- This is a thin wrapper that follows the standard experiment template and delegates the actual computation to `:mod:mathxlab.experiments.prime_suite`.
- Writes reproducible artifacts (`params.json`, `report.md`, and figures).
- Designed to surface patterns *and* “looks-true-until-it-breaks” behavior.

## Goal

This is a thin wrapper that follows the standard experiment template and delegates the actual computation to `:mod:mathxlab.experiments.prime_suite`.

## Background (quick refresher)

- *Prime numbers refresher*

### Research question

Which **prime-related claim, heuristic, or algorithm** breaks first under a clean, controlled computational sweep, and what does the smallest or clearest counterexample (or deviation) look like?

### Why this qualifies as a mathematical experiment

- **Finite procedure:** run a bounded search / sweep with recorded parameters.
- **Observable(s):** counts, gaps, residues, runtime scaling, or first counterexample witnesses.
- **Parameter space:** vary bounds (and sometimes algorithmic choices).
- **Outcome:** plots/tables + “witness objects” for failures.
- **Reproducibility:** outputs saved to `out/e036/` with a parameter snapshot.

### Experiment design

- **Computation:** bounded enumeration / sampling with explicit limits.
- **Outputs:** figures and a short `report.md` summarizing what was found.
- **Artifacts written:**
  - `figures/fig_*.png`
  - `params.json`
  - `report.md`

### How to run

```
make run EXP=e036
```

or:

```
uv run python -m mathxlab.experiments.e036
```

### Notes / pitfalls

- “No counterexample found” only means “none found within the configured bounds”.
- For probabilistic tests (when used), treat outcomes as *evidence*, not proof.

### Extensions

- Increase bounds and rerun (recording runtime and memory).
- Compare alternative heuristics or algorithms on the same parameter grid.
- Turn found deviations into new, tighter conjectures.

### References

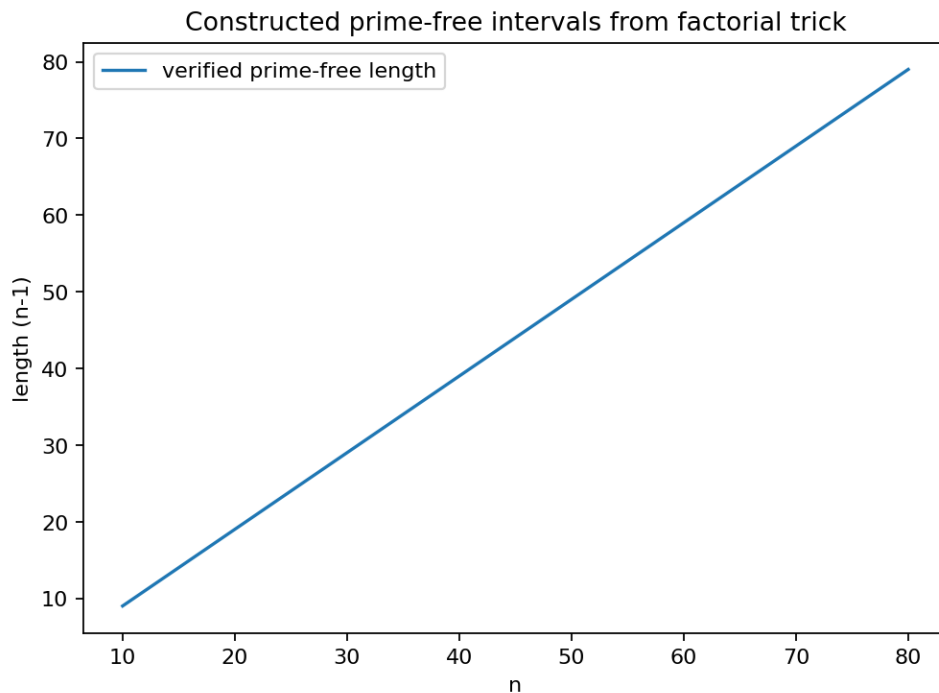
See *References*.

### Related experiments

- *E035: Primes in arithmetic progressions mod  $q$*  (Primes in arithmetic progressions mod  $q$ )
- *E009: Small-factor scan for Mersenne numbers* (Small-factor scan for Mersenne numbers)
- *E042: Repunit primes (small  $k$  scan)* (Repunit primes (small  $k$  scan))
- *E076: Chebyshev  $(x;q,a)$ : weighted prime counts in progressions.* (Chebyshev  $(x;q,a)$ : weighted prime counts in progressions.)

- *E002: Even Perfect Numbers — Generator and Growth* (Even Perfect Numbers — Generator and Growth)

### 3.3.37 E037: Prime-free intervals via factorial construction



Tags: number-theory, quantitative-exploration, visualization See: [Valid Tags](#).

#### Highlights

- This is a thin wrapper that follows the standard experiment template and delegates the actual computation to `:mod:mathxlab.experiments.prime_suite`.
- Writes reproducible artifacts (`params.json`, `report.md`, and figures).
- Designed to surface patterns *and* “looks-true-until-it-breaks” behavior.

#### Goal

This is a thin wrapper that follows the standard experiment template and delegates the actual computation to `:mod:mathxlab.experiments.prime_suite`.

#### Background (quick refresher)

- *Prime numbers refresher*

#### Research question

Which **prime-related claim, heuristic, or algorithm** breaks first under a clean, controlled computational sweep, and what does the smallest or clearest counterexample (or deviation) look like?

#### Why this qualifies as a mathematical experiment

- **Finite procedure:** run a bounded search / sweep with recorded parameters.
- **Observable(s):** counts, gaps, residues, runtime scaling, or first counterexample witnesses.
- **Parameter space:** vary bounds (and sometimes algorithmic choices).
- **Outcome:** plots/tables + “witness objects” for failures.

- **Reproducibility:** outputs saved to `out/e037/` with a parameter snapshot.

### Experiment design

- **Computation:** bounded enumeration / sampling with explicit limits.
- **Outputs:** figures and a short `report.md` summarizing what was found.
- **Artifacts written:**
  - `figures/fig_*.png`
  - `params.json`
  - `report.md`

### How to run

```
make run EXP=e037
```

or:

```
uv run python -m mathxlab.experiments.e037
```

### Notes / pitfalls

- “No counterexample found” only means “none found within the configured bounds”.
- For probabilistic tests (when used), treat outcomes as *evidence*, not proof.

### Extensions

- Increase bounds and rerun (recording runtime and memory).
- Compare alternative heuristics or algorithms on the same parameter grid.
- Turn found deviations into new, tighter conjectures.

### References

See *References*.

### Related experiments

- *E002: Even Perfect Numbers — Generator and Growth* (Even Perfect Numbers — Generator and Growth)
- *E003: Abundancy Index Landscape* (Abundancy Index Landscape)
- *E007: Mersenne growth (bits and digits)* (Mersenne growth (bits and digits))
- *E008: Lucas–Lehmer scan (prime exponents)* (Lucas–Lehmer scan (prime exponents))
- *E009: Small-factor scan for Mersenne numbers* (Small-factor scan for Mersenne numbers)

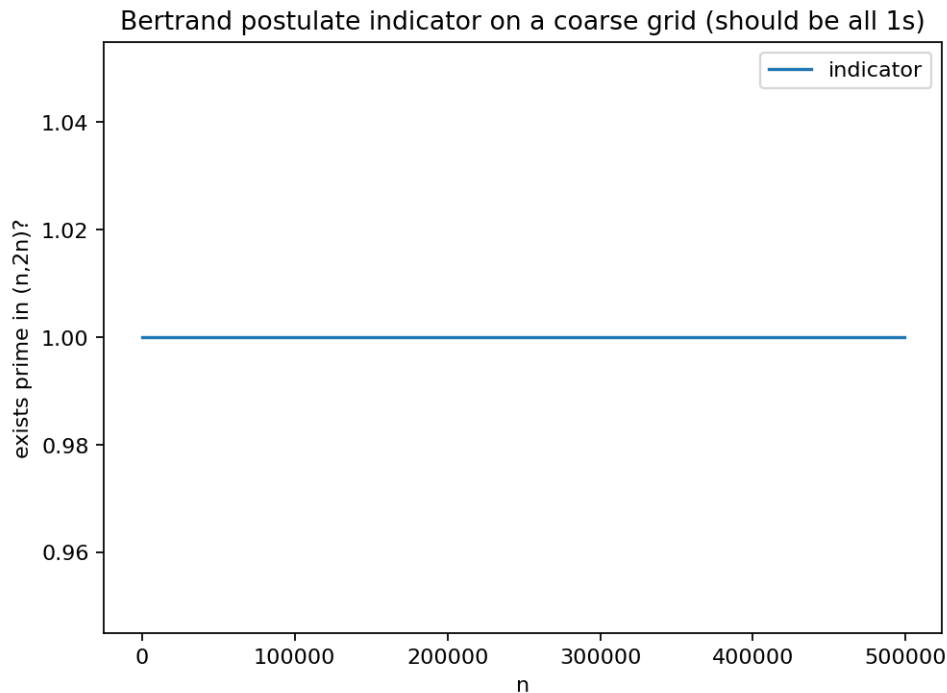
## 3.3.38 E038: Bertrand’s postulate (computational verification)

**Tags:** number-theory, quantitative-exploration, visualization See: *Valid Tags*.

### Highlights

- This is a thin wrapper that follows the standard experiment template and delegates the actual computation to `:mod:mathxlab.experiments.prime_suite`.
- Writes reproducible artifacts (`params.json`, `report.md`, and figures).
- Designed to surface patterns *and* “looks-true-until-it-breaks” behavior.





### Goal

This is a thin wrapper that follows the standard experiment template and delegates the actual computation to `:mod:mathxlab.experiments.prime_suite`.

### Background (quick refresher)

- *Prime numbers refresher*

### Research question

Which **prime-related claim, heuristic, or algorithm** breaks first under a clean, controlled computational sweep, and what does the smallest or clearest counterexample (or deviation) look like?

### Why this qualifies as a mathematical experiment

- **Finite procedure:** run a bounded search / sweep with recorded parameters.
- **Observable(s):** counts, gaps, residues, runtime scaling, or first counterexample witnesses.
- **Parameter space:** vary bounds (and sometimes algorithmic choices).
- **Outcome:** plots/tables + “witness objects” for failures.
- **Reproducibility:** outputs saved to `out/e038/` with a parameter snapshot.

### Experiment design

- **Computation:** bounded enumeration / sampling with explicit limits.
- **Outputs:** figures and a short `report.md` summarizing what was found.
- **Artifacts written:**
  - `figures/fig_*.png`
  - `params.json`
  - `report.md`

## How to run

```
make run EXP=e038
```

or:

```
uv run python -m mathxlab.experiments.e038
```

## Notes / pitfalls

- “No counterexample found” only means “none found within the configured bounds”.
- For probabilistic tests (when used), treat outcomes as *evidence*, not proof.

## Extensions

- Increase bounds and rerun (recording runtime and memory).
- Compare alternative heuristics or algorithms on the same parameter grid.
- Turn found deviations into new, tighter conjectures.

## References

See *References*.

## Related experiments

- *E002: Even Perfect Numbers — Generator and Growth* (Even Perfect Numbers — Generator and Growth)
- *E003: Abundancy Index Landscape* (Abundancy Index Landscape)
- *E007: Mersenne growth (bits and digits)* (Mersenne growth (bits and digits))
- *E008: Lucas–Lehmer scan (prime exponents)* (Lucas–Lehmer scan (prime exponents))
- *E009: Small-factor scan for Mersenne numbers* (Small-factor scan for Mersenne numbers)

### 3.3.39 E039: Sophie Germain and safe primes

**Tags:** number-theory, quantitative-exploration, visualization See: *Valid Tags*.

## Highlights

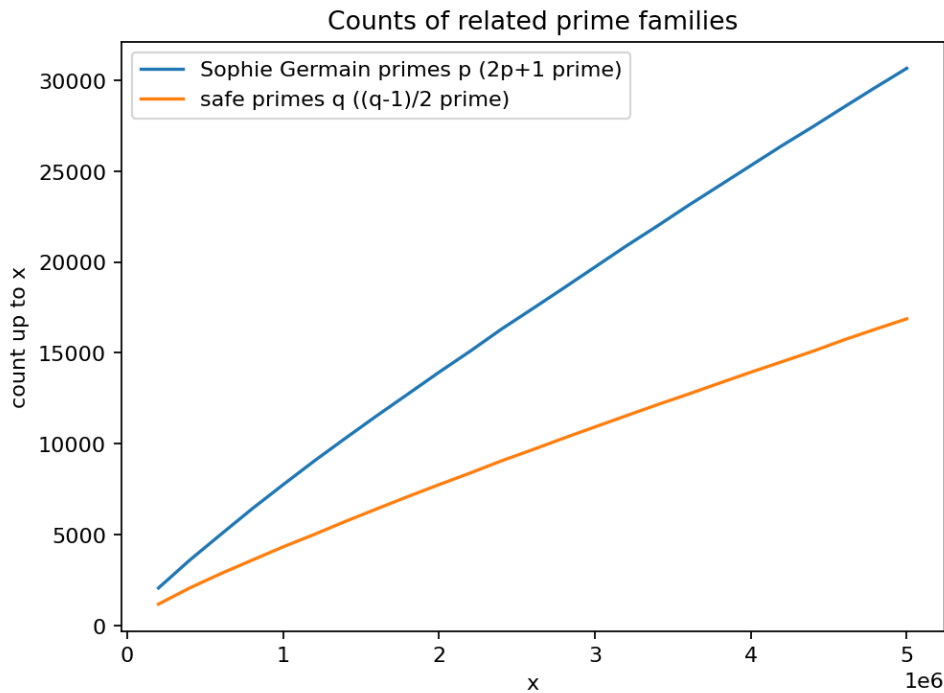
- This is a thin wrapper that follows the standard experiment template and delegates the actual computation to `:mod:mathxlab.experiments.prime_suite`.
- Writes reproducible artifacts (`params.json`, `report.md`, and figures).
- Designed to surface patterns *and* “looks-true-until-it-breaks” behavior.

## Goal

This is a thin wrapper that follows the standard experiment template and delegates the actual computation to `:mod:mathxlab.experiments.prime_suite`.

## Background (quick refresher)

- *Prime numbers refresher*



### Research question

Which **prime-related claim, heuristic, or algorithm** breaks first under a clean, controlled computational sweep, and what does the smallest or clearest counterexample (or deviation) look like?

### Why this qualifies as a mathematical experiment

- **Finite procedure:** run a bounded search / sweep with recorded parameters.
- **Observable(s):** counts, gaps, residues, runtime scaling, or first counterexample witnesses.
- **Parameter space:** vary bounds (and sometimes algorithmic choices).
- **Outcome:** plots/tables + “witness objects” for failures.
- **Reproducibility:** outputs saved to `out/e039/` with a parameter snapshot.

### Experiment design

- **Computation:** bounded enumeration / sampling with explicit limits.
- **Outputs:** figures and a short `report.md` summarizing what was found.
- **Artifacts written:**
  - `figures/fig_*.png`
  - `params.json`
  - `report.md`

### How to run

```
make run EXP=e039
```

or:

```
uv run python -m mathxlab.experiments.e039
```

### Notes / pitfalls

- “No counterexample found” only means “none found within the configured bounds”.
- For probabilistic tests (when used), treat outcomes as *evidence*, not proof.

### Extensions

- Increase bounds and rerun (recording runtime and memory).
- Compare alternative heuristics or algorithms on the same parameter grid.
- Turn found deviations into new, tighter conjectures.

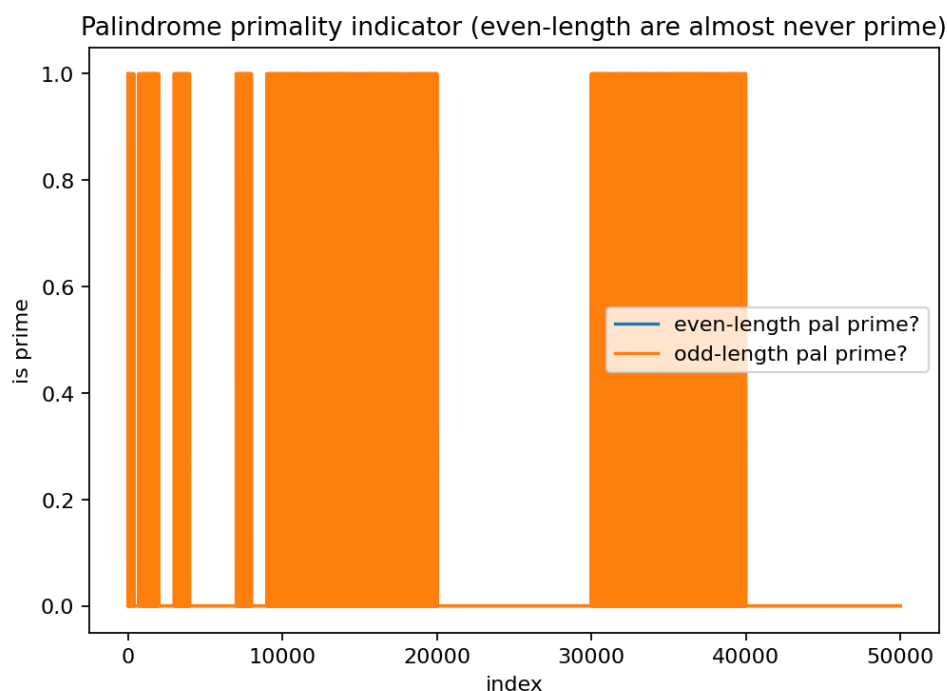
### References

See [References](#).

### Related experiments

- *E002: Even Perfect Numbers — Generator and Growth* (Even Perfect Numbers — Generator and Growth)
- *E003: Abundancy Index Landscape* (Abundancy Index Landscape)
- *E007: Mersenne growth (bits and digits)* (Mersenne growth (bits and digits))
- *E008: Lucas–Lehmer scan (prime exponents)* (Lucas–Lehmer scan (prime exponents))
- *E009: Small-factor scan for Mersenne numbers* (Small-factor scan for Mersenne numbers)

## 3.3.40 E040: Palindromic primes and the ‘11 trap’



Tags: number-theory, quantitative-exploration, visualization See: [Valid Tags](#).

## Highlights

- This is a thin wrapper that follows the standard experiment template and delegates the actual computation to `:mod:mathxlab.experiments.prime_suite`.
- Writes reproducible artifacts (`params.json`, `report.md`, and figures).
- Designed to surface patterns *and* “looks-true-until-it-breaks” behavior.

## Goal

This is a thin wrapper that follows the standard experiment template and delegates the actual computation to `:mod:mathxlab.experiments.prime_suite`.

## Background (quick refresher)

- *Prime numbers refresher*

## Research question

Which **prime-related claim, heuristic, or algorithm** breaks first under a clean, controlled computational sweep, and what does the smallest or clearest counterexample (or deviation) look like?

## Why this qualifies as a mathematical experiment

- **Finite procedure:** run a bounded search / sweep with recorded parameters.
- **Observable(s):** counts, gaps, residues, runtime scaling, or first counterexample witnesses.
- **Parameter space:** vary bounds (and sometimes algorithmic choices).
- **Outcome:** plots/tables + “witness objects” for failures.
- **Reproducibility:** outputs saved to `out/e040/` with a parameter snapshot.

## Experiment design

- **Computation:** bounded enumeration / sampling with explicit limits.
- **Outputs:** figures and a short `report.md` summarizing what was found.
- **Artifacts written:**
  - `figures/fig_*.png`
  - `params.json`
  - `report.md`

## How to run

```
make run EXP=e040
```

or:

```
uv run python -m mathxlab.experiments.e040
```

## Notes / pitfalls

- “No counterexample found” only means “none found within the configured bounds”.
- For probabilistic tests (when used), treat outcomes as *evidence*, not proof.

## Extensions

- Increase bounds and rerun (recording runtime and memory).
- Compare alternative heuristics or algorithms on the same parameter grid.
- Turn found deviations into new, tighter conjectures.

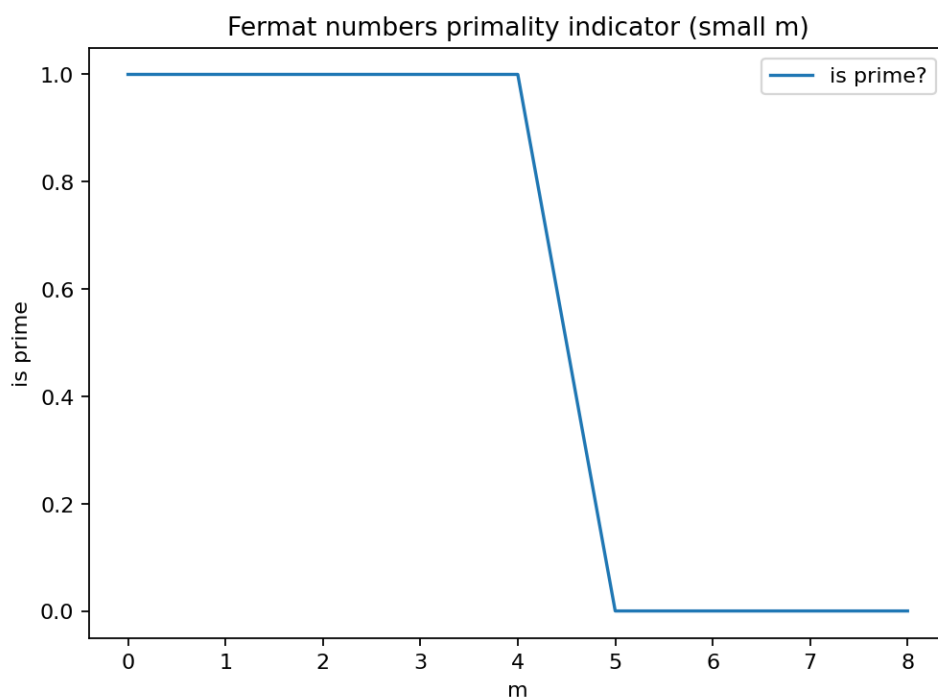
## References

See *References*.

## Related experiments

- *E002: Even Perfect Numbers — Generator and Growth* (Even Perfect Numbers — Generator and Growth)
- *E003: Abundancy Index Landscape* (Abundancy Index Landscape)
- *E007: Mersenne growth (bits and digits)* (Mersenne growth (bits and digits))
- *E008: Lucas–Lehmer scan (prime exponents)* (Lucas–Lehmer scan (prime exponents))
- *E009: Small-factor scan for Mersenne numbers* (Small-factor scan for Mersenne numbers)

### 3.3.41 E041: Fermat numbers: not all prime



Tags: number-theory, quantitative-exploration, visualization See: *Valid Tags*.

## Highlights

- This is a thin wrapper that follows the standard experiment template and delegates the actual computation to `:mod:mathxlab.experiments.prime_suite`.
- Writes reproducible artifacts (`params.json`, `report.md`, and figures).
- Designed to surface patterns *and* “looks-true-until-it-breaks” behavior.

## Goal

This is a thin wrapper that follows the standard experiment template and delegates the actual computation to `:mod:mathxlab.experiments.prime_suite`.

## Background (quick refresher)

- *Prime numbers refresher*

## Research question

Which **prime-related claim, heuristic, or algorithm** breaks first under a clean, controlled computational sweep, and what does the smallest or clearest counterexample (or deviation) look like?

## Why this qualifies as a mathematical experiment

- **Finite procedure:** run a bounded search / sweep with recorded parameters.
- **Observable(s):** counts, gaps, residues, runtime scaling, or first counterexample witnesses.
- **Parameter space:** vary bounds (and sometimes algorithmic choices).
- **Outcome:** plots/tables + “witness objects” for failures.
- **Reproducibility:** outputs saved to `out/e041/` with a parameter snapshot.

## Experiment design

- **Computation:** bounded enumeration / sampling with explicit limits.
- **Outputs:** figures and a short `report.md` summarizing what was found.
- **Artifacts written:**
  - `figures/fig_*.png`
  - `params.json`
  - `report.md`

## How to run

```
make run EXP=e041
```

or:

```
uv run python -m mathxlab.experiments.e041
```

## Notes / pitfalls

- “No counterexample found” only means “none found within the configured bounds”.
- For probabilistic tests (when used), treat outcomes as *evidence*, not proof.

## Extensions

- Increase bounds and rerun (recording runtime and memory).
- Compare alternative heuristics or algorithms on the same parameter grid.
- Turn found deviations into new, tighter conjectures.

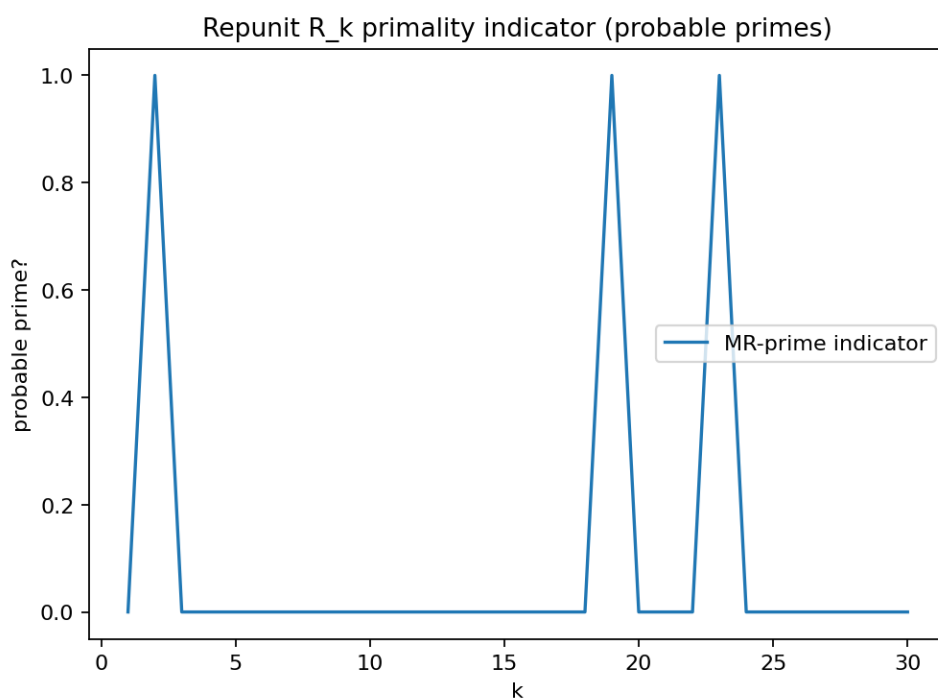
## References

See *References*.

## Related experiments

- *E048: Carmichael numbers: Korselt scan + Fermat counterexamples* (Carmichael numbers: Korselt scan + Fermat counterexamples)
- *E012: Fermat pseudoprimes and Carmichael numbers (counterexamples)* (Fermat pseudoprimes and Carmichael numbers (counterexamples))
- *E047: Fermat numbers: Pépin test + factor witnesses* (Fermat numbers: Pépin test + factor witnesses)
- *E002: Even Perfect Numbers — Generator and Growth* (Even Perfect Numbers — Generator and Growth)
- *E003: Abundancy Index Landscape* (Abundancy Index Landscape)

### 3.3.42 E042: Repunit primes (small k scan)



Tags: number-theory, quantitative-exploration, visualization See: *Valid Tags*.

## Highlights

- This is a thin wrapper that follows the standard experiment template and delegates the actual computation to `:mod:mathxlab.experiments.prime_suite`.
- Writes reproducible artifacts (`params.json`, `report.md`, and figures).
- Designed to surface patterns *and* “looks-true-until-it-breaks” behavior.

## Goal

This is a thin wrapper that follows the standard experiment template and delegates the actual computation to `:mod:mathxlab.experiments.prime_suite`.



## Background (quick refresher)

- *Prime numbers refresher*

## Research question

Which **prime-related claim, heuristic, or algorithm** breaks first under a clean, controlled computational sweep, and what does the smallest or clearest counterexample (or deviation) look like?

## Why this qualifies as a mathematical experiment

- **Finite procedure:** run a bounded search / sweep with recorded parameters.
- **Observable(s):** counts, gaps, residues, runtime scaling, or first counterexample witnesses.
- **Parameter space:** vary bounds (and sometimes algorithmic choices).
- **Outcome:** plots/tables + “witness objects” for failures.
- **Reproducibility:** outputs saved to `out/e042/` with a parameter snapshot.

## Experiment design

- **Computation:** bounded enumeration / sampling with explicit limits.
- **Outputs:** figures and a short `report.md` summarizing what was found.
- **Artifacts written:**
  - `figures/fig_*.png`
  - `params.json`
  - `report.md`

## How to run

```
make run EXP=e042
```

or:

```
uv run python -m mathxlab.experiments.e042
```

## Notes / pitfalls

- “No counterexample found” only means “none found within the configured bounds”.
- For probabilistic tests (when used), treat outcomes as *evidence*, not proof.

## Extensions

- Increase bounds and rerun (recording runtime and memory).
- Compare alternative heuristics or algorithms on the same parameter grid.
- Turn found deviations into new, tighter conjectures.

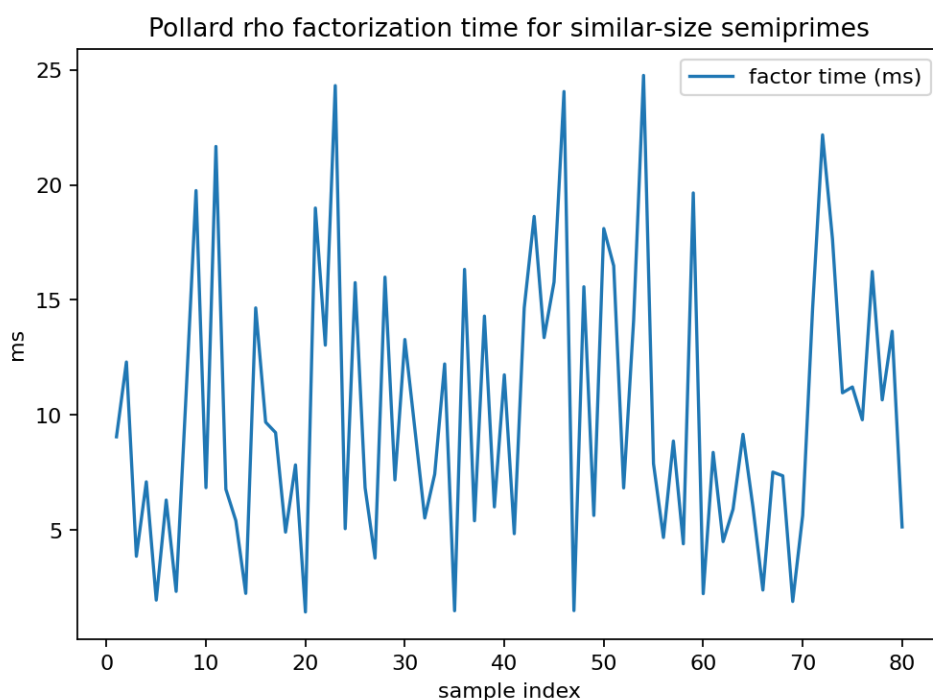
## References

See *References*.

## Related experiments

- *E009: Small-factor scan for Mersenne numbers* (Small-factor scan for Mersenne numbers)
- *E008: Lucas–Lehmer scan (prime exponents)* (Lucas–Lehmer scan (prime exponents))
- *E036: Prime arithmetic progressions (small search)* (Prime arithmetic progressions (small search))
- *E048: Carmichael numbers: Korselt scan + Fermat counterexamples* (Carmichael numbers: Korselt scan + Fermat counterexamples)
- *E049: Wieferich primes (base 2): scan and quotient visualization* (Wieferich primes (base 2): scan and quotient visualization)

### 3.3.43 E043: Pollard rho runtime variability



Tags: number-theory, quantitative-exploration, visualization See: [Valid Tags](#).

## Highlights

- This is a thin wrapper that follows the standard experiment template and delegates the actual computation to `:mod:mathxlab.experiments.prime_suite`.
- Writes reproducible artifacts (`params.json`, `report.md`, and figures).
- Designed to surface patterns *and* “looks-true-until-it-breaks” behavior.

## Goal

This is a thin wrapper that follows the standard experiment template and delegates the actual computation to `:mod:mathxlab.experiments.prime_suite`.

## Background (quick refresher)

- *Prime numbers refresher*

### Research question

Which **prime-related claim, heuristic, or algorithm** breaks first under a clean, controlled computational sweep, and what does the smallest or clearest counterexample (or deviation) look like?

### Why this qualifies as a mathematical experiment

- **Finite procedure:** run a bounded search / sweep with recorded parameters.
- **Observable(s):** counts, gaps, residues, runtime scaling, or first counterexample witnesses.
- **Parameter space:** vary bounds (and sometimes algorithmic choices).
- **Outcome:** plots/tables + “witness objects” for failures.
- **Reproducibility:** outputs saved to `out/e043/` with a parameter snapshot.

### Experiment design

- **Computation:** bounded enumeration / sampling with explicit limits.
- **Outputs:** figures and a short `report.md` summarizing what was found.
- **Artifacts written:**
  - `figures/fig_*.png`
  - `params.json`
  - `report.md`

### How to run

```
make run EXP=e043
```

or:

```
uv run python -m mathxlab.experiments.e043
```

### Notes / pitfalls

- “No counterexample found” only means “none found within the configured bounds”.
- For probabilistic tests (when used), treat outcomes as *evidence*, not proof.

### Extensions

- Increase bounds and rerun (recording runtime and memory).
- Compare alternative heuristics or algorithms on the same parameter grid.
- Turn found deviations into new, tighter conjectures.

### References

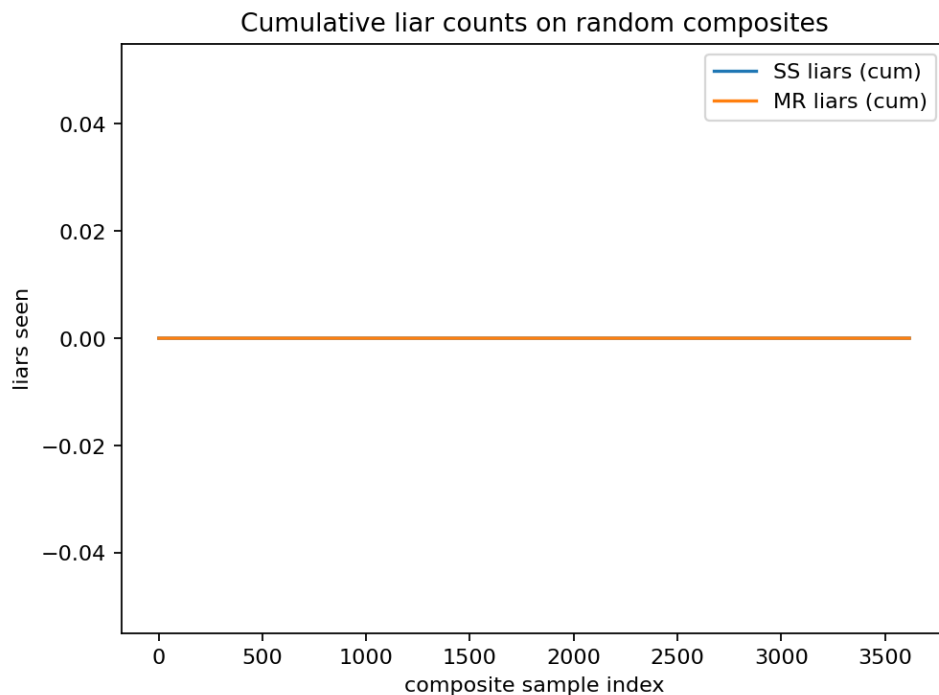
See *References*.

### Related experiments

- *E002: Even Perfect Numbers — Generator and Growth* (Even Perfect Numbers — Generator and Growth)
- *E003: Abundancy Index Landscape* (Abundancy Index Landscape)
- *E007: Mersenne growth (bits and digits)* (Mersenne growth (bits and digits))
- *E008: Lucas–Lehmer scan (prime exponents)* (Lucas–Lehmer scan (prime exponents))

- *E009: Small-factor scan for Mersenne numbers* (Small-factor scan for Mersenne numbers)

### 3.3.44 E044: Solovay–Strassen vs. Miller–Rabin (liars)



Tags: number-theory, quantitative-exploration, visualization See: [Valid Tags](#).

#### Highlights

- This is a thin wrapper that follows the standard experiment template and delegates the actual computation to `:mod:mathxlab.experiments.prime_suite`.
- Writes reproducible artifacts (`params.json`, `report.md`, and figures).
- Designed to surface patterns *and* “looks-true-until-it-breaks” behavior.

#### Goal

This is a thin wrapper that follows the standard experiment template and delegates the actual computation to `:mod:mathxlab.experiments.prime_suite`.

#### Background (quick refresher)

- *Prime numbers refresher*

#### Research question

Which **prime-related claim, heuristic, or algorithm** breaks first under a clean, controlled computational sweep, and what does the smallest or clearest counterexample (or deviation) look like?

#### Why this qualifies as a mathematical experiment

- **Finite procedure:** run a bounded search / sweep with recorded parameters.
- **Observable(s):** counts, gaps, residues, runtime scaling, or first counterexample witnesses.
- **Parameter space:** vary bounds (and sometimes algorithmic choices).
- **Outcome:** plots/tables + “witness objects” for failures.

- **Reproducibility:** outputs saved to `out/e044/` with a parameter snapshot.

### Experiment design

- **Computation:** bounded enumeration / sampling with explicit limits.
- **Outputs:** figures and a short `report.md` summarizing what was found.
- **Artifacts written:**
  - `figures/fig_*.png`
  - `params.json`
  - `report.md`

### How to run

```
make run EXP=e044
```

or:

```
uv run python -m mathxlab.experiments.e044
```

### Notes / pitfalls

- “No counterexample found” only means “none found within the configured bounds”.
- For probabilistic tests (when used), treat outcomes as *evidence*, not proof.

### Extensions

- Increase bounds and rerun (recording runtime and memory).
- Compare alternative heuristics or algorithms on the same parameter grid.
- Turn found deviations into new, tighter conjectures.

### References

See *References*.

### Related experiments

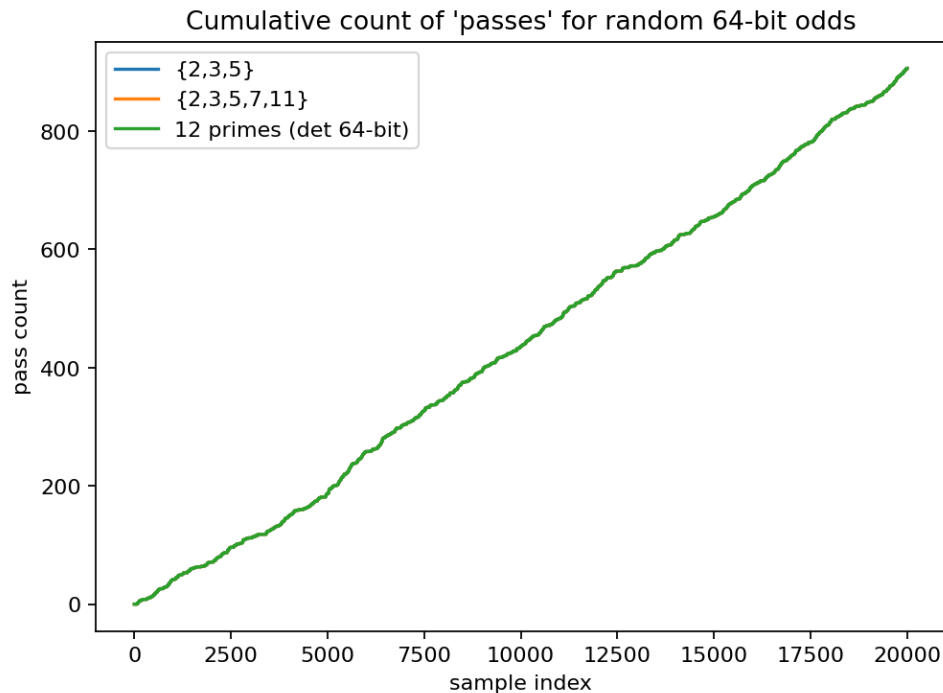
- *E016: Trial division vs. Miller–Rabin scaling* (Trial division vs. Miller–Rabin scaling)
- *E018: Miller–Rabin base choice counterexamples* (Miller–Rabin base choice counterexamples)
- *E002: Even Perfect Numbers — Generator and Growth* (Even Perfect Numbers — Generator and Growth)
- *E003: Abundancy Index Landscape* (Abundancy Index Landscape)
- *E007: Mersenne growth (bits and digits)* (Mersenne growth (bits and digits))

### 3.3.45 E045: Deterministic 64-bit MR base sets

**Tags:** number-theory, quantitative-exploration, visualization See: *Valid Tags*.

### Highlights

- This is a thin wrapper that follows the standard experiment template and delegates the actual computation to `:mod:mathxlab.experiments.prime_suite`.
- Writes reproducible artifacts (`params.json`, `report.md`, and figures).
- Designed to surface patterns *and* “looks-true-until-it-breaks” behavior.



### Goal

This is a thin wrapper that follows the standard experiment template and delegates the actual computation to `:mod:mathxlab.experiments.prime_suite`.

### Background (quick refresher)

- *Prime numbers refresher*

### Research question

Which **prime-related claim, heuristic, or algorithm** breaks first under a clean, controlled computational sweep, and what does the smallest or clearest counterexample (or deviation) look like?

### Why this qualifies as a mathematical experiment

- **Finite procedure:** run a bounded search / sweep with recorded parameters.
- **Observable(s):** counts, gaps, residues, runtime scaling, or first counterexample witnesses.
- **Parameter space:** vary bounds (and sometimes algorithmic choices).
- **Outcome:** plots/tables + “witness objects” for failures.
- **Reproducibility:** outputs saved to `out/e045/` with a parameter snapshot.

### Experiment design

- **Computation:** bounded enumeration / sampling with explicit limits.
- **Outputs:** figures and a short `report.md` summarizing what was found.
- **Artifacts written:**
  - `figures/fig_*.png`
  - `params.json`
  - `report.md`

## How to run

```
make run EXP=e045
```

or:

```
uv run python -m mathxlab.experiments.e045
```

## Notes / pitfalls

- “No counterexample found” only means “none found within the configured bounds”.
- For probabilistic tests (when used), treat outcomes as *evidence*, not proof.

## Extensions

- Increase bounds and rerun (recording runtime and memory).
- Compare alternative heuristics or algorithms on the same parameter grid.
- Turn found deviations into new, tighter conjectures.

## References

See *References*.

## Related experiments

- *E018: Miller–Rabin base choice counterexamples* (Miller–Rabin base choice counterexamples)
- *E049: Wieferich primes (base 2): scan and quotient visualization* (Wieferich primes (base 2): scan and quotient visualization)
- *E002: Even Perfect Numbers — Generator and Growth* (Even Perfect Numbers — Generator and Growth)
- *E003: Abundancy Index Landscape* (Abundancy Index Landscape)
- *E007: Mersenne growth (bits and digits)* (Mersenne growth (bits and digits))

### 3.3.46 E046: Prime-testing pipeline and tuning pitfalls

**Tags:** number-theory, quantitative-exploration, visualization See: *Valid Tags*.

## Highlights

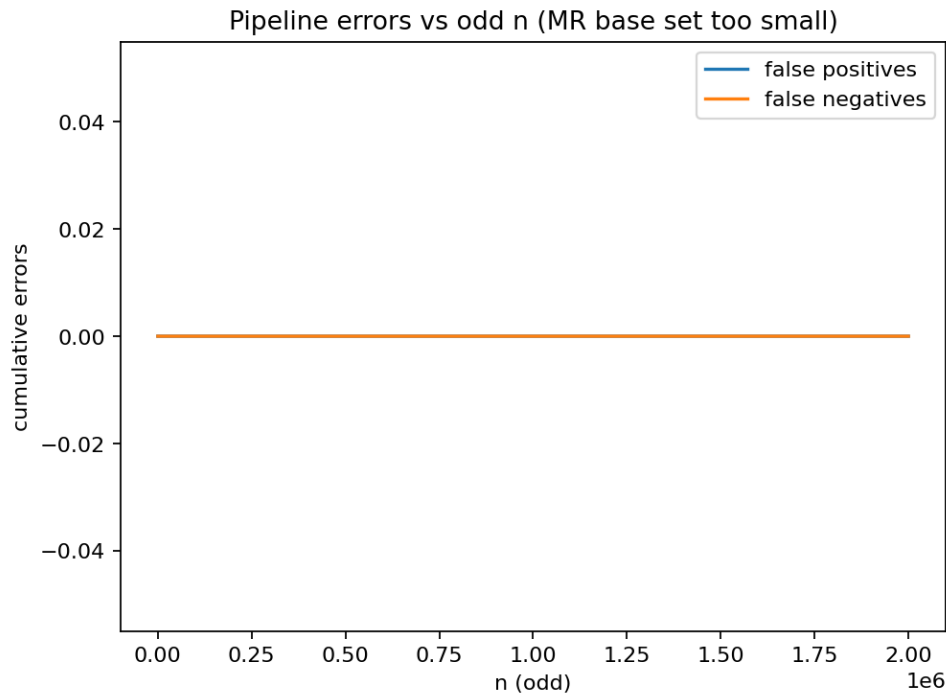
- This is a thin wrapper that follows the standard experiment template and delegates the actual computation to `:mod:mathxlab.experiments.prime_suite`.
- Writes reproducible artifacts (`params.json`, `report.md`, and figures).
- Designed to surface patterns *and* “looks-true-until-it-breaks” behavior.

## Goal

This is a thin wrapper that follows the standard experiment template and delegates the actual computation to `:mod:mathxlab.experiments.prime_suite`.

## Background (quick refresher)

- *Prime numbers refresher*



### Research question

Which **prime-related claim, heuristic, or algorithm** breaks first under a clean, controlled computational sweep, and what does the smallest or clearest counterexample (or deviation) look like?

### Why this qualifies as a mathematical experiment

- **Finite procedure:** run a bounded search / sweep with recorded parameters.
- **Observable(s):** counts, gaps, residues, runtime scaling, or first counterexample witnesses.
- **Parameter space:** vary bounds (and sometimes algorithmic choices).
- **Outcome:** plots/tables + “witness objects” for failures.
- **Reproducibility:** outputs saved to `out/e046/` with a parameter snapshot.

### Experiment design

- **Computation:** bounded enumeration / sampling with explicit limits.
- **Outputs:** figures and a short `report.md` summarizing what was found.
- **Artifacts written:**
  - `figures/fig_*.png`
  - `params.json`
  - `report.md`

### How to run

```
make run EXP=e046
```

or:

```
uv run python -m mathxlab.experiments.e046
```



### Notes / pitfalls

- “No counterexample found” only means “none found within the configured bounds”.
- For probabilistic tests (when used), treat outcomes as *evidence*, not proof.

### Extensions

- Increase bounds and rerun (recording runtime and memory).
- Compare alternative heuristics or algorithms on the same parameter grid.
- Turn found deviations into new, tighter conjectures.

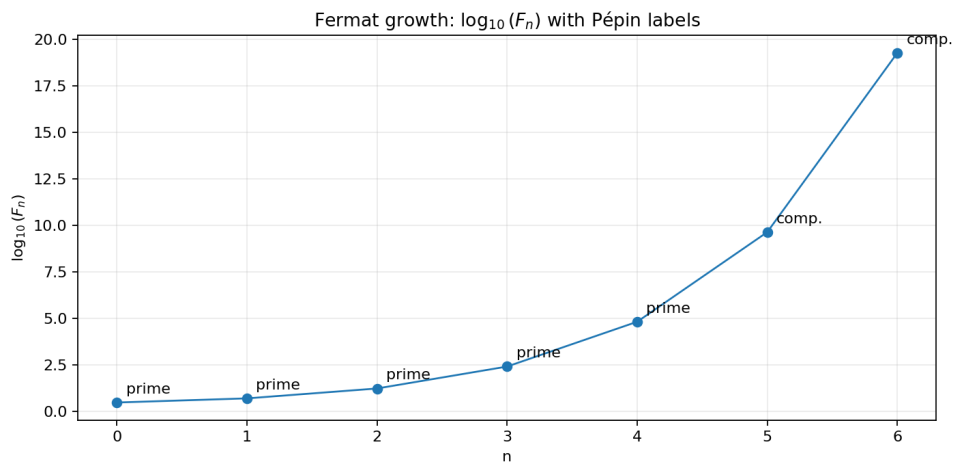
### References

See [References](#).

### Related experiments

- *E005: Odd Perfect Numbers — Constraint Filter Pipeline* (Odd Perfect Numbers — Constraint Filter Pipeline)
- *E002: Even Perfect Numbers — Generator and Growth* (Even Perfect Numbers — Generator and Growth)
- *E003: Abundancy Index Landscape* (Abundancy Index Landscape)
- *E007: Mersenne growth (bits and digits)* (Mersenne growth (bits and digits))
- *E008: Lucas–Lehmer scan (prime exponents)* (Lucas–Lehmer scan (prime exponents))

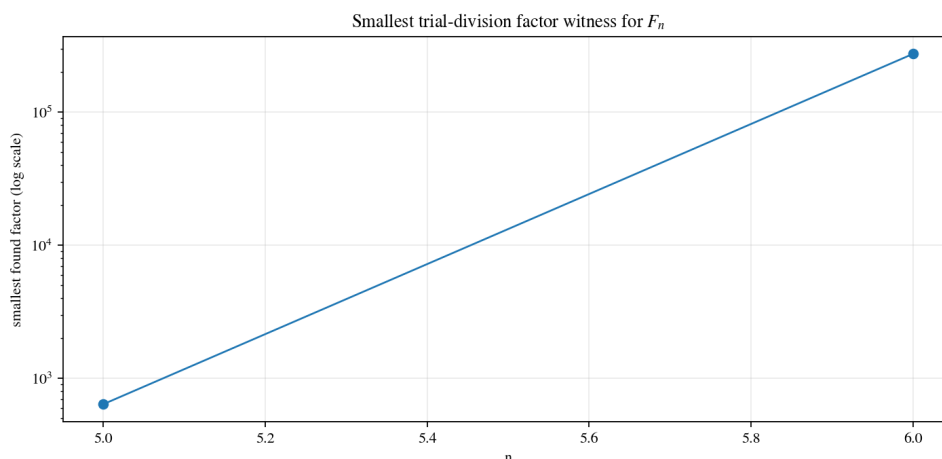
### 3.3.47 E047: Fermat numbers: Pépin test + factor witnesses



**Tags:** number-theory, counterexample-search, visualization, fermat See: [Valid Tags](#).

### Highlights

- Runs Pépin’s test for  $F_n = 2^{2^n} + 1$  (definitive for Fermat numbers).
- Finds small, explicit factor witnesses by bounded trial division.
- Visualizes how explosively  $F_n$  grows and where the first counterexample appears.



## Goal

Demonstrate (computationally) why the statement “all Fermat numbers are prime” fails, and produce concrete witnesses.

## Background (quick refresher)

- *Fermat numbers*

## Research question

How quickly do Fermat numbers become composite in practice, and how easy is it to exhibit a *witness* factor for the first failures?

## Experiment design

- Compute  $F_n$  for  $n \leq n_{max}$ .
- Apply Pépin’s test (for  $n \geq 1$ ) to classify prime vs. composite.
- If composite, attempt to find a small factor by trial division up to a configurable bound.
- Plot  $\log_{10}(F_n)$  and the smallest discovered factor (if any).

## Reproducibility

- `params.json` records the run settings.
- `report.md` summarizes the key findings.
- `figures/*.png` contains the plots for the run.

## Interpreting the results

- If Pépin says “composite”, the number is definitively composite (not a probabilistic claim).
- The factor search is bounded: “no factor found” does not mean “prime”.
- The classic factor for  $F_5$  (and often for  $F_6$ ) appears quickly under the default bounds.

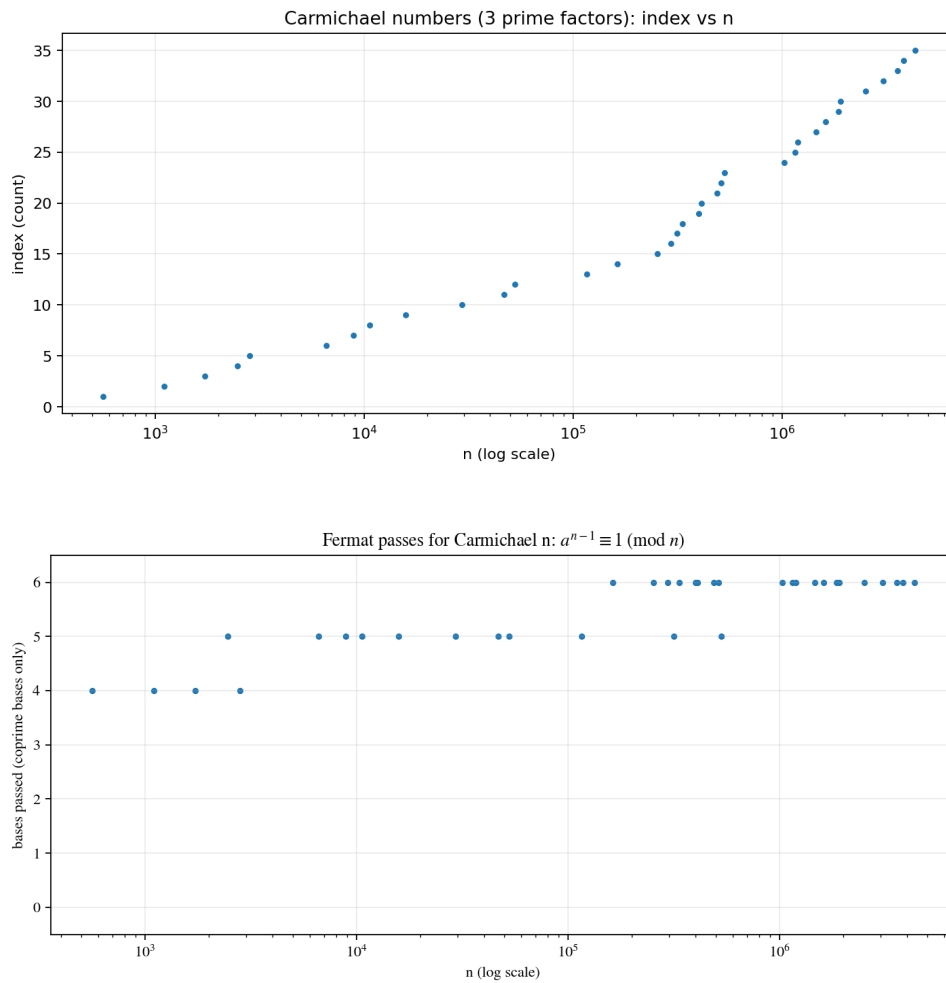
## References

See Kř\'izek *et al.* [2013], Wikipedia contributors [2025], The OEIS Foundation Inc. [2025].

## Related experiments

- *E012: Fermat pseudoprimes and Carmichael numbers (counterexamples)* (Fermat pseudoprimes and Carmichael numbers (counterexamples))
- *E048: Carmichael numbers: Korselt scan + Fermat counterexamples* (Carmichael numbers: Korselt scan + Fermat counterexamples)
- *E009: Small-factor scan for Mersenne numbers* (Small-factor scan for Mersenne numbers)
- *E015: Wilson test infeasibility* (Wilson test infeasibility)
- *E041: Fermat numbers: not all prime* (Fermat numbers: not all prime)

### 3.3.48 E048: Carmichael numbers: Korselt scan + Fermat counterexamples



**Tags:** number-theory, counterexample-search, quantitative-exploration, visualization, carmichael, pseudoprime See: [Valid Tags](#).

## Highlights

- Enumerates many small Carmichael numbers by scanning squarefree products of three primes.
- Verifies that these composites pass Fermat's test for several common bases.
- Visualizes the distribution of discovered Carmichael numbers on a log scale.

## Goal

Show that Fermat’s primality test can fail spectacularly, and produce a dataset of explicit counterexamples.

## Background (quick refresher)

- *Carmichael numbers*
- *Prime numbers refresher*

## Research question

Within a moderate bound  $N$ , how many Carmichael numbers can we find by a simple Korselt scan, and how many Fermat bases do they fool?

## Experiment design

- Enumerate candidates  $n = pqr \leq N$  with distinct primes  $p < q < r$ .
- Apply Korselt’s criterion:  $(p - 1) \mid (n - 1)$  for each prime factor  $p \mid n$ .
- For each Carmichael  $n$ , test Fermat congruence for a small base set (skipping non-coprime bases).
- Plot index vs.  $n$  and “bases passed” vs.  $n$ .

## Reproducibility

- `params.json` records the run settings.
- `report.md` summarizes the key findings.
- `figures/*.png` contains the plots for the run.

## Interpreting the results

- All numbers found are composite by construction, yet they pass Fermat congruences for *all* coprime bases (in theory).
- The scan is restricted to 3-prime-factor Carmichael numbers; larger-factor Carmichaels exist too.
- Seeing “all bases passed” in the plot is a strong reminder: Fermat alone is not a reliable primality test.

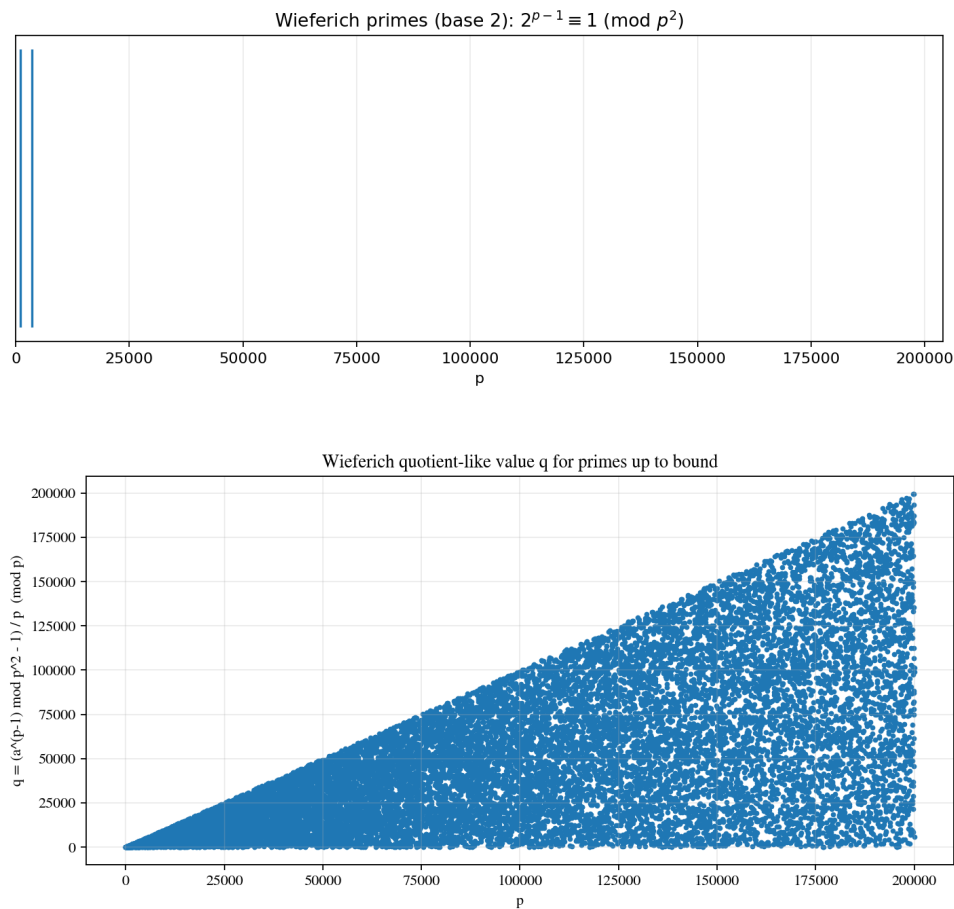
## References

See Alford *et al.* [1994], Carmichael [1912], The OEIS Foundation Inc. [2025], Wikipedia contributors [2025].

## Related experiments

- *E012: Fermat pseudoprimes and Carmichael numbers (counterexamples)* (Fermat pseudoprimes and Carmichael numbers (counterexamples))
- *E013: Prime-polynomial counterexamples (Euler’s  $n^2 + n + 41$ )* (Prime-polynomial counterexamples (Euler’s  $n^2 + n + 41$ ))
- *E014: Primorial  $\pm 1$  counterexamples* (Primorial  $\pm 1$  counterexamples)
- *E018: Miller–Rabin base choice counterexamples* (Miller–Rabin base choice counterexamples)
- *E047: Fermat numbers: Pépin test + factor witnesses* (Fermat numbers: Pépin test + factor witnesses)

### 3.3.49 E049: Wieferich primes (base 2): scan and quotient visualization



**Tags:** number-theory, counterexample-search, visualization, wieferich See: *Valid Tags*.

#### Highlights

- Scans primes up to a bound for the Wieferich condition  $2^{p-1} \equiv 1 \pmod{p^2}$ .
- Recovers the known small hits (1093 and 3511) under default settings.
- Visualizes a quotient-like value that is exactly zero for Wieferich primes.

#### Goal

Explore a rare strengthening of Fermat's congruence and visualize how exceptional Wieferich primes are.

#### Background (quick refresher)

- *Wieferich primes*
- *Prime numbers refresher*

#### Research question

Up to a bound  $B$ , which primes satisfy the Wieferich condition (base 2), and how does the quotient-like statistic vary across primes?

### Experiment design

- Generate all primes  $p \leq B$  (excluding  $p = 2$  for base 2).
- Compute  $r = 2^{p-1} \bmod p^2$  and derive  $q = (r - 1)/p \pmod{p}$ .
- Mark primes with  $q = 0$  as Wieferich hits.
- Plot hit positions and a scatter of  $q$  values.

### Reproducibility

- `params.json` records the run settings.
- `report.md` summarizes the key findings.
- `figures/*.png` contains the plots for the run.

### Interpreting the results

- Hits at 1093 and 3511 confirm the implementation.
- The quotient scatter is purely exploratory; it's a compact way to see “how far” a prime is from the Wieferich condition.
- Increasing the bound quickly becomes compute-heavy (but remains feasible for moderate bounds in pure Python).

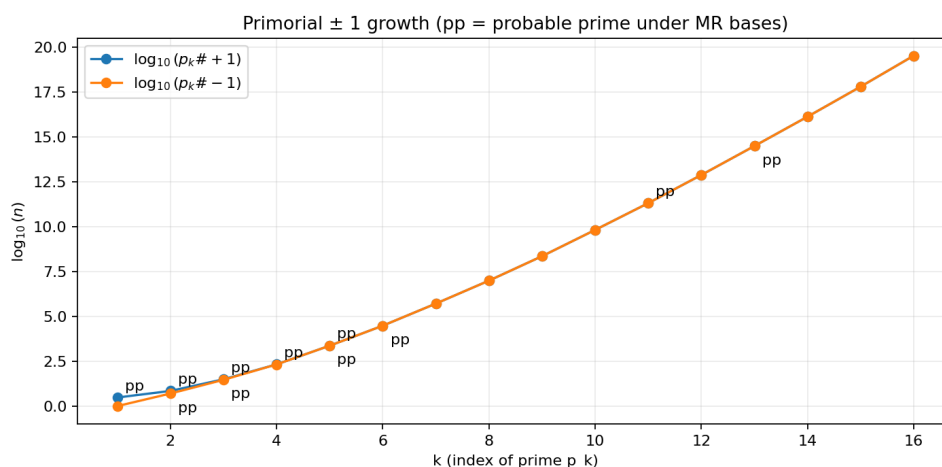
### References

See Wieferich [1909], The OEIS Foundation Inc. [2025], Wikipedia contributors [2025].

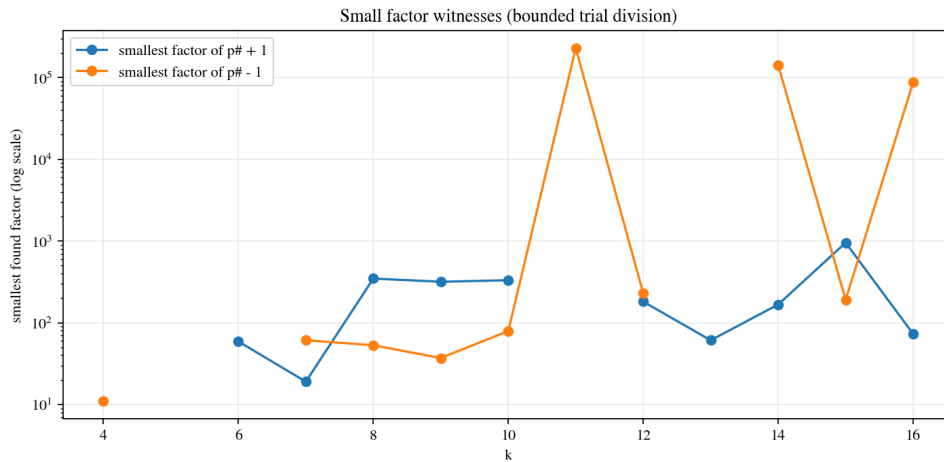
### Related experiments

- *E018: Miller–Rabin base choice counterexamples* (Miller–Rabin base choice counterexamples)
- *E048: Carmichael numbers: Korselt scan + Fermat counterexamples* (Carmichael numbers: Korselt scan + Fermat counterexamples)
- *E008: Lucas–Lehmer scan (prime exponents)* (Lucas–Lehmer scan (prime exponents))
- *E009: Small-factor scan for Mersenne numbers* (Small-factor scan for Mersenne numbers)
- *E042: Repunit primes (small k scan)* (Repunit primes (small k scan))

## 3.3.50 E050: Primorials and Euclid numbers: $p\# \pm 1$ are usually composite



**Tags:** number-theory, counterexample-search, quantitative-exploration, visualization, primorial See: *Valid Tags*.



## Highlights

- Computes primorials  $p_k\# = \prod_{i \leq k} p_i$  and Euclid-style numbers  $p_k\# \pm 1$ .
- Uses a probable-prime test to show “often composite” behavior early.
- Finds small factor witnesses via bounded trial division for many early  $k$ .

## Goal

Demonstrate that Euclid-style numbers are coprime to small primes but are not reliably prime, and produce explicit factor witnesses.

## Background (quick refresher)

- *Primorials*
- *Prime numbers refresher*

## Research question

For small  $k$ , how often are  $p_k\# \pm 1$  prime (or probable prime), and how easy is it to find a small factor when they are composite?

## Experiment design

- Compute primorials for  $k \leq k_{max}$ .
- Test  $p_k\# \pm 1$  with Miller–Rabin (probable prime).
- If composite under MR, try to find a small trial-division factor as a concrete witness.
- Plot  $\log_{10}(p_k\# \pm 1)$  and smallest factor witnesses (log scale).

## Reproducibility

- `params.json` records the run settings.
- `report.md` summarizes the key findings.
- `figures/*.png` contains the plots for the run.

## Interpreting the results

- “Probable prime” is not a proof; it’s a fast filter for this exploratory experiment.
- Factor witnesses come from bounded trial division: not finding a factor is expected for some  $k$ .

- You should see early composite examples such as  $p_6\# + 1 = 30031$  (with a small factor) under default bounds.

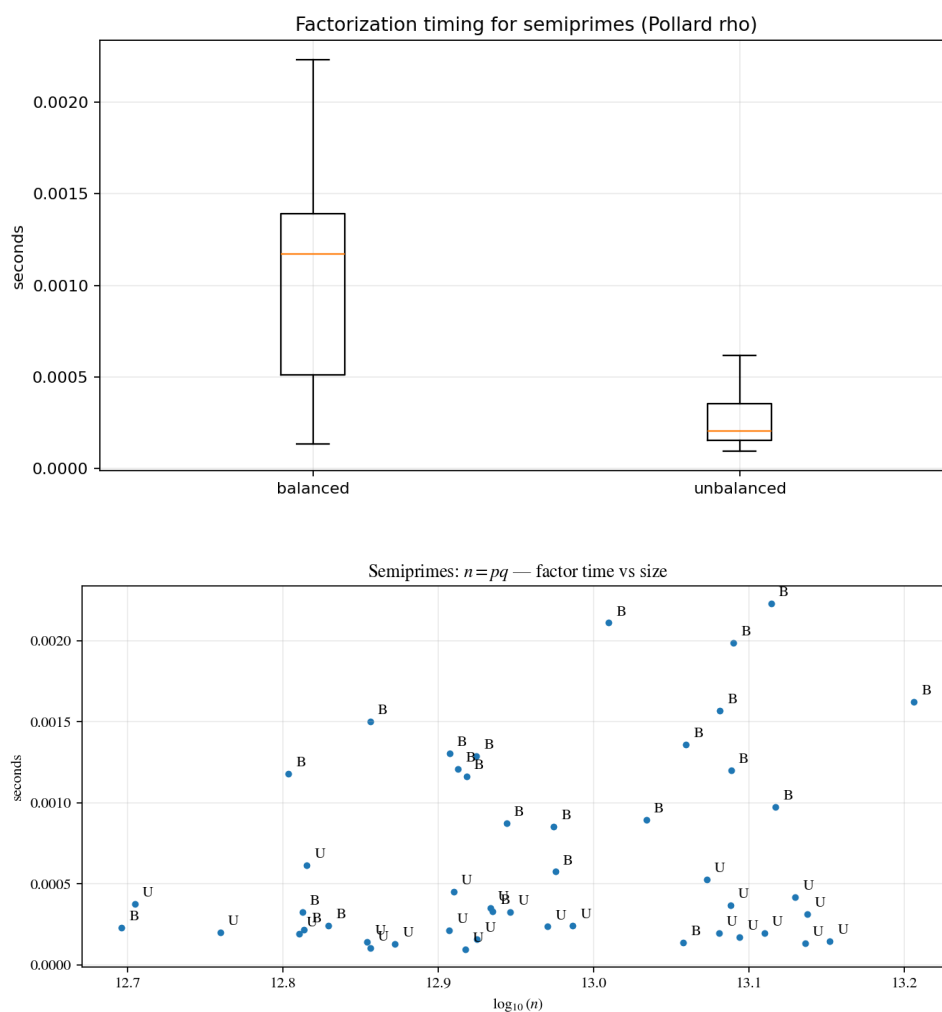
## References

See Hardy and Wright [2008], The OEIS Foundation Inc. [2025], Prime Pages (UTM) [2025], Wikipedia contributors [2025].

## Related experiments

- *E048: Carmichael numbers: Korselt scan + Fermat counterexamples* (Carmichael numbers: Korselt scan + Fermat counterexamples)
- *E005: Odd Perfect Numbers — Constraint Filter Pipeline* (Odd Perfect Numbers — Constraint Filter Pipeline)
- *E012: Fermat pseudoprimes and Carmichael numbers (counterexamples)* (Fermat pseudoprimes and Carmichael numbers (counterexamples))
- *E013: Prime-polynomial counterexamples (Euler's  $n^2 + n + 41$ )* (Prime-polynomial counterexamples (Euler's  $n^2 + n + 41$ ))
- *E014: Primorial  $\pm 1$  counterexamples* (Primorial  $\pm 1$  counterexamples)

### 3.3.51 E051: Semiprimes: balanced vs. unbalanced factorization timing



**Tags:** number-theory, quantitative-exploration, visualization, factorization, semiprime See: *Valid Tags*.



## Highlights

- Generates small semiprimes  $n = pq$  in balanced and unbalanced regimes.
- Factors them using trial division + Pollard’s rho and records timings.
- Visualizes time distributions to make “balanced is harder” tangible.

## Goal

Compare how factorization difficulty changes when one prime factor is much smaller than the other.

## Background (quick refresher)

- *Semiprimes*
- *Prime numbers refresher*

## Research question

For semiprimes of similar overall size, how does Pollard-rho timing differ between balanced ( $p \approx q$ ) and unbalanced ( $p \ll q$ ) cases?

## Experiment design

- Generate balanced semiprimes with roughly equal factor bit lengths.
- Generate unbalanced semiprimes with a fixed small factor size.
- Factor each using trial division pre-pass + Pollard’s rho, measuring wall-clock time.
- Plot boxplots and a size-vs-time scatter.

## Reproducibility

- `params.json` records the run settings.
- `report.md` summarizes the key findings.
- `figures/*.png` contains the plots for the run.

## Interpreting the results

- Timings are noisy: random instance structure matters a lot at this size range.
- The trend should still show unbalanced instances often factoring faster (small factor is easier to find).
- This is an educational-scale experiment, not a cryptographic benchmark.

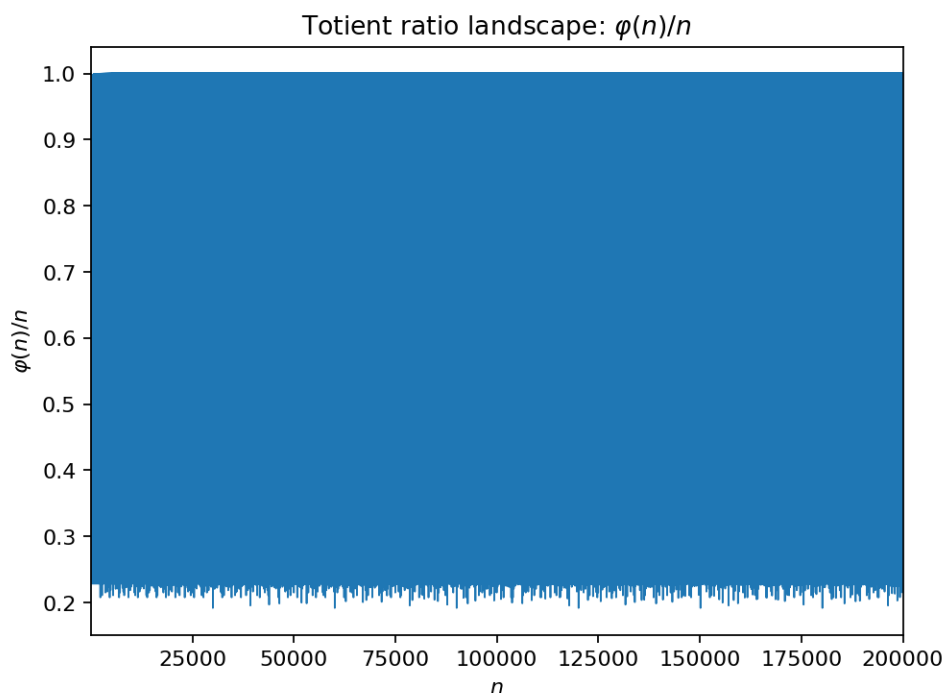
## References

See Rivest *et al.* [1978], The OEIS Foundation Inc. [2025].

## Related experiments

- *E004: Computing  $\sigma(n)$  at Scale — Sieve vs. Factorization* (Computing  $\sigma(n)$  at Scale — Sieve vs. Factorization)
- *E002: Even Perfect Numbers — Generator and Growth* (Even Perfect Numbers — Generator and Growth)
- *E003: Abundancy Index Landscape* (Abundancy Index Landscape)
- *E007: Mersenne growth (bits and digits)* (Mersenne growth (bits and digits))
- *E008: Lucas–Lehmer scan (prime exponents)* (Lucas–Lehmer scan (prime exponents))

### 3.3.52 E052: Totient ratio landscape



**Tags:** number-theory, quantitative-exploration, visualization, arithmetic-functions, totient, multiplicative See: *Valid Tags*.

#### Highlights

- Plot  $\varphi(n)/n$  for a range of  $n$  and annotate record lows.
- Relate deep dips to integers with many small prime factors (primorial-like structure).

#### Goal

Make the multiplicative structure of the totient function visible, especially the role of small primes.

#### Background (quick refresher)

- *Arithmetic functions refresher*
- *Euler’s totient function  $\varphi(n)$  refresher*
- *Primorials*

#### Research question

Which  $n$  minimize  $\varphi(n)/n$  up to a bound, and how do they relate to primorial products?

#### Method

- Compute  $\varphi(n)$  for  $n \leq N$  (sieve / SPF-based).
- Plot the ratio  $\varphi(n)/n$  and track record minima.
- Optionally compare record minima to primorials and nearby multiples.

### How to run

- make run EXP=e052
- uv run python -m mathxlab.experiments.e052

### Outputs

This experiment follows the standard output contract:

- out/e052/figures/ — generated figures (PNG)
- out/e052/report.md — short narrative report
- out/e052/manifest.json — snapshot metadata for the gallery

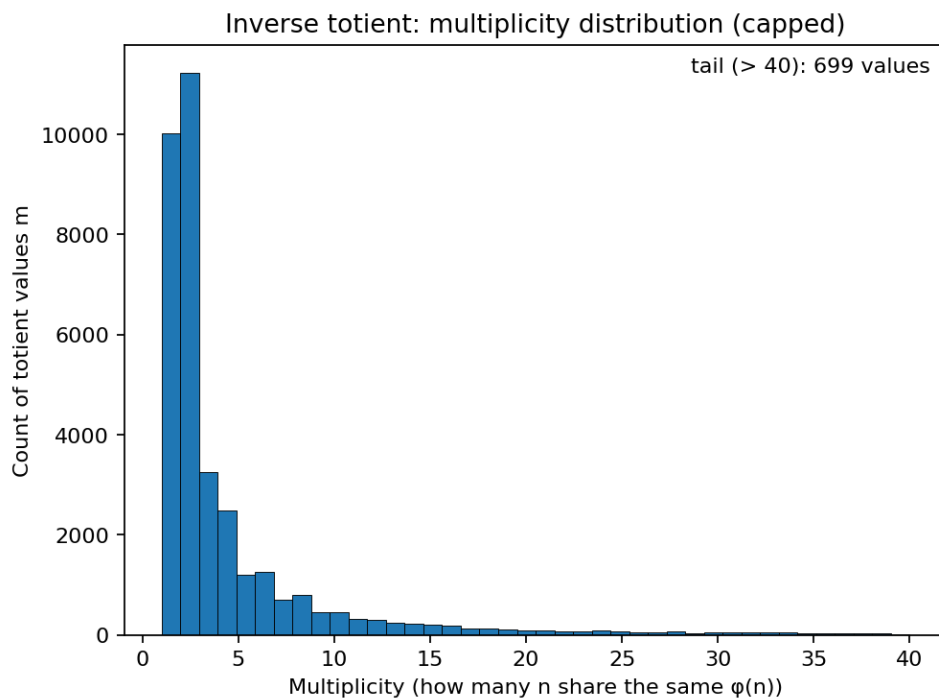
### References

See Apostol [1976], Tenenbaum [2015], Niven *et al.* [1991].

### Related experiments

- *E060: Jordan totients* (Jordan totients)
- *E053: Inverse totient multiplicities* (Inverse totient multiplicities)
- *E119 — Summatory totient  $\Phi(x)$  scaling check* (E119 — Summatory totient  $\Phi(x)$  scaling check)
- *E102 — Dirichlet convolution identity zoo* (E102 — Dirichlet convolution identity zoo)
- *E062: Carmichael  $(n)$  vs.  $(n)$*  (Carmichael  $(n)$  vs.  $(n)$ )

### 3.3.53 E053: Inverse totient multiplicities



**Tags:** number-theory, quantitative-exploration, visualization, arithmetic-functions, totient, search See: [Valid Tags](#).

## Highlights

- Compute  $\varphi(n)$  for  $n \leq N$  and count how many preimages each value has.
- Plot the histogram of multiplicities and list the top collisions.

## Goal

Empirically study the many-to-one nature of the totient function.

## Background (quick refresher)

- *Euler's totient function  $\varphi(n)$  refresher*
- *Arithmetic functions refresher*

## Research question

For  $n \leq N$ , how large can the multiplicity of a totient value get, and what do the maximizers look like?

## Method

- Compute  $\varphi(n)$  on a range and build a frequency table for values.
- Visualize multiplicities (histogram) and inspect the most frequent values.

## How to run

- `make run EXP=e053`
- `uv run python -m mathxlab.experiments.e053`

## Outputs

This experiment follows the standard output contract:

- `out/e053/figures/` — generated figures (PNG)
- `out/e053/report.md` — short narrative report
- `out/e053/manifest.json` — snapshot metadata for the gallery

## References

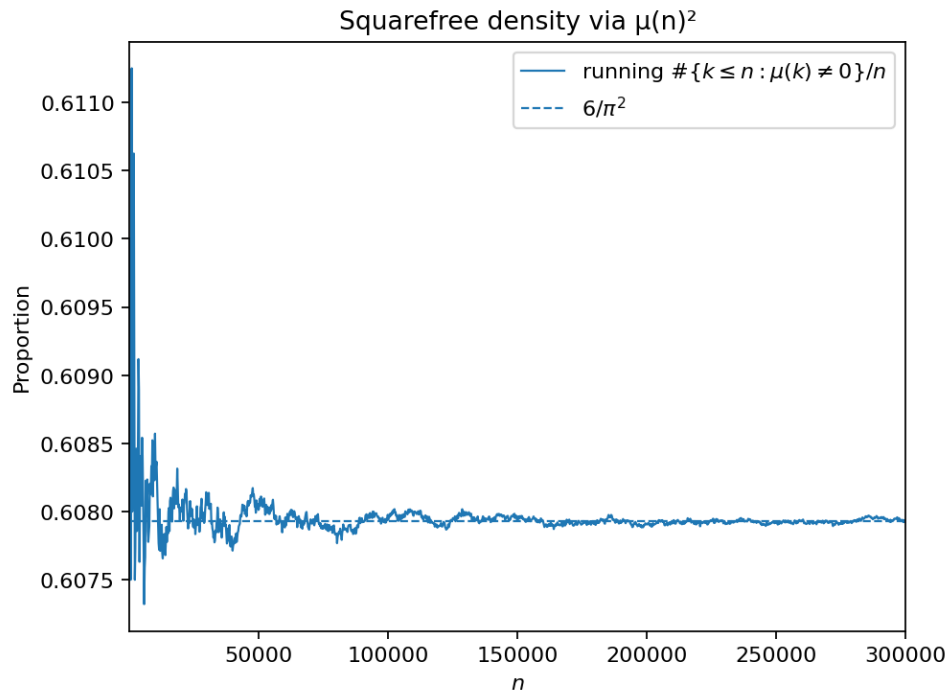
See Apostol [1976], Niven *et al.* [1991].

## Related experiments

- *E052: Totient ratio landscape* (Totient ratio landscape)
- *E119 — Summatory totient  $\Phi(x)$  scaling check* (E119 — Summatory totient  $\Phi(x)$  scaling check)
- *E005: Odd Perfect Numbers — Constraint Filter Pipeline* (Odd Perfect Numbers — Constraint Filter Pipeline)
- *E060: Jordan totients* (Jordan totients)
- *E062: Carmichael  $(n)$  vs.  $(n)$*  (Carmichael  $(n)$  vs.  $(n)$ )

### 3.3.54 E054: Squarefree density via Möbius

**Tags:** number-theory, quantitative-exploration, visualization, arithmetic-functions, mobius, summatory See: *Valid Tags*.



### Highlights

- Use  $\mu(n)^2$  as an indicator for squarefree integers.
- Plot the running density and compare to  $6/\pi^2$ .

### Goal

Show how an arithmetic function encodes a classic density result.

### Background (quick refresher)

- *Möbius function  $\mu(n)$  and Mertens function  $M(x)$  refresher*
- *Arithmetic functions refresher*

### Research question

How quickly does the empirical density of squarefree numbers approach  $6/\pi^2$ ?

### Method

- Compute  $\mu(n)$  up to  $N$  and accumulate  $\sum_{n \leq x} \mu(n)^2$ .
- Plot the ratio  $\frac{1}{x} \sum_{n \leq x} \mu(n)^2$  with a reference line at  $6/\pi^2$ .

### How to run

- `make run EXP=e054`
- `uv run python -m mathxlab.experiments.e054`

### Outputs

This experiment follows the standard output contract:

- `out/e054/figures/` — generated figures (PNG)
- `out/e054/report.md` — short narrative report

- `out/e054/manifest.json` — snapshot metadata for the gallery

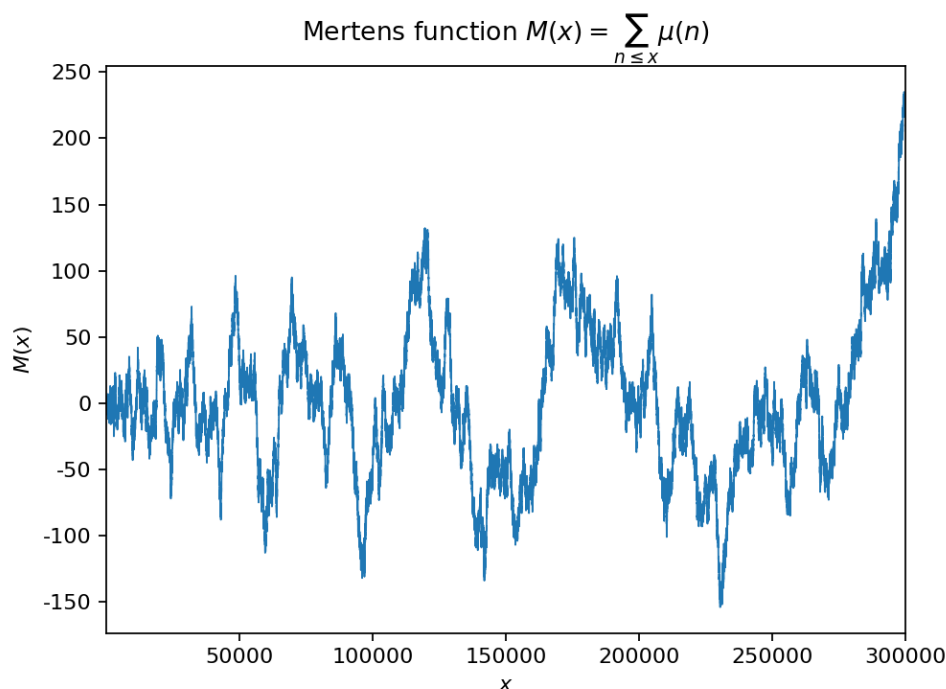
## References

See Apostol [1976], Tenenbaum [2015].

## Related experiments

- *E056: Liouville vs. Möbius walks* (Liouville vs. Möbius walks)
- *E055: Mertens function walk* (Mertens function walk)
- *E095 — Squarefree filter:  $\omega(n)=\Omega(n)$  when  $\omega(n)=0$*  (E095 — Squarefree filter:  $\omega(n)=\Omega(n)$  when  $\omega(n)=0$ )
- *E105 — Mertens  $M(x)$ : scaling views* (E105 — Mertens  $M(x)$ : scaling views)
- *E092 —  $1/(s)$  via the Möbius Dirichlet series* (E092 —  $1/(s)$  via the Möbius Dirichlet series)

### 3.3.55 E055: Mertens function walk



**Tags:** number-theory, quantitative-exploration, visualization, arithmetic-functions, mobius, summatory, model-checking See: *Valid Tags*.

## Highlights

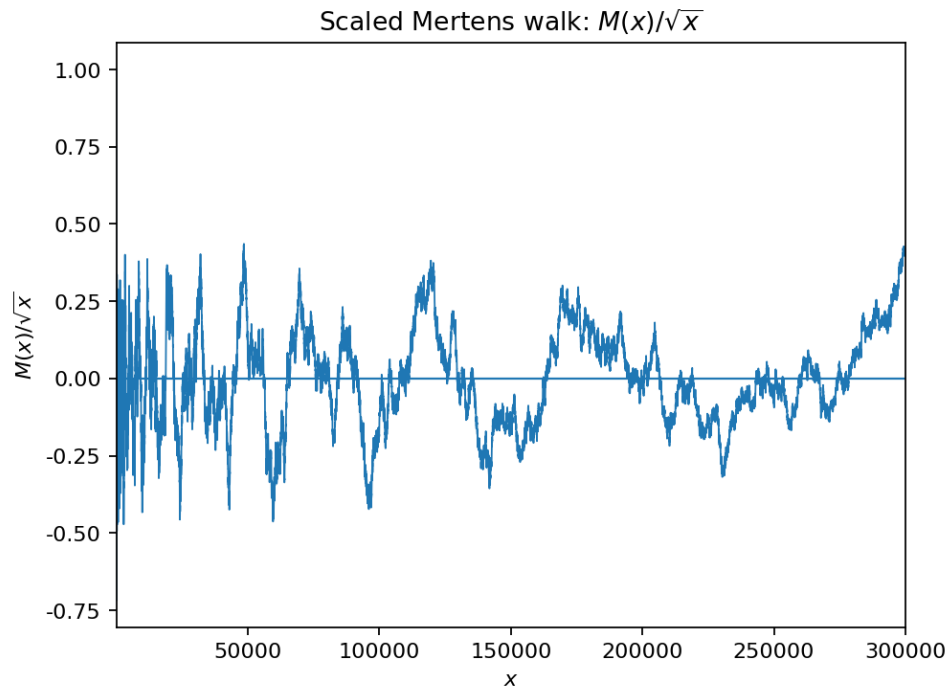
- Plot  $M(x)$  and scaled variants such as  $M(x)/\sqrt{x}$ .
- Compare qualitative behavior to a simple random-walk heuristic (without overclaiming).

## Goal

Build intuition for summatory Möbius behavior and typical fluctuation scales.

## Background (quick refresher)

- *Möbius function  $\mu(n)$  and Mertens function  $M(x)$  refresher*
- *Average orders and the Erdős–Kac viewpoint*



### Research question

What does  $M(x)/\sqrt{x}$  look like numerically over moderate ranges?

### Method

- Compute  $\mu(n)$  up to  $N$  and cumulative sums  $M(x)$ .
- Plot  $M(x)$  and  $M(x)/\sqrt{x}$ ; optionally add moving-window statistics.

### How to run

- `make run EXP=e055`
- `uv run python -m mathxlab.experiments.e055`

### Outputs

This experiment follows the standard output contract:

- `out/e055/figures/` — generated figures (PNG)
- `out/e055/report.md` — short narrative report
- `out/e055/manifest.json` — snapshot metadata for the gallery

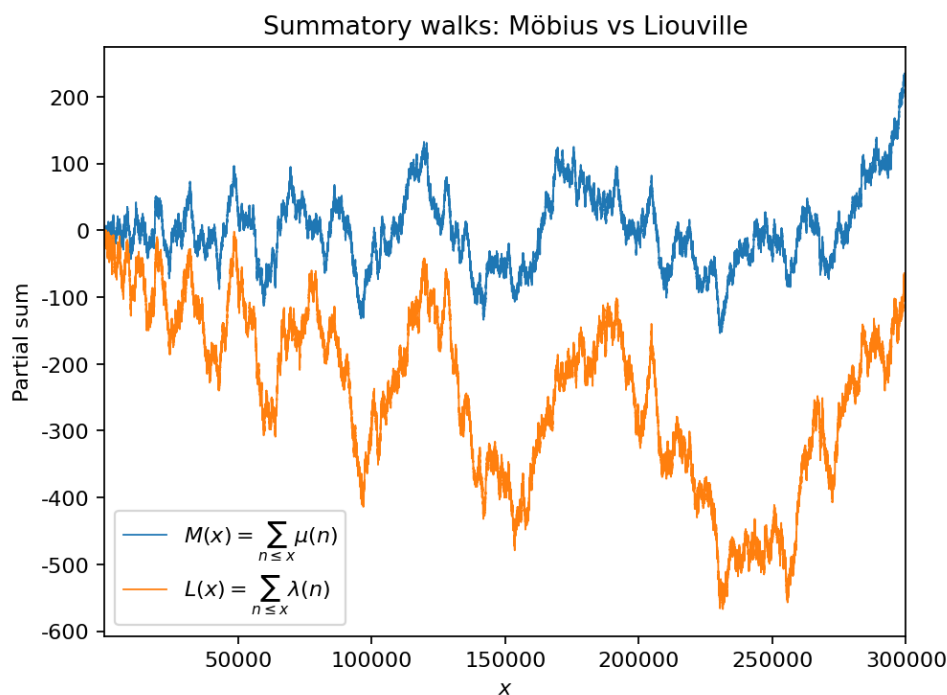
### References

See Tenenbaum [2015], Odlyzko and te Riele [1985].

### Related experiments

- *E105 — Mertens  $M(x)$ : scaling views* (E105 — Mertens  $M(x)$ : scaling views)
- *E054: Squarefree density via Möbius* (Squarefree density via Möbius)
- *E056: Liouville vs. Möbius walks* (Liouville vs. Möbius walks)
- *E095 — Squarefree filter:  $(n)=\Omega(n)$  when  $(n) \neq 0$*  (E095 — Squarefree filter:  $(n)=\Omega(n)$  when  $(n) \neq 0$ )
- *E121 — Möbius inversion as convolution undo* (E121 — Möbius inversion as convolution undo)

### 3.3.56 E056: Liouville vs. Möbius walks



**Tags:** number-theory, quantitative-exploration, visualization, arithmetic-functions, summatory, liouville, mobius See: [Valid Tags](#).

#### Highlights

- Plot partial sums of  $\lambda(n)$  and  $\mu(n)$  side by side.
- Compare scaled versions to see how the inclusion/exclusion of squareful terms changes the walk.

#### Goal

Contrast two closely related  $\pm 1/0$  sequences arising from prime factorizations.

#### Background (quick refresher)

- *Liouville function* `\lambda(n)` *refresher*
- *Möbius function* `\mu(n)` *and Mertens function*  $M(x)$  *refresher*
- *Prime-factor counting:* `\omega(n)` *and* `\Omega(n)` *refresher*

#### Research question

How do the fluctuations of  $\sum_{n \leq x} \lambda(n)$  compare to  $\sum_{n \leq x} \mu(n)$ ?

#### Method

- Compute  $\Omega(n)$ , then  $\lambda(n) = (-1)^{\Omega(n)}$ ; compute  $\mu(n)$ .
- Plot partial sums and scaled partial sums (e.g., divide by  $\sqrt{x}$ ).

#### How to run

- `make run EXP=e056`
- `uv run python -m mathxlab.experiments.e056`



## Outputs

This experiment follows the standard output contract:

- `out/e056/figures/` — generated figures (PNG)
- `out/e056/report.md` — short narrative report
- `out/e056/manifest.json` — snapshot metadata for the gallery

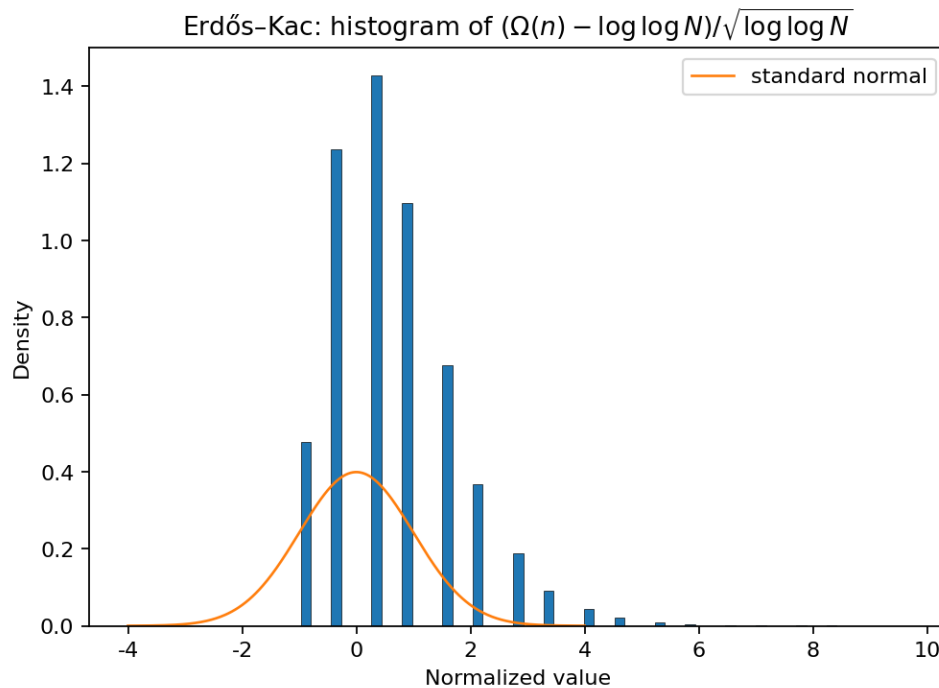
## References

See Tenenbaum [2015].

## Related experiments

- *E054: Squarefree density via Möbius* (Squarefree density via Möbius)
- *E120 — Liouville (n): partial sums and parity* (E120 — Liouville (n): partial sums and parity)
- *E055: Mertens function walk* (Mertens function walk)
- *E105 — Mertens  $M(x)$ : scaling views* (E105 — Mertens  $M(x)$ : scaling views)
- *E092 —  $1/(s)$  via the Möbius Dirichlet series* (E092 —  $1/(s)$  via the Möbius Dirichlet series)

### 3.3.57 E057: Erdős–Kac in practice



**Tags:** number-theory, quantitative-exploration, visualization, arithmetic-functions, omega, heuristics See: [Valid Tags](#).

## Highlights

- Compute  $\Omega(n)$  for  $n \leq N$  and plot its histogram.
- Optionally normalize to compare with a Gaussian curve (Erdős–Kac heuristic).

## Goal

Visualize the probabilistic number-theory flavor of prime-factor counts.

## Background (quick refresher)

- *Prime-factor counting:  $\omega(n)$  and  $\Omega(n)$  refresher*
- *Average orders and the Erdős–Kac viewpoint*

## Research question

How close does the distribution of  $\Omega(n)$  (after normalization) look to a normal curve for moderate  $N$ ?

## Method

- Compute  $\Omega(n)$  using an SPF sieve.
- Plot histogram; optionally overlay a normal density with matching mean/variance approximations.

## How to run

- `make run EXP=e057`
- `uv run python -m mathxlab.experiments.e057`

## Outputs

This experiment follows the standard output contract:

- `out/e057/figures/` — generated figures (PNG)
- `out/e057/report.md` — short narrative report
- `out/e057/manifest.json` — snapshot metadata for the gallery

## References

See Erdős and Kac [1940], Tenenbaum [2015].

## Related experiments

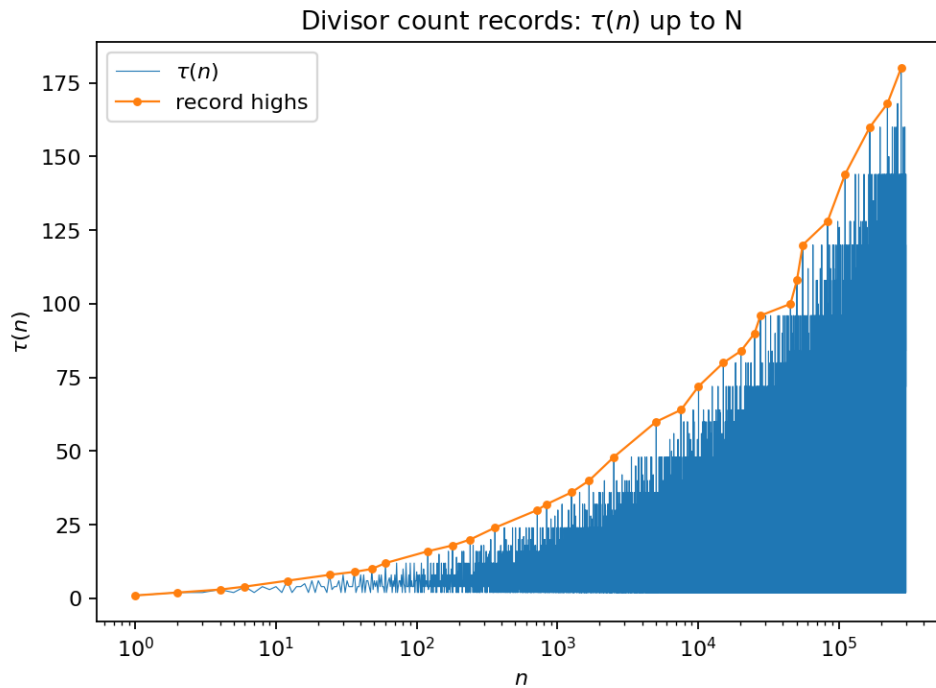
- *E094 —  $(n)$  vs.  $\Omega(n)$ : Erdős–Kac normalization* (E094 —  $(n)$  vs.  $\Omega(n)$ : Erdős–Kac normalization)
- *E095 — Squarefree filter:  $(n)=\Omega(n)$  when  $(n) \leq 0$*  (E095 — Squarefree filter:  $(n)=\Omega(n)$  when  $(n) \leq 0$ )
- *E118 — Chebyshev bias: lead-time statistics* (E118 — Chebyshev bias: lead-time statistics)
- *E120 — Liouville  $(n)$ : partial sums and parity* (E120 — Liouville  $(n)$ : partial sums and parity)
- *E052: Totient ratio landscape* (Totient ratio landscape)

## 3.3.58 E058: Divisor-count record highs

**Tags:** number-theory, quantitative-exploration, visualization, arithmetic-functions, divisor-function, bounds See: *Valid Tags*.

## Highlights

- Track record highs of  $\tau(n)$  as  $n$  grows.
- Inspect factorizations of record-holders (many small primes with decreasing exponents).



### Goal

Make the record phenomenon behind highly composite numbers tangible.

### Background (quick refresher)

- *Divisor functions  $d(n)$  and  $\sigma_k(n)$  refresher*
- *Arithmetic functions refresher*

### Research question

How do record values of  $\tau(n)$  grow, and what structure do record-holders share?

### Method

- Compute  $\tau(n)$  up to  $N$  and track record positions.
- Plot record curve; list factorizations of top records.

### How to run

- `make run EXP=e058`
- `uv run python -m mathxlab.experiments.e058`

### Outputs

This experiment follows the standard output contract:

- `out/e058/figures/` — generated figures (PNG)
- `out/e058/report.md` — short narrative report
- `out/e058/manifest.json` — snapshot metadata for the gallery

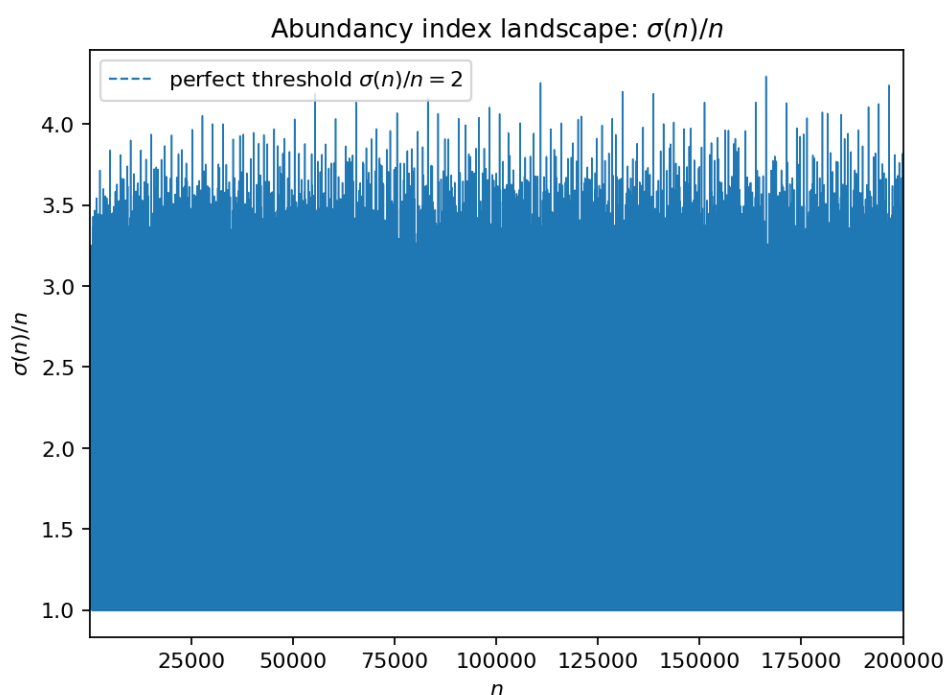
## References

See Ramanujan [1915], Tenenbaum [2015].

## Related experiments

- *E096 — Record-holders for  $\sigma(n)$*  (E096 — Record-holders for  $\sigma(n)$ )
- *E059: Abundancy index landscape* (Abundancy index landscape)
- *E119 — Summatory totient  $\Phi(x)$  scaling check* (E119 — Summatory totient  $\Phi(x)$  scaling check)
- *E117 — Prime-counting approximations:  $li(x)$  and friends* (E117 — Prime-counting approximations:  $li(x)$  and friends)
- *E123 —  $(x;q,a)$  vs. a simple baseline* (E123 —  $(x;q,a)$  vs. a simple baseline)

### 3.3.59 E059: Abundancy index landscape



**Tags:** number-theory, quantitative-exploration, visualization, arithmetic-functions, sigma, divisor-function, perfect See: [Valid Tags](#).

## Highlights

- Plot  $\sigma(n)/n$  and mark the threshold 2 (perfect numbers).
- Highlight spikes caused by many small prime factors.

## Goal

Connect a divisor-sum function to the perfect/abundant classification visually.

## Background (quick refresher)

- *Divisor functions  $d(n)$  and  $\sigma_k(n)$  refresher*
- *Perfect numbers refresher*

### Research question

What shapes and spikes appear in  $\sigma(n)/n$  over moderate ranges, and where do perfect numbers sit?

### Method

- Compute  $\sigma(n)$  up to  $N$  using factorization via SPF.
- Plot  $\sigma(n)/n$  and annotate known perfect numbers in range.

### How to run

- `make run EXP=e059`
- `uv run python -m mathxlab.experiments.e059`

### Outputs

This experiment follows the standard output contract:

- `out/e059/figures/` — generated figures (PNG)
- `out/e059/report.md` — short narrative report
- `out/e059/manifest.json` — snapshot metadata for the gallery

### References

See Apostol [1976], Niven *et al.* [1991].

### Related experiments

- *E097 —  $(n)/n$  landscape: deficient, perfect, abundant* (E097 —  $(n)/n$  landscape: deficient, perfect, abundant)
- *E003: Abundancy Index Landscape* (Abundancy Index Landscape)
- *E058: Divisor-count record highs* (Divisor-count record highs)
- *E096 — Record-holders for  $(n)$*  (E096 — Record-holders for  $(n)$ )
- *E052: Totient ratio landscape* (Totient ratio landscape)

## 3.3.60 E060: Jordan totients

**Tags:** number-theory, quantitative-exploration, visualization, arithmetic-functions, totient, multiplicative See: *Valid Tags*.

### Highlights

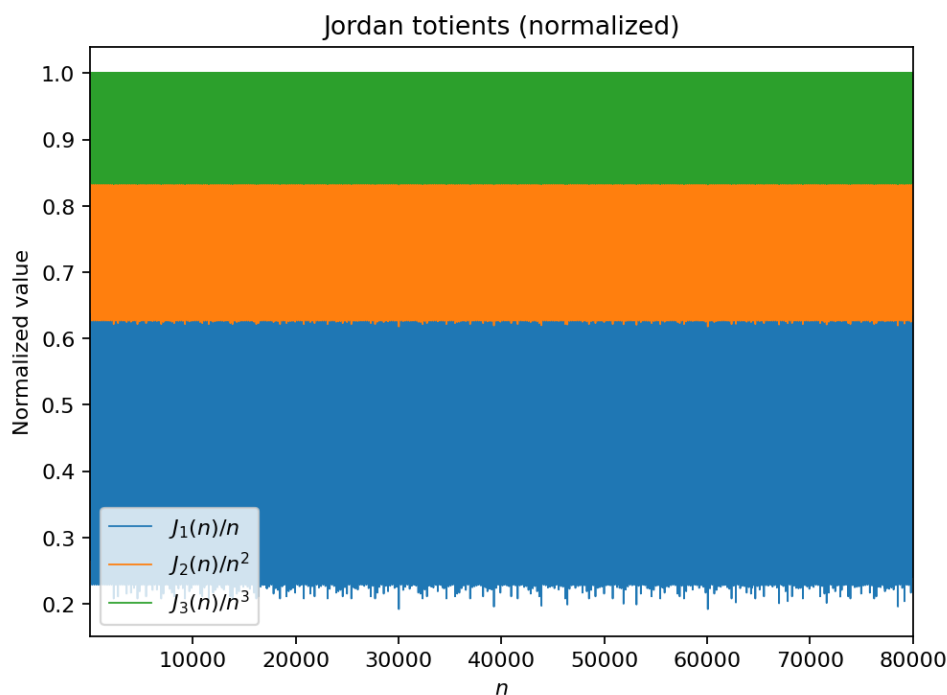
- Compute  $J_k(n)$  for  $k=1,2,3$  and plot  $J_k(n)/n^k$ .
- Compare how the small-prime structure changes as  $k$  increases.

### Goal

Show a clean generalization of  $(n)$  and how normalization reveals multiplicative structure.

### Background (quick refresher)

- *Jordan totient  $J_k(n)$  refresher*
- *Euler's totient function  $\varphi(n)$  refresher*



### Research question

How do the normalized Jordan totients  $J_k(n)/n^k$  behave for  $k=1..3$ ?

### Method

- Compute  $J_k(n) = n^k \prod_{p|n} (1 - p^{-k})$  via prime factors.
- Plot normalized values for  $k=1..3$ .

### How to run

- `make run EXP=e060`
- `uv run python -m mathxlab.experiments.e060`

### Outputs

This experiment follows the standard output contract:

- `out/e060/figures/` — generated figures (PNG)
- `out/e060/report.md` — short narrative report
- `out/e060/manifest.json` — snapshot metadata for the gallery

### References

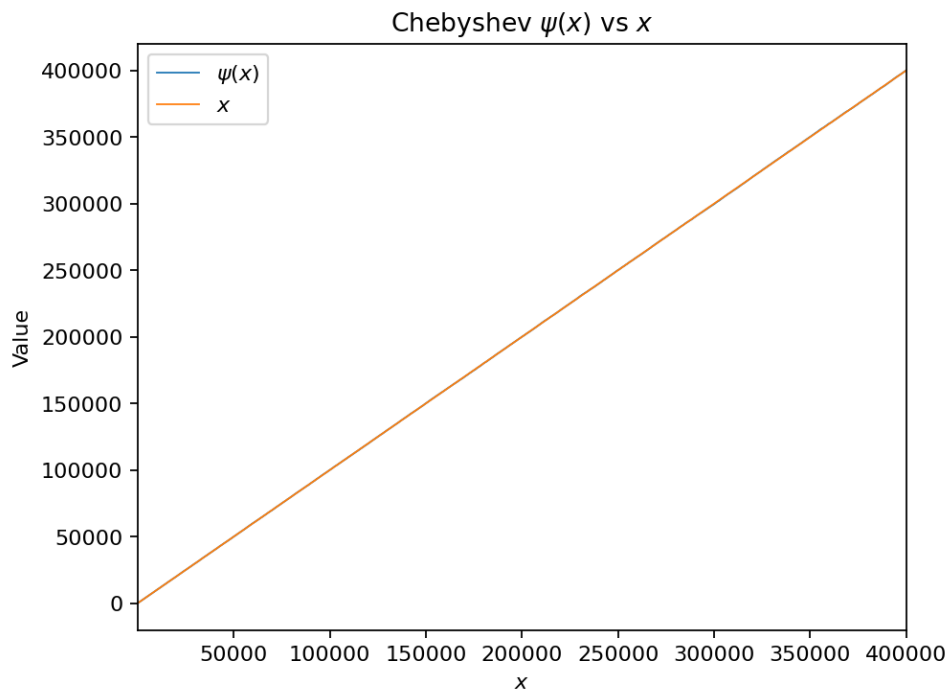
See Apostol [1976].

### Related experiments

- *E099 — Jordan totients  $J_k$ : identities and ratios* (E099 — Jordan totients  $J_k$ : identities and ratios)
- *E052: Totient ratio landscape* (Totient ratio landscape)
- *E102 — Dirichlet convolution identity zoo* (E102 — Dirichlet convolution identity zoo)
- *E053: Inverse totient multiplicities* (Inverse totient multiplicities)

- *E062: Carmichael  $\psi(n)$  vs.  $\phi(n)$  (Carmichael  $\psi(n)$  vs.  $\phi(n)$ )*

### 3.3.61 E061: Chebyshev $\psi(x)$ and prime powers



**Tags:** number-theory, quantitative-exploration, visualization, arithmetic-functions, mangoldt, pnt, summatory See: [Valid Tags](#).

#### Highlights

- Plot  $\psi(x)$  and the error  $\psi(x) - x$ .
- Make the role of prime powers visible (jumps at  $p^k$ ).

#### Goal

Build intuition for Chebyshev functions as “smoothed” prime counters.

#### Background (quick refresher)

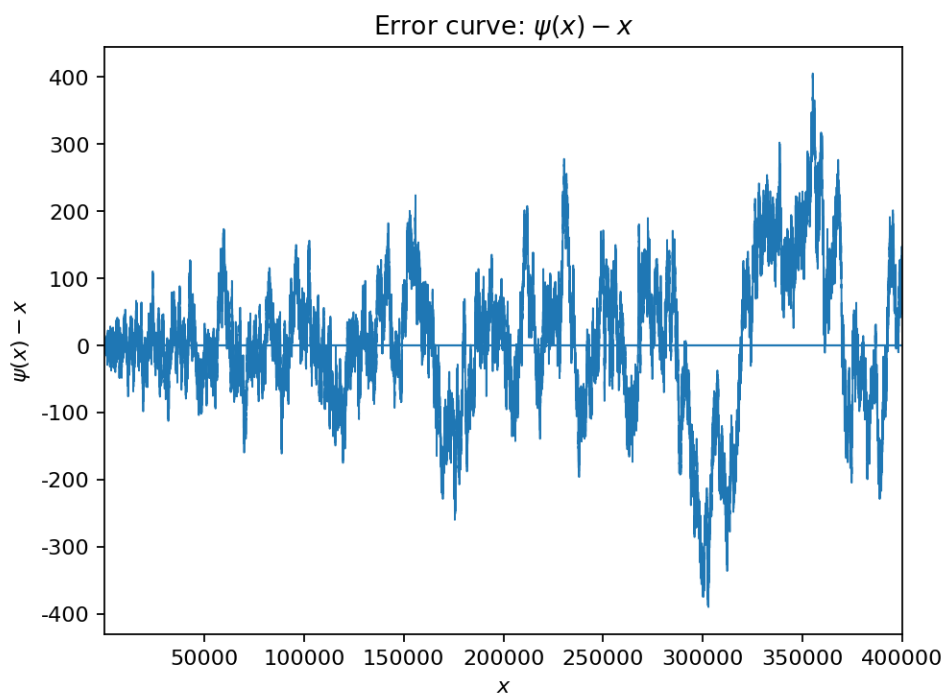
- *von Mangoldt  $\Lambda(n)$  and Chebyshev functions refresher*
- *Prime numbers refresher*

#### Research question

Over a moderate range, what does  $\psi(x) - x$  look like and where do the main jumps occur?

#### Method

- Compute  $\Lambda(n)$  (log  $p$  on prime powers) and cumulative  $\psi(x)$ .
- Plot  $\psi(x)$  and  $\psi(x) - x$ ; optionally annotate largest jumps.



### How to run

- `make run EXP=e061`
- `uv run python -m mathxlab.experiments.e061`

### Outputs

This experiment follows the standard output contract:

- `out/e061/figures/` — generated figures (PNG)
- `out/e061/report.md` — short narrative report
- `out/e061/manifest.json` — snapshot metadata for the gallery

### References

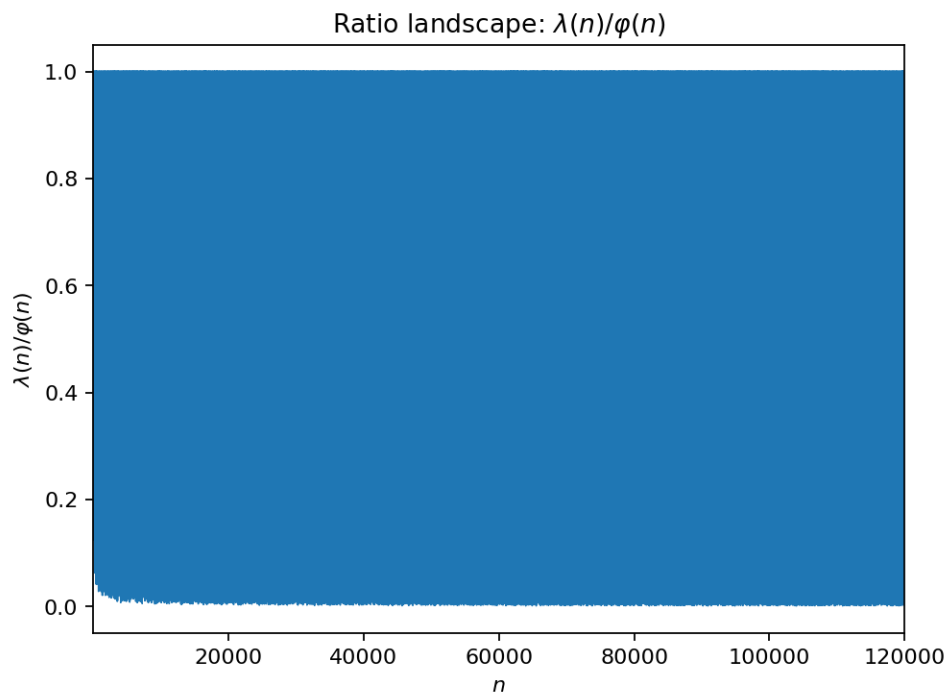
See Montgomery and Vaughan [2006], Apostol [1976].

### Related experiments

- *E103 — Chebyshev (x): prime powers drive jumps* (E103 — Chebyshev (x): prime powers drive jumps)
- *E104 — von Mangoldt  $\Lambda(n)$ : support and statistics* (E104 — von Mangoldt  $\Lambda(n)$ : support and statistics)
- *E117 — Prime-counting approximations:  $\text{li}(x)$  and friends* (E117 — Prime-counting approximations:  $\text{li}(x)$  and friends)
- *E054: Squarefree density via Möbius* (Squarefree density via Möbius)
- *E055: Mertens function walk* (Mertens function walk)



### 3.3.62 E062: Carmichael $\lambda(n)$ vs. $\varphi(n)$



**Tags:** number-theory, quantitative-exploration, visualization, arithmetic-functions, carmichael-lambda, totient See: [Valid Tags](#).

#### Highlights

- Plot or histogram the ratio  $\varphi(n)/\lambda(n)$ .
- Show how 2-adic structure affects  $\lambda(n)$  (notably for powers of 2).

#### Goal

See how the exponent of the multiplicative group mod  $n$  differs from its size.

#### Background (quick refresher)

- *Carmichael's  $\lambda(n)$  function refresher*
- *Euler's totient function  $\varphi(n)$  refresher*

#### Research question

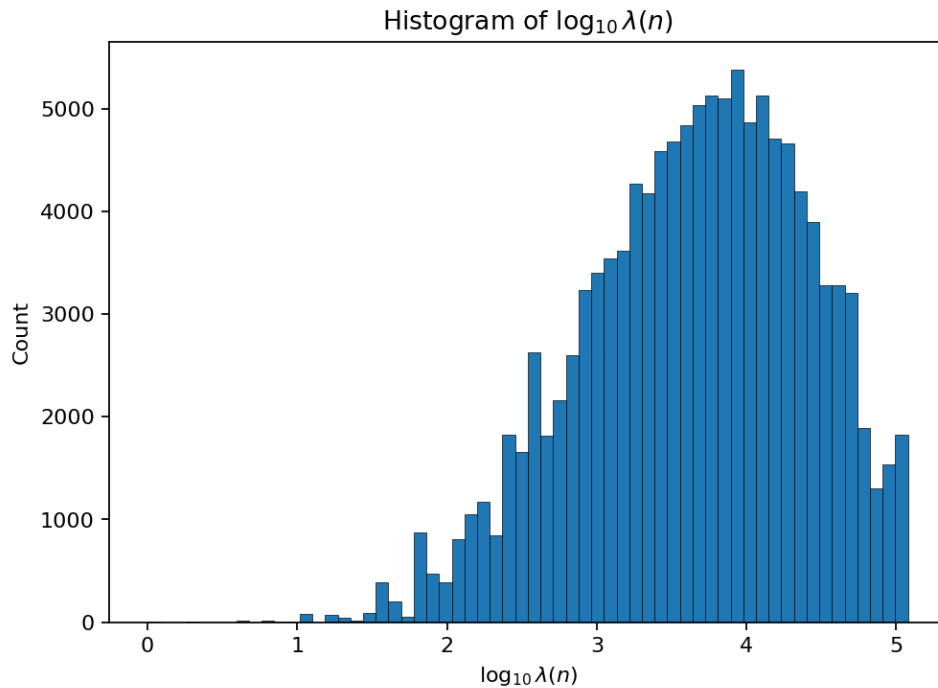
How large can  $\varphi(n)/\lambda(n)$  get for  $n \leq N$ , and what kinds of  $n$  cause large gaps?

#### Method

- Compute  $\lambda(n)$  and  $\varphi(n)$  from prime-power formulas and lcm composition.
- Visualize ratios (histograms, quantiles) and list top outliers.

#### How to run

- `make run EXP=e062`
- `uv run python -m mathxlab.experiments.e062`



## Outputs

This experiment follows the standard output contract:

- `out/e062/figures/` — generated figures (PNG)
- `out/e062/report.md` — short narrative report
- `out/e062/manifest.json` — snapshot metadata for the gallery

## References

See Erdős *et al.* [1991], Montgomery and Vaughan [2006].

## Related experiments

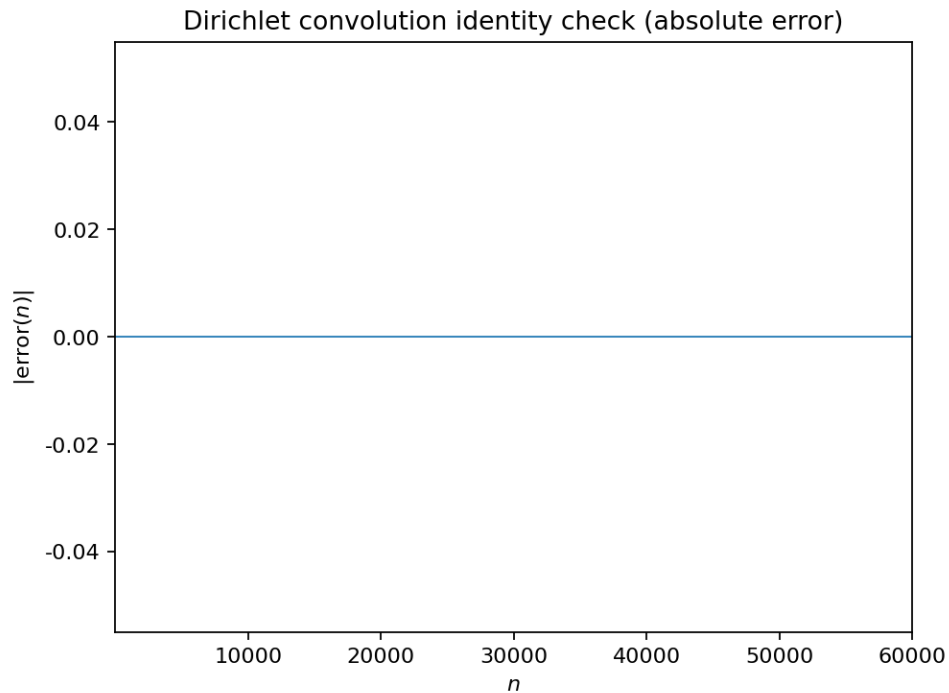
- *E100 — Carmichael (n) vs. Euler (n)* (E100 — Carmichael (n) vs. Euler (n))
- *E052: Totient ratio landscape* (Totient ratio landscape)
- *E053: Inverse totient multiplicities* (Inverse totient multiplicities)
- *E060: Jordan totients* (Jordan totients)
- *E099 — Jordan totients  $J_k$ : identities and ratios* (E099 — Jordan totients  $J_k$ : identities and ratios)

### 3.3.63 E063: Dirichlet convolution playground

**Tags:** number-theory, quantitative-exploration, visualization, arithmetic-functions, dirichlet-convolution, model-checking See: *Valid Tags*.

## Highlights

- Compute Dirichlet convolutions numerically on a finite range.
- Check exact identities and report any mismatches (should be zero).



### Goal

Turn algebraic identities of arithmetic functions into concrete computed checks.

### Background (quick refresher)

- *Dirichlet convolution refresher*
- *Arithmetic functions refresher*
- *Euler's totient function  $\varphi(n)$  refresher*

### Research question

Can we numerically verify standard Dirichlet-convolution identities up to a bound without mistakes?

### Method

- Implement divisor-sum convolution on  $[1..N]$ .
- Compute  $1$  and  $\text{id}$  and compare to  $\text{id}$  and  $\text{id}$  respectively.

### How to run

- `make run EXP=e063`
- `uv run python -m mathxlab.experiments.e063`

### Outputs

This experiment follows the standard output contract:

- `out/e063/figures/` — generated figures (PNG)
- `out/e063/report.md` — short narrative report
- `out/e063/manifest.json` — snapshot metadata for the gallery

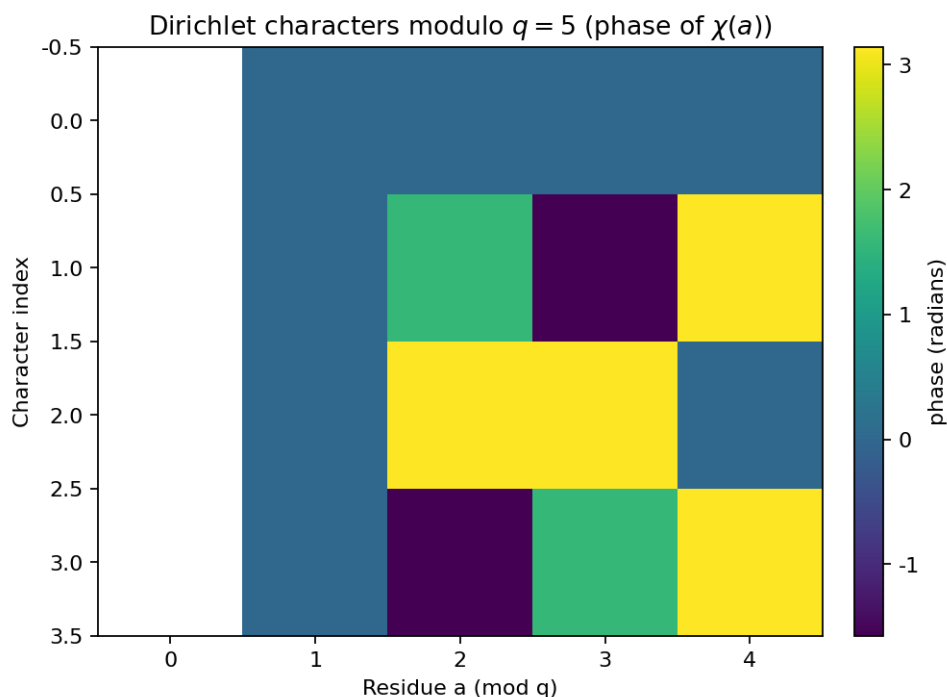
## References

See Apostol [1976].

## Related experiments

- *E102 — Dirichlet convolution identity zoo* (E102 — Dirichlet convolution identity zoo)
- *E121 — Möbius inversion as convolution undo* (E121 — Möbius inversion as convolution undo)
- *E111 — Euler product vs. Dirichlet series for  $L(s, \chi)$*  (E111 — Euler product vs. Dirichlet series for  $L(s, \chi)$ )
- *E055: Mertens function walk* (Mertens function walk)
- *E095 — Squarefree filter:  $\Omega(n) = \Omega(n)$  when  $\chi(n) \neq 0$*  (E095 — Squarefree filter:  $\Omega(n) = \Omega(n)$  when  $\chi(n) \neq 0$ )

### 3.3.64 E064: Dirichlet character tables (phase view).



Tags: number-theory, quantitative-exploration, visualization, dirichlet-characters

## Highlights

- Constructs full character tables for a small modulus  $q$ .
- Visualizes values as phases / real parts to spot structure and symmetries.
- Checks orthogonality and basic sanity constraints numerically.

## What this experiment does

This experiment enumerates all Dirichlet characters modulo a small modulus  $q$  and visualizes the values  $\chi(a)$  for residues  $a=0..q-1$ .

The implementation focuses on a compact, reproducible numerical workflow: deterministic parameter defaults, structured output folders, and one or more figures saved for the gallery.

## Outputs

This experiment writes into out/e064/:

- figures/fig\_01\_character\_phases.png

## How to run

```
make run EXP=e064
```

## Notes

- The gallery preview figure shipped with the documentation uses conservative cutoffs so builds stay fast. If you run the experiment locally, increase the cutoffs to see the asymptotic regime more clearly.
- Prime-race plots depend on the chosen sampling of  $x$  (linear vs. log grid). The qualitative “who leads” picture can change when you zoom in.

## References

Davenport [2000], Niven *et al.* [1991]

## Related experiments

- *E065: Orthogonality matrix for Dirichlet characters.* (Orthogonality matrix for Dirichlet characters.)
- *E066: Character partial sums: cancellation profiles.* (Character partial sums: cancellation profiles.)
- *E068: Dirichlet  $L(s, \cdot)$ : series vs. Euler product (partial approximations).* (Dirichlet  $L(s, \cdot)$ : series vs. Euler product (partial approximations).)
- *E077: Indicator via character orthogonality (sanity check).* (Indicator via character orthogonality (sanity check).)
- *E122 — Character averages over primes* (E122 — Character averages over primes)

### 3.3.65 E065: Orthogonality matrix for Dirichlet characters.

**Tags:** number-theory, quantitative-exploration, visualization, dirichlet-characters

## Highlights

- Constructs full character tables for a small modulus  $q$ .
- Visualizes values as phases / real parts to spot structure and symmetries.
- Checks orthogonality and basic sanity constraints numerically.

## What this experiment does

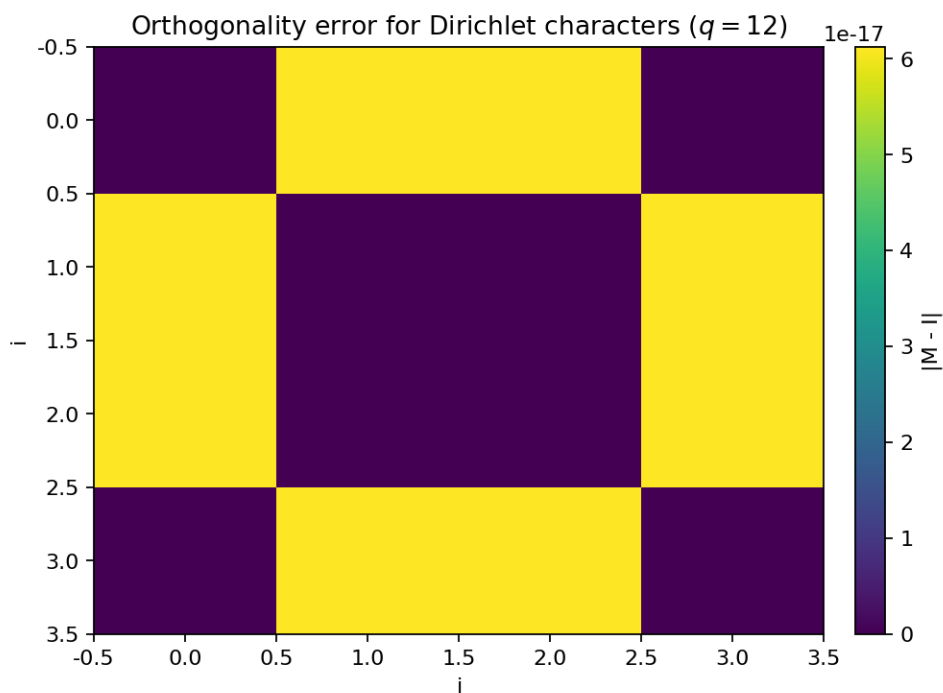
For characters  $\chi$  modulo  $q$ , orthogonality says:

The implementation focuses on a compact, reproducible numerical workflow: deterministic parameter defaults, structured output folders, and one or more figures saved for the gallery.

## Outputs

This experiment writes into out/e065/:

- figures/fig\_01\_orthogonality\_error.png



### How to run

```
make run EXP=e065
```

### Notes

- The gallery preview figure shipped with the documentation uses conservative cutoffs so builds stay fast. If you run the experiment locally, increase the cutoffs to see the asymptotic regime more clearly.
- Prime-race plots depend on the chosen sampling of  $x$  (linear vs. log grid). The qualitative “who leads” picture can change when you zoom in.

### References

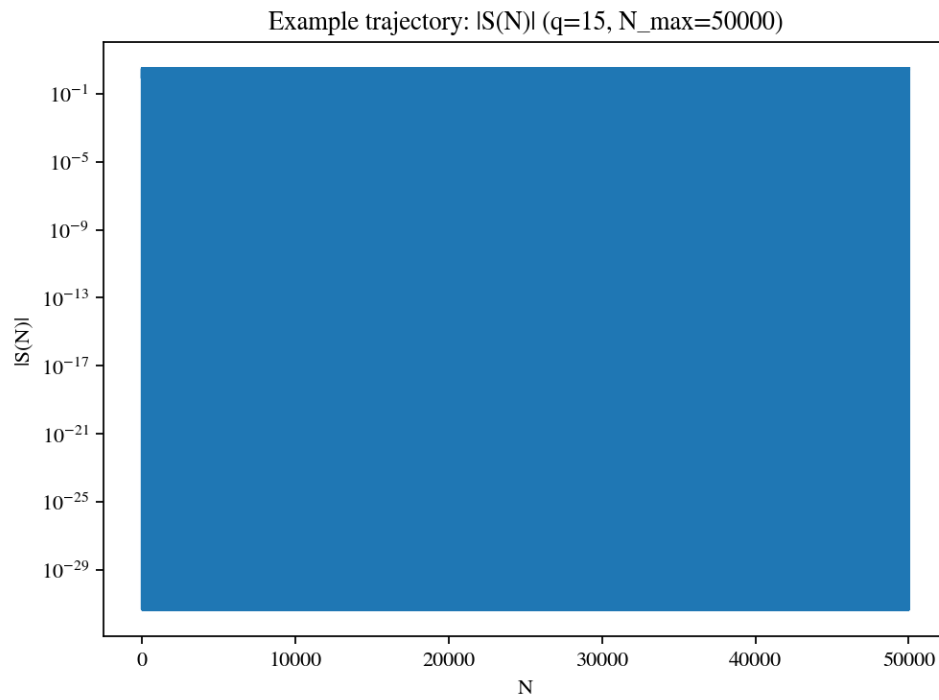
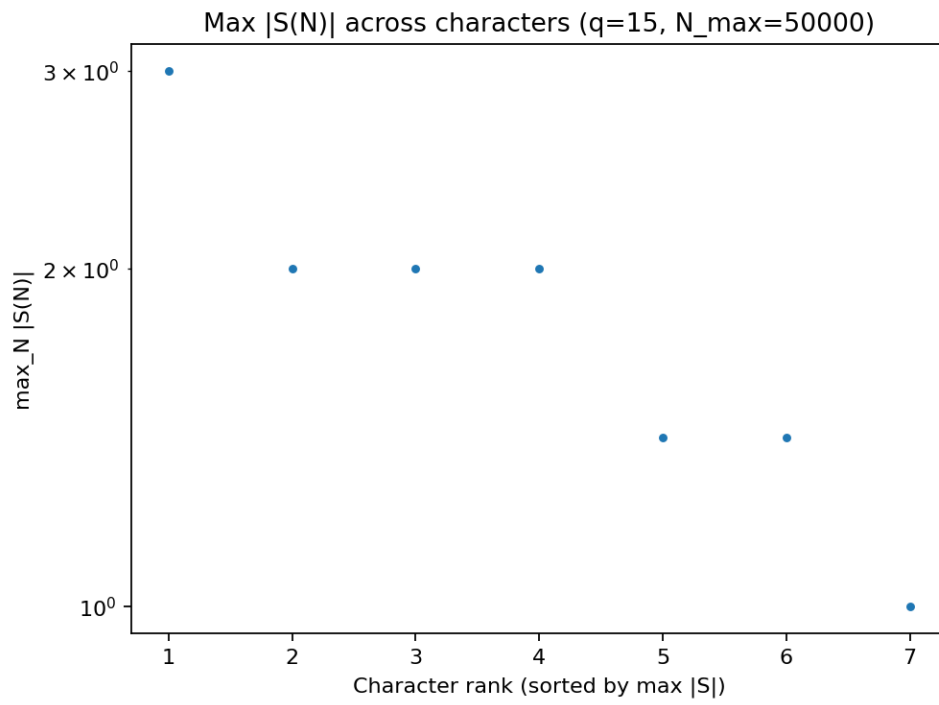
Davenport [2000], Niven *et al.* [1991]

### Related experiments

- *E108 — Orthogonality heatmap for characters* (E108 — Orthogonality heatmap for characters)
- *E064: Dirichlet character tables (phase view)*. (Dirichlet character tables (phase view).)
- *E068: Dirichlet  $L(s, \chi)$ : series vs. Euler product (partial approximations)*. (Dirichlet  $L(s, \chi)$ : series vs. Euler product (partial approximations).)
- *E077: Indicator via character orthogonality (sanity check)*. (Indicator via character orthogonality (sanity check).)
- *E078: Max partial sums across characters*. (Max partial sums across characters.)

### 3.3.66 E066: Character partial sums: cancellation profiles.

Tags: number-theory, quantitative-exploration, visualization, dirichlet-characters



## Highlights

- Constructs full character tables for a small modulus  $q$ .
- Visualizes values as phases / real parts to spot structure and symmetries.
- Checks orthogonality and basic sanity constraints numerically.

## What this experiment does

For a Dirichlet character modulo  $q$ , consider partial sums:

The implementation focuses on a compact, reproducible numerical workflow: deterministic parameter defaults, structured output folders, and one or more figures saved for the gallery.

## Outputs

This experiment writes into `out/e066/`:

- `figures/fig_01_max_partial_sums.png`
- `figures/fig_02_example_partial_sum.png`

## How to run

```
make run EXP=e066
```

## Notes

- The gallery preview figure shipped with the documentation uses conservative cutoffs so builds stay fast. If you run the experiment locally, increase the cutoffs to see the asymptotic regime more clearly.
- Prime-race plots depend on the chosen sampling of  $x$  (linear vs. log grid). The qualitative “who leads” picture can change when you zoom in.

## References

Davenport [2000], Niven *et al.* [1991]

## Related experiments

- *E078: Max partial sums across characters.* (Max partial sums across characters.)
- *E110 — Dirichlet L-series partial sums at  $s=1$  and  $s=1/2$*  (E110 — Dirichlet L-series partial sums at  $s=1$  and  $s=1/2$ )
- *E120 — Liouville  $(n)$ : partial sums and parity* (E120 — Liouville  $(n)$ : partial sums and parity)
- *E064: Dirichlet character tables (phase view).* (Dirichlet character tables (phase view).)
- *E067: Gauss sums: magnitude vs.  $\sqrt{q}$ .* (Gauss sums: magnitude vs.  $\sqrt{q}$ .)

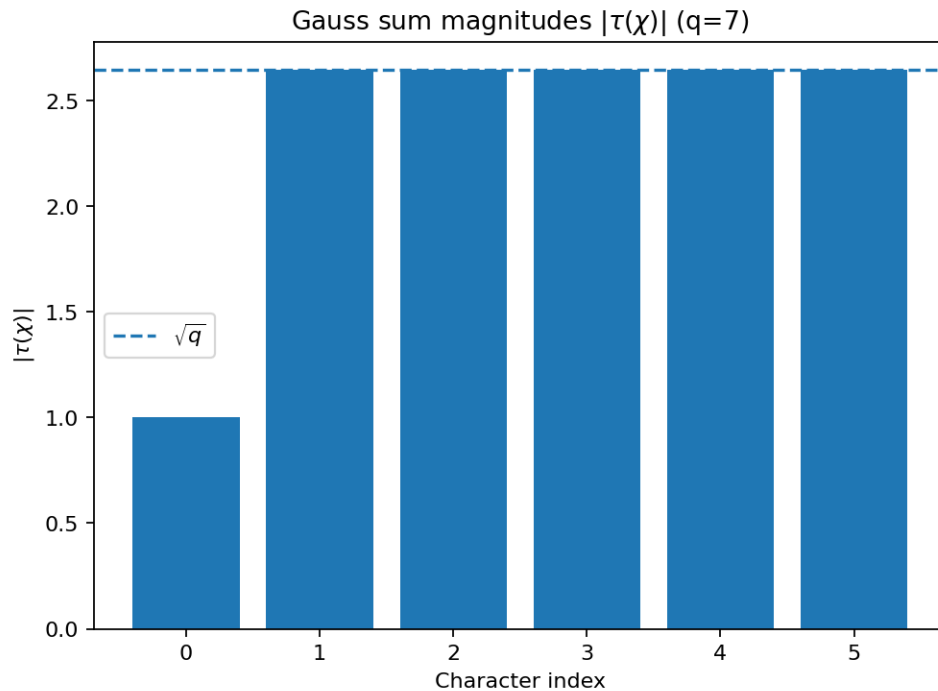
### 3.3.67 E067: Gauss sums: magnitude vs. $\sqrt{q}$ .

Tags: number-theory, quantitative-exploration, visualization, dirichlet-characters

## Highlights

- Constructs full character tables for a small modulus  $q$ .
- Visualizes values as phases / real parts to spot structure and symmetries.
- Checks orthogonality and basic sanity constraints numerically.





### What this experiment does

For a Dirichlet character modulo  $q$ , define the Gauss sum

The implementation focuses on a compact, reproducible numerical workflow: deterministic parameter defaults, structured output folders, and one or more figures saved for the gallery.

### Outputs

This experiment writes into `out/e067/`:

- `figures/fig_01_gauss_sum_magnitudes.png`

### How to run

```
make run EXP=e067
```

### Notes

- The gallery preview figure shipped with the documentation uses conservative cutoffs so builds stay fast. If you run the experiment locally, increase the cutoffs to see the asymptotic regime more clearly.
- Prime-race plots depend on the chosen sampling of  $x$  (linear vs. log grid). The qualitative “who leads” picture can change when you zoom in.

### References

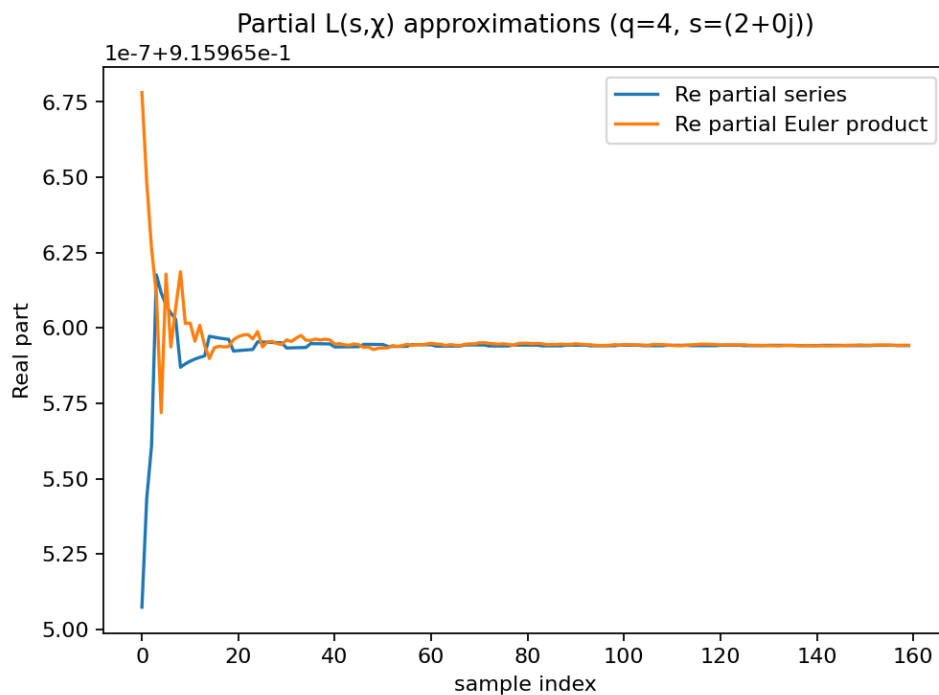
Davenport [2000], Niven *et al.* [1991]

### Related experiments

- *E109 — Gauss sums: magnitude patterns* (E109 — Gauss sums: magnitude patterns)
- *E066: Character partial sums: cancellation profiles.* (Character partial sums: cancellation profiles.)
- *E078: Max partial sums across characters.* (Max partial sums across characters.)

- *E110 — Dirichlet L-series partial sums at  $s=1$  and  $s=1/2$*  (E110 — Dirichlet L-series partial sums at  $s=1$  and  $s=1/2$ )
- *E120 — Liouville ( $n$ ): partial sums and parity* (E120 — Liouville ( $n$ ): partial sums and parity)

### 3.3.68 E068: Dirichlet $L(s, \chi)$ : series vs. Euler product (partial approximations).



Tags: number-theory, quantitative-exploration, visualization, dirichlet-characters, l-functions

#### Highlights

- Computes truncated Dirichlet L-series for a nontrivial character.
- Compares series truncation against Euler-product truncation (where meaningful).
- Shows convergence behavior (fast for  $s>1$ , slow at  $s=1$ ).

#### What this experiment does

For  $\text{Re}(s) > 1$ , Dirichlet L-functions admit both:

The implementation focuses on a compact, reproducible numerical workflow: deterministic parameter defaults, structured output folders, and one or more figures saved for the gallery.

#### Outputs

This experiment writes into `out/e068/`:

- `figures/fig_01_series_vs_euler.png`

#### How to run

```
make run EXP=e068
```

## Notes

- The gallery preview figure shipped with the documentation uses conservative cutoffs so builds stay fast. If you run the experiment locally, increase the cutoffs to see the asymptotic regime more clearly.
- Prime-race plots depend on the chosen sampling of  $x$  (linear vs. log grid). The qualitative “who leads” picture can change when you zoom in.

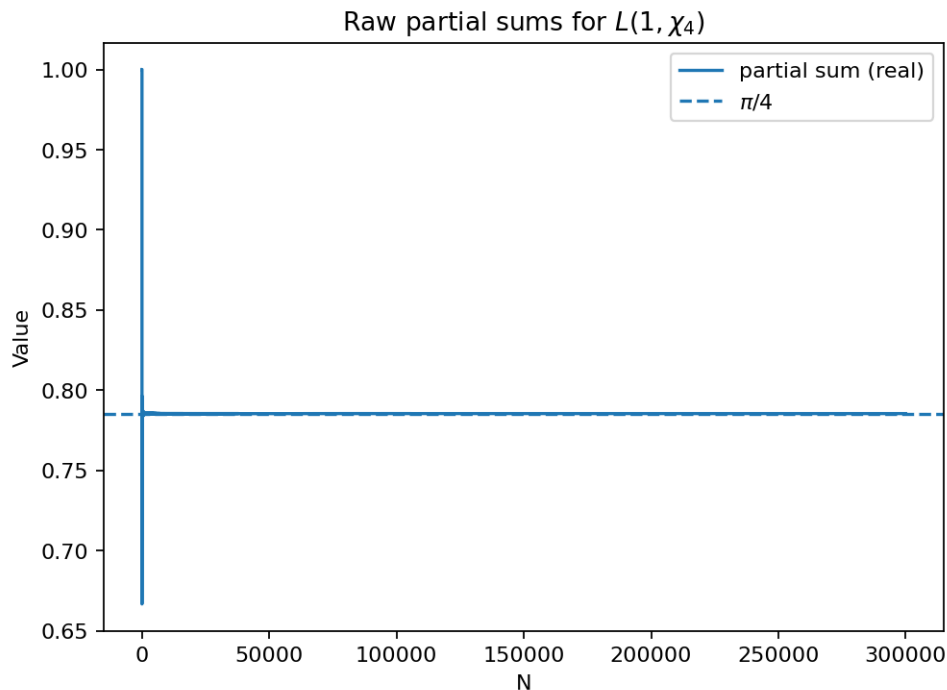
## References

Apostol [1976], Montgomery and Vaughan [2006]

## Related experiments

- *E111 — Euler product vs. Dirichlet series for  $L(s, \chi)$*  (E111 — Euler product vs. Dirichlet series for  $L(s, \chi)$ )
- *E110 — Dirichlet L-series partial sums at  $s=1$  and  $s=1/2$*  (E110 — Dirichlet L-series partial sums at  $s=1$  and  $s=1/2$ )
- *E083 — Series vs. Euler product* (E083 — Series vs. Euler product)
- *E091 — Partial Euler products on the critical line* (E091 — Partial Euler products on the critical line)
- *E092 —  $1/\zeta(s)$  via the Möbius Dirichlet series* (E092 —  $1/\zeta(s)$  via the Möbius Dirichlet series)

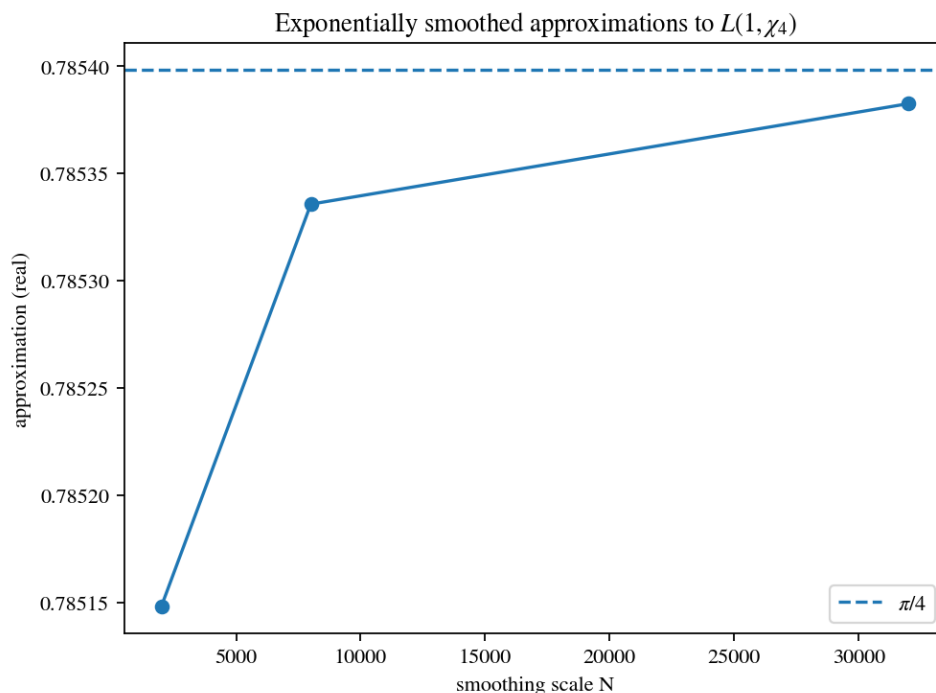
### 3.3.69 E069: $L(1, \chi_4)$ : slow convergence and smoothing.



**Tags:** number-theory, quantitative-exploration, visualization, dirichlet-characters, l-functions

## Highlights

- Computes truncated Dirichlet L-series for a nontrivial character.
- Compares series truncation against Euler-product truncation (where meaningful).
- Shows convergence behavior (fast for  $s>1$ , slow at  $s=1$ ).



### What this experiment does

At  $s=1$ , the Dirichlet series for  $L(1, \chi_4)$  converges very slowly. For the nontrivial character modulo 4:

The implementation focuses on a compact, reproducible numerical workflow: deterministic parameter defaults, structured output folders, and one or more figures saved for the gallery.

### Outputs

This experiment writes into `out/e069/`:

- `figures/fig_01_l1_partial_sums.png`
- `figures/fig_02_l1_smoothed.png`

### How to run

```
make run EXP=e069
```

### Notes

- The gallery preview figure shipped with the documentation uses conservative cutoffs so builds stay fast. If you run the experiment locally, increase the cutoffs to see the asymptotic regime more clearly.
- Prime-race plots depend on the chosen sampling of  $x$  (linear vs. log grid). The qualitative “who leads” picture can change when you zoom in.

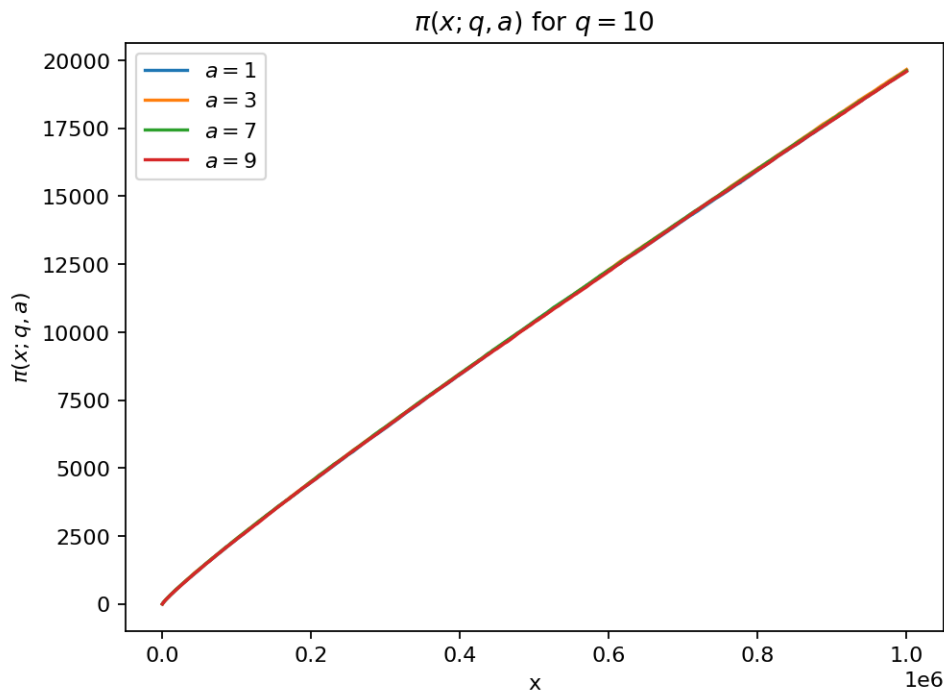
### References

Apostol [1976], Montgomery and Vaughan [2006]

### Related experiments

- *E068: Dirichlet  $L(s, \cdot)$ : series vs. Euler product (partial approximations).* (Dirichlet  $L(s, \cdot)$ : series vs. Euler product (partial approximations).)
- *E107 — Conductor: primitive vs. induced characters* (E107 — Conductor: primitive vs. induced characters)
- *E122 — Character averages over primes* (E122 — Character averages over primes)
- *E110 — Dirichlet L-series partial sums at  $s=1$  and  $s=1/2$*  (E110 — Dirichlet L-series partial sums at  $s=1$  and  $s=1/2$ )
- *E111 — Euler product vs. Dirichlet series for  $L(s, \cdot)$*  (E111 — Euler product vs. Dirichlet series for  $L(s, \cdot)$ )

### 3.3.70 E070: Primes in residue classes: $\pi(x; q, a)$ .



Tags: number-theory, quantitative-exploration, visualization, aps

#### Highlights

- Counts primes in residue classes  $a \pmod q$  for several  $a$ .
- Compares empirical counts to the first-order prediction  $\text{li}(x)/\phi(q)$ .
- Tracks a simple error proxy across  $x$  to visualize deviation patterns.

#### What this experiment does

This experiment counts primes in selected reduced residue classes modulo  $q$ :

The implementation focuses on a compact, reproducible numerical workflow: deterministic parameter defaults, structured output folders, and one or more figures saved for the gallery.

## Outputs

This experiment writes into out/e070/:

- figures/fig\_01\_pi\_x\_q\_a.png

## How to run

```
make run EXP=e070
```

## Notes

- The gallery preview figure shipped with the documentation uses conservative cutoffs so builds stay fast. If you run the experiment locally, increase the cutoffs to see the asymptotic regime more clearly.
- Prime-race plots depend on the chosen sampling of  $x$  (linear vs. log grid). The qualitative “who leads” picture can change when you zoom in.

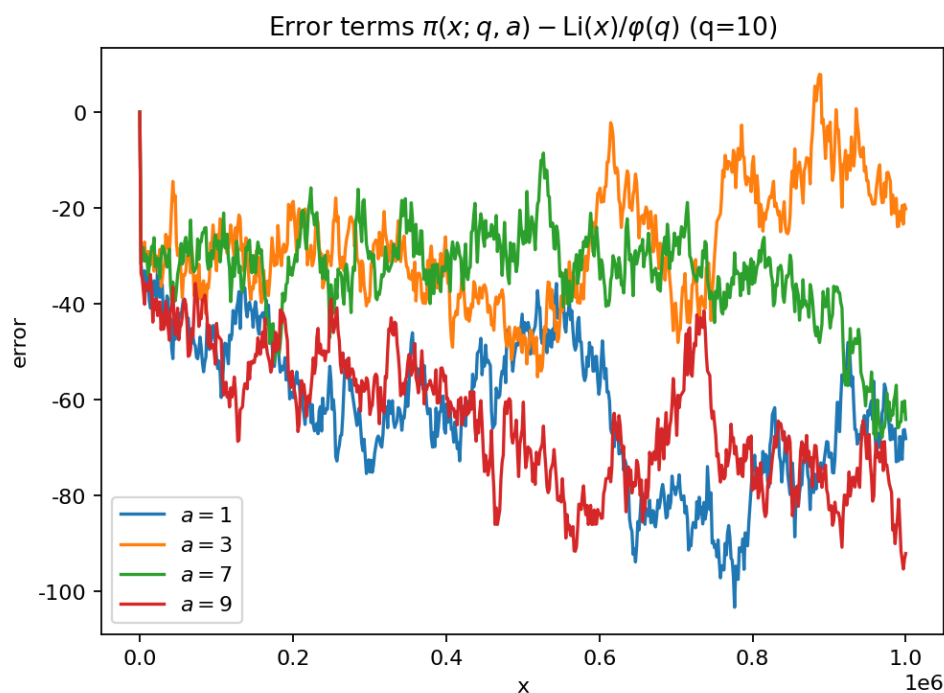
## References

Apostol [1976], Davenport [2000]

## Related experiments

- *E113 — First prime in each residue class* (E113 — First prime in each residue class)
- *E023: Residue class distribution mod  $q$*  (Residue class distribution mod  $q$ )
- *E071: PNT(AP) numerics:  $\pi(x;q,a) - \text{Li}(x)/\phi(q)$ .* (PNT(AP) numerics:  $\pi(x;q,a) - \text{Li}(x)/\phi(q)$ .)
- *E072: Prime race mod 4:  $\pi(x;4,3)$  vs.  $\pi(x;4,1)$ .* (Prime race mod 4:  $\pi(x;4,3)$  vs.  $\pi(x;4,1)$ .)
- *E073: Prime race mod 3:  $\pi(x;3,2)$  vs.  $\pi(x;3,1)$ .* (Prime race mod 3:  $\pi(x;3,2)$  vs.  $\pi(x;3,1)$ .)

### 3.3.71 E071: PNT(AP) numerics: $\pi(x;q,a) - \text{Li}(x)/\phi(q)$ .



Tags: number-theory, quantitative-exploration, visualization, aps

## Highlights

- Counts primes in residue classes  $a \bmod q$  for several  $a$ .
- Compares empirical counts to the first-order prediction  $\text{li}(x)/q$ .
- Tracks a simple error proxy across  $x$  to visualize deviation patterns.

## What this experiment does

The prime number theorem in arithmetic progressions suggests:

The implementation focuses on a compact, reproducible numerical workflow: deterministic parameter defaults, structured output folders, and one or more figures saved for the gallery.

## Outputs

This experiment writes into `out/e071/`:

- `figures/fig_01_error_terms.png`

## How to run

```
make run EXP=e071
```

## Notes

- The gallery preview figure shipped with the documentation uses conservative cutoffs so builds stay fast. If you run the experiment locally, increase the cutoffs to see the asymptotic regime more clearly.
- Prime-race plots depend on the chosen sampling of  $x$  (linear vs. log grid). The qualitative “who leads” picture can change when you zoom in.

## References

Apostol [1976], Davenport [2000]

## Related experiments

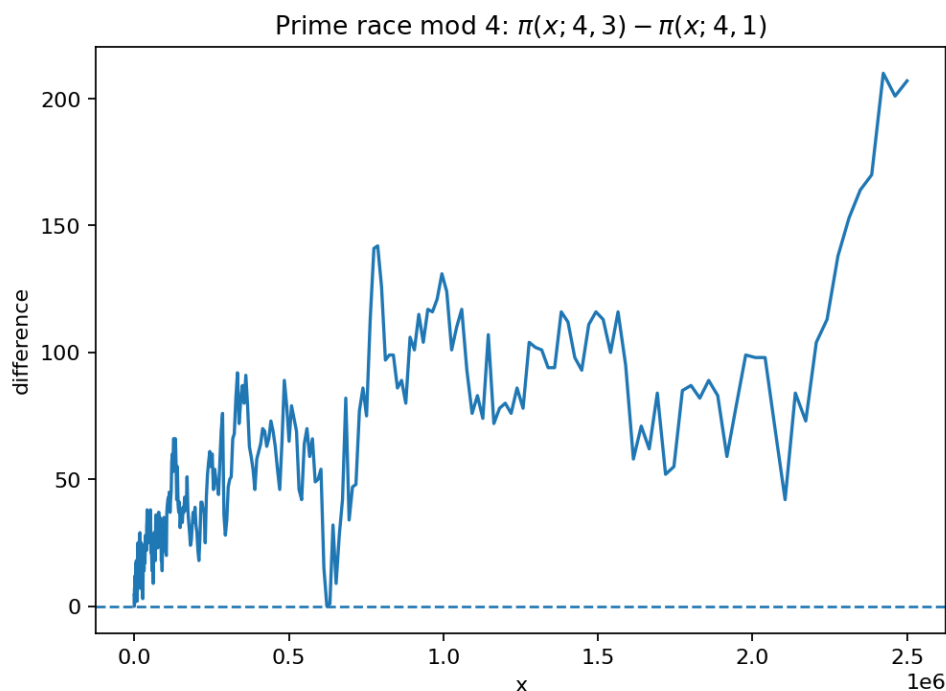
- *E070: Primes in residue classes:  $\pi(x; q, a)$ .* (Primes in residue classes:  $\pi(x; q, a)$ .)
- *E072: Prime race mod 4:  $\pi(x; 4, 3)$  vs.  $\pi(x; 4, 1)$ .* (Prime race mod 4:  $\pi(x; 4, 3)$  vs.  $\pi(x; 4, 1)$ .)
- *E073: Prime race mod 3:  $\pi(x; 3, 2)$  vs.  $\pi(x; 3, 1)$ .* (Prime race mod 3:  $\pi(x; 3, 2)$  vs.  $\pi(x; 3, 1)$ .)
- *E074: Prime race mod 8: leaderboard among 1,3,5,7.* (Prime race mod 8: leaderboard among 1,3,5,7.)
- *E075: Prime race statistic: distribution on a log-grid.* (Prime race statistic: distribution on a log-grid.)

### 3.3.72 E072: Prime race mod 4: $\pi(x; 4, 3)$ vs. $\pi(x; 4, 1)$ .

Tags: number-theory, quantitative-exploration, visualization, prime-races, aps

## Highlights

- Computes prime-race differences  $\pi(x; q, a) - \pi(x; q, b)$  on a grid of  $x$  values.
- Visualizes lead changes and the size of fluctuations as  $x$  grows.
- Adds a derived statistic (normalization / -variant) to compare behaviors.



### What this experiment does

A classical “prime race” compares how often one residue class leads another in prime counts. The most famous is mod 4:

The implementation focuses on a compact, reproducible numerical workflow: deterministic parameter defaults, structured output folders, and one or more figures saved for the gallery.

### Outputs

This experiment writes into `out/e072/`:

- `figures/fig_01_race_mod4_diff.png`

### How to run

```
make run EXP=e072
```

### Notes

- The gallery preview figure shipped with the documentation uses conservative cutoffs so builds stay fast. If you run the experiment locally, increase the cutoffs to see the asymptotic regime more clearly.
- Prime-race plots depend on the chosen sampling of  $x$  (linear vs. log grid). The qualitative “who leads” picture can change when you zoom in.

### References

Granville and Martin [2006], Rubinstein and Sarnak [1994]

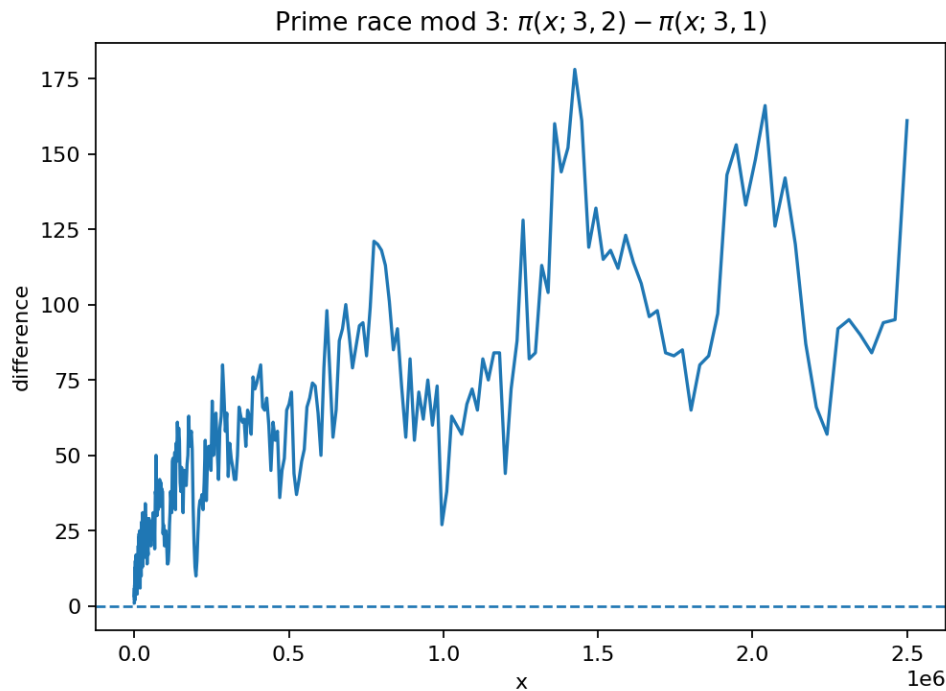
### Related experiments

- *E073: Prime race mod 3:  $\pi(x; 3, 2)$  vs.  $\pi(x; 3, 1)$ .* (Prime race mod 3:  $\pi(x; 3, 2)$  vs.  $\pi(x; 3, 1)$ .)
- *E074: Prime race mod 8: leaderboard among 1,3,5,7.* (Prime race mod 8: leaderboard among 1,3,5,7.)



- *E075: Prime race statistic: distribution on a log-grid.* (Prime race statistic: distribution on a log-grid.)
- *E112 — Prime race:  $(x;q,a) - (x;q,b)$*  (E112 — Prime race:  $(x;q,a) - (x;q,b)$ )
- *E081: Prime race sign changes: first crossings table.* (Prime race sign changes: first crossings table.)

### 3.3.73 E073: Prime race mod 3: $\pi(x;3,2)$ vs. $\pi(x;3,1)$ .



Tags: number-theory, quantitative-exploration, visualization, prime-races, aps

#### Highlights

- Computes prime-race differences  $(x;q,a) - (x;q,b)$  on a grid of  $x$  values.
- Visualizes lead changes and the size of fluctuations as  $x$  grows.
- Adds a derived statistic (normalization / -variant) to compare behaviors.

#### What this experiment does

Another small prime race compares the two reduced residue classes modulo 3:

The implementation focuses on a compact, reproducible numerical workflow: deterministic parameter defaults, structured output folders, and one or more figures saved for the gallery.

#### Outputs

This experiment writes into `out/e073/`:

- `figures/fig_01_race_mod3_diff.png`

#### How to run

```
make run EXP=e073
```

## Notes

- The gallery preview figure shipped with the documentation uses conservative cutoffs so builds stay fast. If you run the experiment locally, increase the cutoffs to see the asymptotic regime more clearly.
- Prime-race plots depend on the chosen sampling of  $x$  (linear vs. log grid). The qualitative “who leads” picture can change when you zoom in.

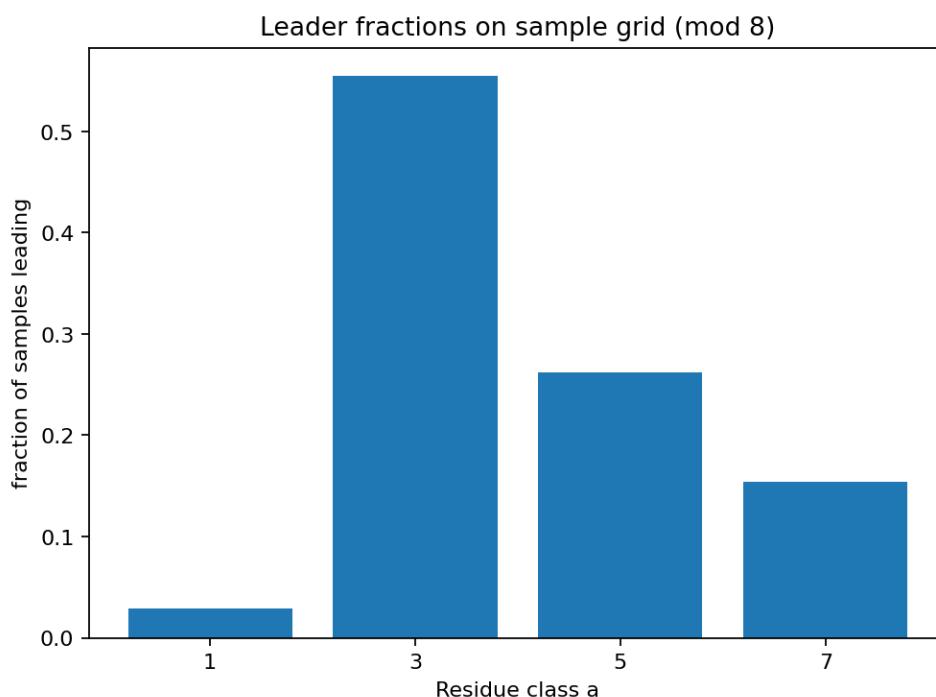
## References

Granville and Martin [2006], Rubinstein and Sarnak [1994]

## Related experiments

- *E072: Prime race mod 4:  $\pi(x;4,3)$  vs.  $\pi(x;4,1)$ .* (Prime race mod 4:  $\pi(x;4,3)$  vs.  $\pi(x;4,1)$ .)
- *E074: Prime race mod 8: leaderboard among 1,3,5,7.* (Prime race mod 8: leaderboard among 1,3,5,7.)
- *E075: Prime race statistic: distribution on a log-grid.* (Prime race statistic: distribution on a log-grid.)
- *E112 — Prime race:  $(x;q,a) - (x;q,b)$*  (E112 — Prime race:  $(x;q,a) - (x;q,b)$ )
- *E081: Prime race sign changes: first crossings table.* (Prime race sign changes: first crossings table.)

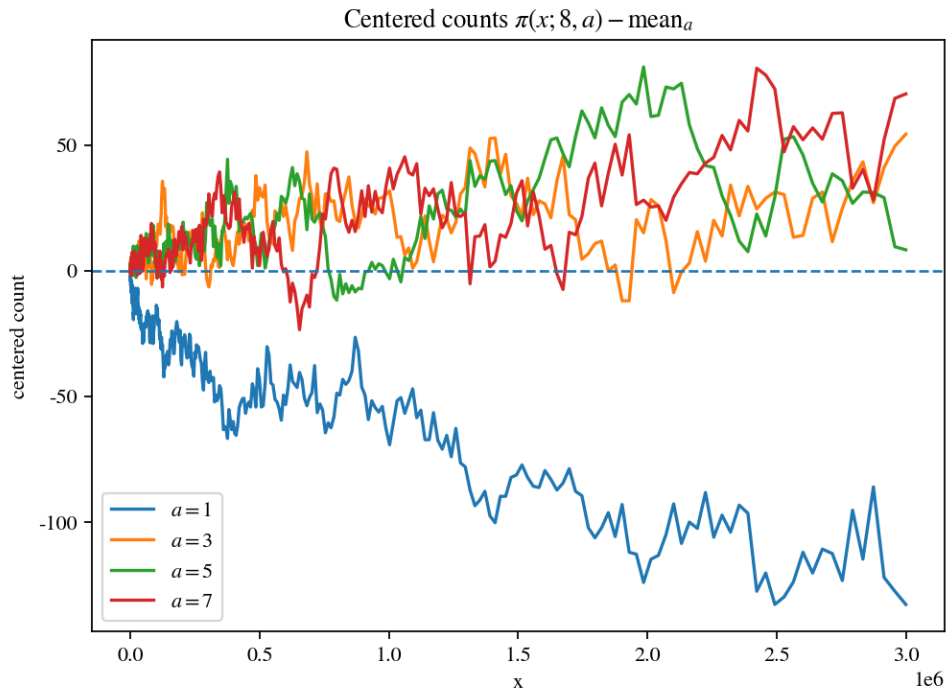
### 3.3.74 E074: Prime race mod 8: leaderboard among 1,3,5,7.



Tags: number-theory, quantitative-exploration, visualization, prime-races, aps

## Highlights

- Computes prime-race differences  $(x;q,a) - (x;q,b)$  on a grid of  $x$  values.
- Visualizes lead changes and the size of fluctuations as  $x$  grows.
- Adds a derived statistic (normalization / -variant) to compare behaviors.



### What this experiment does

This experiment tracks four residue classes modulo 8:

The implementation focuses on a compact, reproducible numerical workflow: deterministic parameter defaults, structured output folders, and one or more figures saved for the gallery.

### Outputs

This experiment writes into `out/e074/`:

- `figures/fig_01_leader_fractions.png`
- `figures/fig_02_counts_minus_mean.png`

### How to run

```
make run EXP=e074
```

### Notes

- The gallery preview figure shipped with the documentation uses conservative cutoffs so builds stay fast. If you run the experiment locally, increase the cutoffs to see the asymptotic regime more clearly.
- Prime-race plots depend on the chosen sampling of  $x$  (linear vs. log grid). The qualitative “who leads” picture can change when you zoom in.

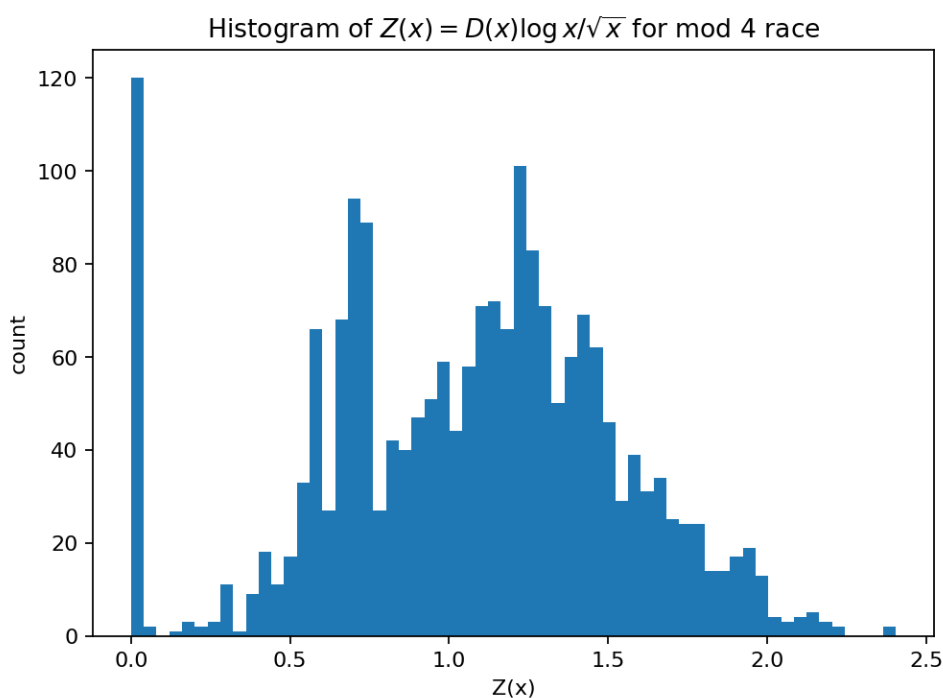
### References

Granville and Martin [2006], Rubinstein and Sarnak [1994]

## Related experiments

- *E072: Prime race mod 4:  $\pi(x;4,3)$  vs.  $\pi(x;4,1)$ .* (Prime race mod 4:  $\pi(x;4,3)$  vs.  $\pi(x;4,1)$ .)
- *E073: Prime race mod 3:  $\pi(x;3,2)$  vs.  $\pi(x;3,1)$ .* (Prime race mod 3:  $\pi(x;3,2)$  vs.  $\pi(x;3,1)$ .)
- *E075: Prime race statistic: distribution on a log-grid.* (Prime race statistic: distribution on a log-grid.)
- *E112 — Prime race:  $(x;q,a) - (x;q,b)$*  (E112 — Prime race:  $(x;q,a) - (x;q,b)$ )
- *E081: Prime race sign changes: first crossings table.* (Prime race sign changes: first crossings table.)

### 3.3.75 E075: Prime race statistic: distribution on a log-grid.



**Tags:** number-theory, quantitative-exploration, visualization, prime-races, aps

## Highlights

- Computes prime-race differences  $(x;q,a) - (x;q,b)$  on a grid of  $x$  values.
- Visualizes lead changes and the size of fluctuations as  $x$  grows.
- Adds a derived statistic (normalization / -variant) to compare behaviors.

## What this experiment does

For a prime race difference  $D(x)$ , a common heuristic normalization is:

The implementation focuses on a compact, reproducible numerical workflow: deterministic parameter defaults, structured output folders, and one or more figures saved for the gallery.

## Outputs

This experiment writes into `out/e075/`:

- `figures/fig_01_statistic_hist.png`

## How to run

```
make run EXP=e075
```

## Notes

- The gallery preview figure shipped with the documentation uses conservative cutoffs so builds stay fast. If you run the experiment locally, increase the cutoffs to see the asymptotic regime more clearly.
- Prime-race plots depend on the chosen sampling of  $x$  (linear vs. log grid). The qualitative “who leads” picture can change when you zoom in.

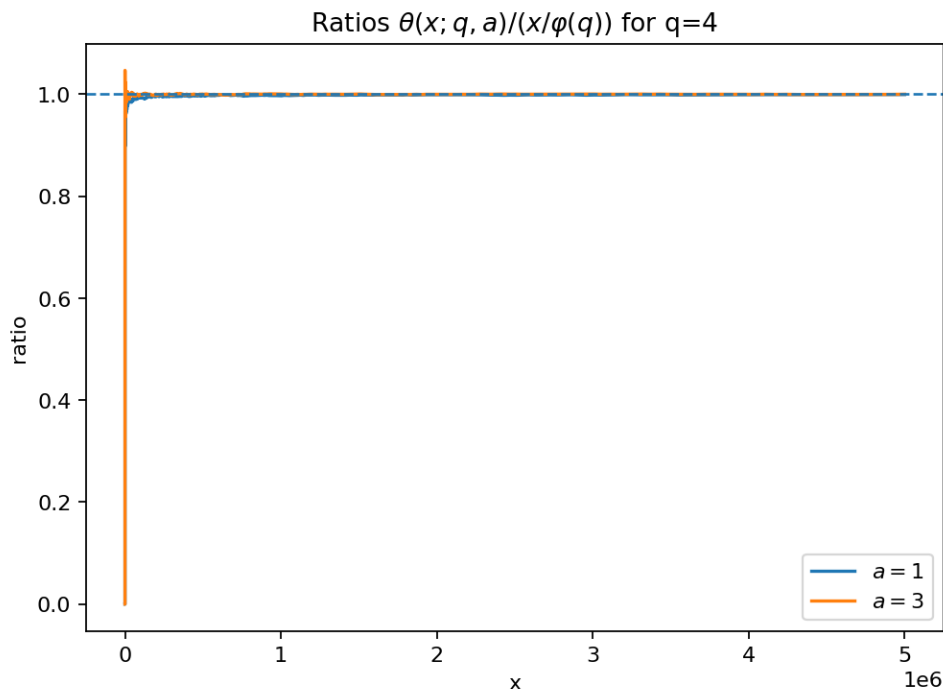
## References

Granville and Martin [2006], Rubinstein and Sarnak [1994]

## Related experiments

- *E072: Prime race mod 4:  $\pi(x;4,3)$  vs.  $\pi(x;4,1)$ .* (Prime race mod 4:  $\pi(x;4,3)$  vs.  $\pi(x;4,1)$ .)
- *E073: Prime race mod 3:  $\pi(x;3,2)$  vs.  $\pi(x;3,1)$ .* (Prime race mod 3:  $\pi(x;3,2)$  vs.  $\pi(x;3,1)$ .)
- *E074: Prime race mod 8: leaderboard among 1,3,5,7.* (Prime race mod 8: leaderboard among 1,3,5,7.)
- *E112 — Prime race:  $(x;q,a) - (x;q,b)$*  (E112 — Prime race:  $(x;q,a) - (x;q,b)$ )
- *E081: Prime race sign changes: first crossings table.* (Prime race sign changes: first crossings table.)

### 3.3.76 E076: Chebyshev $(x;q,a)$ : weighted prime counts in progressions.



Tags: number-theory, quantitative-exploration, visualization, prime-races, aps

## Highlights

- Computes prime-race differences  $(x;q,a) - (x;q,b)$  on a grid of  $x$  values.
- Visualizes lead changes and the size of fluctuations as  $x$  grows.
- Adds a derived statistic (normalization / -variant) to compare behaviors.

## What this experiment does

Define the Chebyshev theta function in a residue class:

The implementation focuses on a compact, reproducible numerical workflow: deterministic parameter defaults, structured output folders, and one or more figures saved for the gallery.

## Outputs

This experiment writes into `out/e076/`:

- `figures/fig_01_theta_ratios.png`

## How to run

```
make run EXP=e076
```

## Notes

- The gallery preview figure shipped with the documentation uses conservative cutoffs so builds stay fast. If you run the experiment locally, increase the cutoffs to see the asymptotic regime more clearly.
- Prime-race plots depend on the chosen sampling of  $x$  (linear vs. log grid). The qualitative “who leads” picture can change when you zoom in.

## References

Granville and Martin [2006], Rubinstein and Sarnak [1994]

## Related experiments

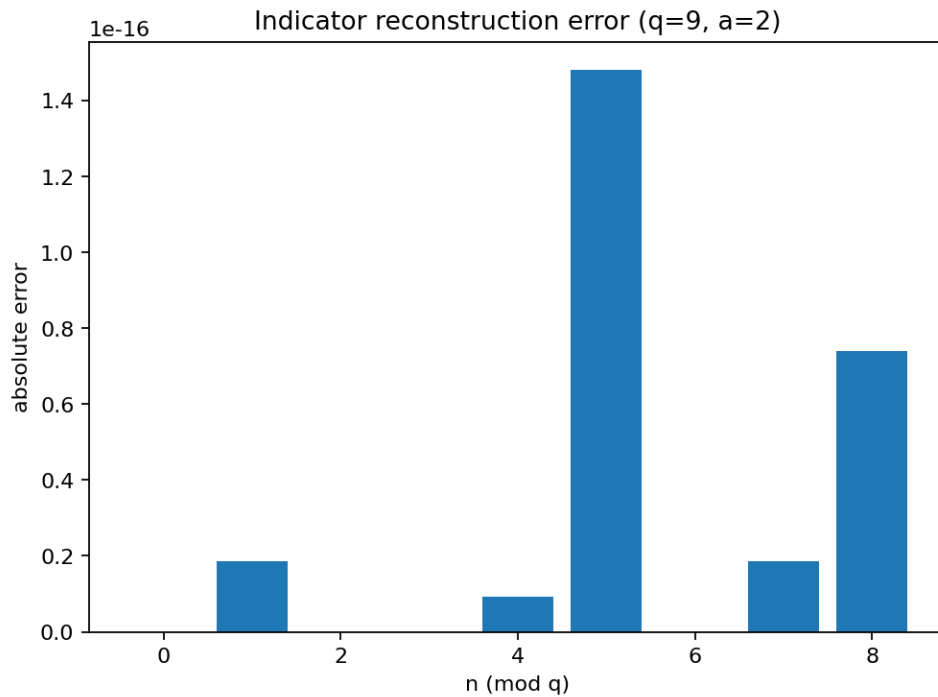
- *E080: Chebyshev bias: leader fraction vs.  $x$ .* (Chebyshev bias: leader fraction vs.  $x$ .)
- *E118 — Chebyshev bias: lead-time statistics* (E118 — Chebyshev bias: lead-time statistics)
- *E072: Prime race mod 4:  $\pi(x;4,3)$  vs.  $\pi(x;4,1)$ .* (Prime race mod 4:  $\pi(x;4,3)$  vs.  $\pi(x;4,1)$ .)
- *E073: Prime race mod 3:  $\pi(x;3,2)$  vs.  $\pi(x;3,1)$ .* (Prime race mod 3:  $\pi(x;3,2)$  vs.  $\pi(x;3,1)$ .)
- *E074: Prime race mod 8: leaderboard among 1,3,5,7.* (Prime race mod 8: leaderboard among 1,3,5,7.)

### 3.3.77 E077: Indicator via character orthogonality (sanity check).

**Tags:** number-theory, quantitative-exploration, visualization, dirichlet-characters

## Highlights

- Uses character orthogonality to reconstruct residue indicators.
- Measures reconstruction error / maximal partial sums across characters.
- Explores how primitive characters vary with the modulus.



### What this experiment does

A standard identity expresses an indicator of a residue class using Dirichlet characters. For  $\gcd(a,q)=1$  and  $\gcd(n,q)=1$ :

The implementation focuses on a compact, reproducible numerical workflow: deterministic parameter defaults, structured output folders, and one or more figures saved for the gallery.

### Outputs

This experiment writes into `out/e077/`:

- `figures/fig_01_indicator_error.png`

### How to run

```
make run EXP=e077
```

### Notes

- The gallery preview figure shipped with the documentation uses conservative cutoffs so builds stay fast. If you run the experiment locally, increase the cutoffs to see the asymptotic regime more clearly.
- Prime-race plots depend on the chosen sampling of  $x$  (linear vs. log grid). The qualitative “who leads” picture can change when you zoom in.

### References

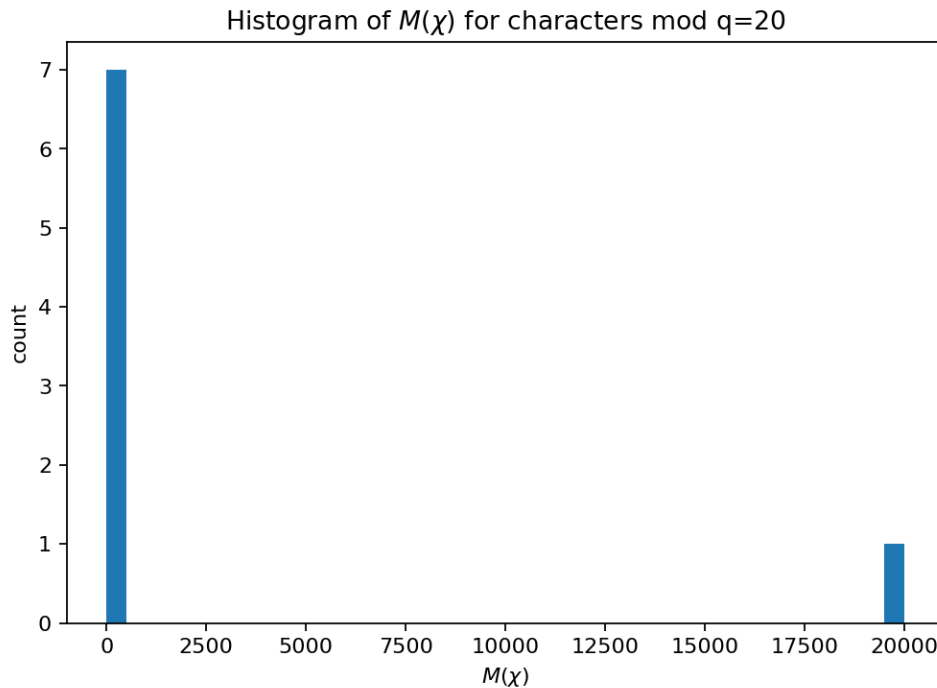
Davenport [2000], Niven *et al.* [1991]

### Related experiments

- *E064: Dirichlet character tables (phase view)*. (Dirichlet character tables (phase view).)
- *E065: Orthogonality matrix for Dirichlet characters*. (Orthogonality matrix for Dirichlet characters.)

- *E066: Character partial sums: cancellation profiles.* (Character partial sums: cancellation profiles.)
- *E122 — Character averages over primes* (E122 — Character averages over primes)
- *E106 — Character gallery: real vs. complex* (E106 — Character gallery: real vs. complex)

### 3.3.78 E078: Max partial sums across characters.



Tags: number-theory, quantitative-exploration, visualization, dirichlet-characters

#### Highlights

- Uses character orthogonality to reconstruct residue indicators.
- Measures reconstruction error / maximal partial sums across characters.
- Explores how primitive characters vary with the modulus.

#### What this experiment does

For each Dirichlet character modulo  $q$ , define

The implementation focuses on a compact, reproducible numerical workflow: deterministic parameter defaults, structured output folders, and one or more figures saved for the gallery.

#### Outputs

This experiment writes into `out/e078/`:

- `figures/fig_01_max_sums_hist.png`

#### How to run

```
make run EXP=e078
```



## Notes

- The gallery preview figure shipped with the documentation uses conservative cutoffs so builds stay fast. If you run the experiment locally, increase the cutoffs to see the asymptotic regime more clearly.
- Prime-race plots depend on the chosen sampling of  $x$  (linear vs. log grid). The qualitative “who leads” picture can change when you zoom in.

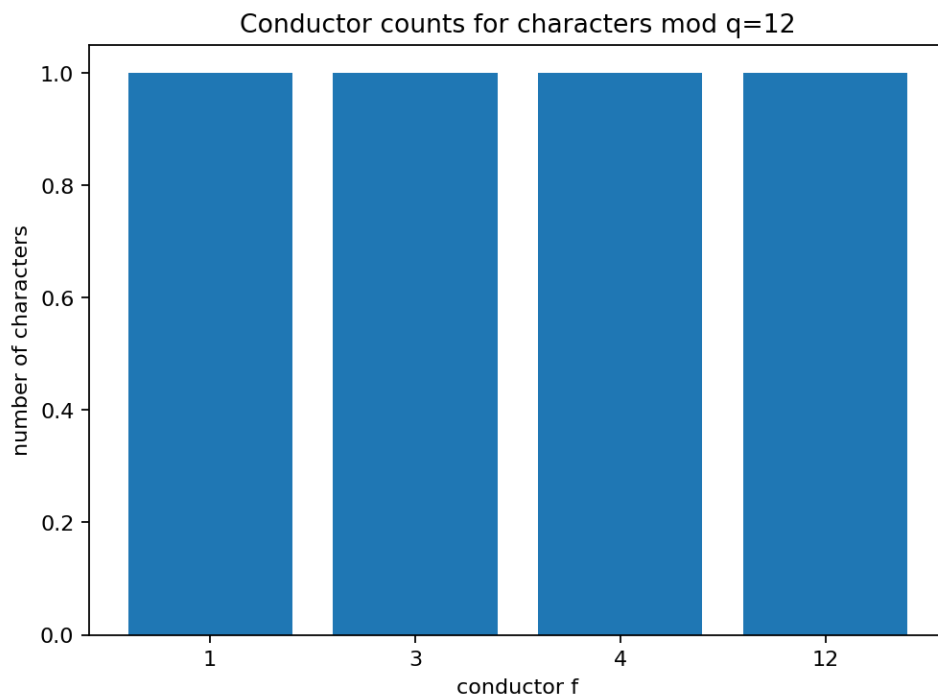
## References

Davenport [2000], Niven *et al.* [1991]

## Related experiments

- *E066: Character partial sums: cancellation profiles.* (Character partial sums: cancellation profiles.)
- *E110 — Dirichlet L-series partial sums at  $s=1$  and  $s=1/2$*  (E110 — Dirichlet L-series partial sums at  $s=1$  and  $s=1/2$ )
- *E120 — Liouville  $(n)$ : partial sums and parity* (E120 — Liouville  $(n)$ : partial sums and parity)
- *E065: Orthogonality matrix for Dirichlet characters.* (Orthogonality matrix for Dirichlet characters.)
- *E067: Gauss sums: magnitude vs.  $\sqrt{q}$ .* (Gauss sums: magnitude vs.  $\sqrt{q}$ .)

### 3.3.79 E079: Primitive vs. imprimitive characters: conductors.



Tags: number-theory, quantitative-exploration, visualization, dirichlet-characters

## Highlights

- Uses character orthogonality to reconstruct residue indicators.
- Measures reconstruction error / maximal partial sums across characters.
- Explores how primitive characters vary with the modulus.

### What this experiment does

A character modulo  $q$  may factor through a smaller modulus  $f \mid q$ . The smallest such modulus is the **conductor** of the character.

The implementation focuses on a compact, reproducible numerical workflow: deterministic parameter defaults, structured output folders, and one or more figures saved for the gallery.

### Outputs

This experiment writes into `out/e079/`:

- `figures/fig_01_conductor_counts.png`

### How to run

```
make run EXP=e079
```

### Notes

- The gallery preview figure shipped with the documentation uses conservative cutoffs so builds stay fast. If you run the experiment locally, increase the cutoffs to see the asymptotic regime more clearly.
- Prime-race plots depend on the chosen sampling of  $x$  (linear vs. log grid). The qualitative “who leads” picture can change when you zoom in.

### References

Davenport [2000], Niven *et al.* [1991]

### Related experiments

- *E107 — Conductor: primitive vs. induced characters* (E107 — Conductor: primitive vs. induced characters)
- *E065: Orthogonality matrix for Dirichlet characters.* (Orthogonality matrix for Dirichlet characters.)
- *E078: Max partial sums across characters.* (Max partial sums across characters.)
- *E108 — Orthogonality heatmap for characters* (E108 — Orthogonality heatmap for characters)
- *E064: Dirichlet character tables (phase view).* (Dirichlet character tables (phase view).)

## 3.3.80 E080: Chebyshev bias: leader fraction vs. $x$ .

**Tags:** number-theory, quantitative-exploration, visualization, prime-races

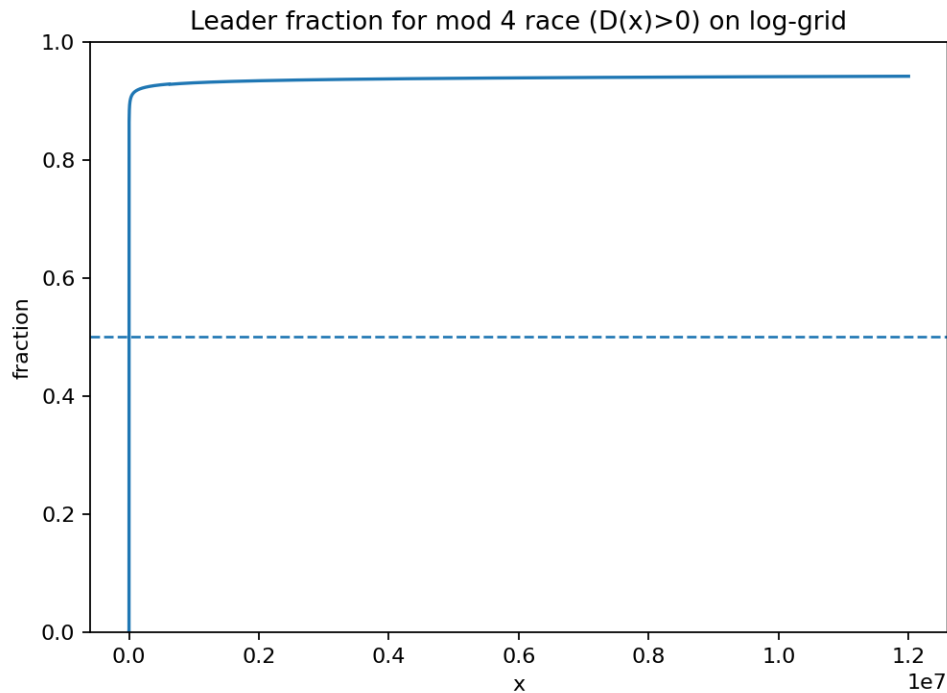
### Highlights

- Estimates the fraction of  $x$  where one residue class leads another.
- Plots a running bias estimate over log-sampled  $x$  values.
- Detects coarse sign-change neighborhoods for the race difference.

### What this experiment does

For the mod 4 race  $D(x) = (x; 4, 3) - (x; 4, 1)$ , define the empirical leader fraction:

The implementation focuses on a compact, reproducible numerical workflow: deterministic parameter defaults, structured output folders, and one or more figures saved for the gallery.



## Outputs

This experiment writes into out/e080/:

- figures/fig\_01\_bias\_fraction.png

## How to run

```
make run EXP=e080
```

## Notes

- The gallery preview figure shipped with the documentation uses conservative cutoffs so builds stay fast. If you run the experiment locally, increase the cutoffs to see the asymptotic regime more clearly.
- Prime-race plots depend on the chosen sampling of  $x$  (linear vs. log grid). The qualitative “who leads” picture can change when you zoom in.

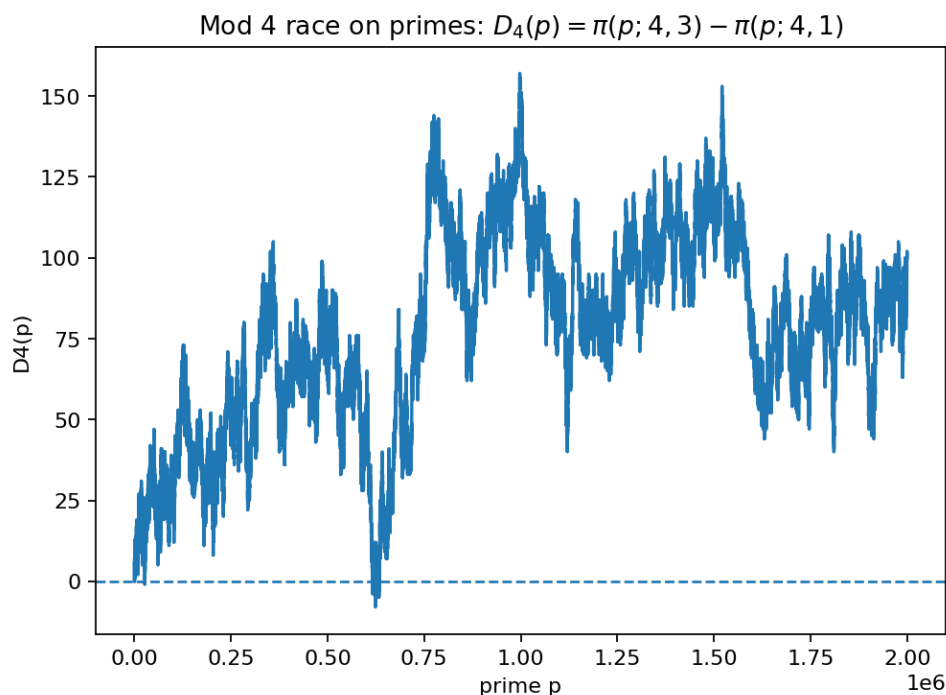
## References

Granville and Martin [2006], Rubinstein and Sarnak [1994]

## Related experiments

- *E118 — Chebyshev bias: lead-time statistics* (E118 — Chebyshev bias: lead-time statistics)
- *E076: Chebyshev  $(x;q,a)$ : weighted prime counts in progressions.* (Chebyshev  $(x;q,a)$ : weighted prime counts in progressions.)
- *E061: Chebyshev  $(x)$  and prime powers* (Chebyshev  $(x)$  and prime powers)
- *E103 — Chebyshev  $(x)$ : prime powers drive jumps* (E103 — Chebyshev  $(x)$ : prime powers drive jumps)
- *E072: Prime race mod 4:  $\pi(x;4,3)$  vs.  $\pi(x;4,1)$ .* (Prime race mod 4:  $\pi(x;4,3)$  vs.  $\pi(x;4,1)$ .)

### 3.3.81 E081: Prime race sign changes: first crossings table.



**Tags:** number-theory, quantitative-exploration, visualization, prime-races

#### Highlights

- Estimates the fraction of  $x$  where one residue class leads another.
- Plots a running bias estimate over log-sampled  $x$  values.
- Detects coarse sign-change neighborhoods for the race difference.

#### What this experiment does

A prime race difference  $D(x)$  only changes when  $x$  passes a prime. We can track sign changes by iterating primes in order.

The implementation focuses on a compact, reproducible numerical workflow: deterministic parameter defaults, structured output folders, and one or more figures saved for the gallery.

#### Outputs

This experiment writes into `out/e081/`:

- `figures/fig_01_mod4_diff_on_primes.png`

#### How to run

```
make run EXP=e081
```

#### Notes

- The gallery preview figure shipped with the documentation uses conservative cutoffs so builds stay fast. If you run the experiment locally, increase the cutoffs to see the asymptotic regime more clearly.
- Prime-race plots depend on the chosen sampling of  $x$  (linear vs. log grid). The qualitative “who leads” picture can change when you zoom in.

## References

Granville and Martin [2006], Rubinstein and Sarnak [1994]

## Related experiments

- *E115 — Hardy Z: sign changes and zero bracketing* (E115 — Hardy Z: sign changes and zero bracketing)
- *E072: Prime race mod 4:  $\pi(x;4,3)$  vs.  $\pi(x;4,1)$ .* (Prime race mod 4:  $\pi(x;4,3)$  vs.  $\pi(x;4,1)$ .)
- *E073: Prime race mod 3:  $\pi(x;3,2)$  vs.  $\pi(x;3,1)$ .* (Prime race mod 3:  $\pi(x;3,2)$  vs.  $\pi(x;3,1)$ .)
- *E074: Prime race mod 8: leaderboard among 1,3,5,7.* (Prime race mod 8: leaderboard among 1,3,5,7.)
- *E075: Prime race statistic: distribution on a log-grid.* (Prime race statistic: distribution on a log-grid.)

### 3.3.82 E082 — Zeta(s) series convergence

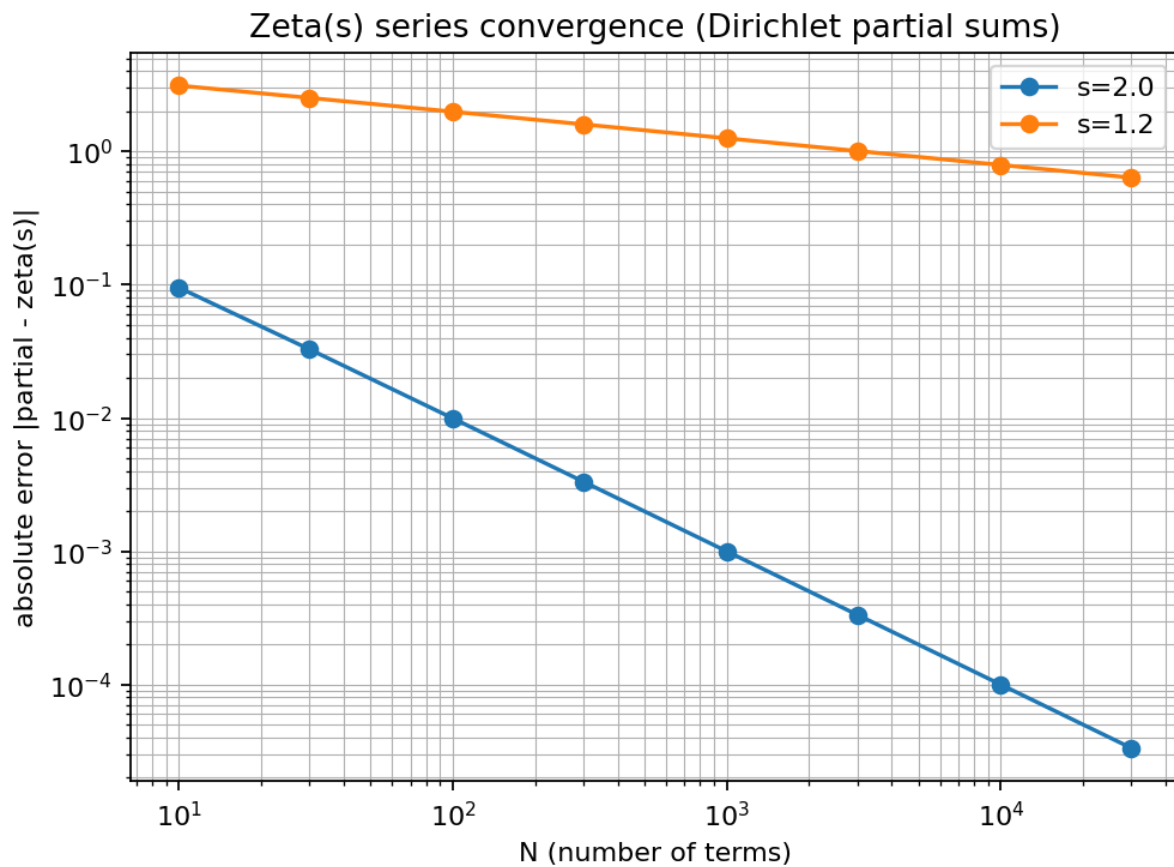


Fig. 1: E082 — Zeta(s) series convergence

Tags: analysis, quantitative-exploration, visualization, riemann-zeta, numerics

## Highlights

- Focused numeric experiment with a single main figure.
- Parameters saved to `params.json` for reproducibility.
- Lightweight computation suitable for CI “slow” suite (small defaults).

### What is computed

- A parameterized numeric evaluation related to the Riemann zeta function.
- A visualization summarizing the main phenomenon for the chosen parameter range.

### Algorithm sketch

1. Build the numeric grid / sampling points.
2. Evaluate the target quantity with controlled truncation.
3. Render the figure and write a short report.

### Outputs

- `report.md` — short narrative summary
- `params.json` — experiment parameters snapshot
- `figures/fig_01_series_convergence.png` — main figure

### References

Edwards [1974], Titchmarsh [1986]

### Related experiments

- *E083 — Series vs. Euler product ( )* (E083 — Series vs. Euler product ( ))
- *E092 —  $1/(s)$  via the Möbius Dirichlet series* (E092 —  $1/(s)$  via the Möbius Dirichlet series)
- *E093 —  $-(s)/(s)$  via the von Mangoldt series* (E093 —  $-(s)/(s)$  via the von Mangoldt series)
- *E084 —  $| (1/2+it) |$  growth snapshots* (E084 —  $| (1/2+it) |$  growth snapshots)
- *E085 — Dirichlet eta acceleration for (s)* (E085 — Dirichlet eta acceleration for (s))

## 3.3.83 E083 — Series vs. Euler product ( )

Tags: analysis, quantitative-exploration, visualization, riemann-zeta, numerics

### Highlights

- Focused numeric experiment with a single main figure.
- Parameters saved to `params.json` for reproducibility.
- Lightweight computation suitable for CI “slow” suite (small defaults).

### What is computed

- A parameterized numeric evaluation related to the Riemann zeta function.
- A visualization summarizing the main phenomenon for the chosen parameter range.

### Algorithm sketch

1. Build the numeric grid / sampling points.
2. Evaluate the target quantity with controlled truncation.
3. Render the figure and write a short report.

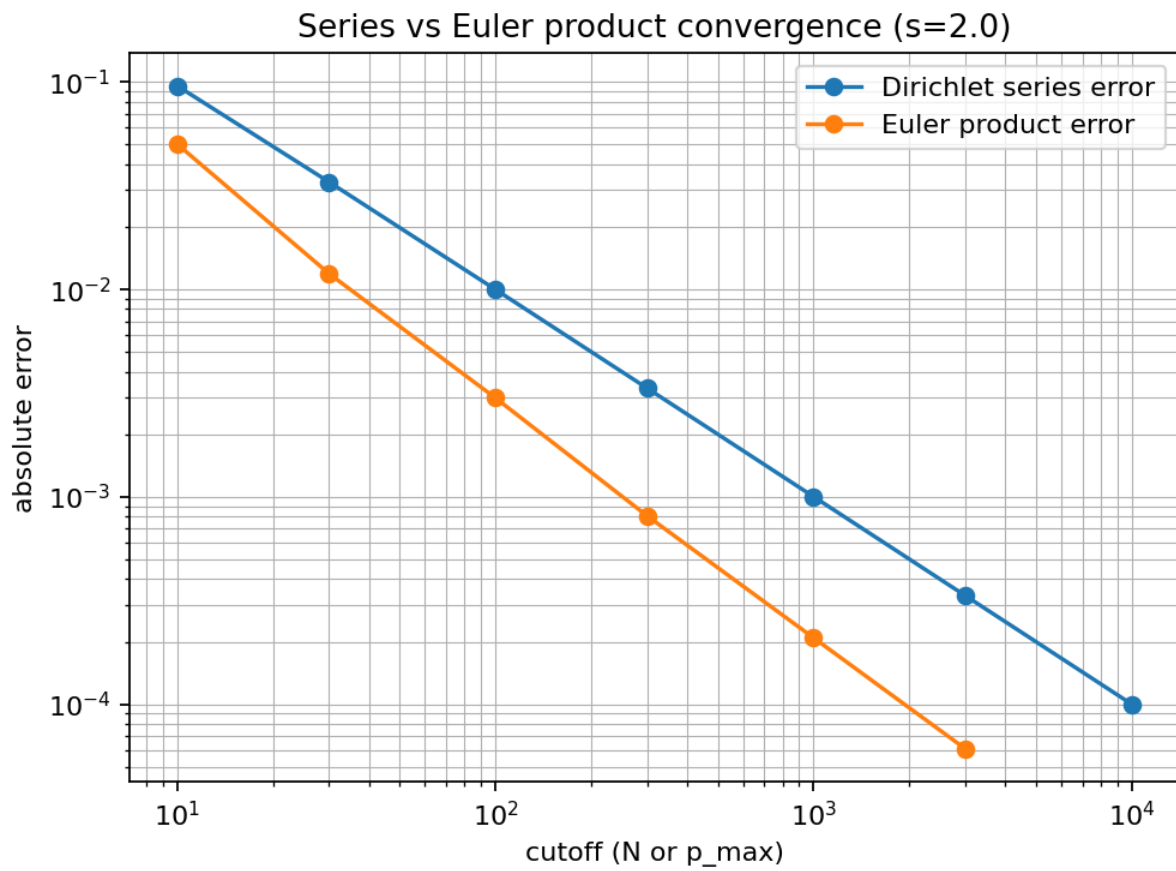


Fig. 2: E083 — Series vs. Euler product ( )

## Outputs

- `report.md` — short narrative summary
- `params.json` — experiment parameters snapshot
- `figures/fig_01_series_vs_euler_product.png` — main figure

## References

Edwards [1974], Titchmarsh [1986]

## Related experiments

- *E111 — Euler product vs. Dirichlet series for  $L(s, \chi)$*  (E111 — Euler product vs. Dirichlet series for  $L(s, \chi)$ )
- *E082 — Zeta(s) series convergence* (E082 — Zeta(s) series convergence)
- *E091 — Partial Euler products on the critical line* (E091 — Partial Euler products on the critical line)
- *E092 —  $1/\zeta(s)$  via the Möbius Dirichlet series* (E092 —  $1/\zeta(s)$  via the Möbius Dirichlet series)
- *E093 —  $-\zeta'(s)/\zeta(s)$  via the von Mangoldt series* (E093 —  $-\zeta'(s)/\zeta(s)$  via the von Mangoldt series)

### 3.3.84 E084 — $|\zeta(1/2+it)|$ growth snapshots

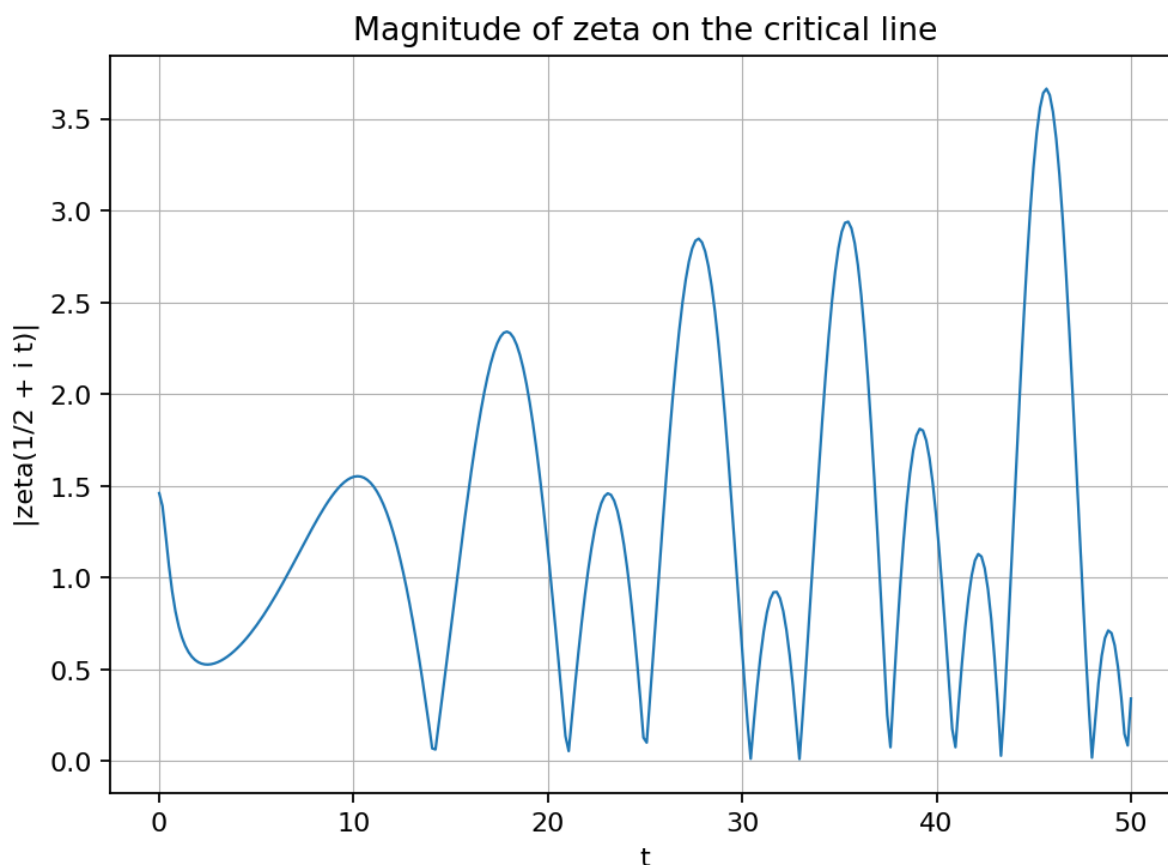


Fig. 3: E084 —  $|\zeta(1/2+it)|$  growth snapshots

**Tags:** analysis, quantitative-exploration, visualization, riemann-zeta, critical-line, numerics



### Highlights

- Focused numeric experiment with a single main figure.
- Parameters saved to `params.json` for reproducibility.
- Lightweight computation suitable for CI “slow” suite (small defaults).

### What is computed

- A parameterized numeric evaluation related to the Riemann zeta function.
- A visualization summarizing the main phenomenon for the chosen parameter range.

### Algorithm sketch

1. Build the numeric grid / sampling points.
2. Evaluate the target quantity with controlled truncation.
3. Render the figure and write a short report.

### Outputs

- `report.md` — short narrative summary
- `params.json` — experiment parameters snapshot
- `figures/fig_01_critical_line_logabs.png` — main figure

### References

Edwards [1974], Titchmarsh [1986]

### Related experiments

- *E086 — Hardy  $Z(t)$  near zeros* (E086 — Hardy  $Z(t)$  near zeros)
- *E114 —  $\text{via}$  : stability map on the critical line* (E114 —  $\text{via}$  : stability map on the critical line)
- *E082 — Zeta( $s$ ) series convergence* (E082 — Zeta( $s$ ) series convergence)
- *E083 — Series vs. Euler product ( )* (E083 — Series vs. Euler product ( ))
- *E085 — Dirichlet eta acceleration for (s)* (E085 — Dirichlet eta acceleration for (s))

## 3.3.85 E085 — Dirichlet eta acceleration for (s)

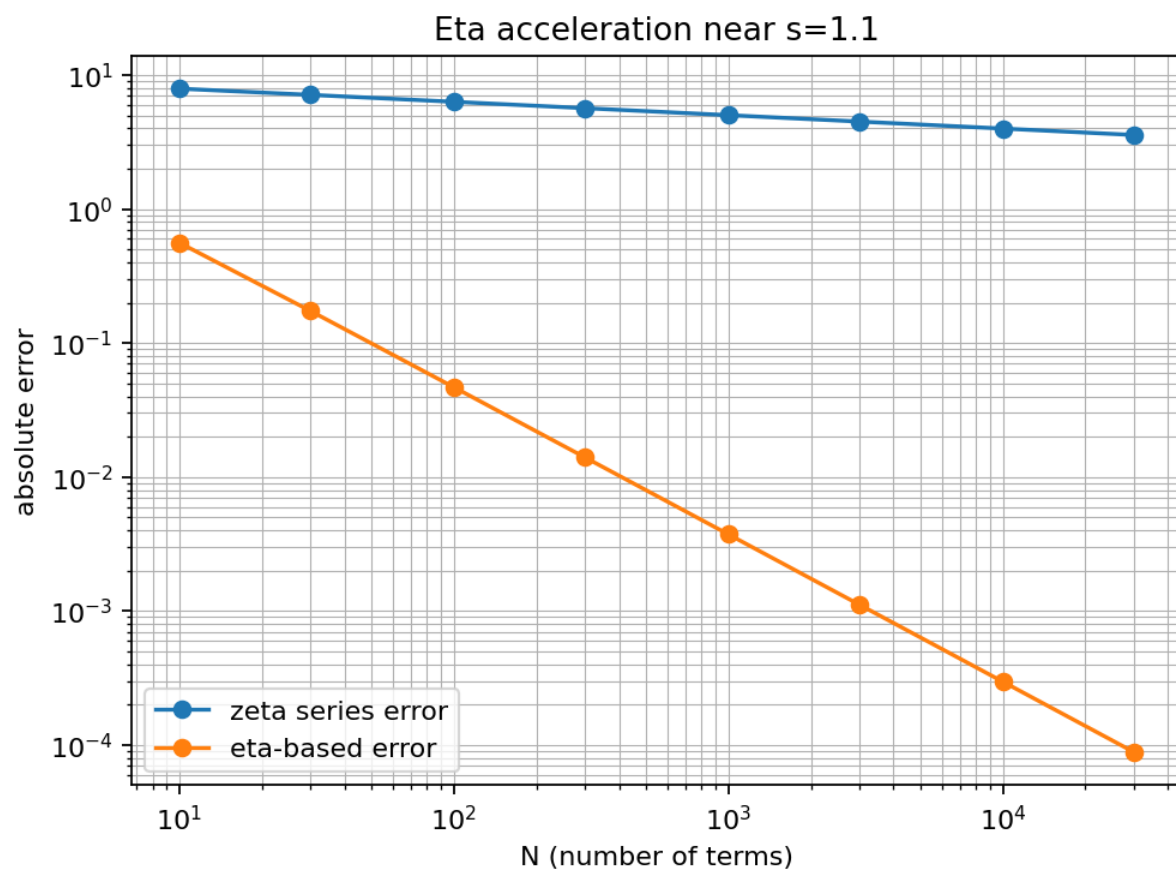
Tags: analysis, quantitative-exploration, visualization, riemann-zeta, numerics

### Highlights

- Focused numeric experiment with a single main figure.
- Parameters saved to `params.json` for reproducibility.
- Lightweight computation suitable for CI “slow” suite (small defaults).

### What is computed

- A parameterized numeric evaluation related to the Riemann zeta function.
- A visualization summarizing the main phenomenon for the chosen parameter range.

Fig. 4: E085 — Dirichlet eta acceleration for  $(s)$

### Algorithm sketch

1. Build the numeric grid / sampling points.
2. Evaluate the target quantity with controlled truncation.
3. Render the figure and write a short report.

### Outputs

- `report.md` — short narrative summary
- `params.json` — experiment parameters snapshot
- `figures/fig_01_eta_acceleration.png` — main figure

### References

Edwards [1974], Titchmarsh [1986]

### Related experiments

- *E092 —  $1/(s)$  via the Möbius Dirichlet series* (E092 —  $1/(s)$  via the Möbius Dirichlet series)
- *E082 — Zeta(s) series convergence* (E082 — Zeta(s) series convergence)
- *E083 — Series vs. Euler product ( )* (E083 — Series vs. Euler product ( ))
- *E084 —  $| (1/2+it) |$  growth snapshots* (E084 —  $| (1/2+it) |$  growth snapshots)
- *E086 — Hardy  $Z(t)$  near zeros* (E086 — Hardy  $Z(t)$  near zeros)

## 3.3.86 E086 — Hardy $Z(t)$ near zeros

**Tags:** analysis, quantitative-exploration, visualization, riemann-zeta, zeta-zeros, critical-line, numerics

### Highlights

- Focused numeric experiment with a single main figure.
- Parameters saved to `params.json` for reproducibility.
- Lightweight computation suitable for CI “slow” suite (small defaults).

### What is computed

- A parameterized numeric evaluation related to the Riemann zeta function.
- A visualization summarizing the main phenomenon for the chosen parameter range.

### Algorithm sketch

1. Build the numeric grid / sampling points.
2. Evaluate the target quantity with controlled truncation.
3. Render the figure and write a short report.

### Outputs

- `report.md` — short narrative summary
- `params.json` — experiment parameters snapshot
- `figures/fig_01_hardy_Z_near_zeros.png` — main figure

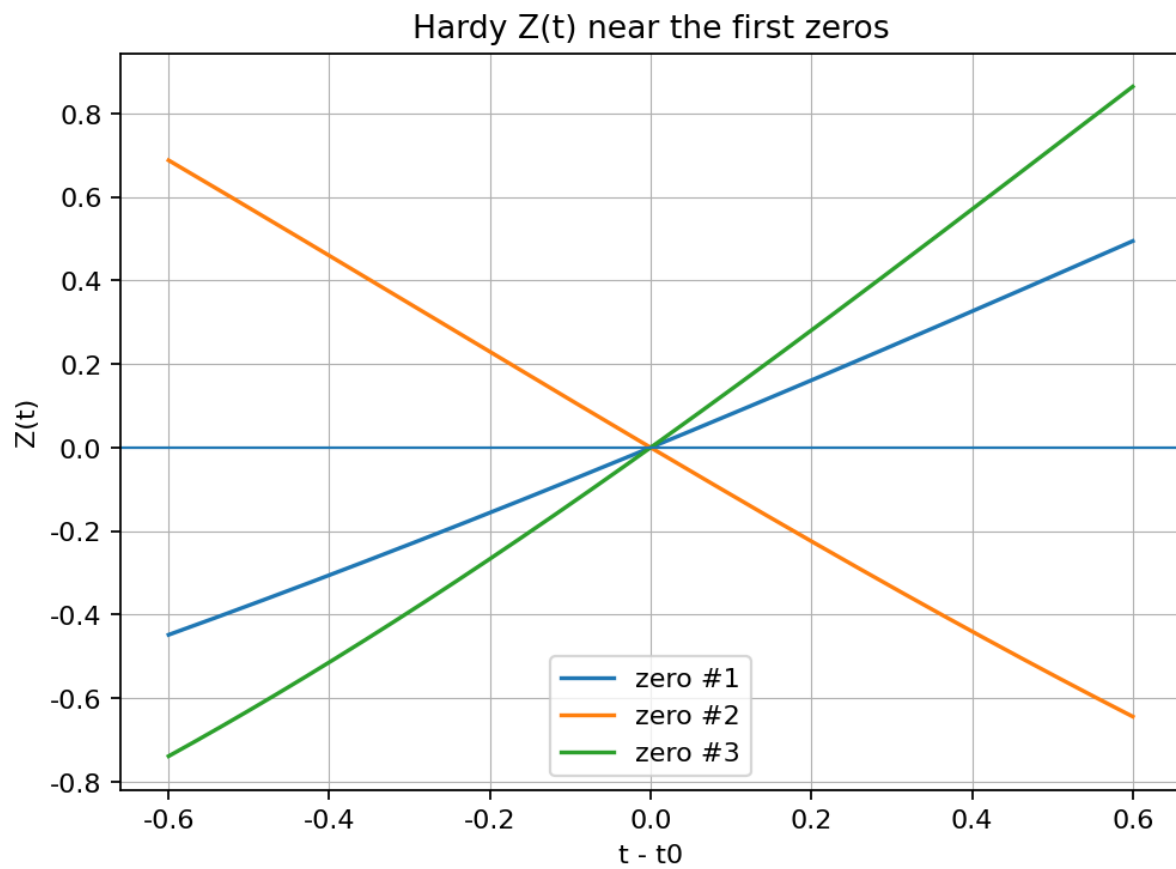


Fig. 5: E086 — Hardy  $Z(t)$  near zeros

## References

Edwards [1974], Titchmarsh [1986]

## Related experiments

- *E115 — Hardy Z: sign changes and zero bracketing* (E115 — Hardy Z: sign changes and zero bracketing)
- *E084 —  $| (1/2+it) |$  growth snapshots* (E084 —  $| (1/2+it) |$  growth snapshots)
- *E114 —  $\text{via}$  : stability map on the critical line* (E114 —  $\text{via}$  : stability map on the critical line)
- *E087 — Gram points and spacing* (E087 — Gram points and spacing)
- *E088 — Zero counting via Riemann-von Mangoldt* (E088 — Zero counting via Riemann-von Mangoldt)

### 3.3.87 E087 — Gram points and spacing

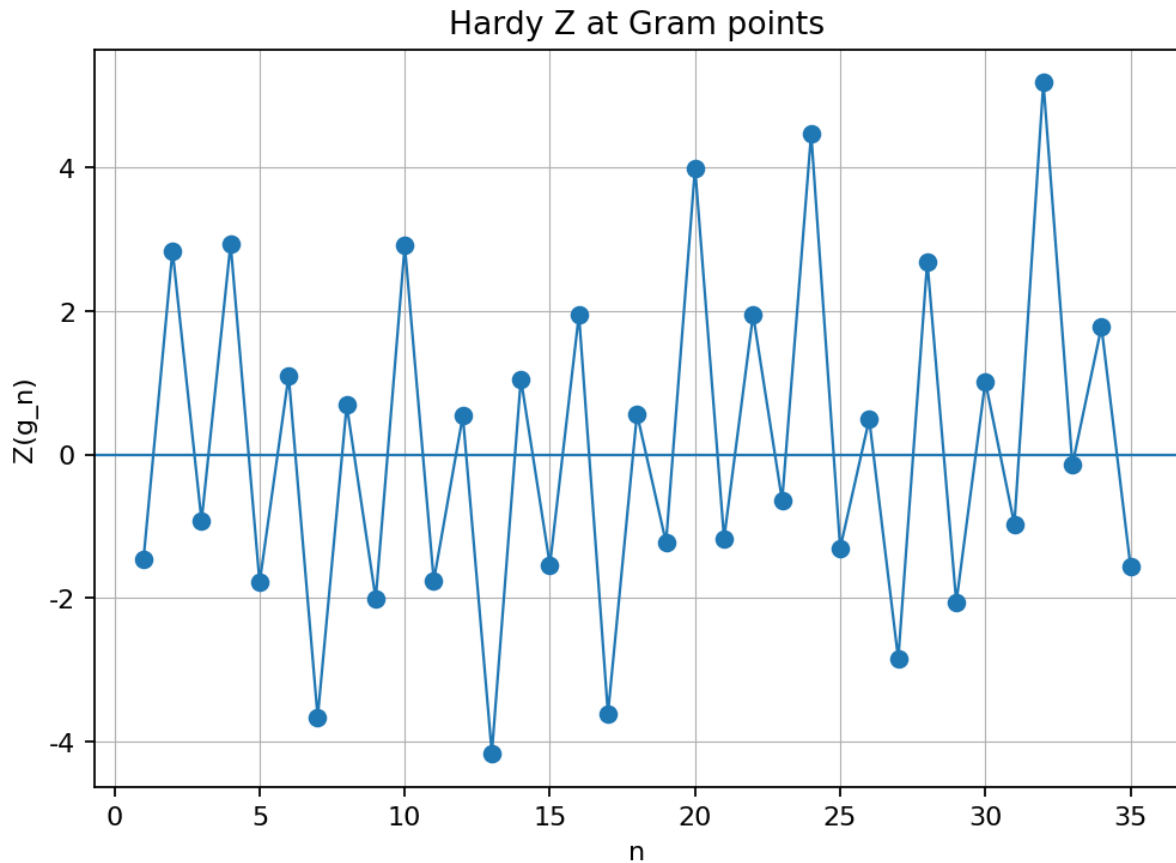


Fig. 6: E087 — Gram points and spacing

**Tags:** analysis, quantitative-exploration, visualization, riemann-zeta, gram-points, zeta-zeros, numerics

## Highlights

- Focused numeric experiment with a single main figure.
- Parameters saved to `params.json` for reproducibility.
- Lightweight computation suitable for CI “slow” suite (small defaults).

### What is computed

- A parameterized numeric evaluation related to the Riemann zeta function.
- A visualization summarizing the main phenomenon for the chosen parameter range.

### Algorithm sketch

1. Build the numeric grid / sampling points.
2. Evaluate the target quantity with controlled truncation.
3. Render the figure and write a short report.

### Outputs

- `report.md` — short narrative summary
- `params.json` — experiment parameters snapshot
- `figures/fig_01_gram_points_spacing.png` — main figure

### References

Edwards [1974], Montgomery [1973], Titchmarsh [1986]

### Related experiments

- *E116 — Gram points and zero-counting heuristics* (E116 — Gram points and zero-counting heuristics)
- *E086 — Hardy  $Z(t)$  near zeros* (E086 — Hardy  $Z(t)$  near zeros)
- *E088 — Zero counting via Riemann–von Mangoldt* (E088 — Zero counting via Riemann–von Mangoldt)
- *E115 — Hardy  $Z$ : sign changes and zero bracketing* (E115 — Hardy  $Z$ : sign changes and zero bracketing)
- *E082 — Zeta(s) series convergence* (E082 — Zeta(s) series convergence)

## 3.3.88 E088 — Zero counting via Riemann–von Mangoldt

Tags: analysis, quantitative-exploration, visualization, riemann-zeta, zeta-zeros, numerics

### Highlights

- Focused numeric experiment with a single main figure.
- Parameters saved to `params.json` for reproducibility.
- Lightweight computation suitable for CI “slow” suite (small defaults).

### What is computed

- A parameterized numeric evaluation related to the Riemann zeta function.
- A visualization summarizing the main phenomenon for the chosen parameter range.

### Algorithm sketch

1. Build the numeric grid / sampling points.
2. Evaluate the target quantity with controlled truncation.
3. Render the figure and write a short report.

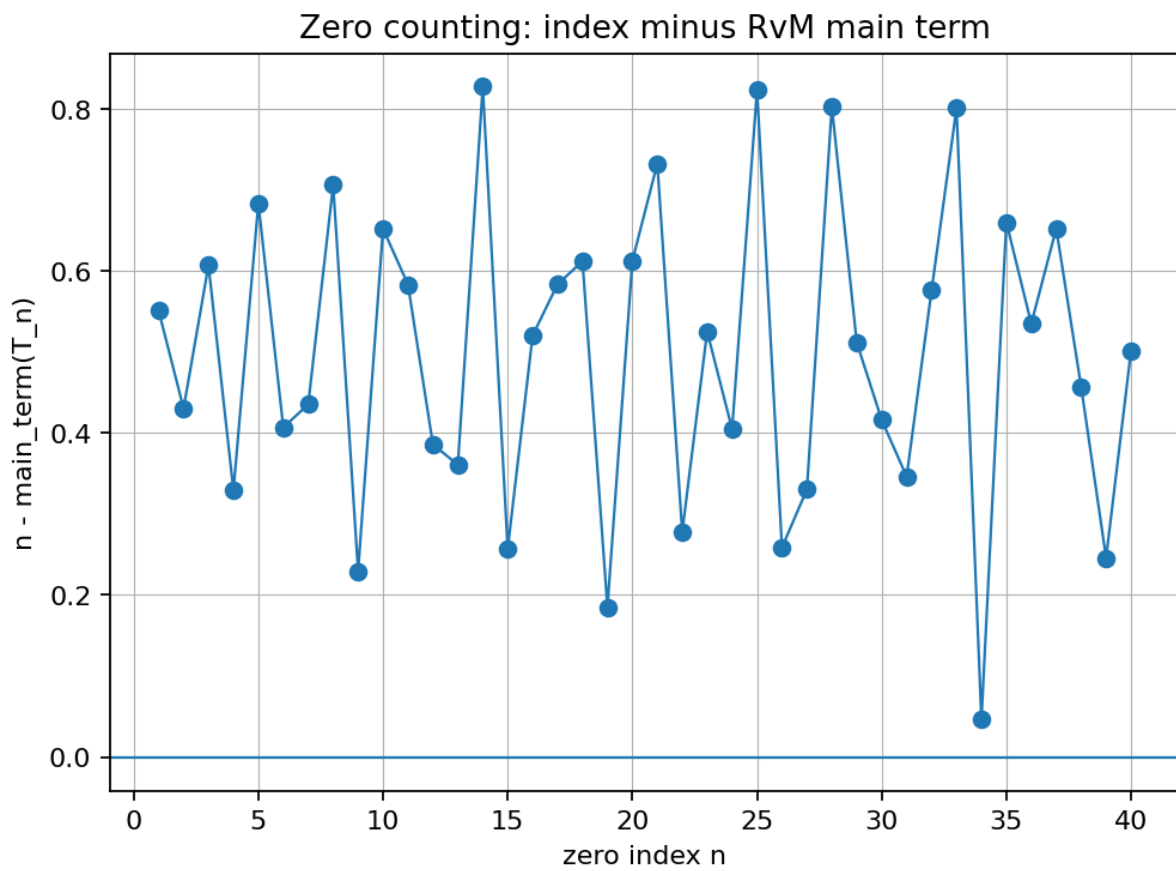


Fig. 7: E088 — Zero counting via Riemann-von Mangoldt

## Outputs

- `report.md` — short narrative summary
- `params.json` — experiment parameters snapshot
- `figures/fig_01_rvm_zero_counting.png` — main figure

## References

Edwards [1974], Montgomery [1973], Titchmarsh [1986]

## Related experiments

- *E116 — Gram points and zero-counting heuristics* (E116 — Gram points and zero-counting heuristics)
- *E115 — Hardy Z: sign changes and zero bracketing* (E115 — Hardy Z: sign changes and zero bracketing)
- *E086 — Hardy  $Z(t)$  near zeros* (E086 — Hardy  $Z(t)$  near zeros)
- *E087 — Gram points and spacing* (E087 — Gram points and spacing)
- *E093 —  $-\frac{(s)}{(s)}$  via the von Mangoldt series* (E093 —  $-\frac{(s)}{(s)}$  via the von Mangoldt series)

### 3.3.89 E089 — $\log|s|$ heatmap

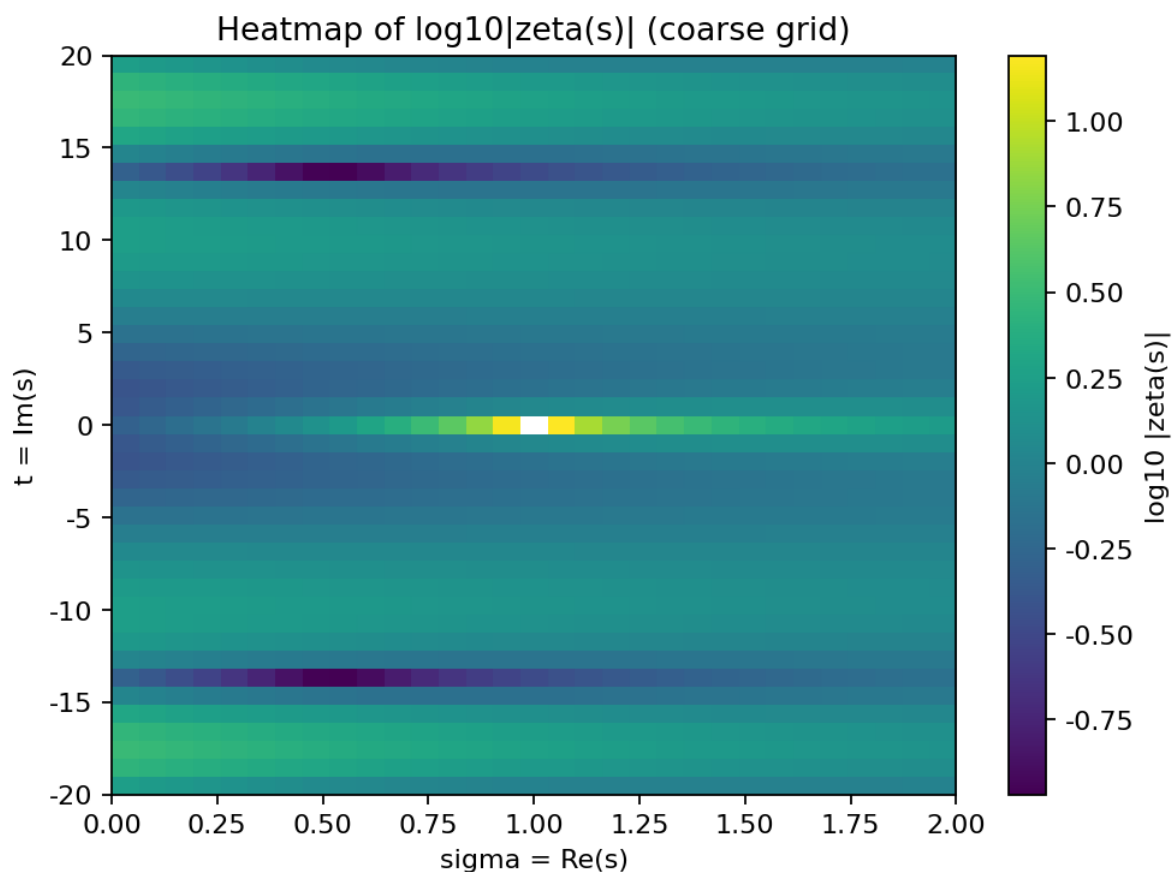


Fig. 8: E089 —  $\log|s|$  heatmap

Tags: analysis, quantitative-exploration, visualization, riemann-zeta, numerics



### Highlights

- Focused numeric experiment with a single main figure.
- Parameters saved to `params.json` for reproducibility.
- Lightweight computation suitable for CI “slow” suite (small defaults).

### What is computed

- A parameterized numeric evaluation related to the Riemann zeta function.
- A visualization summarizing the main phenomenon for the chosen parameter range.

### Algorithm sketch

1. Build the numeric grid / sampling points.
2. Evaluate the target quantity with controlled truncation.
3. Render the figure and write a short report.

### Outputs

- `report.md` — short narrative summary
- `params.json` — experiment parameters snapshot
- `figures/fig_01_zeta_heatmap_logabs.png` — main figure

### References

Edwards [1974], Titchmarsh [1986]

### Related experiments

- *E090 — Functional equation residual heatmap* (E090 — Functional equation residual heatmap)
- *E082 — Zeta(s) series convergence* (E082 — Zeta(s) series convergence)
- *E083 — Series vs. Euler product ( )* (E083 — Series vs. Euler product ( ))
- *E084 —  $| (1/2+it) |$  growth snapshots* (E084 —  $| (1/2+it) |$  growth snapshots)
- *E085 — Dirichlet eta acceleration for (s)* (E085 — Dirichlet eta acceleration for (s))

## 3.3.90 E090 — Functional equation residual heatmap

Tags: analysis, quantitative-exploration, visualization, riemann-zeta, numerics

### Highlights

- Focused numeric experiment with a single main figure.
- Parameters saved to `params.json` for reproducibility.
- Defaults are chosen, so the experiment remains feasible for the CI “slow” suite.

### What is computed

- Evaluate the functional-equation residual ( $R(s)=\zeta(s)-\chi(s)\zeta(1-s)$ ) on a small rectangle on the complex plane.
- Visualize  $\log_{10}(|R(s)| + \text{eps})$  as a heatmap.

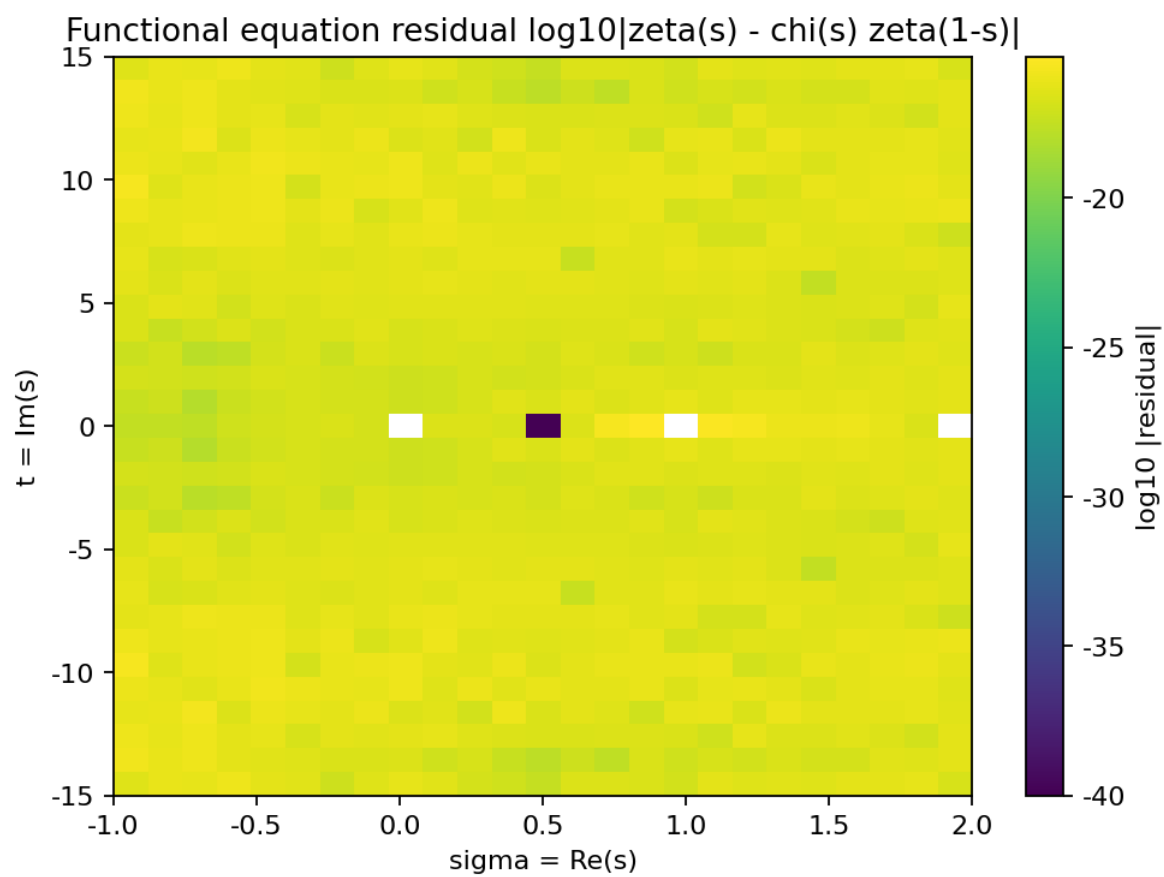


Fig. 9: E090 — Functional equation residual heatmap

## Notes

- This is a sanity-check style plot: away from poles/zeros and with sufficient precision, the residual should be small.
- Parameters control precision and grid size.

## References

- See the zeta / Dirichlet-series references in `refs.bib`.

## Related experiments

- *E089 —  $\log|s|$  heatmap* (E089 —  $\log|s|$  heatmap)
- *E082 — Zeta(s) series convergence* (E082 — Zeta(s) series convergence)
- *E083 — Series vs. Euler product ( )* (E083 — Series vs. Euler product ( ))
- *E084 —  $|1/(2+it)|$  growth snapshots* (E084 —  $|1/(2+it)|$  growth snapshots)
- *E085 — Dirichlet eta acceleration for (s)* (E085 — Dirichlet eta acceleration for (s))

### 3.3.91 E091 — Partial Euler products on the critical line

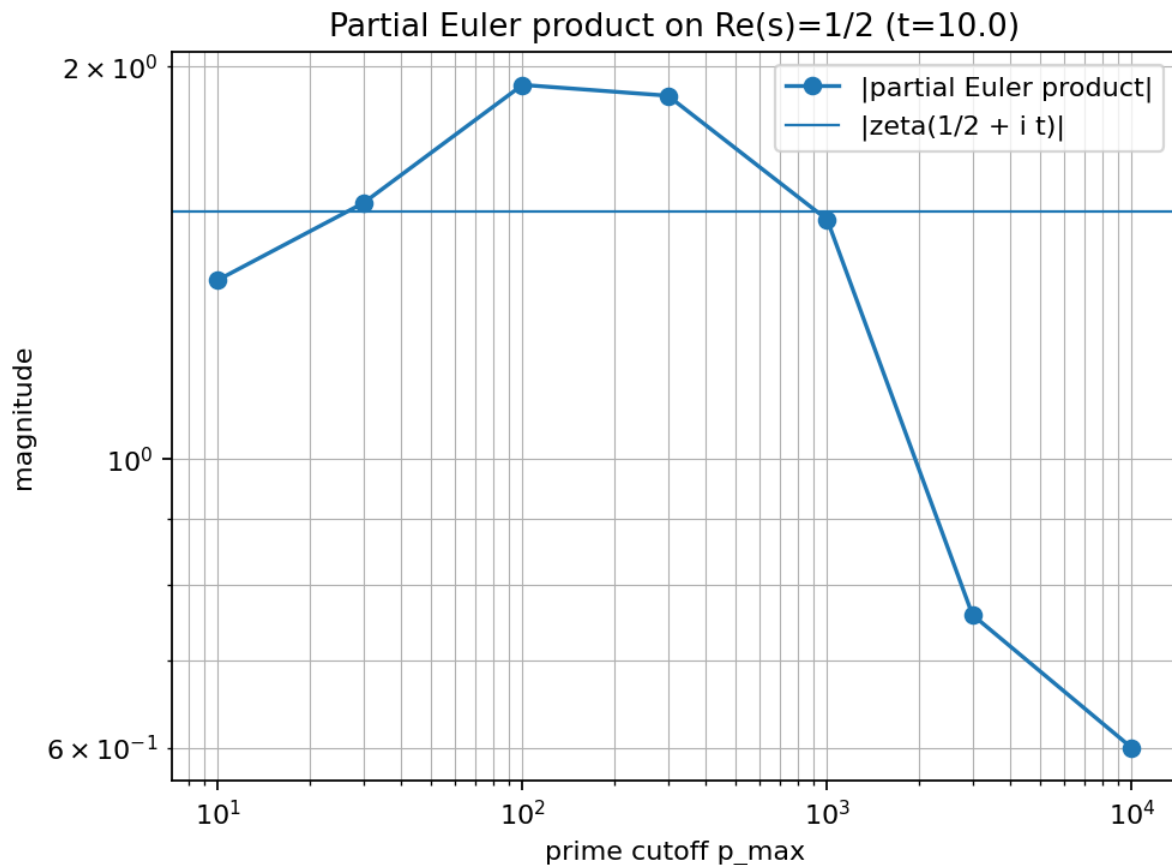


Fig. 10: E091 — Partial Euler products on the critical line

**Tags:** analysis, quantitative-exploration, visualization, riemann-zeta, dirichlet-series, numerics

### Highlights

- Focused numeric experiment with a single main figure.
- Parameters saved to `params.json` for reproducibility.
- Defaults are chosen, so the experiment remains feasible for the CI “slow” suite.

### What is computed

- Compare  $(\zeta(1/2+it))$  to partial Euler products  $(\prod_{p \leq P} (1-p^{-s}))^{-1}$  at a fixed  $t$  while increasing the prime cutoff  $P$ .
- Plot the mismatch to illustrate non-convergence on the critical line.

### Notes

- The Euler product is only convergent for  $(\Re(s) > 1)$ ; this experiment visualizes the failure mode at  $(\Re(s) = 1/2)$ .

### References

- See the zeta / Dirichlet-series references in `refs.bib`.

### Related experiments

- *E114 — via : stability map on the critical line* (E114 — via : stability map on the critical line)
- *E092 —  $1/(s)$  via the Möbius Dirichlet series* (E092 —  $1/(s)$  via the Möbius Dirichlet series)
- *E093 —  $-(s)/(s)$  via the von Mangoldt series* (E093 —  $-(s)/(s)$  via the von Mangoldt series)
- *E083 — Series vs. Euler product ( )* (E083 — Series vs. Euler product ( ))
- *E110 — Dirichlet L-series partial sums at  $s=1$  and  $s=1/2$*  (E110 — Dirichlet L-series partial sums at  $s=1$  and  $s=1/2$ )

## 3.3.92 E092 — $1/(s)$ via the Möbius Dirichlet series

Tags: analysis, quantitative-exploration, visualization, riemann-zeta, mobius, dirichlet-series, numerics

### Highlights

- Focused numeric experiment with a single main figure.
- Parameters saved to `params.json` for reproducibility.
- Defaults are chosen, so the experiment remains feasible for the CI “slow” suite.

### What is computed

- For  $(\Re(s) > 1)$ , compare  $(1/\zeta(s))$  to the partial sums  $(\sum_{n \leq N} \mu(n)/n^s)$ .
- Plot convergence as  $N$  increases, and highlight dependence on  $\Re(s)$ .

### Notes

- This is the classic identity  $(\sum_{n \geq 1} \mu(n)/n^s = 1/\zeta(s))$  (absolute convergence for  $(\Re(s) > 1)$ ).

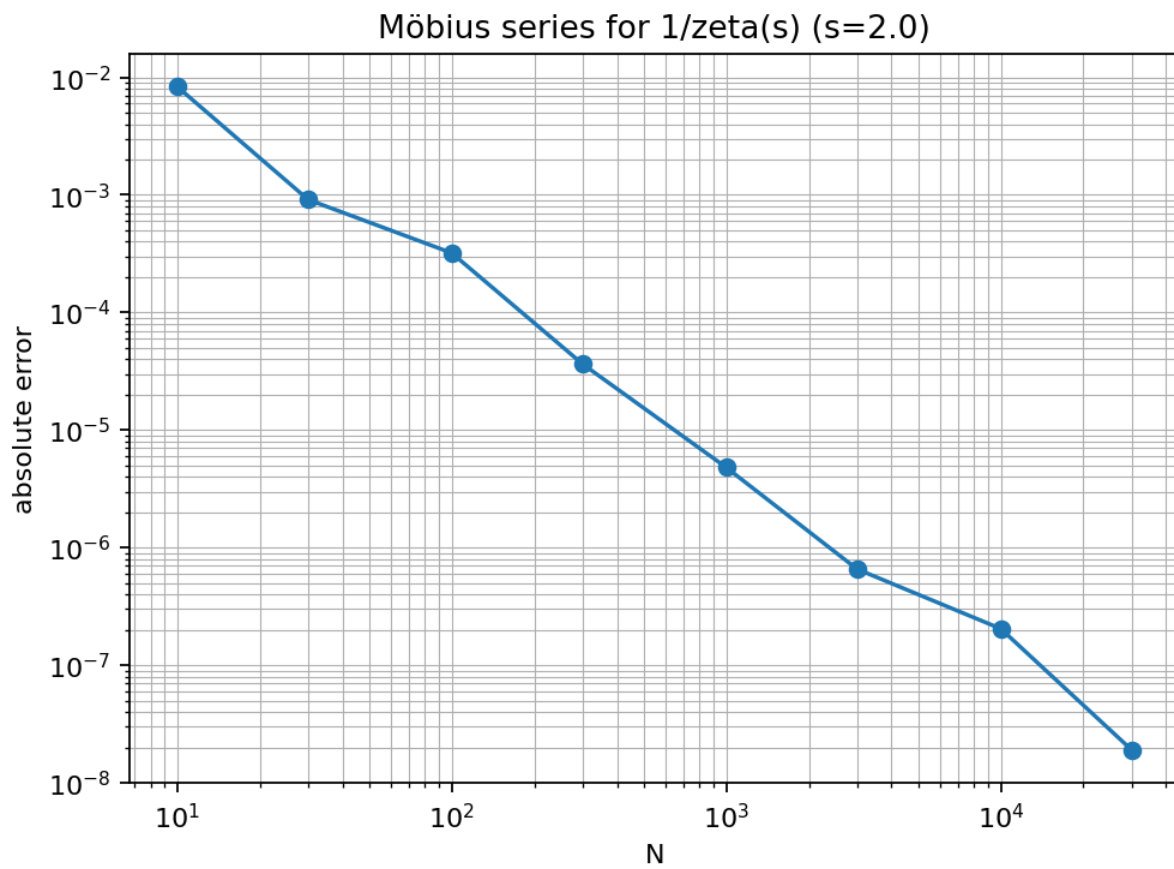


Fig. 11: E092 —  $1/(s)$  via the Möbius Dirichlet series

## References

- See the zeta / Dirichlet-series references in `refs.bib`.

## Related experiments

- *E093 —  $-\zeta'(s)/\zeta(s)$  via the von Mangoldt series* (E093 —  $-\zeta'(s)/\zeta(s)$  via the von Mangoldt series)
- *E110 — Dirichlet L-series partial sums at  $s=1$  and  $s=1/2$*  (E110 — Dirichlet L-series partial sums at  $s=1$  and  $s=1/2$ )
- *E111 — Euler product vs. Dirichlet series for  $L(s, \chi)$*  (E111 — Euler product vs. Dirichlet series for  $L(s, \chi)$ )
- *E091 — Partial Euler products on the critical line* (E091 — Partial Euler products on the critical line)
- *E082 — Zeta(s) series convergence* (E082 — Zeta(s) series convergence)

### 3.3.93 E093 — $-\zeta'(s)/\zeta(s)$ via the von Mangoldt series

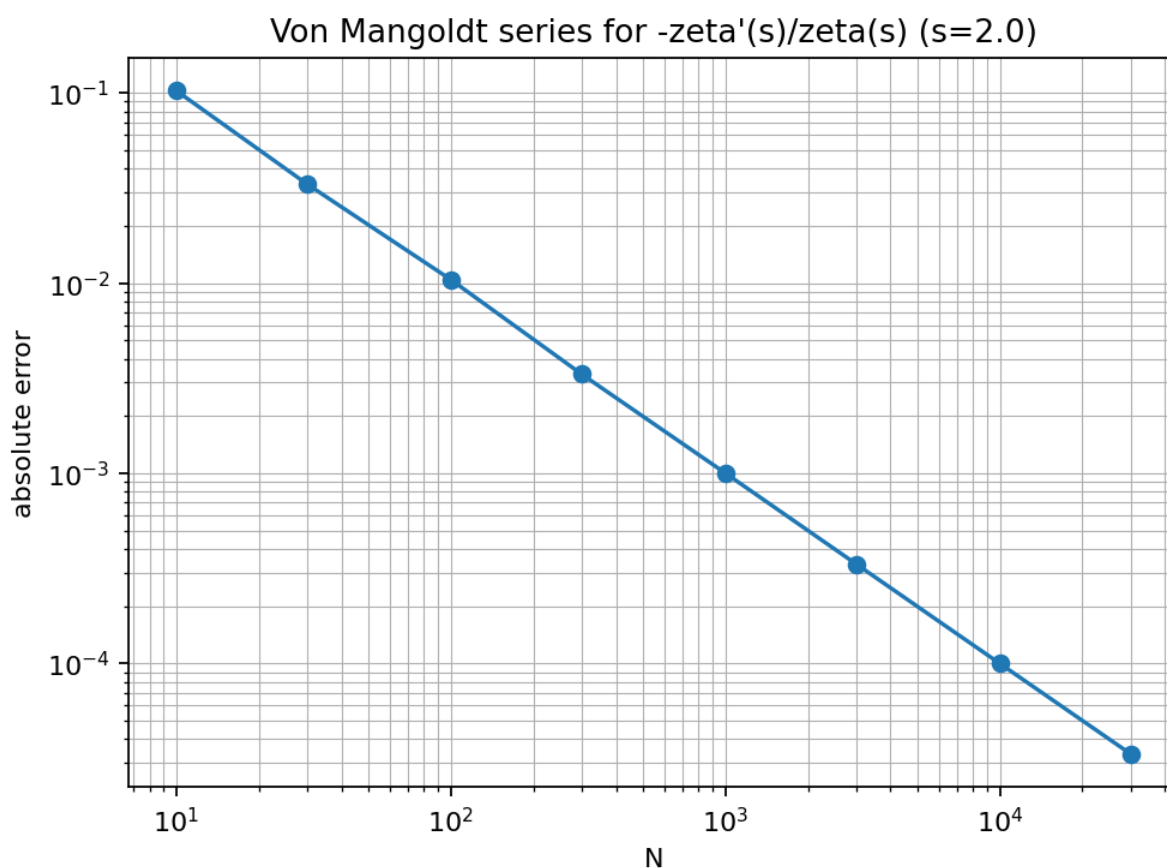


Fig. 12: E093 —  $-\zeta'(s)/\zeta(s)$  via the von Mangoldt series

**Tags:** analysis, quantitative-exploration, visualization, riemann-zeta, dirichlet-series, numerics

## Highlights

- Focused numeric experiment with a single main figure.
- Parameters saved to `params.json` for reproducibility.
- Defaults are chosen, so the experiment remains feasible for the CI “slow” suite.

### What is computed

- For  $(\Re(s) > 1)$ , compare  $(-\zeta'(s)/\zeta(s))$  to the partial sums  $(\sum_{n \leq N} \Lambda(n)/n^s)$ .
- Plot the truncation error as a function of  $N$ .

### Notes

- Uses the von Mangoldt function  $(\Lambda(n))$ ; the series converges absolutely for  $(\Re(s) > 1)$ .

### References

- See the zeta / Dirichlet-series references in `refs.bib`.

### Related experiments

- *E092 —  $1/(s)$  via the Möbius Dirichlet series* (E092 —  $1/(s)$  via the Möbius Dirichlet series)
- *E091 — Partial Euler products on the critical line* (E091 — Partial Euler products on the critical line)
- *E082 — Zeta(s) series convergence* (E082 — Zeta(s) series convergence)
- *E083 — Series vs. Euler product ( )* (E083 — Series vs. Euler product ( ))
- *E088 — Zero counting via Riemann-von Mangoldt* (E088 — Zero counting via Riemann-von Mangoldt)

### 3.3.94 E094 — $\omega(n)$ vs. $\Omega(n)$ : Erdős–Kac normalization

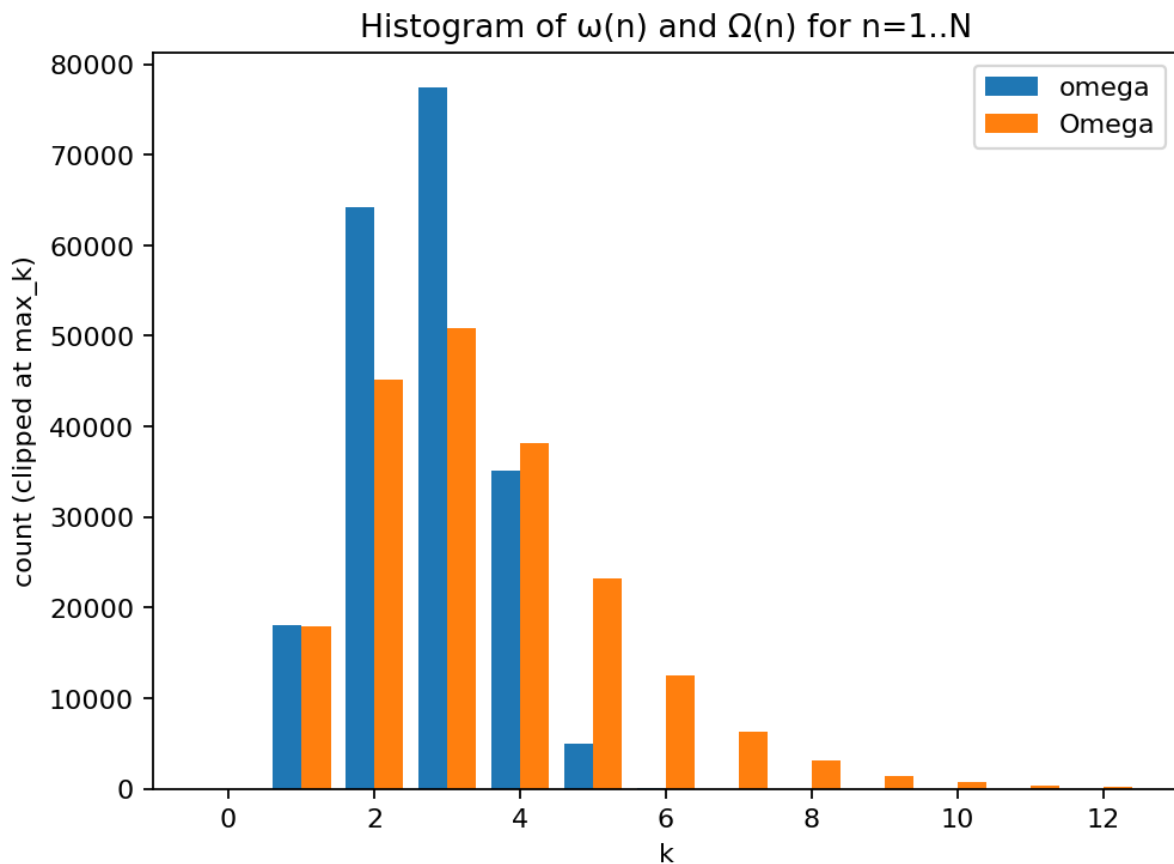


Fig. 13: E094 —  $\omega(n)$  vs.  $\Omega(n)$ : Erdős–Kac normalization

**Tags:** number-theory, quantitative-exploration, visualization, omega, arithmetic-functions, numerics

### Highlights

- Focused numeric experiment with a small set of figures.
- Parameters saved to `params.json` for reproducibility.
- Defaults are chosen, so the experiment remains feasible for the CI “slow” suite.

### What is computed

- Compare  $\phi(n)$  and  $\Omega(n)$  distributions and their Erdős–Kac normalizations.

### Notes

- This page summarizes the intent; see the generated `report.md` in `out/` for concrete outputs.

### References

- See the arithmetic-functions background pages in `docs/background/`.

### Related experiments

- *E057: Erdős–Kac in practice* (Erdős–Kac in practice)
- *E095 — Squarefree filter:  $\phi(n)=\Omega(n)$  when  $\phi(n) \neq 0$*  (E095 — Squarefree filter:  $\phi(n)=\Omega(n)$  when  $\phi(n) \neq 0$ )
- *E120 — Liouville  $\lambda(n)$ : partial sums and parity* (E120 — Liouville  $\lambda(n)$ : partial sums and parity)
- *E052: Totient ratio landscape* (Totient ratio landscape)
- *E053: Inverse totient multiplicities* (Inverse totient multiplicities)

## 3.3.95 E095 — Squarefree filter: $\phi(n)=\Omega(n)$ when $\phi(n) \neq 0$

**Tags:** number-theory, model-checking, visualization, mobius, omega, arithmetic-functions

### Highlights

- Focused numeric experiment with a small set of figures.
- Parameters saved to `params.json` for reproducibility.
- Defaults are chosen, so the experiment remains feasible for the CI “slow” suite.

### What is computed

- Use  $\phi(n)$  as a squarefree indicator and verify  $\phi(n)=\Omega(n)$  on that subset.

### Notes

- This page summarizes the intent; see the generated `report.md` in `out/` for concrete outputs.

### References

- See the arithmetic-functions background pages in `docs/background/`.



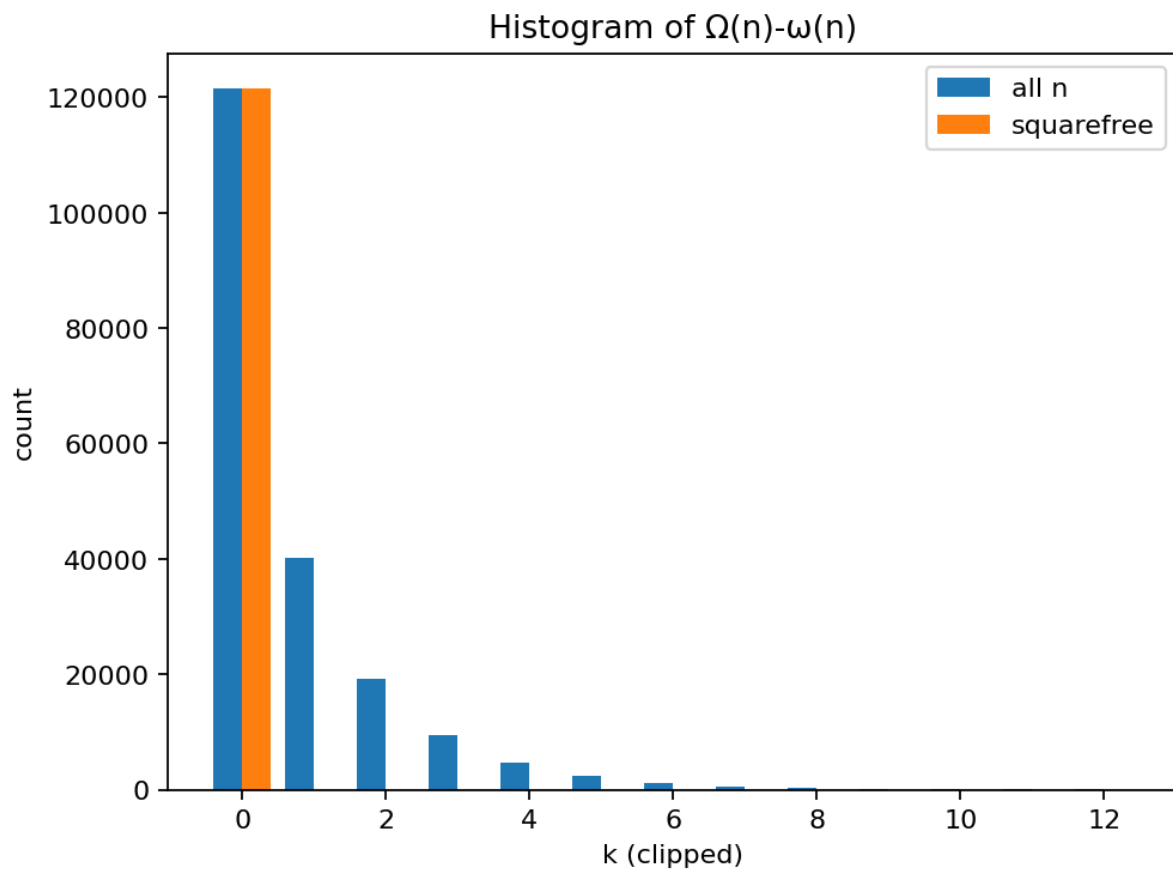


Fig. 14: E095 — Squarefree filter:  $(n)=\Omega(n)$  when  $(n) 0$

### Related experiments

- *E054: Squarefree density via Möbius* (Squarefree density via Möbius)
- *E055: Mertens function walk* (Mertens function walk)
- *E094 —  $(n)$  vs.  $\Omega(n)$ : Erdős–Kac normalization* (E094 —  $(n)$  vs.  $\Omega(n)$ : Erdős–Kac normalization)
- *E057: Erdős–Kac in practice* (Erdős–Kac in practice)
- *E121 — Möbius inversion as convolution undo* (E121 — Möbius inversion as convolution undo)

### 3.3.96 E096 — Record-holders for $(n)$

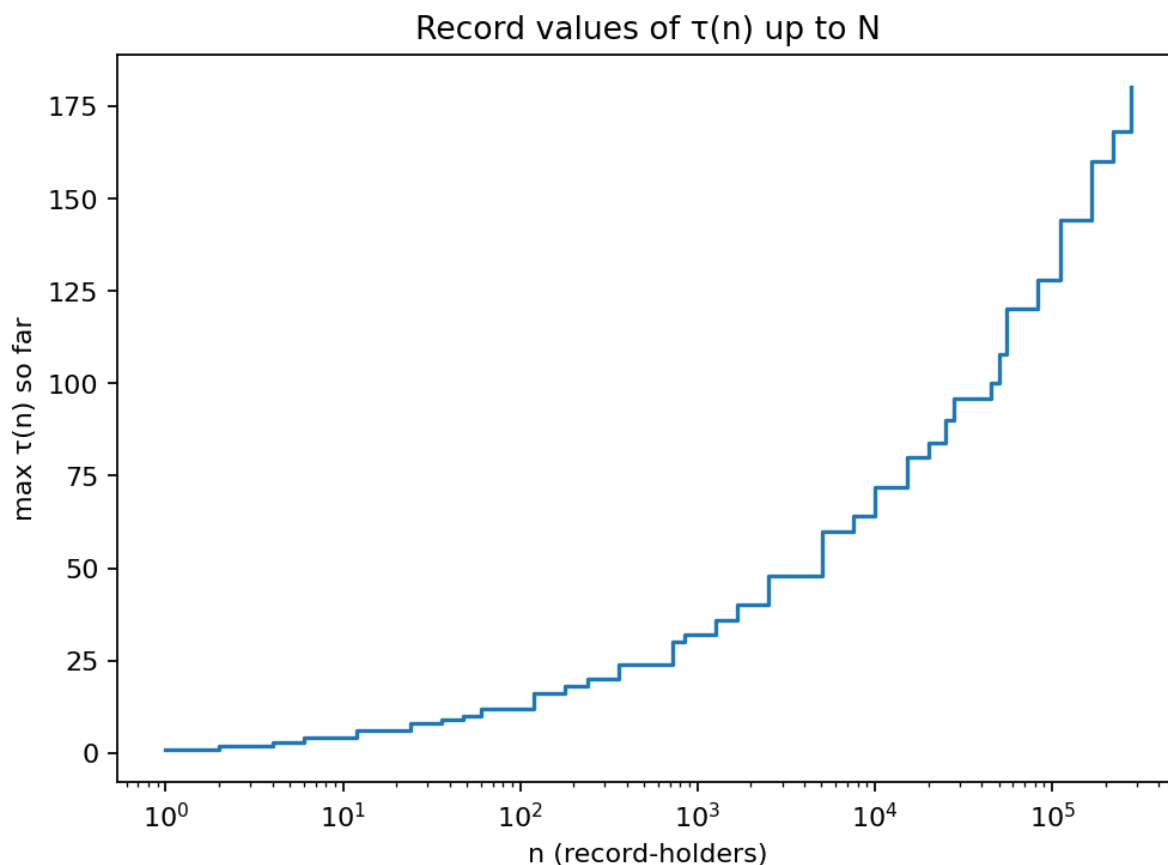


Fig. 15: E096 — Record-holders for  $(n)$

**Tags:** number-theory, quantitative-exploration, visualization, divisor-function, arithmetic-functions

### Highlights

- Focused numeric experiment with a small set of figures.
- Parameters saved to `params.json` for reproducibility.
- Defaults are chosen, so the experiment remains feasible for the CI “slow” suite.

### What is computed

- Track the running maxima of  $(n)$  up to N and visualize the growth of record values.

## Notes

- This page summarizes the intent; see the generated `report.md` in `out/` for concrete outputs.

## References

- See the arithmetic-functions background pages in `docs/background/`.
- See the perfect-numbers and divisor-functions background pages in `docs/background/`.

## Related experiments

- *E058: Divisor-count record highs* (Divisor-count record highs)
- *E059: Abundancy index landscape* (Abundancy index landscape)
- *E027: Record prime gaps vs.  $\log^2$  heuristic* (Record prime gaps vs.  $\log^2$  heuristic)
- *E052: Totient ratio landscape* (Totient ratio landscape)
- *E053: Inverse totient multiplicities* (Inverse totient multiplicities)

### 3.3.97 E097 — $\sigma(n)/n$ landscape: deficient, perfect, abundant

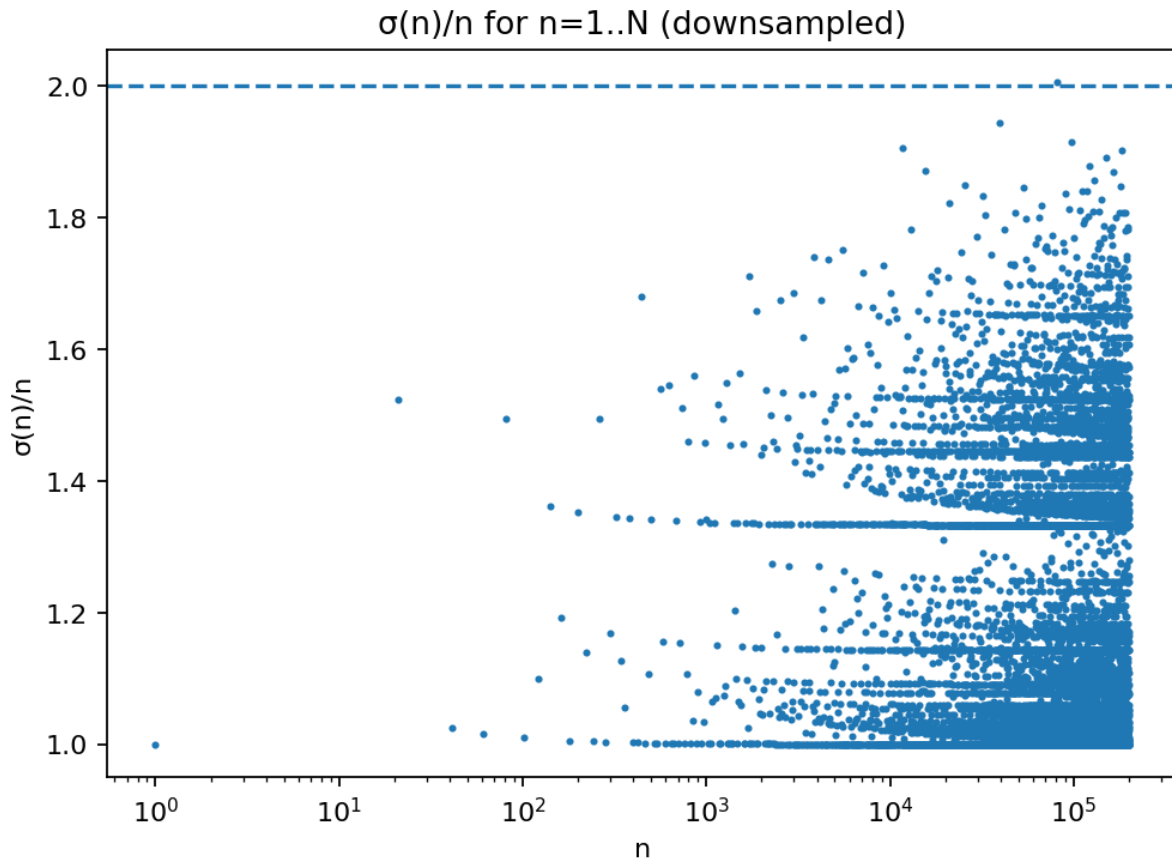


Fig. 16: E097 —  $\sigma(n)/n$  landscape: deficient, perfect, abundant

**Tags:** number-theory, quantitative-exploration, visualization, sigma, perfect, classification

### Highlights

- Focused numeric experiment with a small set of figures.
- Parameters saved to `params.json` for reproducibility.
- Defaults are chosen, so the experiment remains feasible for the CI “slow” suite.

### What is computed

- Visualize  $(n)/n$  and classify integers using  $(n)$  compared to  $2n$ .

### Notes

- This page summarizes the intent; see the generated `report.md` in `out/` for concrete outputs.

### References

- See the arithmetic-functions background pages in `docs/background/`.
- See the perfect-numbers and divisor-functions background pages in `docs/background/`.

### Related experiments

- *E059: Abundancy index landscape* (Abundancy index landscape)
- *E098 — Maximizers of  $(n)/n^{\wedge}$  across* (E098 — Maximizers of  $(n)/n^{\wedge}$  across )
- *E002: Even Perfect Numbers — Generator and Growth* (Even Perfect Numbers — Generator and Growth)
- *E003: Abundancy Index Landscape* (Abundancy Index Landscape)
- *E010: Even perfect numbers from Mersenne primes* (Even perfect numbers from Mersenne primes)

## 3.3.98 E098 — Maximizers of $(n)/n^{\wedge}$ across

Tags: number-theory, quantitative-exploration, visualization, sigma, optimization

### Highlights

- Focused numeric experiment with a small set of figures.
- Parameters saved to `params.json` for reproducibility.
- Defaults are chosen, so the experiment remains feasible for the CI “slow” suite.

### What is computed

- For on a grid, find  $n \in \mathbb{N}$  maximizing  $(n)/n^{\wedge}$  and show regime changes.

### Notes

- This page summarizes the intent; see the generated `report.md` in `out/` for concrete outputs.

### References

- See the arithmetic-functions background pages in `docs/background/`.

### Related experiments

- *E004: Computing  $\sigma(n)$  at Scale — Sieve vs. Factorization* (Computing  $\sigma(n)$  at Scale — Sieve vs. Factorization)
- *E059: Abundancy index landscape* (Abundancy index landscape)

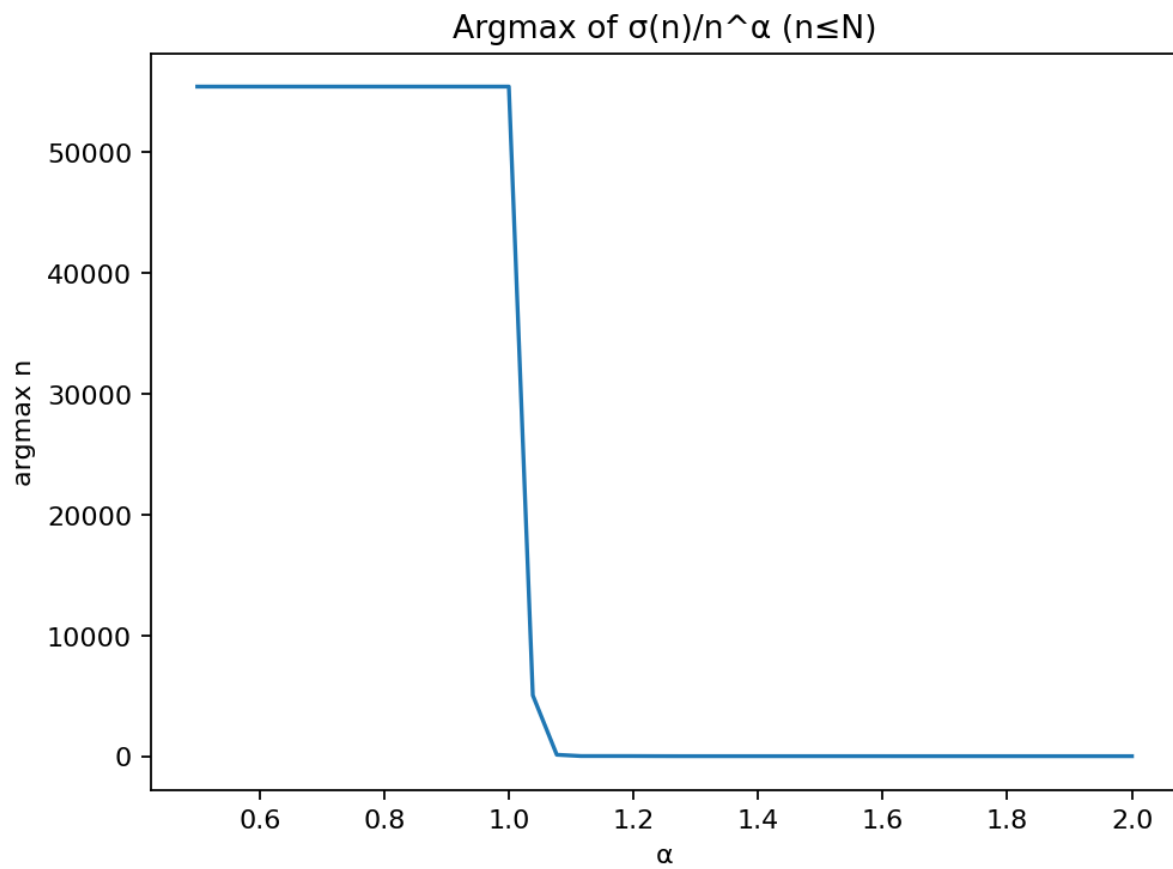


Fig. 17: E098 — Maximizers of  $\sigma(n)/n^\alpha$  across

- *E097 —  $(n)/n$  landscape: deficient, perfect, abundant* (E097 —  $(n)/n$  landscape: deficient, perfect, abundant)
- *E078: Max partial sums across characters.* (Max partial sums across characters.)
- *E002: Even Perfect Numbers — Generator and Growth* (Even Perfect Numbers — Generator and Growth)

### 3.3.99 E099 — Jordan totients $J_k$ : identities and ratios

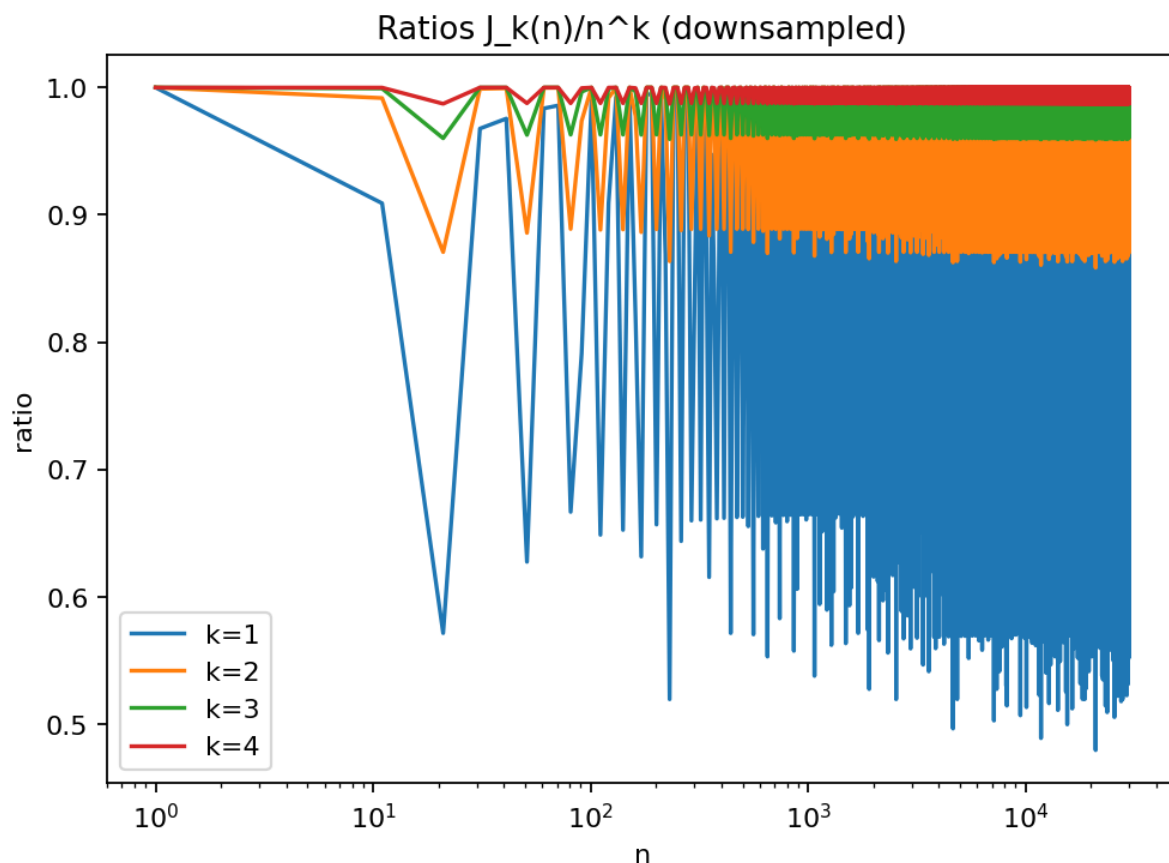


Fig. 18: E099 — Jordan totients  $J_k$ : identities and ratios

**Tags:** number-theory, model-checking, visualization, totient, arithmetic-functions

#### Highlights

- Focused numeric experiment with a small set of figures.
- Parameters saved to `params.json` for reproducibility.
- Defaults are chosen, so the experiment remains feasible for the CI “slow” suite.

#### What is computed

- Compute  $J_k(n)$ , verify  $J_1 = \phi$ , and visualize ratios  $J_k(n)/n^k$ .

#### Notes

- This page summarizes the intent; see the generated `report.md` in `out/` for concrete outputs.

## References

- See the arithmetic-functions background pages in docs/background/.

## Related experiments

- *E060: Jordan totients* (Jordan totients)
- *E101 — Reduced residues modulo  $q$ : concrete structure* (E101 — Reduced residues modulo  $q$ : concrete structure)
- *E052: Totient ratio landscape* (Totient ratio landscape)
- *E053: Inverse totient multiplicities* (Inverse totient multiplicities)
- *E062: Carmichael  $\lambda(n)$  vs.  $\phi(n)$*  (Carmichael  $\lambda(n)$  vs.  $\phi(n)$ )

### 3.3.100 E100 — Carmichael $\lambda(n)$ vs. Euler $\phi(n)$

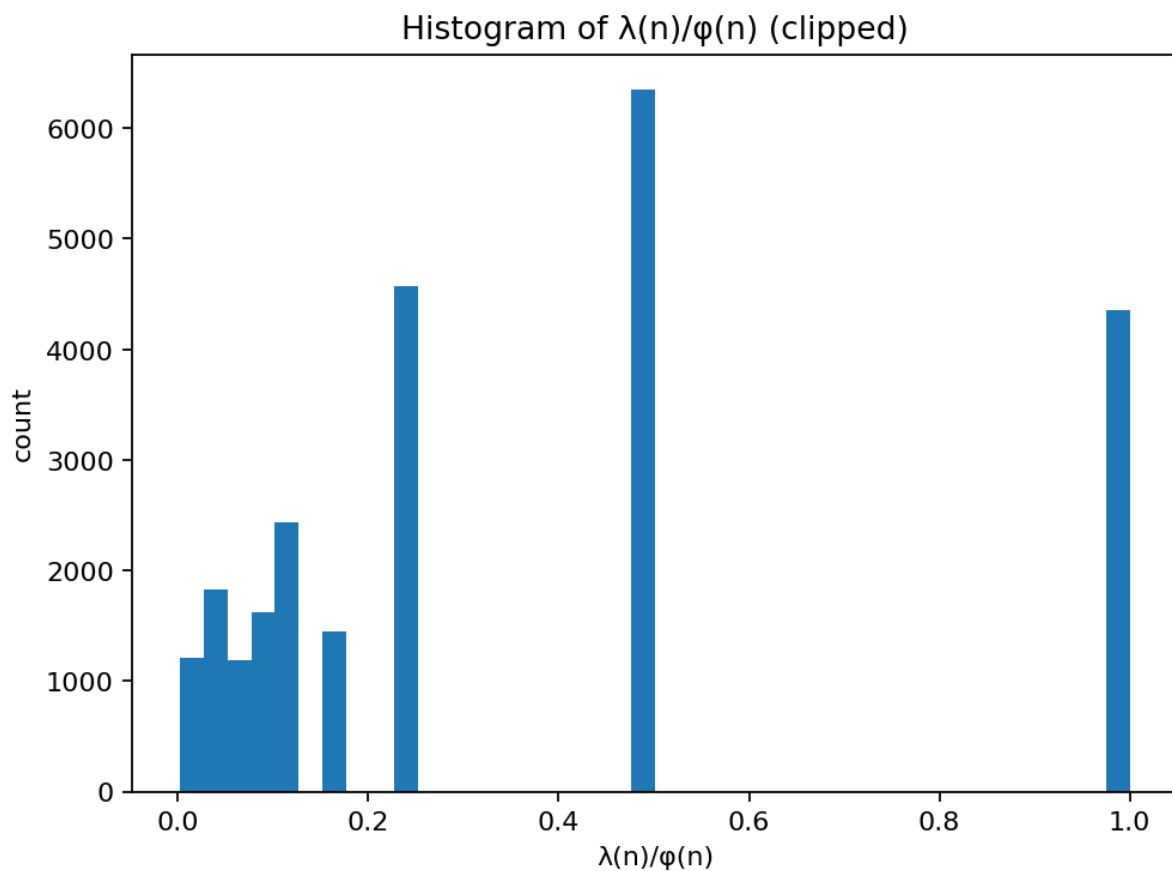


Fig. 19: E100 — Carmichael  $\lambda(n)$  vs. Euler  $\phi(n)$

**Tags:** number-theory, quantitative-exploration, visualization, carmichael-lambda, totient, arithmetic-functions

## Highlights

- Focused numeric experiment with a small set of figures.
- Parameters saved to `params.json` for reproducibility.
- Defaults are chosen, so the experiment remains feasible for the CI “slow” suite.

### What is computed

- Compare  $\phi(n)$  and  $\lambda(n)$  and visualize  $\phi(n)/\lambda(n)$  over a finite range.

### Notes

- This page summarizes the intent; see the generated `report.md` in `out/` for concrete outputs.

### References

- See the arithmetic-functions background pages in `docs/background/`.

### Related experiments

- *E062: Carmichael  $\lambda(n)$  vs.  $\phi(n)$*  (Carmichael  $\lambda(n)$  vs.  $\phi(n)$ )
- *E052: Totient ratio landscape* (Totient ratio landscape)
- *E053: Inverse totient multiplicities* (Inverse totient multiplicities)
- *E060: Jordan totients* (Jordan totients)
- *E099 — Jordan totients  $J_k$ : identities and ratios* (E099 — Jordan totients  $J_k$ : identities and ratios)

### 3.3.101 E101 — Reduced residues modulo $q$ : concrete structure

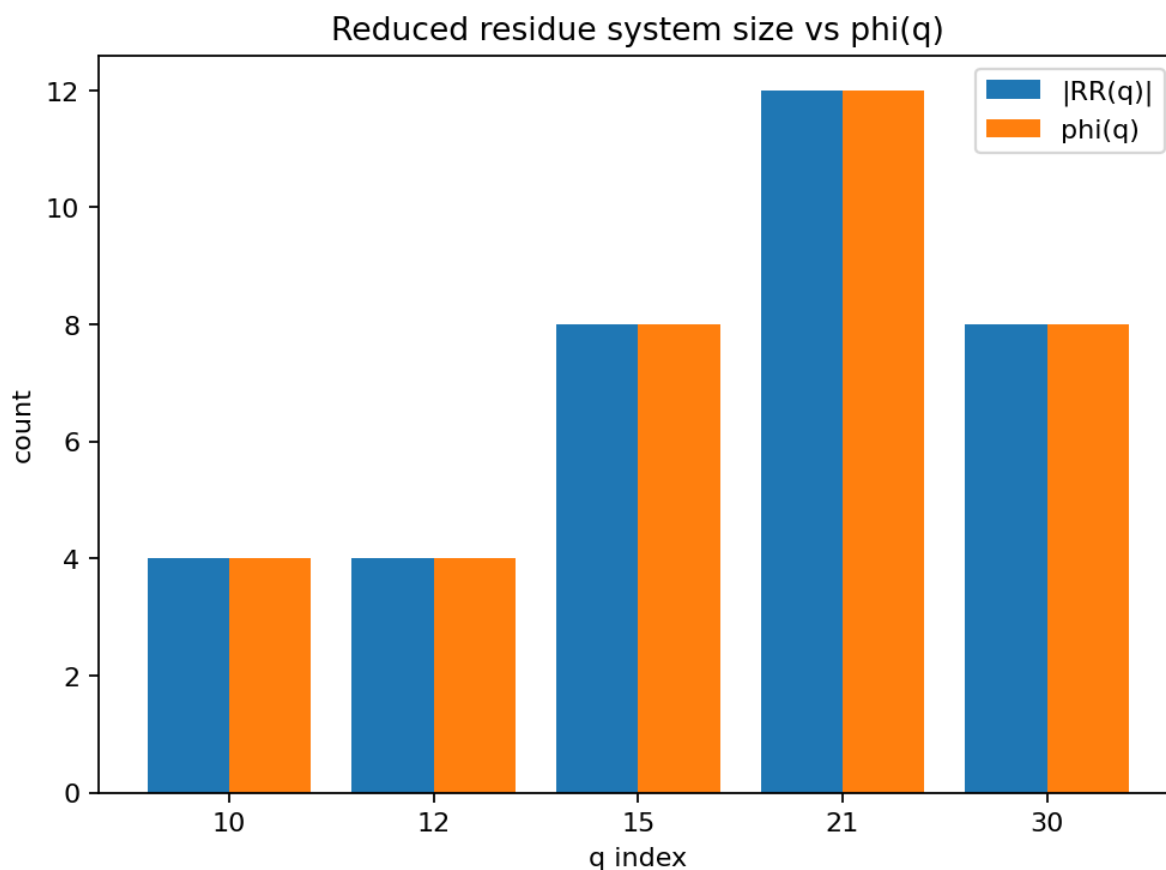


Fig. 20: E101 — Reduced residues modulo  $q$ : concrete structure

Tags: number-theory, dirichlet-characters, model-checking, visualization, totient



### Highlights

- Focused numeric experiment with a small set of figures.
- Parameters saved to `params.json` for reproducibility.
- Defaults are chosen, so the experiment remains feasible for the CI “slow” suite.

### What is computed

- List reduced residues mod  $q$ , verify  $\text{count} = (q)$ , and summarize simple structure checks.

### Notes

- This page summarizes the intent; see the generated `report.md` in `out/` for concrete outputs.

### References

- See the arithmetic-functions background pages in `docs/background/`.
- See `docs/background/dirichlet-characters.md`.

### Related experiments

- *E099 — Jordan totients  $J_k$ : identities and ratios* (E099 — Jordan totients  $J_k$ : identities and ratios)
- *E108 — Orthogonality heatmap for characters* (E108 — Orthogonality heatmap for characters)
- *E052: Totient ratio landscape* (Totient ratio landscape)
- *E053: Inverse totient multiplicities* (Inverse totient multiplicities)
- *E060: Jordan totients* (Jordan totients)

## 3.3.102 E102 — Dirichlet convolution identity zoo

**Tags:** number-theory, model-checking, visualization, dirichlet-convolution, multiplicative, arithmetic-functions

### Highlights

- Focused numeric experiment with a small set of figures.
- Parameters saved to `params.json` for reproducibility.
- Defaults are chosen, so the experiment remains feasible for the CI “slow” suite.

### What is computed

- Verify classic convolution identities numerically on  $1..N$  ( $1=$ ,  $1=\text{id}$ ,  $1*1=$ ).

### Notes

- This page summarizes the intent; see the generated `report.md` in `out/` for concrete outputs.

### References

- See the arithmetic-functions background pages in `docs/background/`.

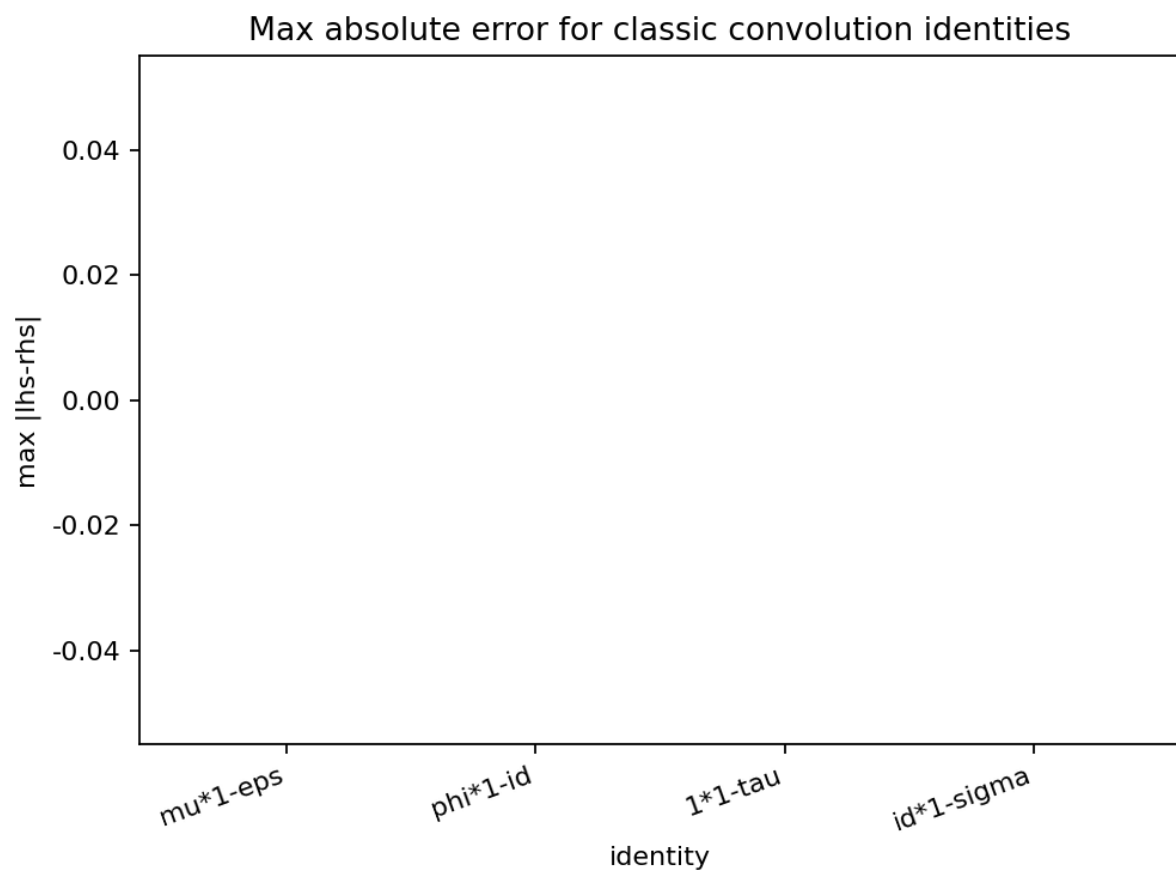


Fig. 21: E102 — Dirichlet convolution identity zoo

### Related experiments

- *E063: Dirichlet convolution playground* (Dirichlet convolution playground)
- *E121 — Möbius inversion as convolution undo* (E121 — Möbius inversion as convolution undo)
- *E108 — Orthogonality heatmap for characters* (E108 — Orthogonality heatmap for characters)
- *E111 — Euler product vs. Dirichlet series for  $L(s, \chi)$*  (E111 — Euler product vs. Dirichlet series for  $L(s, \chi)$ )
- *E052: Totient ratio landscape* (Totient ratio landscape)

### 3.3.103 E103 — Chebyshev $\psi(x)$ : prime powers drive jumps

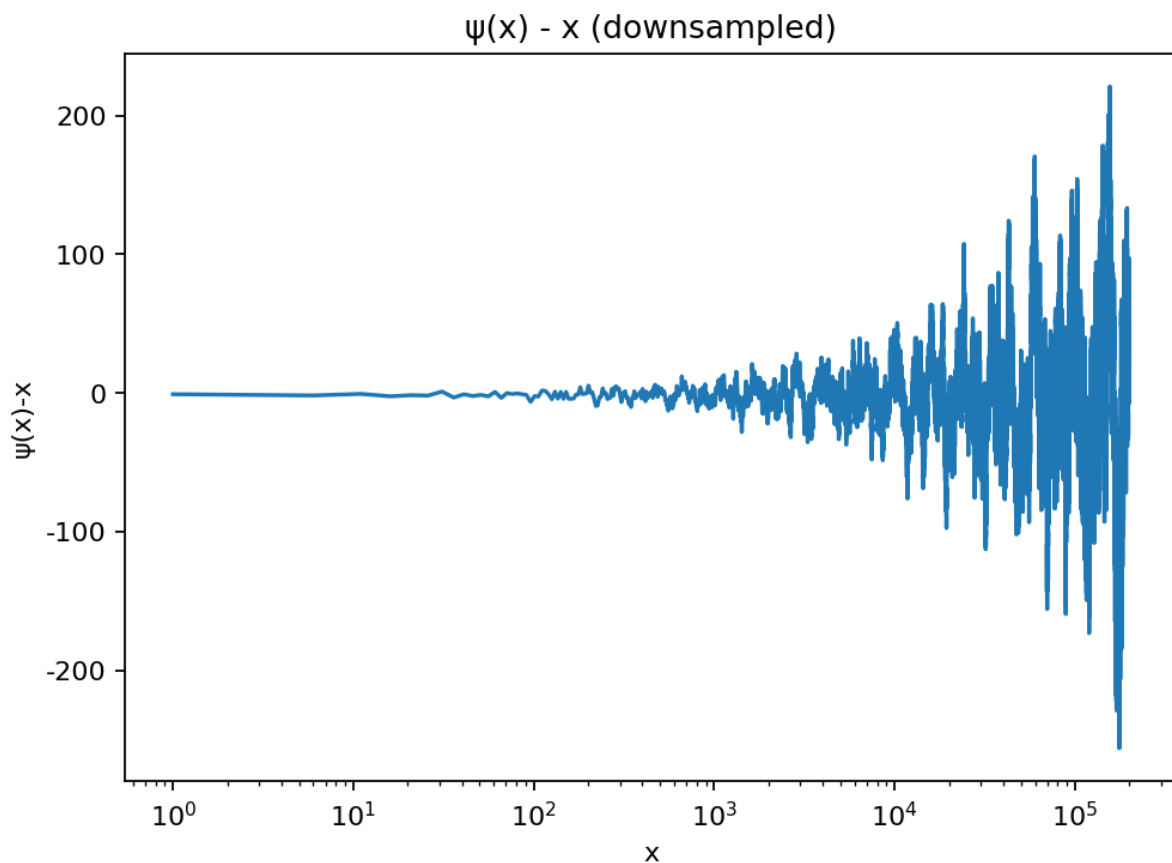


Fig. 22: E103 — Chebyshev  $\psi(x)$ : prime powers drive jumps

**Tags:** number-theory, quantitative-exploration, visualization, mangoldt, explicit-formula

#### Highlights

- Focused numeric experiment with a small set of figures.
- Parameters saved to `params.json` for reproducibility.
- Defaults are chosen, so the experiment remains feasible for the CI “slow” suite.

#### What is computed

- Plot  $\psi(x)$  and highlight jump contributions coming from prime powers.

## Notes

- This page summarizes the intent; see the generated `report.md` in `out/` for concrete outputs.

## References

- See the relevant references in `refs.bib`.

## Related experiments

- *E061: Chebyshev  $(x)$  and prime powers* (Chebyshev  $(x)$  and prime powers)
- *E104 — von Mangoldt  $\Lambda(n)$ : support and statistics* (E104 — von Mangoldt  $\Lambda(n)$ : support and statistics)
- *E076: Chebyshev  $(x;q,a)$ : weighted prime counts in progressions.* (Chebyshev  $(x;q,a)$ : weighted prime counts in progressions.)
- *E080: Chebyshev bias: leader fraction vs.  $x$ .* (Chebyshev bias: leader fraction vs.  $x$ .)
- *E118 — Chebyshev bias: lead-time statistics* (E118 — Chebyshev bias: lead-time statistics)

### 3.3.104 E104 — von Mangoldt $\Lambda(n)$ : support and statistics

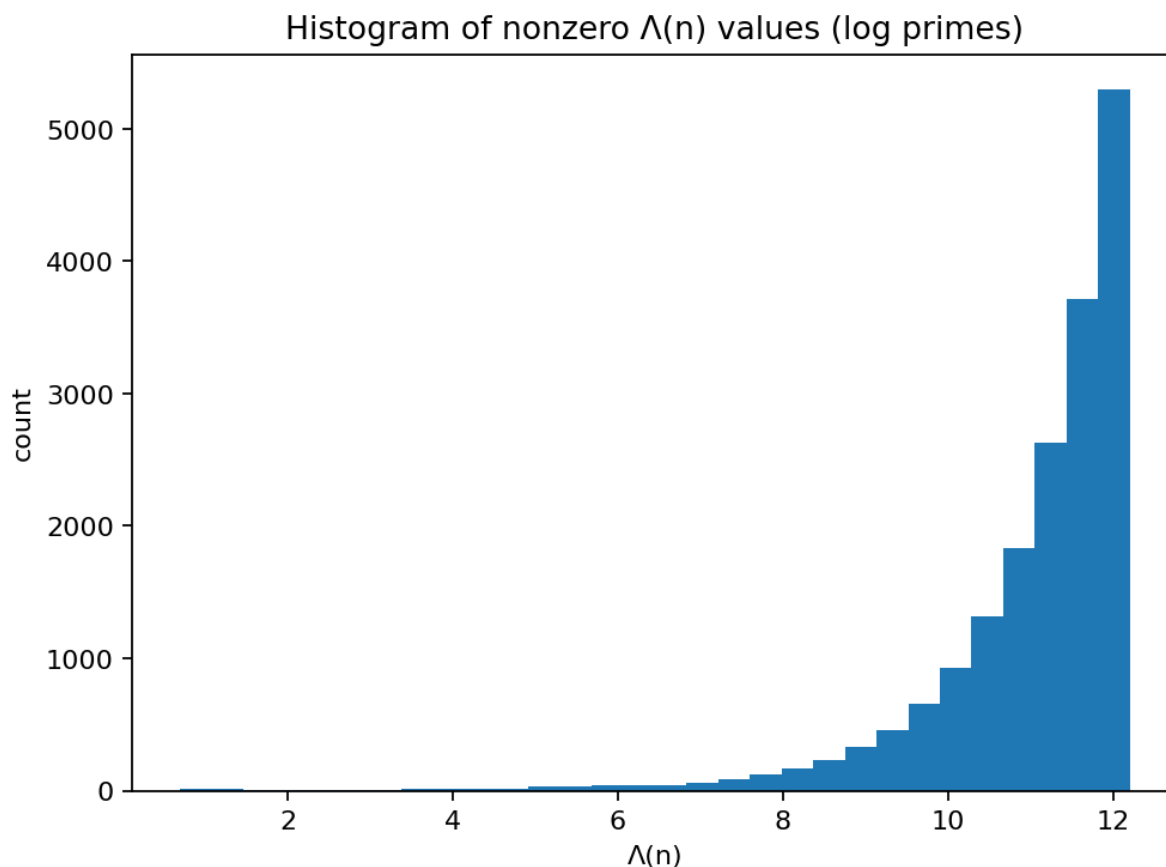


Fig. 23: E104 — von Mangoldt  $\Lambda(n)$ : support and statistics

**Tags:**      number-theory,      quantitative-exploration,      visualization,      mangoldt,  
arithmetic-functions

### Highlights

- Focused numeric experiment with a small set of figures.
- Parameters saved to `params.json` for reproducibility.
- Defaults are chosen, so the experiment remains feasible for the CI “slow” suite.

### What is computed

- Visualize where  $\Lambda(n)$  is nonzero (prime powers) and summarize basic statistics.

### Notes

- This page summarizes the intent; see the generated `report.md` in `out/` for concrete outputs.

### References

- See the arithmetic-functions background pages in `docs/background/`.

### Related experiments

- *E061: Chebyshev (x) and prime powers* (Chebyshev (x) and prime powers)
- *E103 — Chebyshev (x): prime powers drive jumps* (E103 — Chebyshev (x): prime powers drive jumps)
- *E118 — Chebyshev bias: lead-time statistics* (E118 — Chebyshev bias: lead-time statistics)
- *E088 — Zero counting via Riemann–von Mangoldt* (E088 — Zero counting via Riemann–von Mangoldt)
- *E093 — — (s)/ (s) via the von Mangoldt series* (E093 — — (s)/ (s) via the von Mangoldt series)

## 3.3.105 E105 — Mertens $M(x)$ : scaling views

**Tags:** number-theory, quantitative-exploration, visualization, mobius, summatory, open-problems

### Highlights

- Focused numeric experiment with a small set of figures.
- Parameters saved to `params.json` for reproducibility.
- Defaults are chosen, so the experiment remains feasible for the CI “slow” suite.

### What is computed

- Plot  $M(x) = \sum_{n \leq x} \Lambda(n)$  and rescalings such as  $M(x)/\sqrt{x}$ .

### Notes

- This page summarizes the intent; see the generated `report.md` in `out/` for concrete outputs.

### References

- See the arithmetic-functions background pages in `docs/background/`.

### Related experiments

- *E055: Mertens function walk* (Mertens function walk)
- *E054: Squarefree density via Möbius* (Squarefree density via Möbius)
- *E056: Liouville vs. Möbius walks* (Liouville vs. Möbius walks)

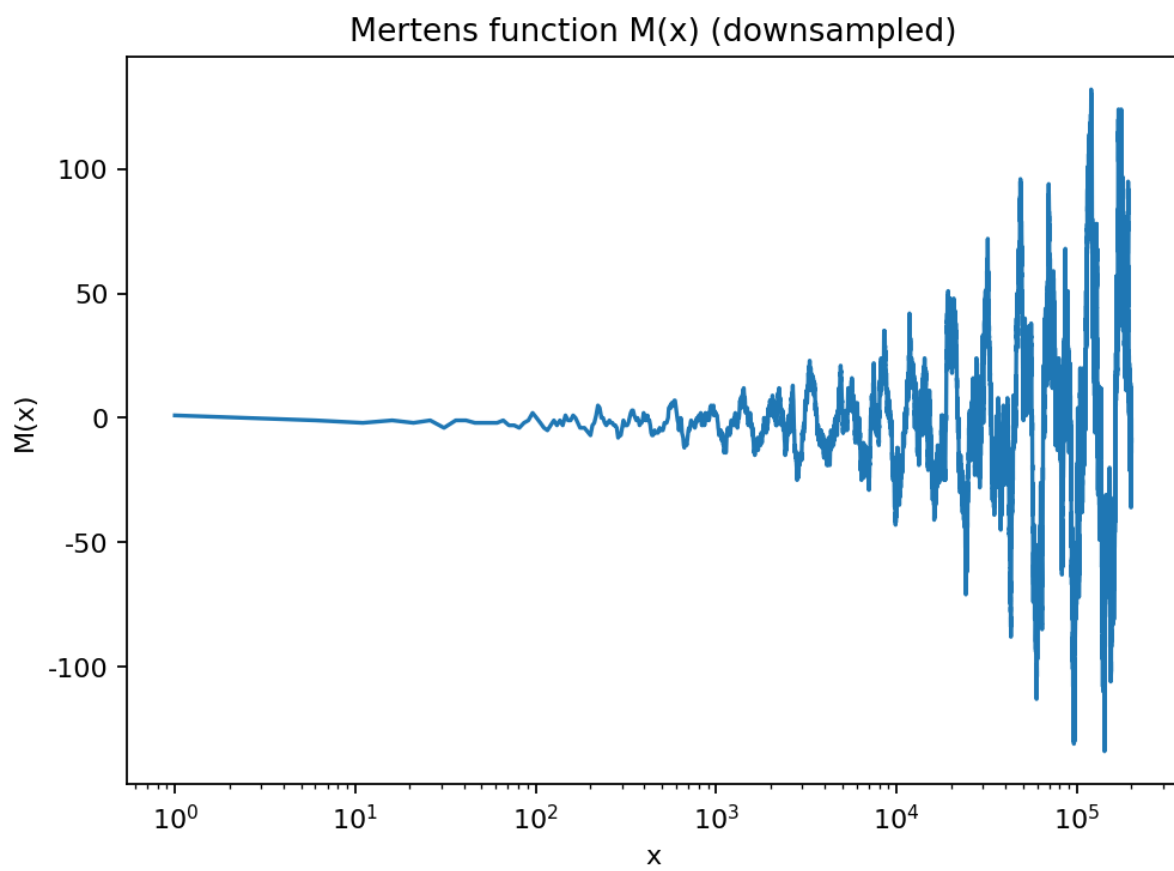
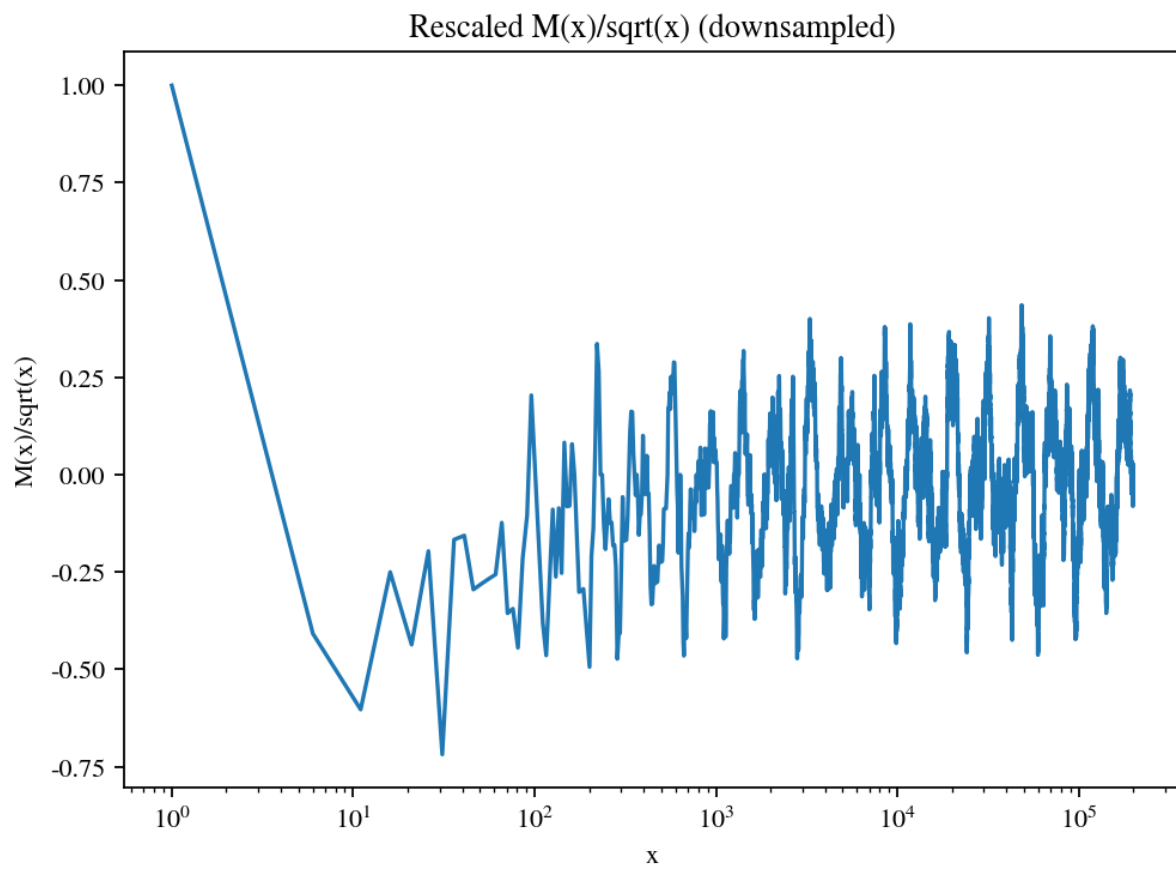


Fig. 24: E105 — Mertens  $M(x)$ : scaling views

Fig. 25: E105 — Mertens  $M(x)$ : scaling views

- *E119 — Summatory totient  $\Phi(x)$  scaling check* (E119 — Summatory totient  $\Phi(x)$  scaling check)
- *E061: Chebyshev  $\psi(x)$  and prime powers* (Chebyshev  $\psi(x)$  and prime powers)

### 3.3.106 E106 — Character gallery: real vs. complex

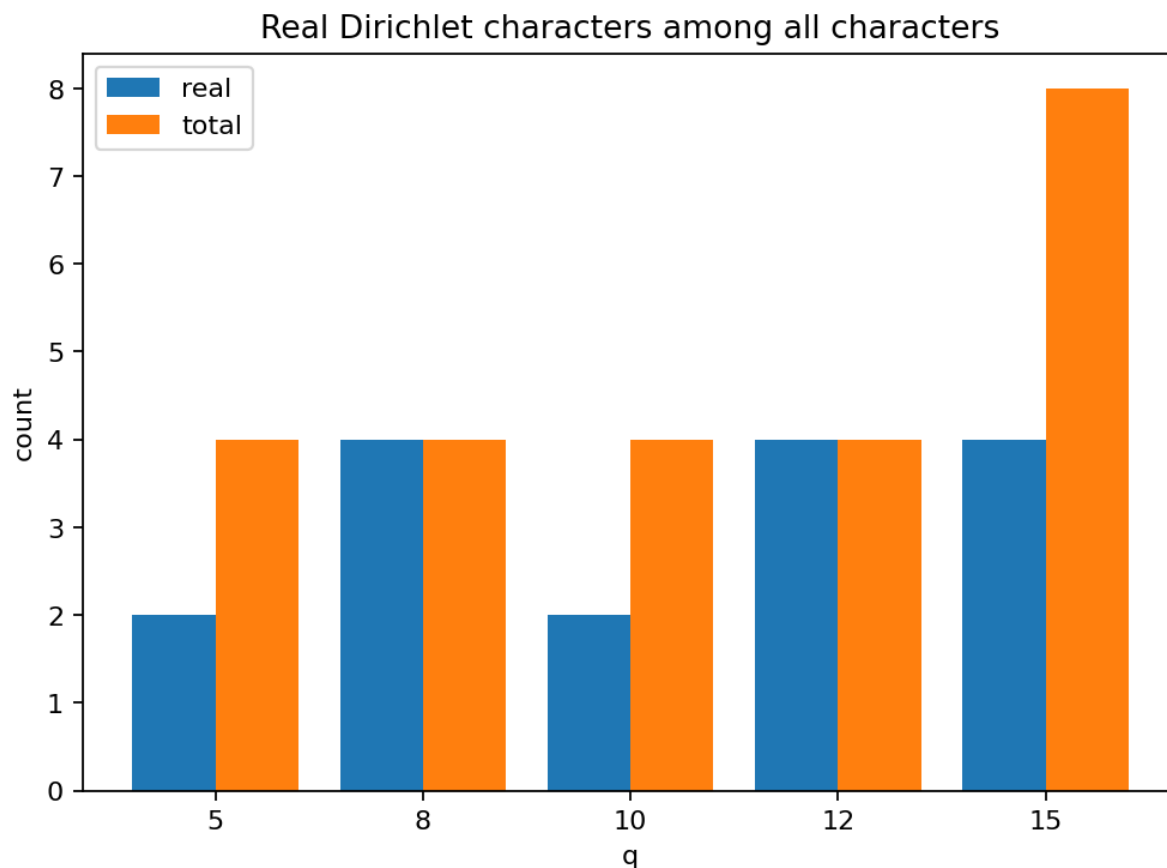


Fig. 26: E106 — Character gallery: real vs. complex

Tags: number-theory, dirichlet-characters, visualization, multiplicative

#### Highlights

- Focused numeric experiment with a small set of figures.
- Parameters saved to `params.json` for reproducibility.
- Defaults are chosen, so the experiment remains feasible for the CI “slow” suite.

#### What is computed

- Compare value-sets of Dirichlet characters and count real-valued characters.

#### Notes

- This page summarizes the intent; see the generated `report.md` in `out/` for concrete outputs.

#### References

- See `docs/background/dirichlet-characters.md`.



### Related experiments

- *E108 — Orthogonality heatmap for characters* (E108 — Orthogonality heatmap for characters)
- *E064: Dirichlet character tables (phase view)*. (Dirichlet character tables (phase view).)
- *E066: Character partial sums: cancellation profiles*. (Character partial sums: cancellation profiles.)
- *E077: Indicator via character orthogonality (sanity check)*. (Indicator via character orthogonality (sanity check).)
- *E122 — Character averages over primes* (E122 — Character averages over primes)

### 3.3.107 E107 — Conductor: primitive vs. induced characters

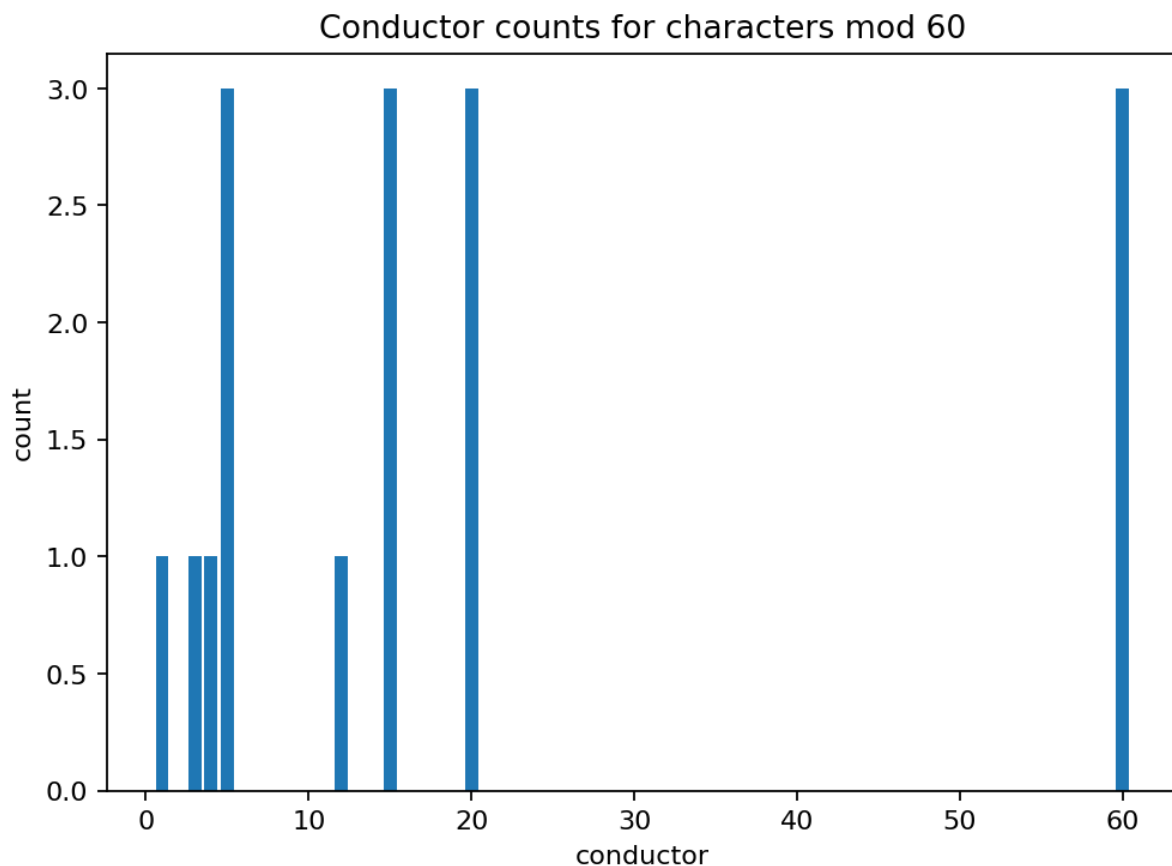


Fig. 27: E107 — Conductor: primitive vs. induced characters

**Tags:** number-theory, dirichlet-characters, quantitative-exploration, visualization, l-functions

### Highlights

- Focused numeric experiment with a small set of figures.
- Parameters saved to `params.json` for reproducibility.
- Defaults are chosen, so the experiment remains feasible for the CI “slow” suite.

### What is computed

- Compute conductors and separate primitive from imprimitive characters.

### Notes

- This page summarizes the intent; see the generated `report.md` in `out/` for concrete outputs.

### References

- See `docs/background/dirichlet-characters.md`.
- See `docs/background/dirichlet-l-functions.md` and `docs/background/dirichlet-convolution.md`.

### Related experiments

- *E079: Primitive vs. imprimitive characters: conductors.* (Primitive vs. imprimitive characters: conductors.)
- *E068: Dirichlet  $L(s, \cdot)$ : series vs. Euler product (partial approximations).* (Dirichlet  $L(s, \cdot)$ : series vs. Euler product (partial approximations).)
- *E069:  $L(1, \cdot)$ : slow convergence and smoothing.* ( $L(1, \cdot)$ : slow convergence and smoothing.)
- *E122 — Character averages over primes* (E122 — Character averages over primes)
- *E065: Orthogonality matrix for Dirichlet characters.* (Orthogonality matrix for Dirichlet characters.)

## 3.3.108 E108 — Orthogonality heatmap for characters

**Tags:** number-theory, dirichlet-characters, model-checking, visualization, multiplicative

### Highlights

- Focused numeric experiment with a small set of figures.
- Parameters saved to `params.json` for reproducibility.
- Defaults are chosen, so the experiment remains feasible for the CI “slow” suite.

### What is computed

- Visualize orthogonality relations via inner-product matrices.

### Notes

- This page summarizes the intent; see the generated `report.md` in `out/` for concrete outputs.

### References

- See `docs/background/dirichlet-characters.md`.

### Related experiments

- *E065: Orthogonality matrix for Dirichlet characters.* (Orthogonality matrix for Dirichlet characters.)
- *E102 — Dirichlet convolution identity zoo* (E102 — Dirichlet convolution identity zoo)
- *E121 — Möbius inversion as convolution undo* (E121 — Möbius inversion as convolution undo)
- *E106 — Character gallery: real vs. complex* (E106 — Character gallery: real vs. complex)
- *E077: Indicator via character orthogonality (sanity check).* (Indicator via character orthogonality (sanity check).)

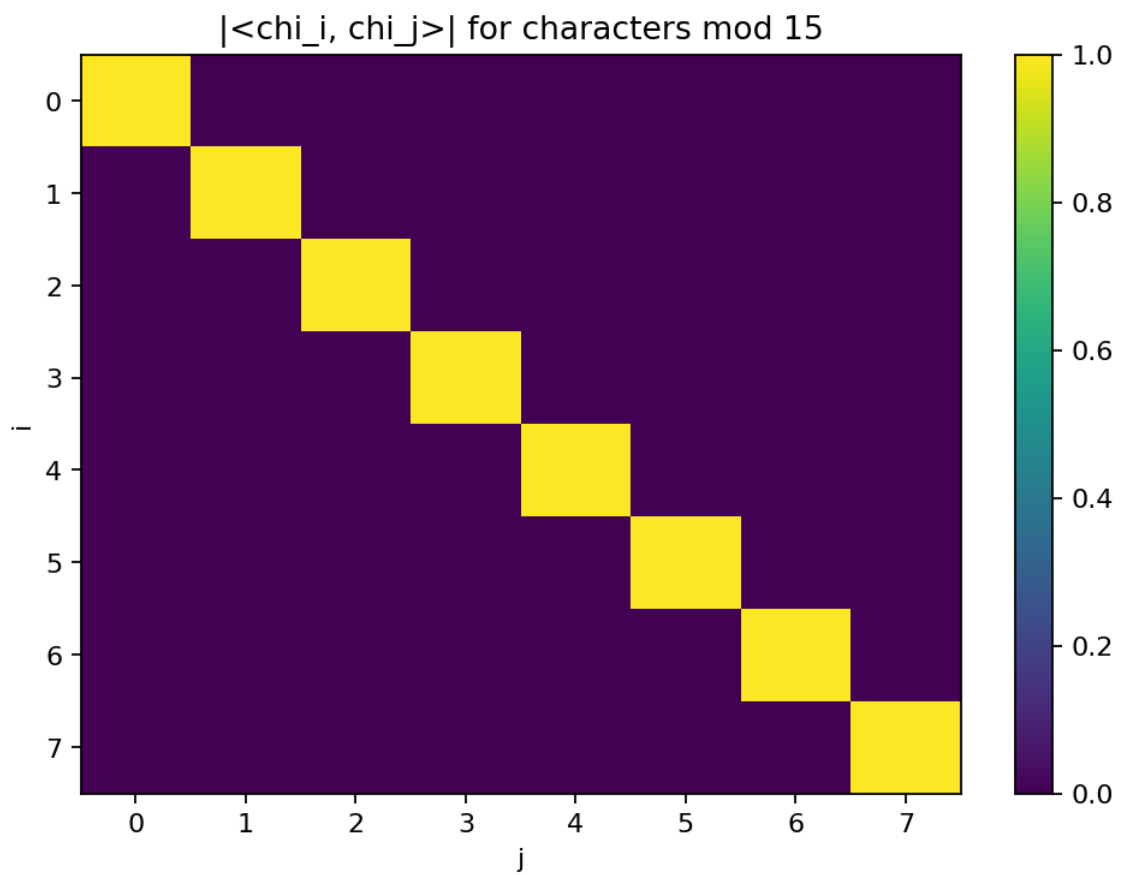


Fig. 28: E108 — Orthogonality heatmap for characters

### 3.3.109 E109 — Gauss sums: magnitude patterns

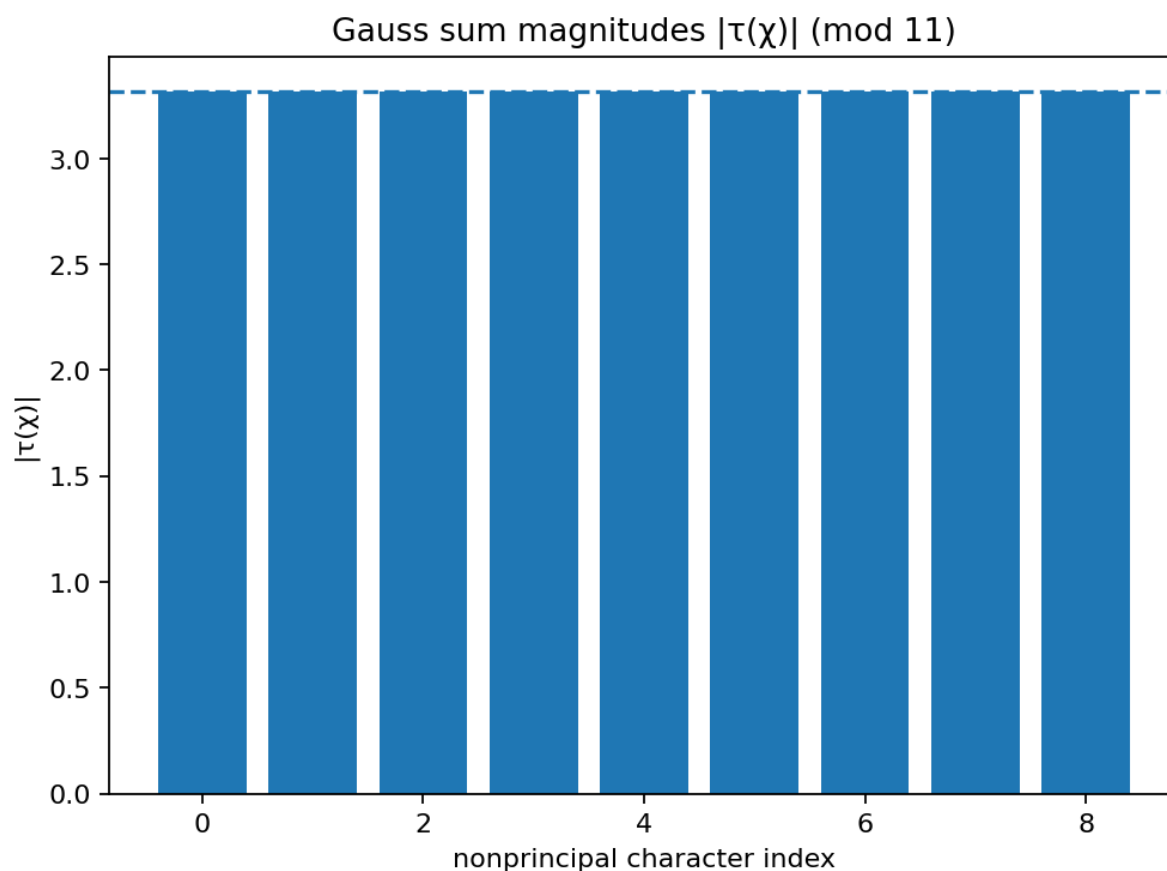


Fig. 29: E109 — Gauss sums: magnitude patterns

**Tags:** number-theory, dirichlet-characters, quantitative-exploration, visualization, numerics

#### Highlights

- Focused numeric experiment with a small set of figures.
- Parameters saved to `params.json` for reproducibility.
- Defaults are chosen, so the experiment remains feasible for the CI “slow” suite.

#### What is computed

- Compute Gauss sums  $\tau(\chi)$  and compare  $|\tau(\chi)|$  to  $\sqrt{q}$ .

#### Notes

- This page summarizes the intent; see the generated `report.md` in `out/` for concrete outputs.

#### References

- See `docs/background/dirichlet-characters.md`.

### Related experiments

- *E067: Gauss sums: magnitude vs.  $\sqrt{q}$ .* (Gauss sums: magnitude vs.  $\sqrt{q}$ .)
- *E066: Character partial sums: cancellation profiles.* (Character partial sums: cancellation profiles.)
- *E078: Max partial sums across characters.* (Max partial sums across characters.)
- *E110 — Dirichlet L-series partial sums at  $s=1$  and  $s=1/2$*  (E110 — Dirichlet L-series partial sums at  $s=1$  and  $s=1/2$ )
- *E120 — Liouville  $(n)$ : partial sums and parity* (E120 — Liouville  $(n)$ : partial sums and parity)

### 3.3.110 E110 — Dirichlet L-series partial sums at $s=1$ and $s=1/2$

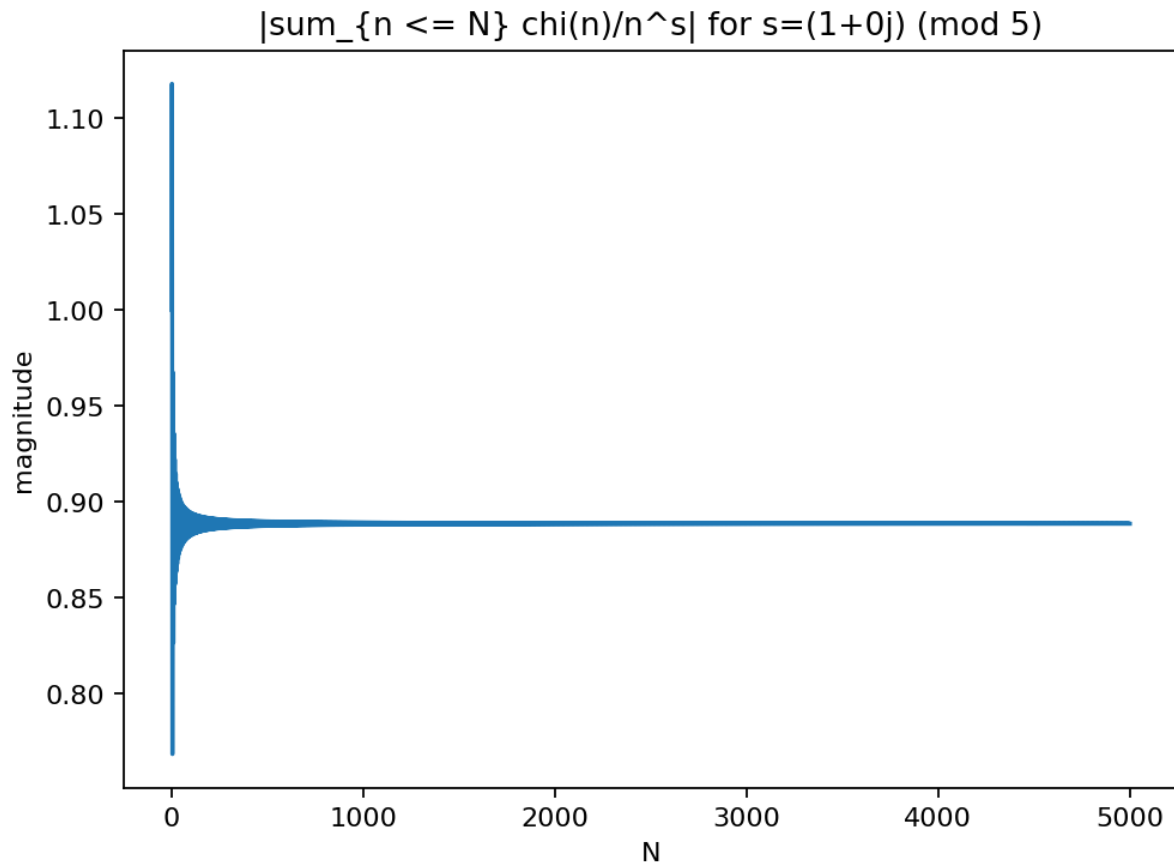


Fig. 30: E110 — Dirichlet L-series partial sums at  $s=1$  and  $s=1/2$

Tags: number-theory, l-functions, quantitative-exploration, visualization, dirichlet-series, numerics

### Highlights

- Focused numeric experiment with a small set of figures.
- Parameters saved to `params.json` for reproducibility.
- Defaults are chosen, so the experiment remains feasible for the CI “slow” suite.

## What is computed

- Compare partial Dirichlet series behavior near  $s=1$  and on the critical line.

## Notes

- This page summarizes the intent; see the generated `report.md` in `out/` for concrete outputs.

## References

- See `docs/background/dirichlet-l-functions.md` and `docs/background/dirichlet-convolution.md`.

## Related experiments

- *E111 — Euler product vs. Dirichlet series for  $L(s, \chi)$*  (E111 — Euler product vs. Dirichlet series for  $L(s, \chi)$ )
- *E068: Dirichlet  $L(s, \chi)$ : series vs. Euler product (partial approximations)*. (Dirichlet  $L(s, \chi)$ : series vs. Euler product (partial approximations).)
- *E092 —  $1/\zeta(s)$  via the Möbius Dirichlet series* (E092 —  $1/\zeta(s)$  via the Möbius Dirichlet series)
- *E091 — Partial Euler products on the critical line* (E091 — Partial Euler products on the critical line)
- *E093 —  $\zeta'(s)/\zeta(s)$  via the von Mangoldt series* (E093 —  $\zeta'(s)/\zeta(s)$  via the von Mangoldt series)

### 3.3.111 E111 — Euler product vs. Dirichlet series for $L(s, \chi)$

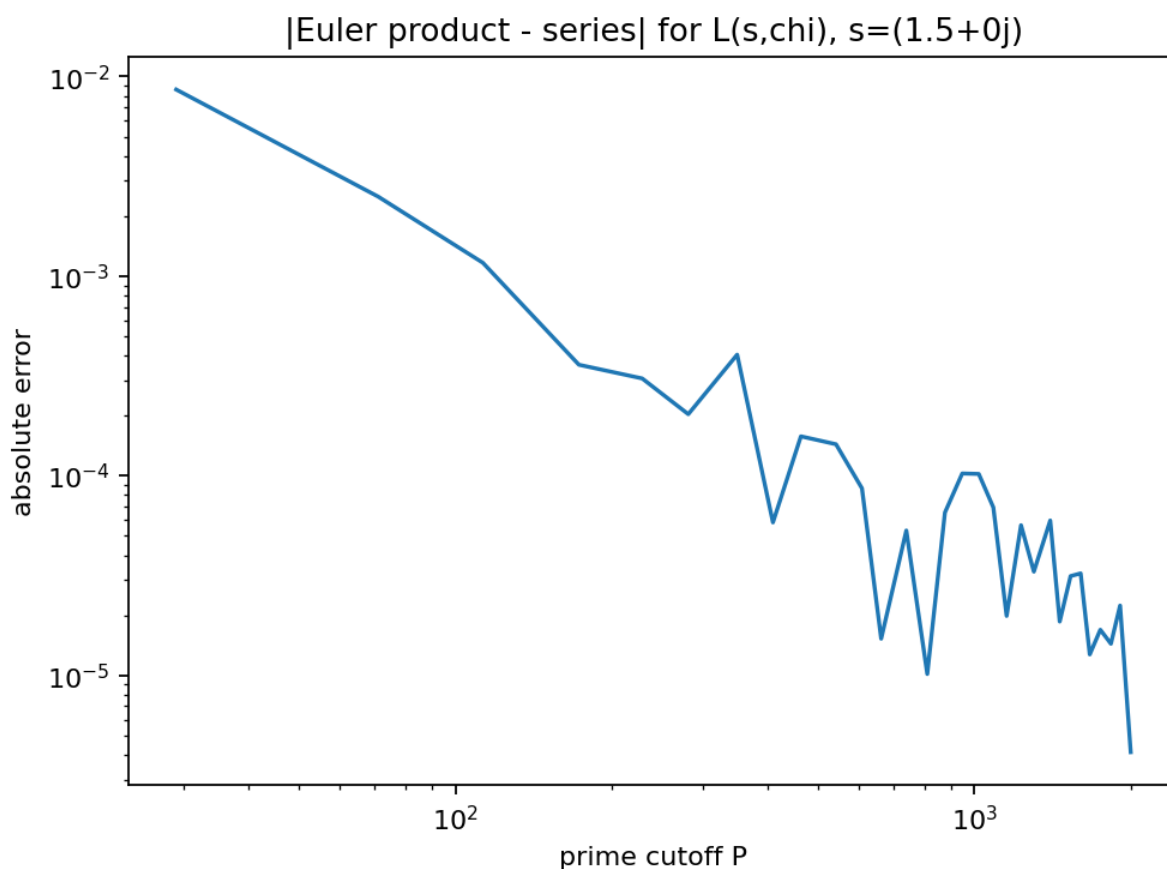


Fig. 31: E111 — Euler product vs. Dirichlet series for  $L(s, \chi)$

Tags: number-theory, l-functions, model-checking, visualization, dirichlet-series, numerics

### Highlights

- Focused numeric experiment with a small set of figures.
- Parameters saved to `params.json` for reproducibility.
- Defaults are chosen, so the experiment remains feasible for the CI “slow” suite.

### What is computed

- Compare truncated Euler products and truncated series for  $L(s, \cdot)$ .

### Notes

- This page summarizes the intent; see the generated `report.md` in `out/` for concrete outputs.

### References

- See `docs/background/dirichlet-l-functions.md` and `docs/background/dirichlet-convolution.md`.

### Related experiments

- *E068: Dirichlet  $L(s, \cdot)$ : series vs. Euler product (partial approximations)*. (Dirichlet  $L(s, \cdot)$ : series vs. Euler product (partial approximations).)
- *E110 — Dirichlet L-series partial sums at  $s=1$  and  $s=1/2$*  (E110 — Dirichlet L-series partial sums at  $s=1$  and  $s=1/2$ )
- *E092 —  $1/(s)$  via the Möbius Dirichlet series* (E092 —  $1/(s)$  via the Möbius Dirichlet series)
- *E083 — Series vs. Euler product ( )* (E083 — Series vs. Euler product ( ))
- *E091 — Partial Euler products on the critical line* (E091 — Partial Euler products on the critical line)

## 3.3.112 E112 — Prime race: $(x;q,a) - (x;q,b)$

Tags: number-theory, prime-races, visualization, aps, primes

### Highlights

- Focused numeric experiment with a small set of figures.
- Parameters saved to `params.json` for reproducibility.
- Defaults are chosen, so the experiment remains feasible for the CI “slow” suite.

### What is computed

- Compute prime counts in residue classes and plot difference curves for representative pairs.

### Notes

- This page summarizes the intent; see the generated `report.md` in `out/` for concrete outputs.

### References

- See `docs/background/primes-in-arithmetic-progressions.md` and `docs/background/prime-number-races.md`.

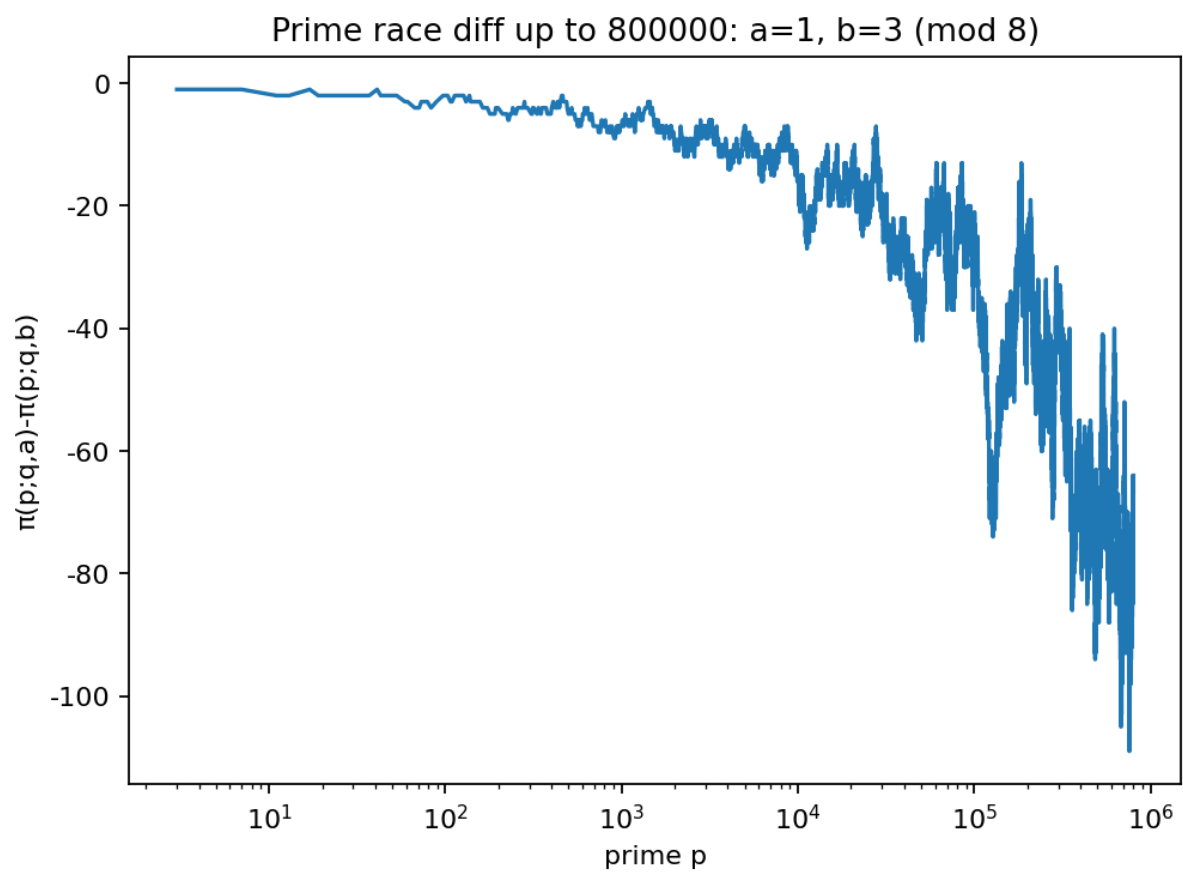


Fig. 32: E112 — Prime race:  $(x;q,a) - (x;q,b)$



### Related experiments

- *E113 — First prime in each residue class* (E113 — First prime in each residue class)
- *E072: Prime race mod 4:  $\pi(x;4,3)$  vs.  $\pi(x;4,1)$ .* (Prime race mod 4:  $\pi(x;4,3)$  vs.  $\pi(x;4,1)$ .)
- *E073: Prime race mod 3:  $\pi(x;3,2)$  vs.  $\pi(x;3,1)$ .* (Prime race mod 3:  $\pi(x;3,2)$  vs.  $\pi(x;3,1)$ .)
- *E074: Prime race mod 8: leaderboard among 1,3,5,7.* (Prime race mod 8: leaderboard among 1,3,5,7.)
- *E075: Prime race statistic: distribution on a log-grid.* (Prime race statistic: distribution on a log-grid.)

#### 3.3.113 E113 — First prime in each residue class

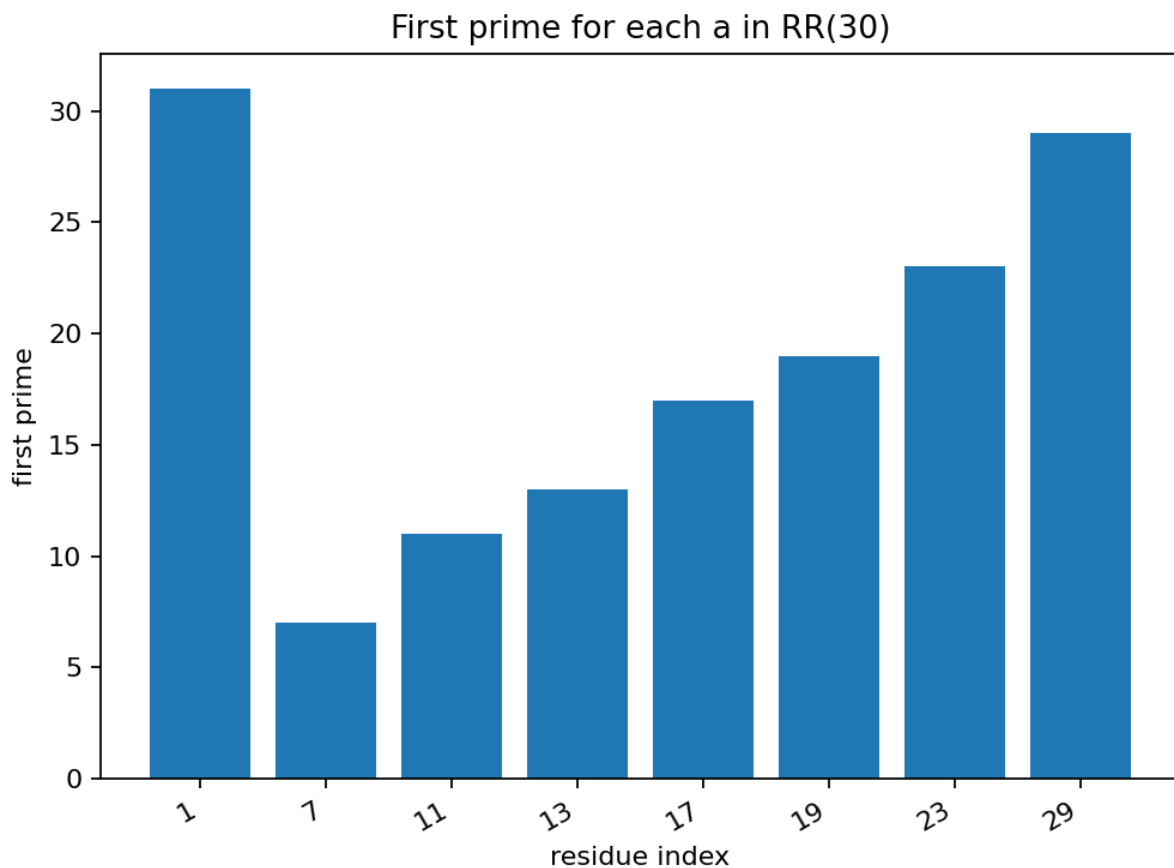


Fig. 33: E113 — First prime in each residue class

Tags: number-theory, prime-races, visualization, aps, primes

### Highlights

- Focused numeric experiment with a small set of figures.
- Parameters saved to `params.json` for reproducibility.
- Defaults are chosen, so the experiment remains feasible for the CI “slow” suite.

## What is computed

- For each reduced residue  $a \pmod{q}$ , find the smallest prime  $p \pmod{q}$  and visualize results.

## Notes

- This page summarizes the intent; see the generated `report.md` in `out/` for concrete outputs.

## References

- See `docs/background/primes-in-arithmetic-progressions.md` and `docs/background/prime-number-races.md`.

## Related experiments

- *E112 — Prime race:  $(x;q,a) - (x;q,b)$*  (E112 — Prime race:  $(x;q,a) - (x;q,b)$ )
- *E023: Residue class distribution mod  $q$*  (Residue class distribution mod  $q$ )
- *E070: Primes in residue classes:  $\pi(x; q, a)$* . (Primes in residue classes:  $\pi(x; q, a)$ .)
- *E081: Prime race sign changes: first crossings table*. (Prime race sign changes: first crossings table.)
- *E072: Prime race mod 4:  $\pi(x;4,3)$  vs.  $\pi(x;4,1)$* . (Prime race mod 4:  $\pi(x;4,3)$  vs.  $\pi(x;4,1)$ .)

### 3.3.114 E114 — via : stability map on the critical line

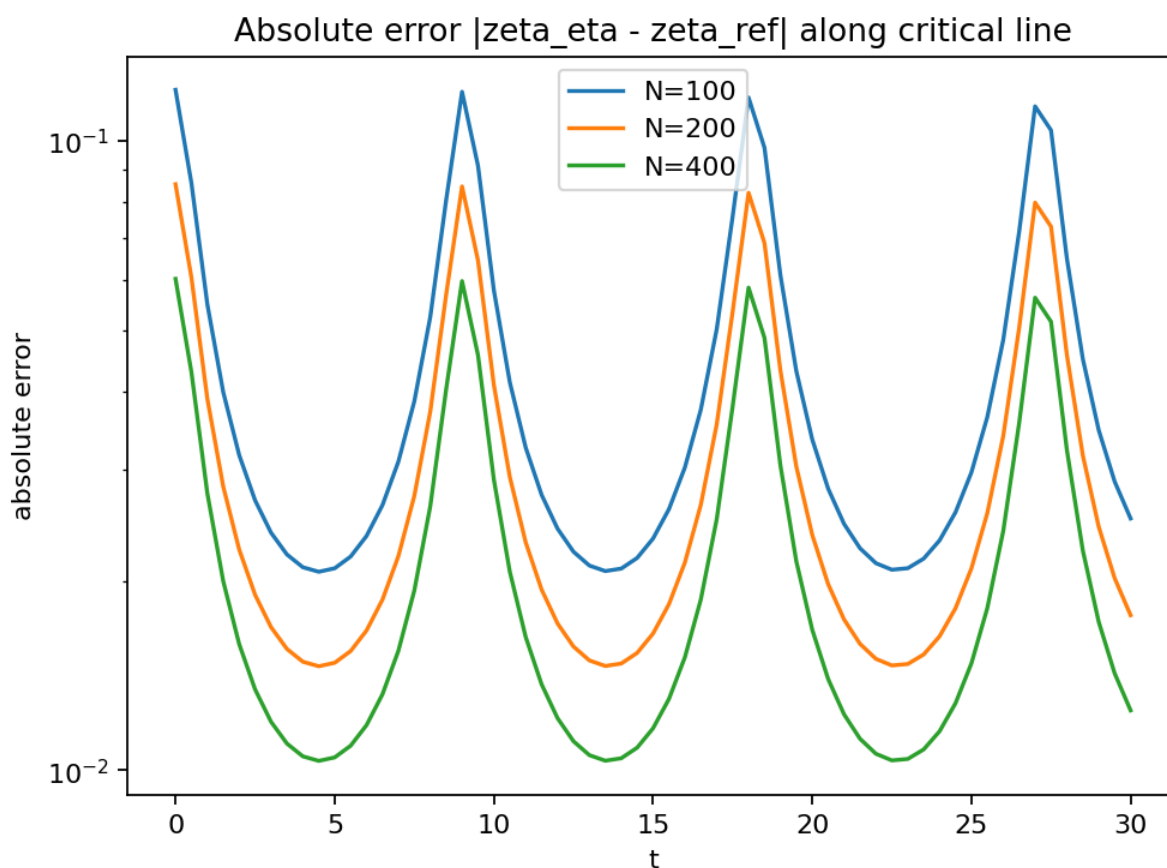


Fig. 34: E114 — via : stability map on the critical line

**Tags:** analysis, quantitative-exploration, visualization, riemann-zeta, critical-line, numerics

### Highlights

- Focused numeric experiment with a small set of figures.
- Parameters saved to `params.json` for reproducibility.
- Defaults are chosen, so the experiment remains feasible for the CI “slow” suite.

### What is computed

- Study stability of  $(1/2+it)$  via  $\gamma$ -acceleration under truncation changes.

### Notes

- This page summarizes the intent; see the generated `report.md` in `out/` for concrete outputs.

### References

- See the zeta / zeros background pages in `docs/background/`.

### Related experiments

- *E091 — Partial Euler products on the critical line* (E091 — Partial Euler products on the critical line)
- *E084 —  $| (1/2+it) |$  growth snapshots* (E084 —  $| (1/2+it) |$  growth snapshots)
- *E086 — Hardy  $Z(t)$  near zeros* (E086 — Hardy  $Z(t)$  near zeros)
- *E082 — Zeta(s) series convergence* (E082 — Zeta(s) series convergence)
- *E083 — Series vs. Euler product ( )* (E083 — Series vs. Euler product ( ))

## 3.3.115 E115 — Hardy Z: sign changes and zero bracketing

Tags: analysis, quantitative-exploration, visualization, riemann-zeta, hardy-z, zeta-zeros

### Highlights

- Focused numeric experiment with a small set of figures.
- Parameters saved to `params.json` for reproducibility.
- Defaults are chosen, so the experiment remains feasible for the CI “slow” suite.

### What is computed

- Scan Hardy  $Z(t)$ , bracket sign changes, and visualize candidate zeros.

### Notes

- This page summarizes the intent; see the generated `report.md` in `out/` for concrete outputs.

### References

- See the zeta / zeros background pages in `docs/background/`.

### Related experiments

- *E086 — Hardy  $Z(t)$  near zeros* (E086 — Hardy  $Z(t)$  near zeros)
- *E088 — Zero counting via Riemann-von Mangoldt* (E088 — Zero counting via Riemann-von Mangoldt)
- *E116 — Gram points and zero-counting heuristics* (E116 — Gram points and zero-counting heuristics)

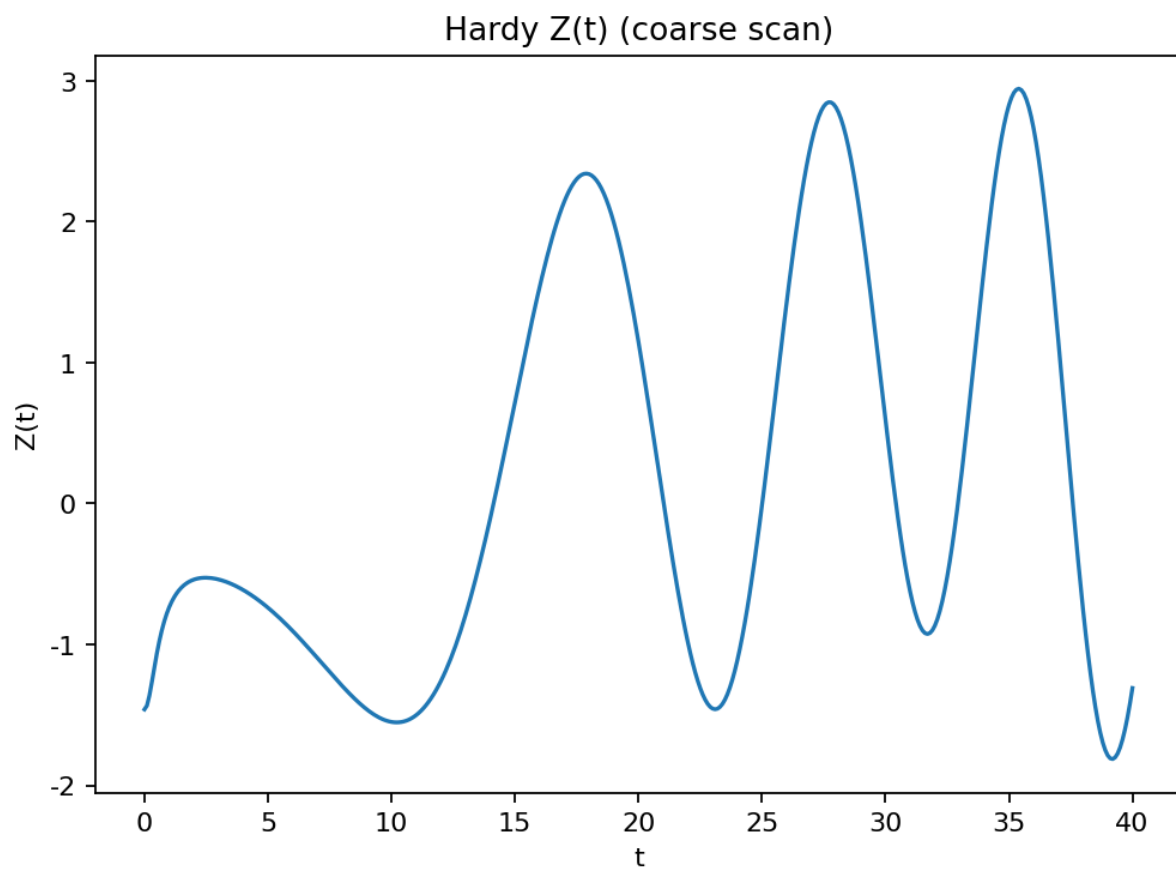


Fig. 35: E115 — Hardy  $Z$ : sign changes and zero bracketing

- *E087 — Gram points and spacing* (E087 — Gram points and spacing)
- *E081: Prime race sign changes: first crossings table.* (Prime race sign changes: first crossings table.)

### 3.3.116 E116 — Gram points and zero-counting heuristics

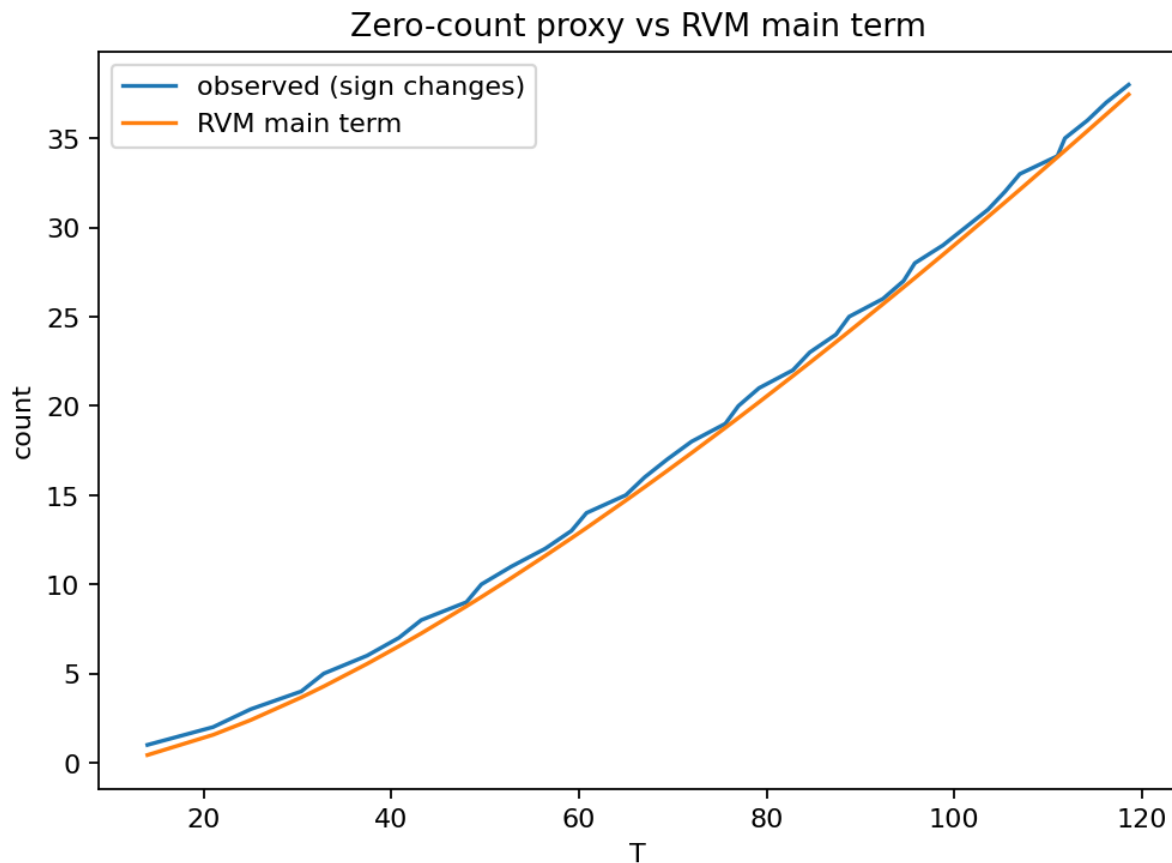


Fig. 36: E116 — Gram points and zero-counting heuristics

Tags: analysis, model-checking, visualization, riemann-zeta, gram-points, zeta-zeros

#### Highlights

- Focused numeric experiment with a small set of figures.
- Parameters saved to `params.json` for reproducibility.
- Defaults are chosen, so the experiment remains feasible for the CI “slow” suite.

#### What is computed

- Compare Gram-point heuristics and simple zero-counting diagnostics over a finite range.

#### Notes

- This page summarizes the intent; see the generated `report.md` in `out/` for concrete outputs.

## References

- See the zeta / zeros background pages in docs/background/.

## Related experiments

- *E087 — Gram points and spacing* (E087 — Gram points and spacing)
- *E088 — Zero counting via Riemann-von Mangoldt* (E088 — Zero counting via Riemann-von Mangoldt)
- *E115 — Hardy Z: sign changes and zero bracketing* (E115 — Hardy Z: sign changes and zero bracketing)
- *E086 — Hardy Z(t) near zeros* (E086 — Hardy Z(t) near zeros)
- *E117 — Prime-counting approximations:  $\text{li}(x)$  and friends* (E117 — Prime-counting approximations:  $\text{li}(x)$  and friends)

### 3.3.117 E117 — Prime-counting approximations: $\text{li}(x)$ and friends

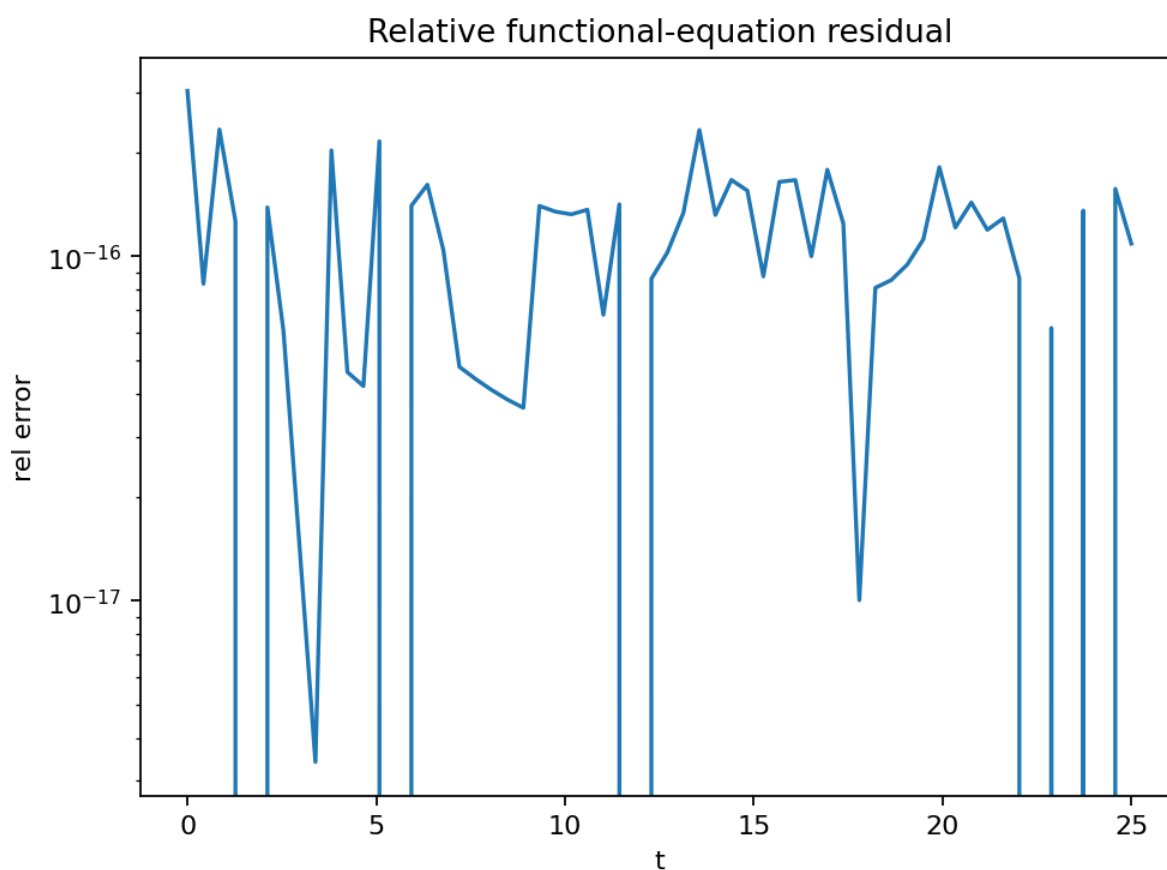


Fig. 37: E117 — Prime-counting approximations:  $\text{li}(x)$  and friends

**Tags:** number-theory, model-checking, visualization, li-x, pnt, bounds

## Highlights

- Focused numeric experiment with a small set of figures.
- Parameters saved to `params.json` for reproducibility.
- Defaults are chosen, so the experiment remains feasible for the CI “slow” suite.

### What is computed

- Compare  $\psi(x)$  to  $\text{li}(x)$  and related approximations and visualize the signed error.

### Notes

- This page summarizes the intent; see the generated `report.md` in `out/` for concrete outputs.

### References

- See the relevant references in `refs.bib`.

### Related experiments

- *E123 —  $(x;q,a)$  vs. a simple baseline* (E123 —  $(x;q,a)$  vs. a simple baseline)
- *E061: Chebyshev  $\psi(x)$  and prime powers* (Chebyshev  $\psi(x)$  and prime powers)
- *E116 — Gram points and zero-counting heuristics* (E116 — Gram points and zero-counting heuristics)
- *E058: Divisor-count record highs* (Divisor-count record highs)
- *E119 — Summatory totient  $\Phi(x)$  scaling check* (E119 — Summatory totient  $\Phi(x)$  scaling check)

### 3.3.118 E118 — Chebyshev bias: lead-time statistics

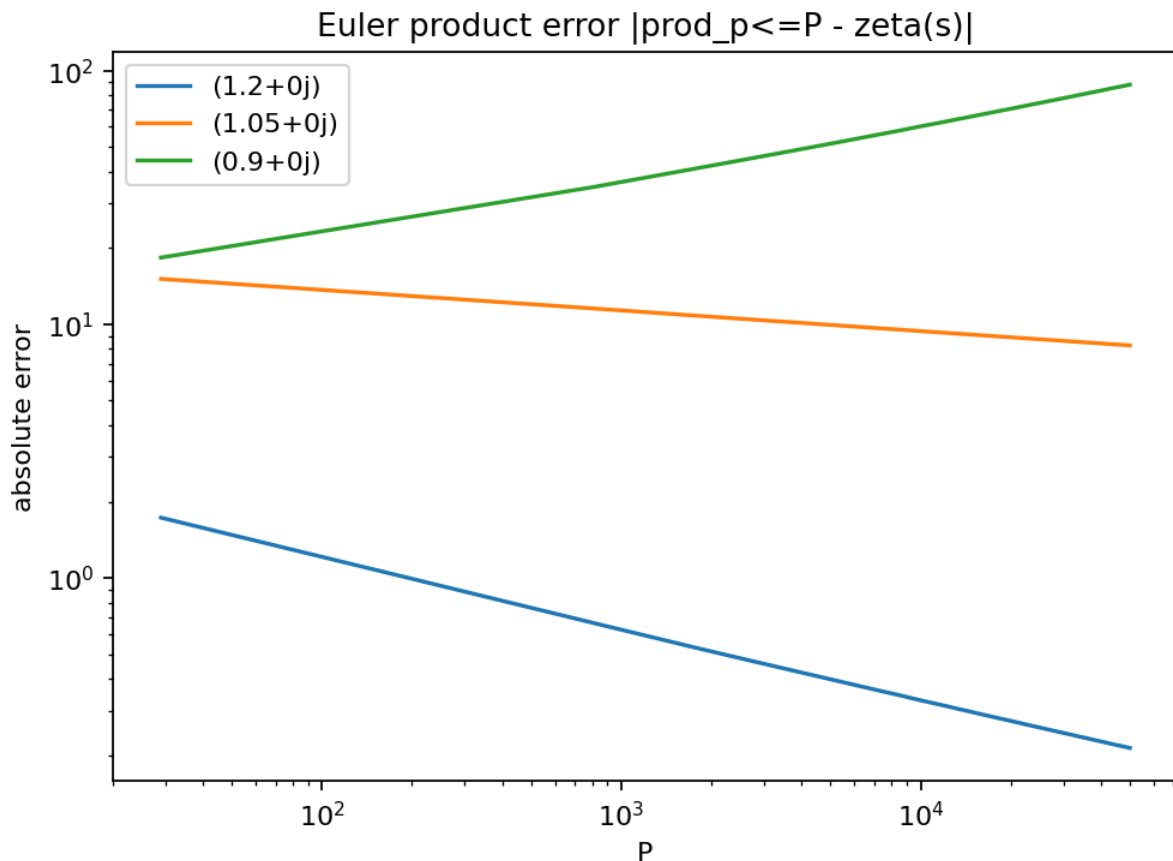


Fig. 38: E118 — Chebyshev bias: lead-time statistics

Tags: number-theory, prime-races, quantitative-exploration, visualization, heuristics

### Highlights

- Focused numeric experiment with a small set of figures.
- Parameters saved to `params.json` for reproducibility.
- Defaults are chosen, so the experiment remains feasible for the CI “slow” suite.

### What is computed

- Measure how often one residue class leads another and visualize lead-time statistics.

### Notes

- This page summarizes the intent; see the generated `report.md` in `out/` for concrete outputs.

### References

- See `docs/background/primes-in-arithmetic-progressions.md` and `docs/background/prime-number-races.md`.

### Related experiments

- *E080: Chebyshev bias: leader fraction vs.  $x$ .* (Chebyshev bias: leader fraction vs.  $x$ .)
- *E057: Erdős–Kac in practice* (Erdős–Kac in practice)
- *E076: Chebyshev  $(x;q,a)$ : weighted prime counts in progressions.* (Chebyshev  $(x;q,a)$ : weighted prime counts in progressions.)
- *E061: Chebyshev  $(x)$  and prime powers* (Chebyshev  $(x)$  and prime powers)
- *E103 — Chebyshev  $(x)$ : prime powers drive jumps* (E103 — Chebyshev  $(x)$ : prime powers drive jumps)

## 3.3.119 E119 — Summatory totient $\Phi(x)$ scaling check

Tags: number-theory, quantitative-exploration, visualization, totient, summatory, bounds

### Highlights

- Focused numeric experiment with a small set of figures.
- Parameters saved to `params.json` for reproducibility.
- Defaults are chosen, so the experiment remains feasible for the CI “slow” suite.

### What is computed

- Plot  $\Phi(x) = \sum_{n \leq x} \phi(n)$  and compare to the main-term scaling  $3x^2/2$ .

### Notes

- This page summarizes the intent; see the generated `report.md` in `out/` for concrete outputs.

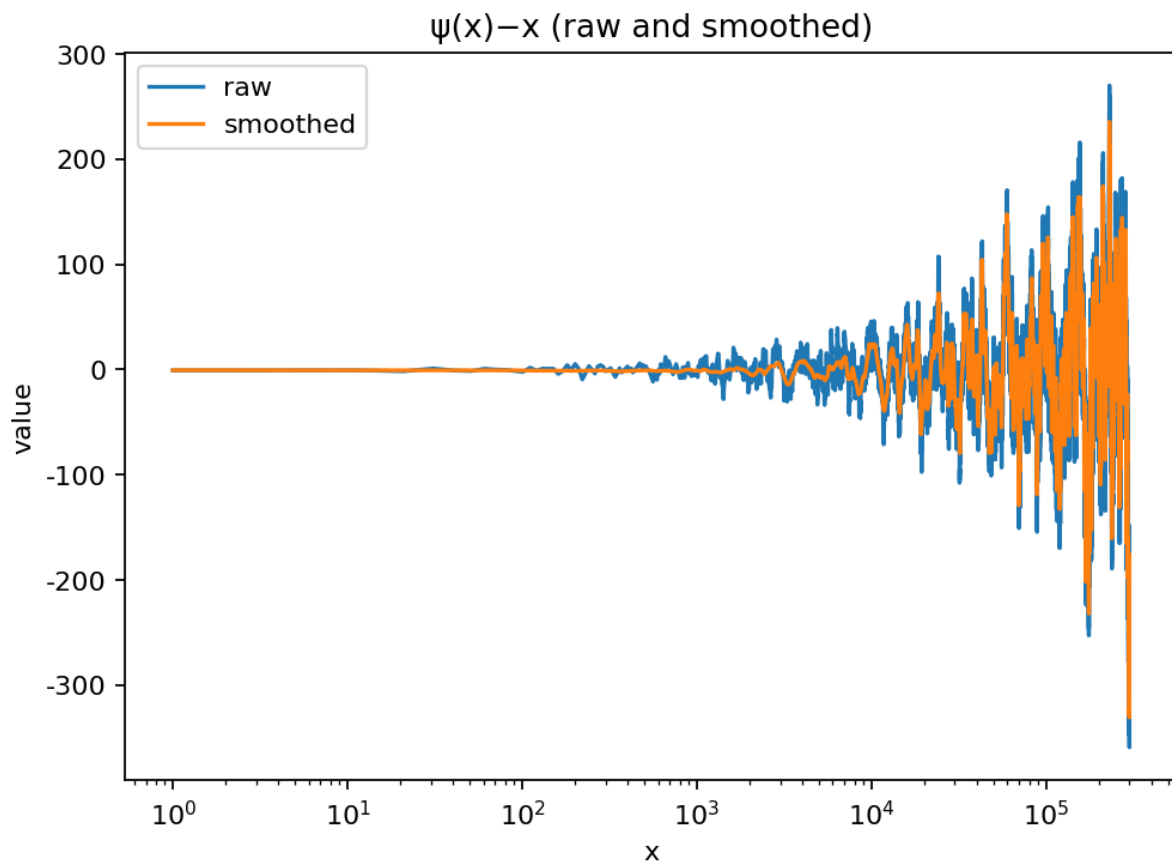
### References

- See the arithmetic-functions background pages in `docs/background/`.

### Related experiments

- *E105 — Mertens  $M(x)$ : scaling views* (E105 — Mertens  $M(x)$ : scaling views)
- *E052: Totient ratio landscape* (Totient ratio landscape)
- *E053: Inverse totient multiplicities* (Inverse totient multiplicities)



Fig. 39: E119 — Summatory totient  $\Phi(x)$  scaling check

- *E058: Divisor-count record highs* (Divisor-count record highs)
- *E117 — Prime-counting approximations:  $li(x)$  and friends* (E117 — Prime-counting approximations:  $li(x)$  and friends)

### 3.3.120 E120 — Liouville (n): partial sums and parity

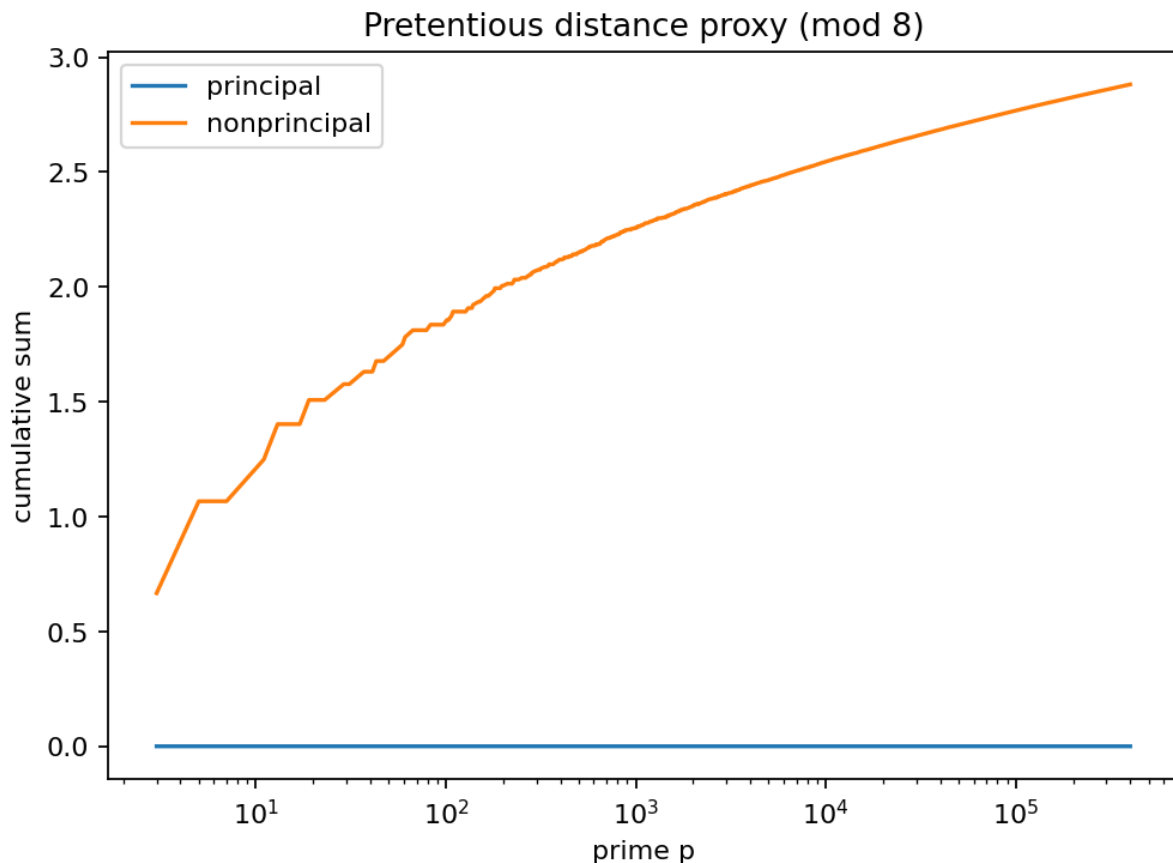


Fig. 40: E120 — Liouville (n): partial sums and parity

Tags: number-theory, quantitative-exploration, visualization, liouville, omega

#### Highlights

- Focused numeric experiment with a small set of figures.
- Parameters saved to `params.json` for reproducibility.
- Defaults are chosen, so the experiment remains feasible for the CI “slow” suite.

#### What is computed

- Explore partial sums of the Liouville function and simple parity correlations.

#### Notes

- This page summarizes the intent; see the generated `report.md` in `out/` for concrete outputs.

## References

- See the arithmetic-functions background pages in docs/background/.

## Related experiments

- *E056: Liouville vs. Möbius walks* (Liouville vs. Möbius walks)
- *E066: Character partial sums: cancellation profiles.* (Character partial sums: cancellation profiles.)
- *E078: Max partial sums across characters.* (Max partial sums across characters.)
- *E110 — Dirichlet L-series partial sums at  $s=1$  and  $s=1/2$*  (E110 — Dirichlet L-series partial sums at  $s=1$  and  $s=1/2$ )
- *E057: Erdős–Kac in practice* (Erdős–Kac in practice)

### 3.3.121 E121 — Möbius inversion as convolution undo

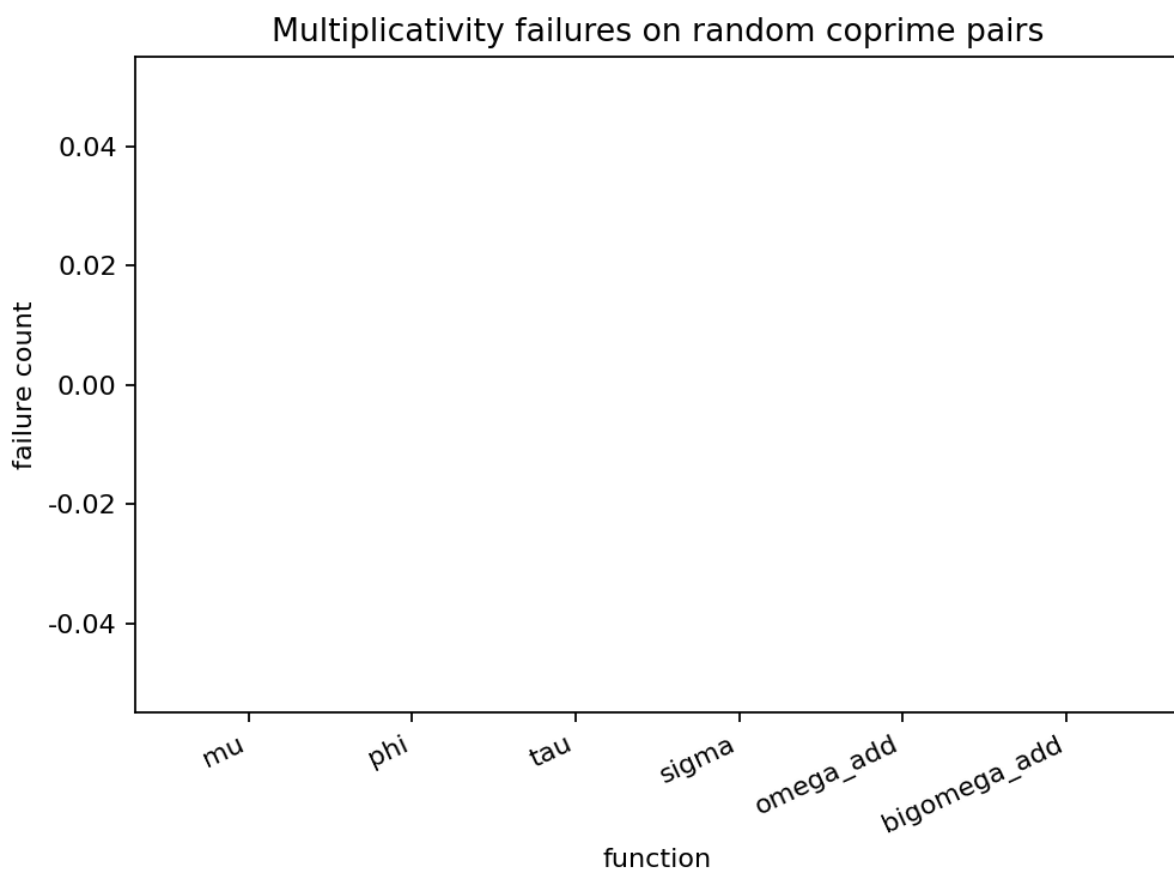


Fig. 41: E121 — Möbius inversion as convolution undo

**Tags:** number-theory, model-checking, visualization, dirichlet-convolution, mobius, multiplicative

## Highlights

- Focused numeric experiment with a small set of figures.
- Parameters saved to `params.json` for reproducibility.
- Defaults are chosen, so the experiment remains feasible for the CI “slow” suite.

### What is computed

- Recover a function from its Dirichlet convolution with 1 via Möbius inversion.

### Notes

- This page summarizes the intent; see the generated `report.md` in `out/` for concrete outputs.

### References

- See the arithmetic-functions background pages in `docs/background/`.

### Related experiments

- *E102 — Dirichlet convolution identity zoo* (E102 — Dirichlet convolution identity zoo)
- *E063: Dirichlet convolution playground* (Dirichlet convolution playground)
- *E054: Squarefree density via Möbius* (Squarefree density via Möbius)
- *E056: Liouville vs. Möbius walks* (Liouville vs. Möbius walks)
- *E092 —  $1/(s)$  via the Möbius Dirichlet series* (E092 —  $1/(s)$  via the Möbius Dirichlet series)

## 3.3.122 E122 — Character averages over primes

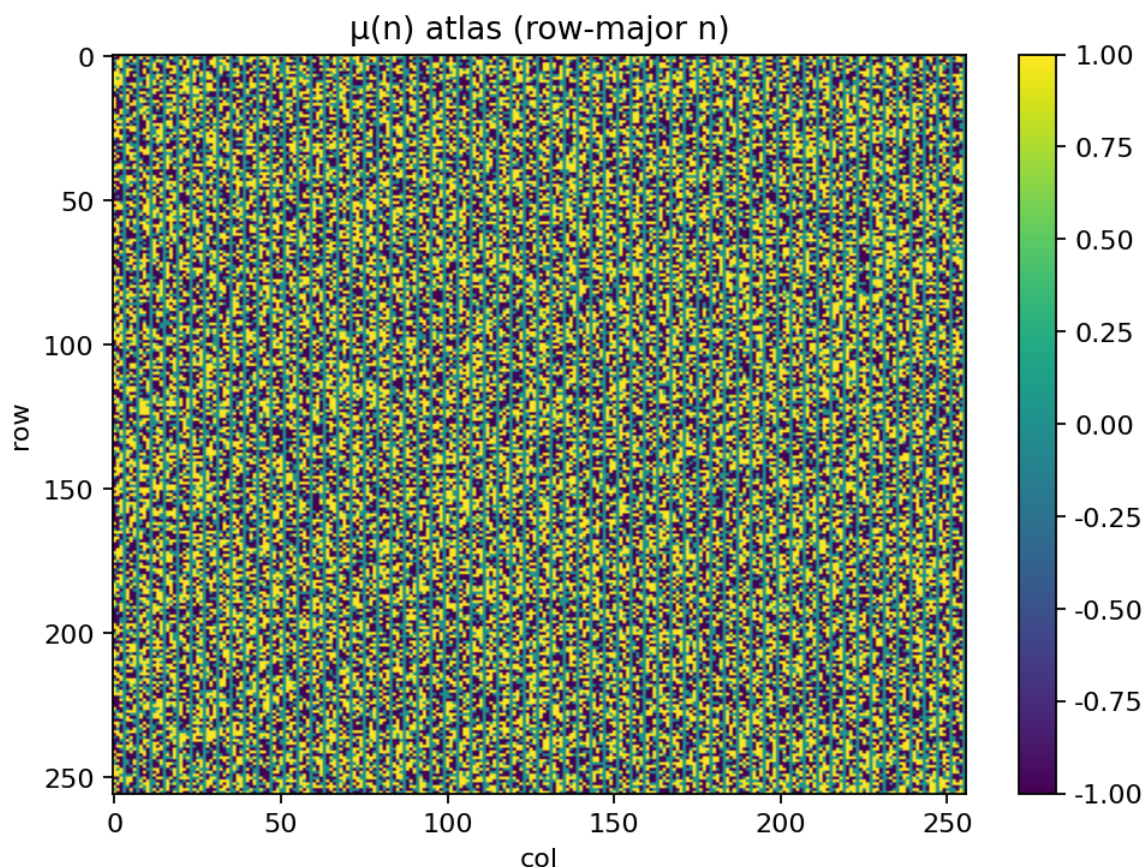


Fig. 42: E122 — Character averages over primes

**Tags:** number-theory, dirichlet-characters, l-functions, quantitative-exploration, visualization, aps

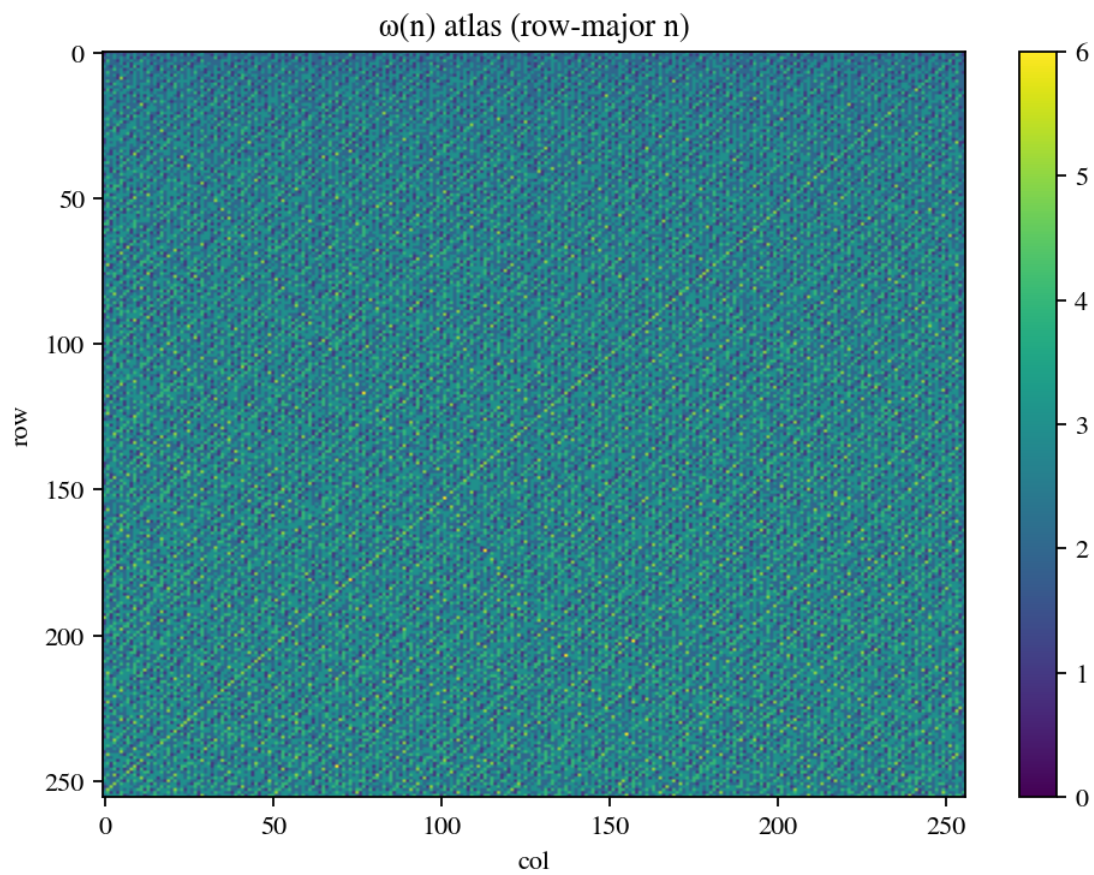


Fig. 43: E122 — Character averages over primes



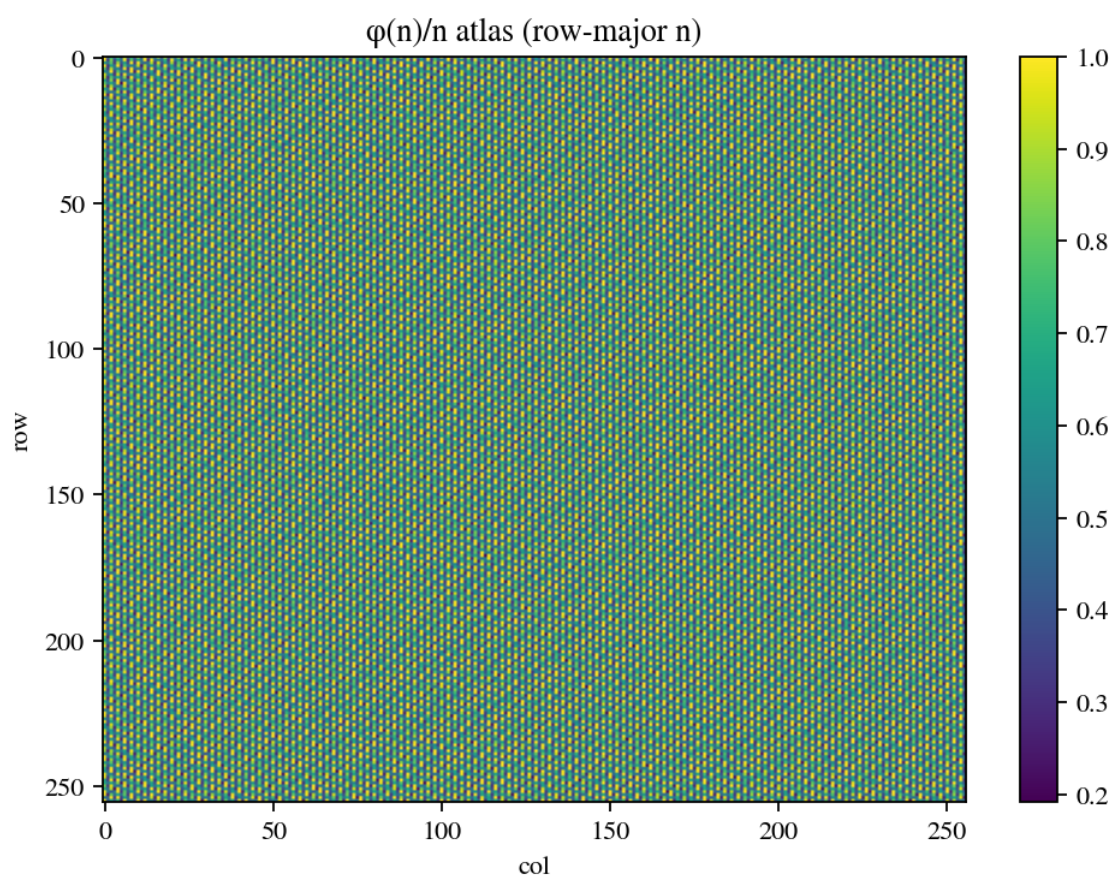


Fig. 44: E122 — Character averages over primes

### Highlights

- Focused numeric experiment with a small set of figures.
- Parameters saved to `params.json` for reproducibility.
- Defaults are chosen, so the experiment remains feasible for the CI “slow” suite.

### What is computed

- Average  $\langle p \rangle$  over primes in arithmetic progressions and compared to equidistribution heuristics.

### Notes

- This page summarizes the intent; see the generated `report.md` in `out/` for concrete outputs.

### References

- See `docs/background/dirichlet-characters.md`.
- See `docs/background/dirichlet-l-functions.md` and `docs/background/dirichlet-convolution.md`.
- See `docs/background/primes-in-arithmetic-progressions.md` and `docs/background/prime-number-races.md`.

### Related experiments

- *E068: Dirichlet  $L(s, \chi)$ : series vs. Euler product (partial approximations)*. (Dirichlet  $L(s, \chi)$ : series vs. Euler product (partial approximations).)
- *E069:  $L(1, \chi)$ : slow convergence and smoothing*. ( $L(1, \chi)$ : slow convergence and smoothing.)
- *E107 — Conductor: primitive vs. induced characters* (E107 — Conductor: primitive vs. induced characters)
- *E064: Dirichlet character tables (phase view)*. (Dirichlet character tables (phase view).)
- *E066: Character partial sums: cancellation profiles*. (Character partial sums: cancellation profiles.)

## 3.3.123 E123 — $(x;q,a)$ vs. a simple baseline

**Tags:** number-theory, prime-races, model-checking, visualization, aps, bounds

### Highlights

- Focused numeric experiment with a small set of figures.
- Parameters saved to `params.json` for reproducibility.
- Defaults are chosen, so the experiment remains feasible for the CI “slow” suite.

### What is computed

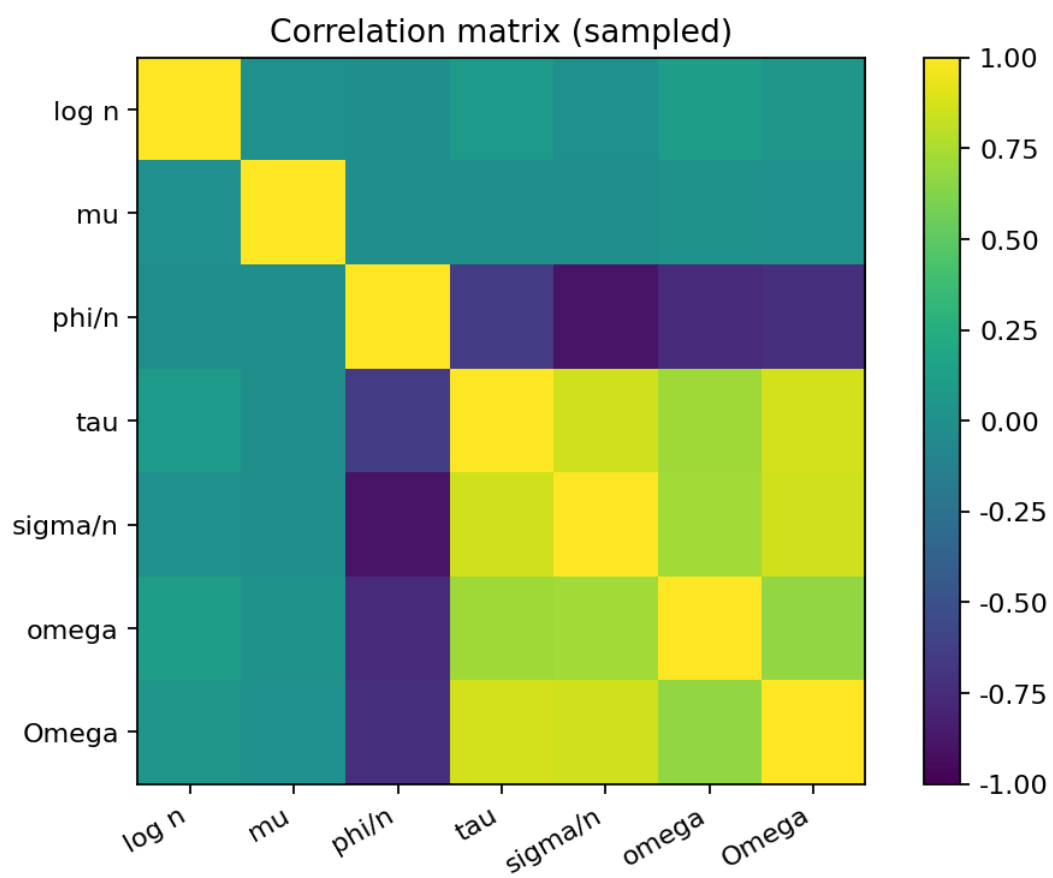
- Compare  $(x;q,a)$  to a baseline  $x/(q \log x)$  and visualize deviations.

### Notes

- This page summarizes the intent; see the generated `report.md` in `out/` for concrete outputs.

### References

- See `docs/background/primes-in-arithmetic-progressions.md` and `docs/background/prime-number-races.md`.

Fig. 45: E123 —  $(x;q,a)$  vs. a simple baseline

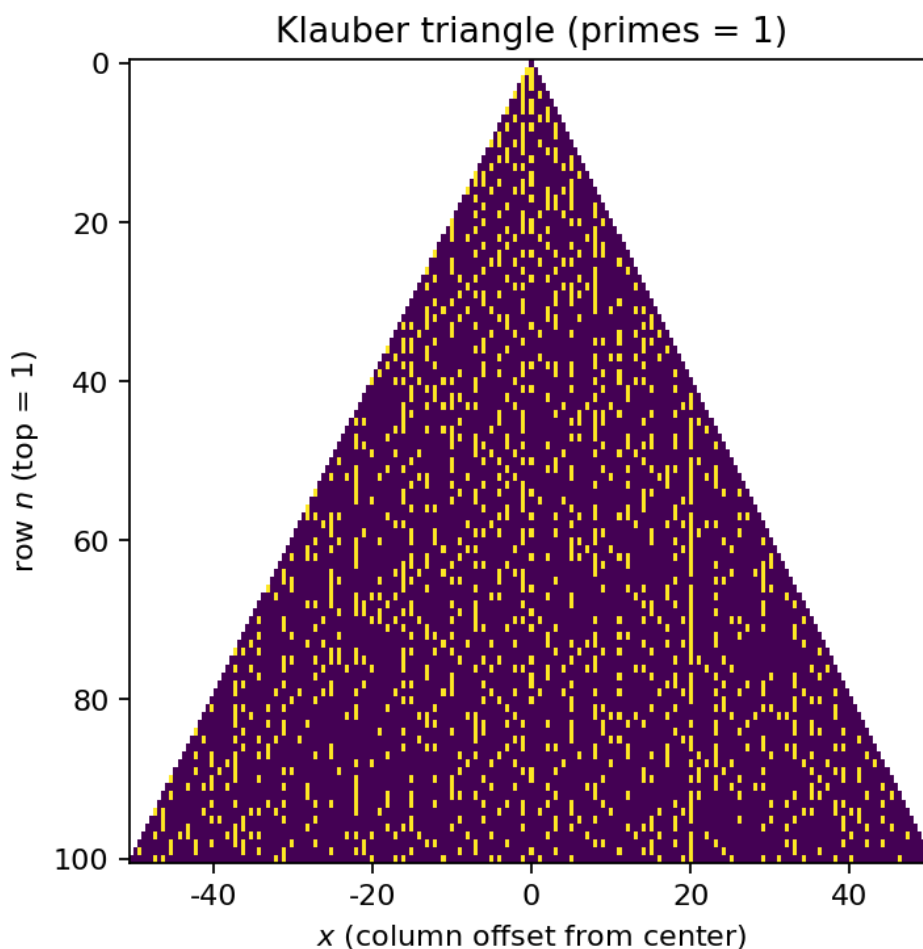


### Related experiments

- *E117 — Prime-counting approximations:  $li(x)$  and friends* (E117 — Prime-counting approximations:  $li(x)$  and friends)
- *E072: Prime race mod 4:  $\pi(x;4,3)$  vs.  $\pi(x;4,1)$ .* (Prime race mod 4:  $\pi(x;4,3)$  vs.  $\pi(x;4,1)$ .)
- *E073: Prime race mod 3:  $\pi(x;3,2)$  vs.  $\pi(x;3,1)$ .* (Prime race mod 3:  $\pi(x;3,2)$  vs.  $\pi(x;3,1)$ .)
- *E074: Prime race mod 8: leaderboard among 1,3,5,7.* (Prime race mod 8: leaderboard among 1,3,5,7.)
- *E075: Prime race statistic: distribution on a log-grid.* (Prime race statistic: distribution on a log-grid.)

### 3.3.124 E124: Klauber triangle structure

size 101: 10,201 numbers and 1,252 primes



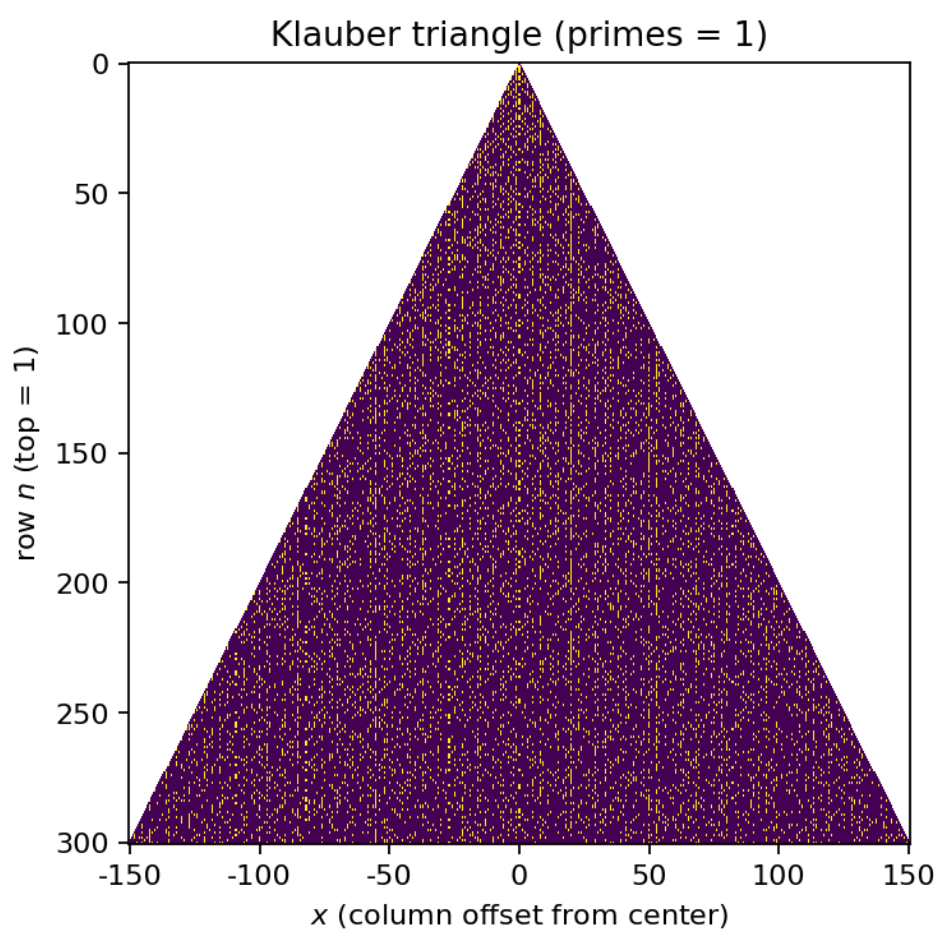
size 301: 90,601 numbers and 8,769 primes

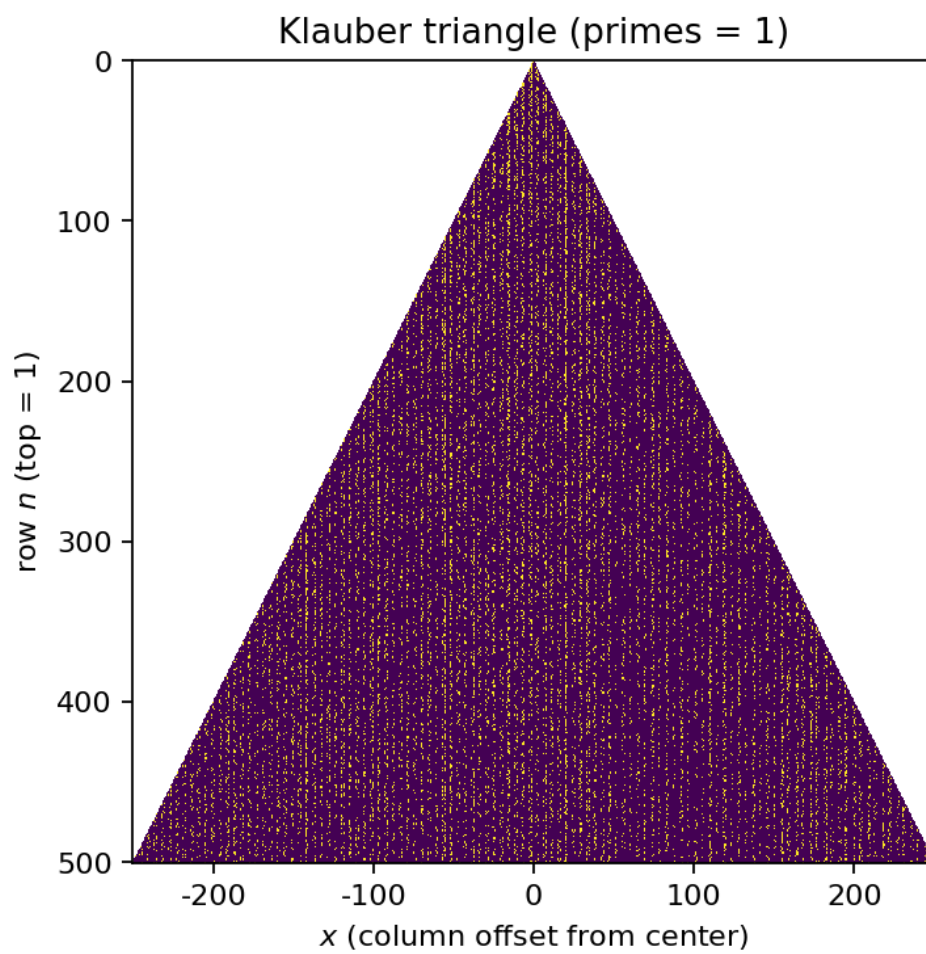
size 501: 251,001 numbers and 22,115 primes

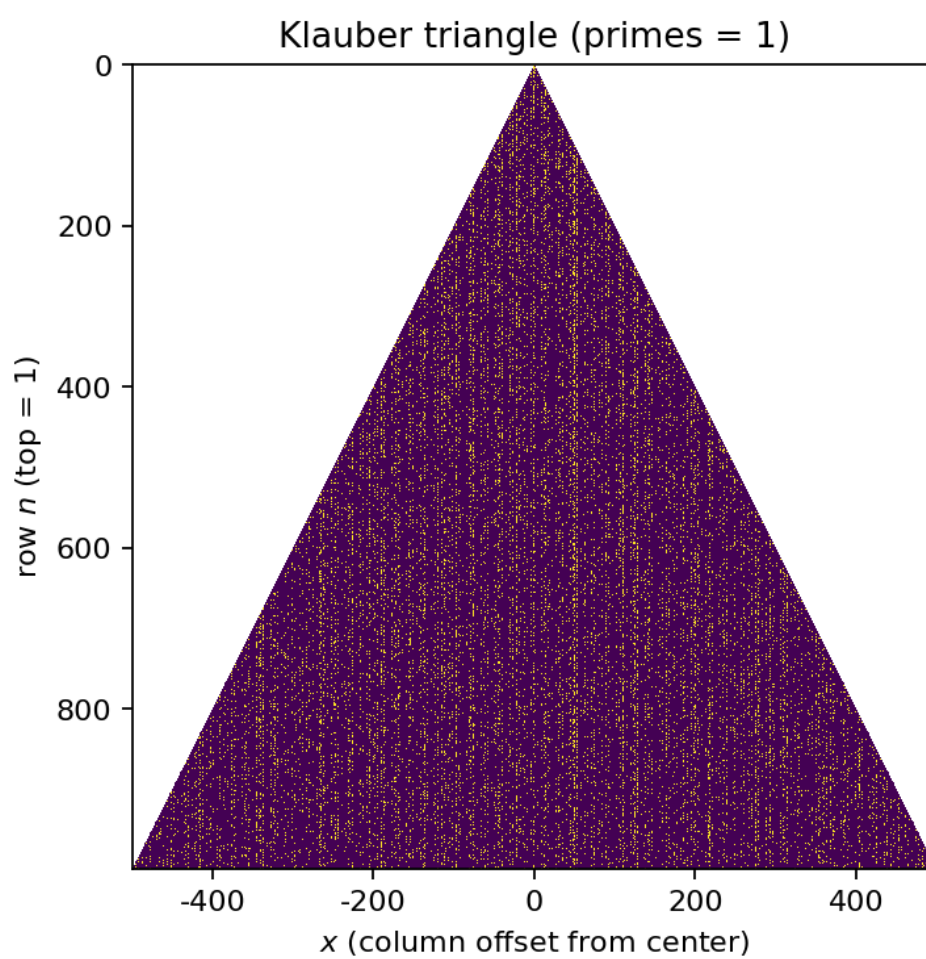
size 999: 998,001 numbers and 78,359 primes

**Tags:** number-theory, quantitative-exploration, visualization

See: *Valid Tags*.







## Highlights

- A triangular “prime map” where row  $m$  contains the integers from  $(m - 1)^2 + 1$  to  $m^2$ .
- Parameterizable run: vary `--size` from the command line (default: 301).
- Writes reproducible artifacts (`params.json`, `report.md`, and figures) into `out/e124/`.

## Goal

Render the **Klauber triangle** (a square-number-based triangular arrangement) and highlight primes to reveal linear “streaks” and other geometric structures, then compare how that structure changes as `size` increases.

## Background (quick refresher)

### What is the Klauber triangle?

The **Klauber triangle** arranges the natural numbers into rows indexed by  $m = 1, 2, 3, \dots$  such that row  $m$  contains the consecutive integers

$$(m - 1)^2 + 1, (m - 1)^2 + 2, \dots, m^2.$$

Each row has  $2m - 1$  entries, and the triangle up to row `size` contains the integers  $1..size^2$ . When primes are highlighted, surprisingly strong visual streaks can appear. See [Hill, 2016].

## Optional background pages

- *Prime numbers refresher*

## Research question

Which prime-rich streaks are visible in the Klauber triangle for a given `size`, and how do those structures change as `size` increases?

## Why this qualifies as a mathematical experiment

- **Finite procedure:** build a finite triangle window containing integers  $1..size^2$  and run a deterministic primality test.
- **Observable(s):** streaks/lines, local clustering, and how visibility changes with `size`.
- **Parameter space:** `size` (and, if you extend it, rendering/annotation choices).
- **Outcome:** figures and a short report capturing the key observation and caveats.
- **Reproducibility:** outputs saved to `out/e124/` with a parameter snapshot.

## Experiment design

- **Computation:** place integers  $1..size^2$  into the Klauber triangle row rule; mark primes.
- **Outputs:** one or more figures showing prime positions in the triangle.
- **Artifacts written:**
  - `figures/fig_*.png`
  - `params.json`
  - `report.md`

## How to run

```
make run EXP=e124
```

Override the triangle size:

```
make run EXP=e124 ARGS="--size 501"
```

Direct invocation (always works):

```
uv run --extra dev python -m mathxlab.experiments.e124 --out out/e124 --size 501
```

## Notes / pitfalls

- Larger `size` means more numbers ( $\text{size}^2$ ) and a longer run time.
- “Looks-true” trap: visual regularity suggests hypotheses but does not prove theorems about primes.
- Rendering choices (marker size, interpolation) can hide or exaggerate streaks.

## Extensions

- Quantify streaks: compute prime density along selected triangle-aligned lines and plot the densities.
- Highlight the square boundaries ( $m^2$ ) or row indices to relate features to the row rule.
- Compare multiple sizes side-by-side (same styling and axis conventions).

## References

- Klauber triangle definition and examples: [Hill, 2016].
- Exposition connecting multiple prime visualizations (includes Klauber and Sacks): [Brockmann, 2019].
- Recreational-math context and “pattern traps”: [Gardner, 1983], [Hoffman, 1989].

See also *References*.

## Related experiments

- *E002: Even Perfect Numbers — Generator and Growth* (Even Perfect Numbers — Generator and Growth)
- *E003: Abundancy Index Landscape* (Abundancy Index Landscape)
- *E024: Ulam spiral structure* (Ulam spiral structure)
- *E125: Sacks spiral structure* (Sacks spiral structure)
- *E126: Hexagonal number spiral structure* (Hexagonal number spiral structure)

## Parameters (example)

- `size`: 301 (implied total integers: 1..90601)

## Recommended size range

The Klauber triangle up to row `size` contains  $\text{size}^2$  integers, so runtime and memory scale roughly with  $\text{size}^2$ .

A practical range that works well for most runs:

- **Minimum (still meaningful):** `size` = 101
- **Default / recommended:** `size` = 301

- **Comfortable upper range (typical laptop):** `size = 801` or `1001`
- **Above `~size = 1501`:** expect noticeably higher runtime (and potentially memory pressure), depending on the primality implementation.

**Rule of thumb:** start with 301, then try 501, 801, 1001. Increase further only if runtime remains acceptable.

### Summary

We render the Klauber triangle containing the first 90,601 positive integers and highlight primes. Even at this moderate scale, visually prominent streaks appear: some triangle-aligned lines contain noticeably more primes than nearby lines.

### Key observations

- Several streaks stand out strongly against a background that still looks relatively “noise-like”.
- The most visible structures shift as `size` increases; some streaks weaken while others become clearer.

### Interpretation

The Klauber triangle is a structured re-indexing of the integers where many visually straight features correspond to simple polynomial or arithmetic progressions in the underlying index. This makes it a useful “pattern detector” for prime-rich sequences, but it is not evidence of a global rule.

### Caveats

- Finite window: patterns can look stronger/weaker depending on the chosen `size` and zoom level.
- Visualization choices matter: marker size and interpolation can hide or exaggerate streaks.
- Different triangle conventions (if you change the row rule) change which streaks appear.

## 3.3.125 E125: Sacks spiral structure

**size 101:** 10,201 numbers and 1,252 primes

**size 301:** 90,601 numbers and 8,769 primes

**size 501:** 251,001 numbers and 22,115 primes

**size 999:** 998,001 numbers and 78,359 primes

**Tags:** number-theory, quantitative-exploration, visualization

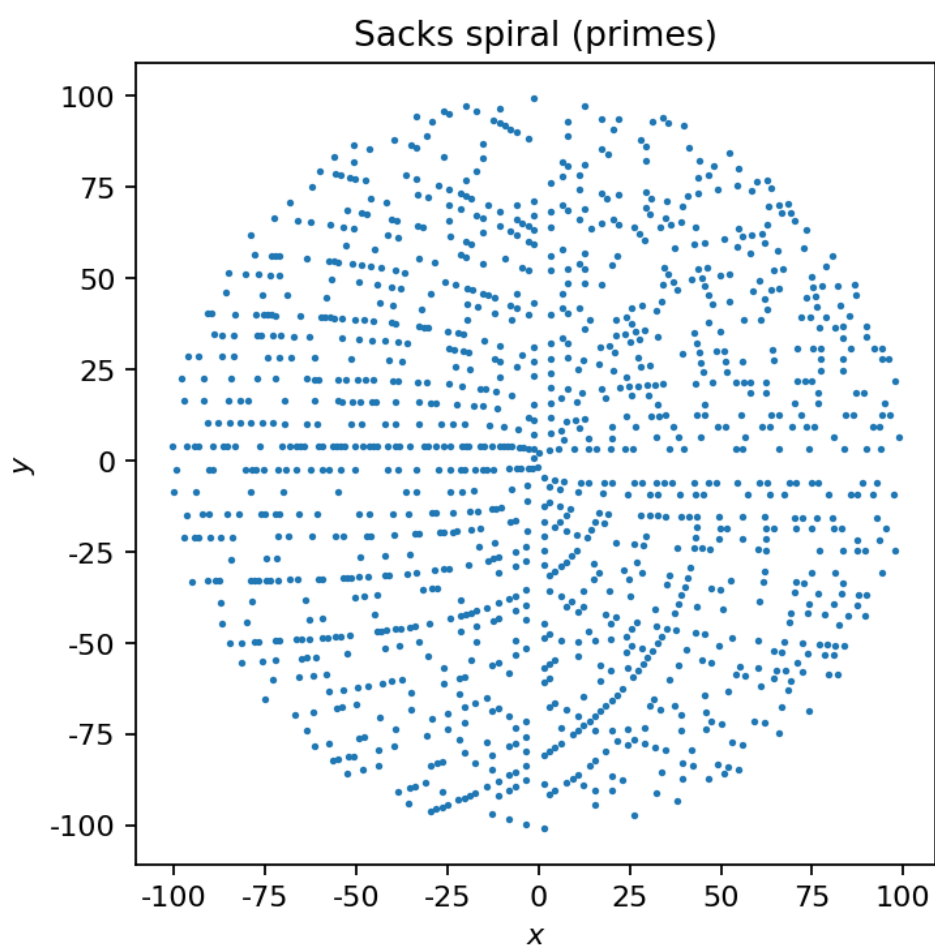
See: *Valid Tags*.

### Highlights

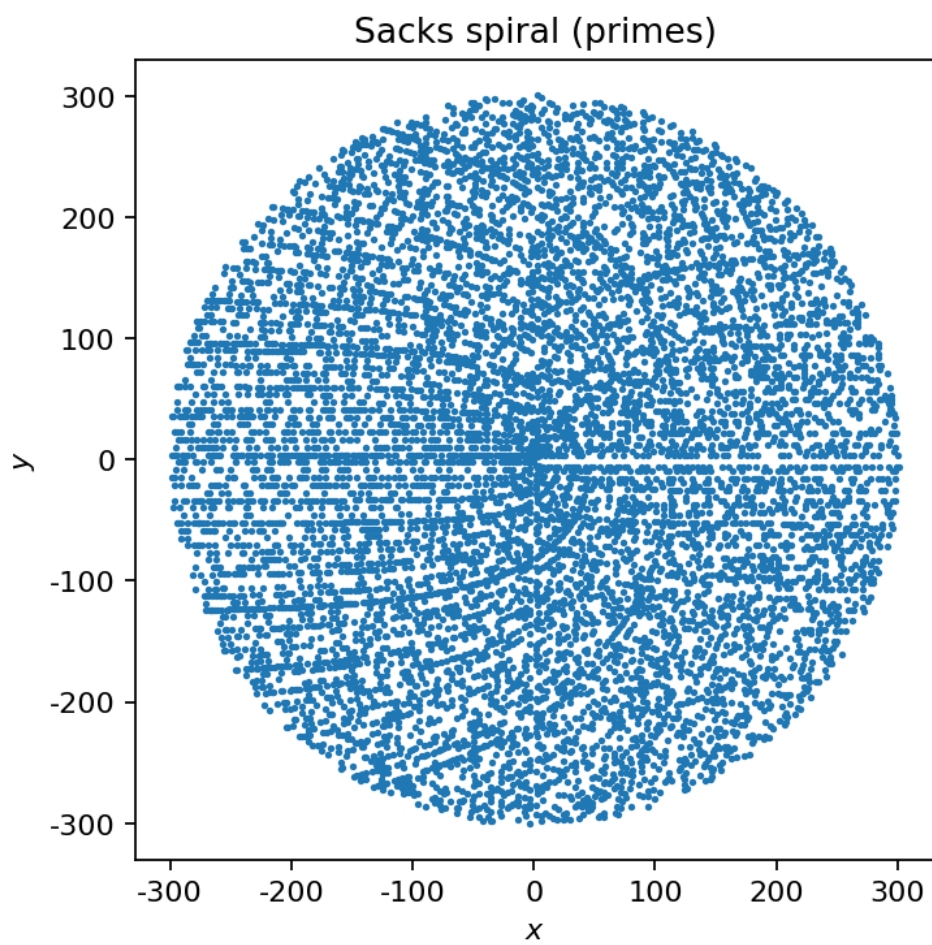
- Places integers on a spiral using polar coordinates  $r = \sqrt{n}$  and  $\theta = 2\pi\sqrt{n}$ .
- Parameterizable run: vary `--size` from the command line (default: 301).
- Writes reproducible artifacts (`params.json`, `report.md`, and figures) into `out/e125/`.

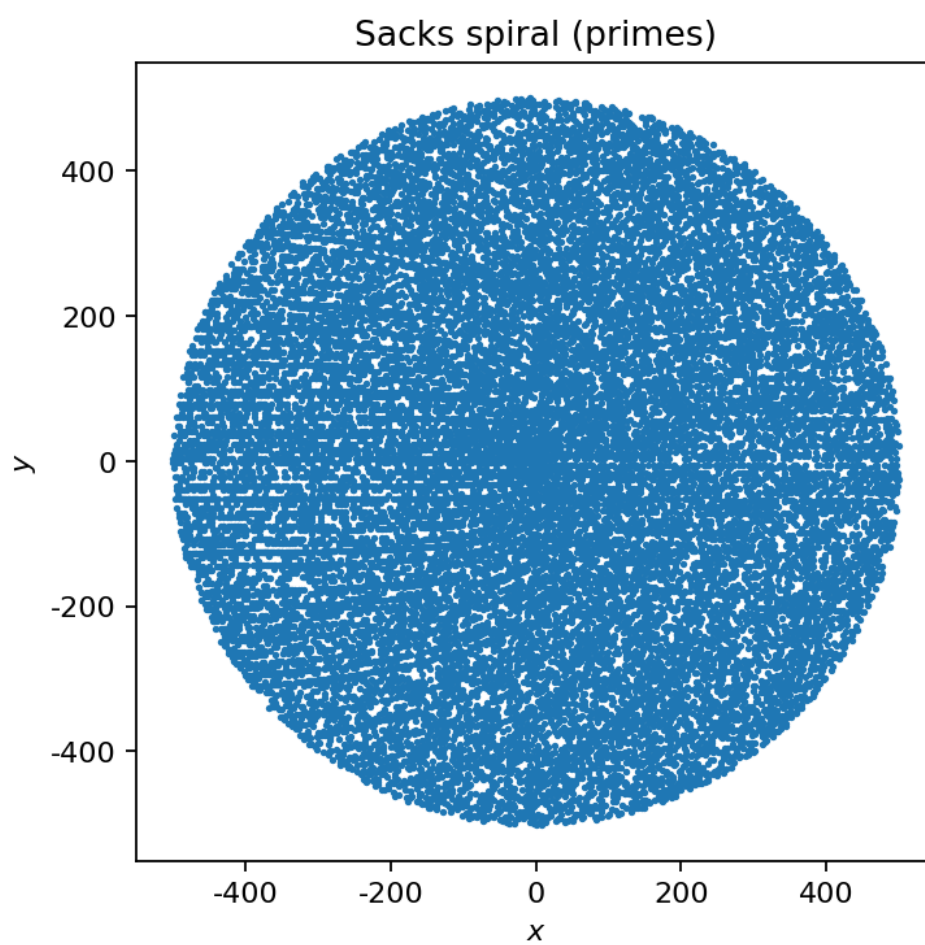
### Goal

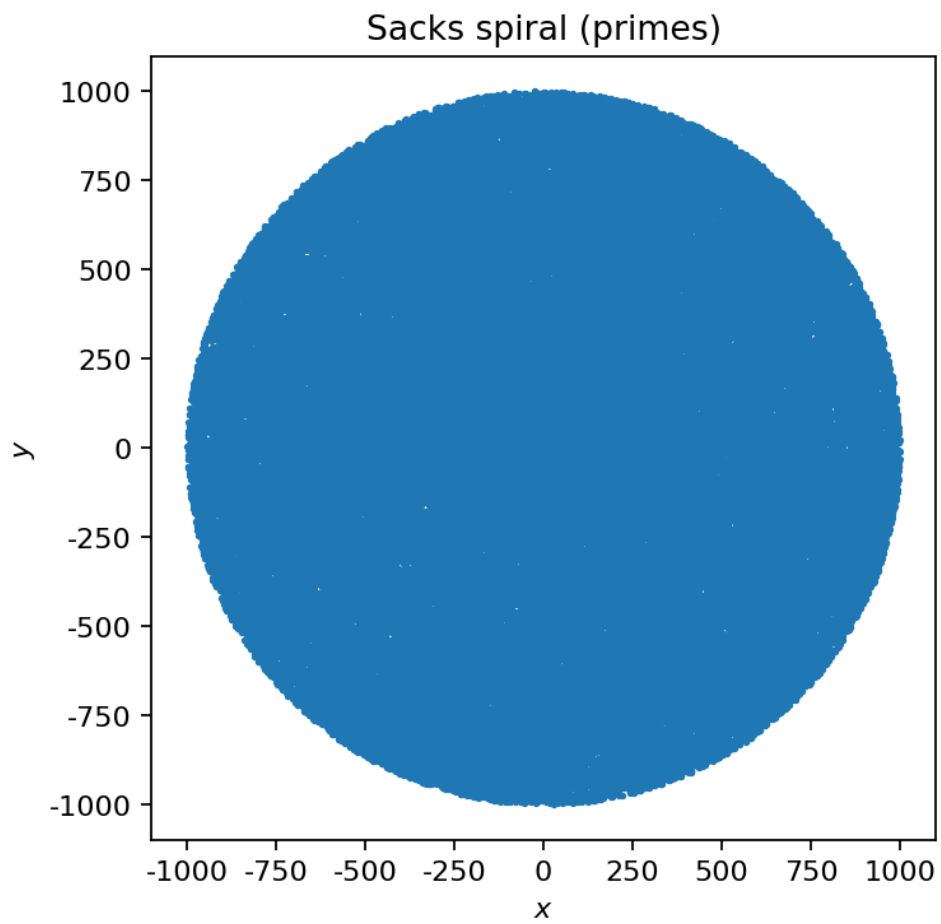
Render the **Sacks spiral** and highlight primes as a point cloud to reveal spiral-aligned structure, then compare how the apparent density and “spiral fragments” change as `size` increases.











## Background (quick refresher)

### What is the Sacks spiral?

In the **Sacks spiral**, each integer  $n$  is mapped to polar coordinates

$$r_n = \sqrt{n}, \quad \theta_n = 2\pi\sqrt{n},$$

and plotted as  $(x, y) = (r \cos \theta, r \sin \theta)$ . Highlighting primes yields striking spiral fragments and density variations. See [Brockmann, 2019].

### Optional background pages

- *Prime numbers refresher*

### Research question

How does the visual “prime spiral density” change as **size** increases (i.e., as we include integers up to  $\text{size}^2$ ), and which spiral-aligned structures persist across scales?

### Why this qualifies as a mathematical experiment

- **Finite procedure:** compute primes up to  $\text{size}^2$  and apply a deterministic mapping  $n \mapsto (x, y)$ .
- **Observable(s):** spiral fragments, density contrasts, and how visibility changes with **size**.
- **Parameter space:** **size** (and, if you extend it, marker size / rendering choices).
- **Outcome:** figures and a short report capturing the key observation and caveats.
- **Reproducibility:** outputs saved to `out/e125/` with a parameter snapshot.

### Experiment design

- **Computation:** for integers  $1..\text{size}^2$ , compute a prime mask; map prime indices to  $(x, y)$  via the Sacks rule.
- **Outputs:** one or more scatter plots of prime points (composites omitted).
- **Artifacts written:**
  - `figures/fig_*.png`
  - `params.json`
  - `report.md`

### How to run

```
make run EXP=e125
```

Override the experiment size:

```
make run EXP=e125 ARGS="--size 501"
```

Direct invocation (always works):

```
uv run --extra dev python -m mathxlab.experiments.e125 --out out/e125 --size 501
```

### Notes / pitfalls

- Scatter plots are sensitive to marker size and resolution: the same data can look different depending on rendering.
- Larger **size** means more numbers ( $\text{size}^2$ ) and a longer run time.
- “Looks-true” trap: a visually strong spiral does not by itself imply a theorem about primes.

### Extensions

- Add a faint background of composite points (with transparency) to compare prime vs. composite geometry.
- Plot multiple sizes on identical axes to compare density and visible spiral fragments.
- Add simple quantitative summaries (e.g., radial density bins for primes vs. all numbers).

### References

- Exposition connecting multiple prime visualizations (includes Sacks spiral): [Brockmann, 2019].
- Recreational-math context and “pattern traps”: [Gardner, 1983], [Hoffman, 1989].

See also *References*.

### Related experiments

- *E002: Even Perfect Numbers — Generator and Growth* (Even Perfect Numbers — Generator and Growth)
- *E003: Abundancy Index Landscape* (Abundancy Index Landscape)
- *E024: Ulam spiral structure* (Ulam spiral structure)
- *E124: Klauber triangle structure* (Klauber triangle structure)
- *E126: Hexagonal number spiral structure* (Hexagonal number spiral structure)

### Parameters (example)

- size: 301 (implied total integers: 1..90601)

### Recommended size range

The Sacks spiral run includes integers  $1..size^2$ , so runtime and memory scale roughly with  $size^2$  (prime classification dominates).

A practical range that works well for most runs:

- **Minimum (still meaningful):** size = 101
- **Default / recommended:** size = 301
- **Comfortable upper range (typical laptop):** size = 801 or 1001
- **Above ~size = 1501:** expect noticeably higher runtime (and potentially memory pressure), depending on the primality implementation.

**Rule of thumb:** start with 301, then try 501, 801, 1001. Increase further only if runtime remains acceptable.

### Summary

We compute primes among the first 90,601 positive integers and plot prime indices using the Sacks polar mapping. Even at this moderate scale, visible spiral-aligned prime fragments emerge, with noticeable density variation across radii.

### Key observations

- Prime points form dense spiral segments rather than appearing uniformly “random” in the plane.
- The visibility of segments depends strongly on marker size and figure resolution.

### Interpretation

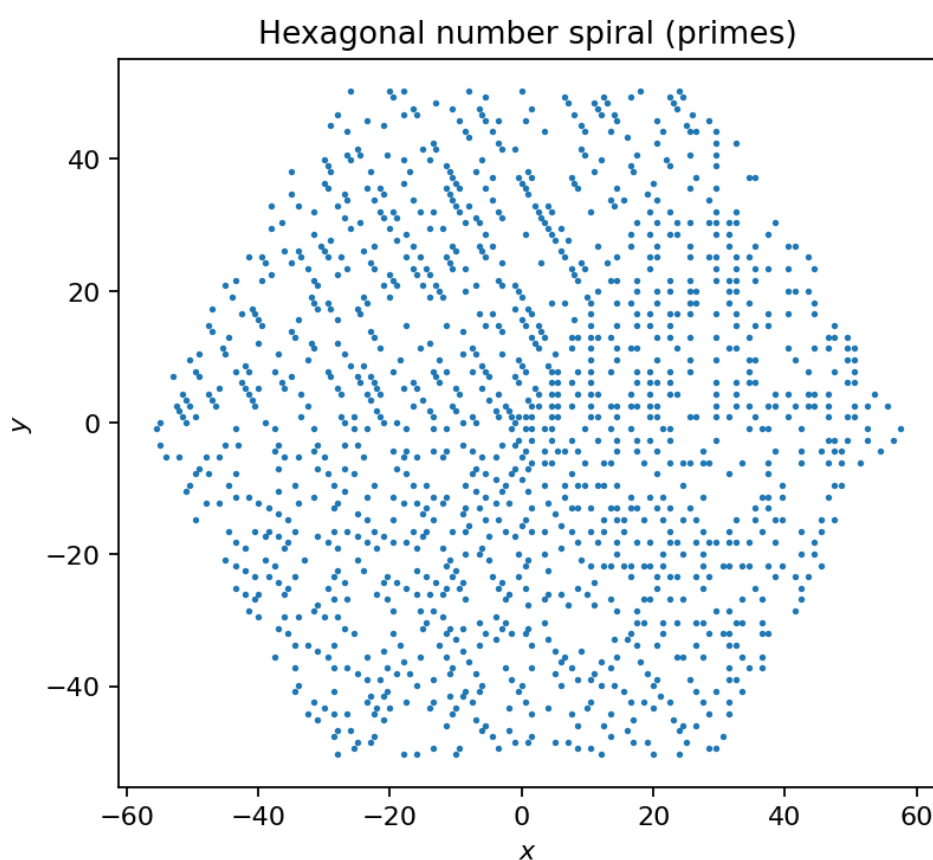
The Sacks mapping is a deterministic geometric transform of the integer line. It tends to visually emphasize arithmetic structure (including sequences with prime-rich behavior), making it a useful exploratory visualization tool — but not a proof technique.

### Caveats

- Rendering matters: marker size, DPI, and axis limits can change perceived structure.
- Finite window: patterns can strengthen or weaken as **size** changes.
- Without quantitative overlays, conclusions remain qualitative.

### 3.3.126 E126: Hexagonal number spiral structure

size 101: 10,201 numbers and 1,252 primes



size 301: 90,601 numbers and 8,769 primes

size 501: 251,001 numbers and 22,115 primes

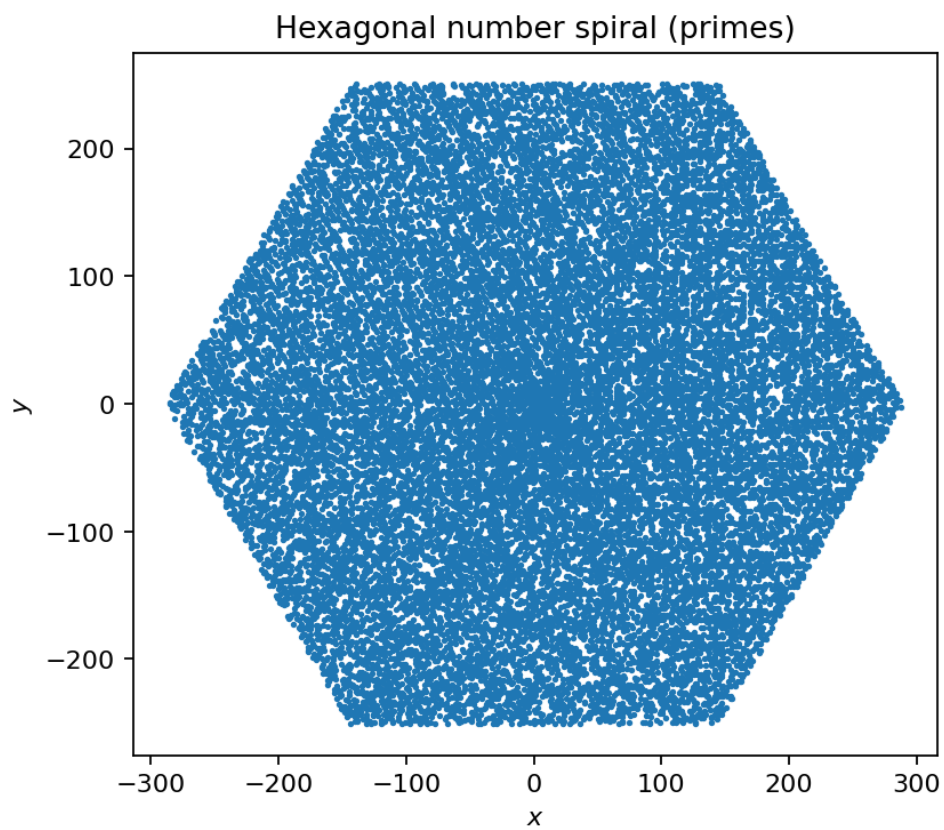
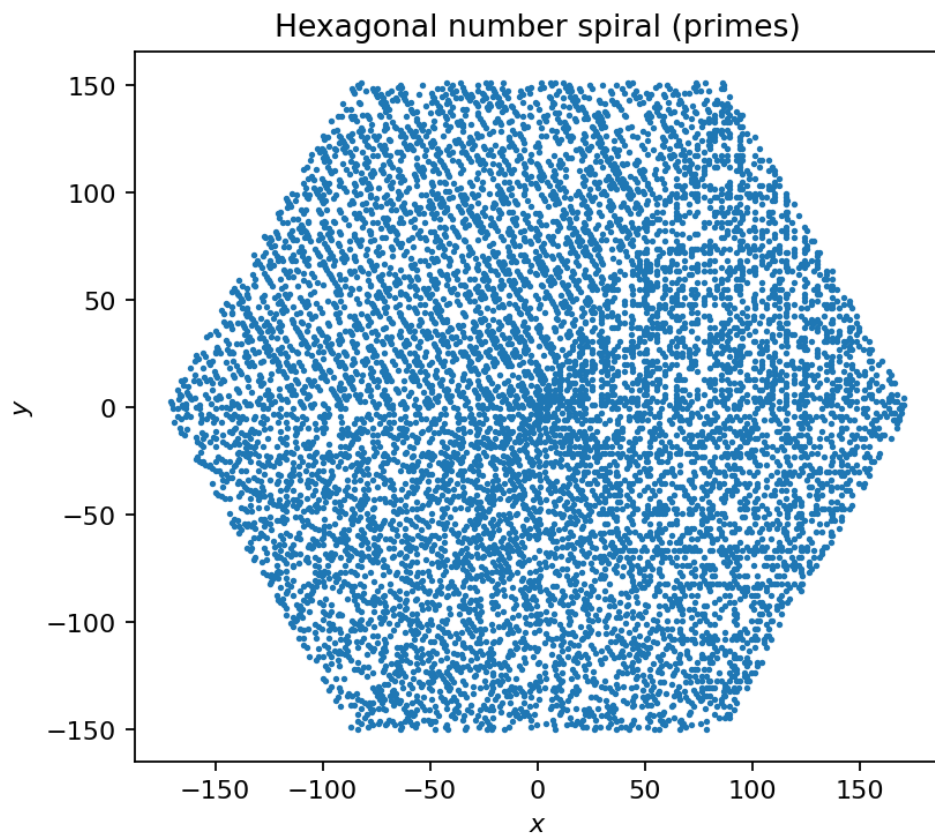
size 999: 998,001 numbers and 78,359 primes

**Tags:** number-theory, quantitative-exploration, visualization

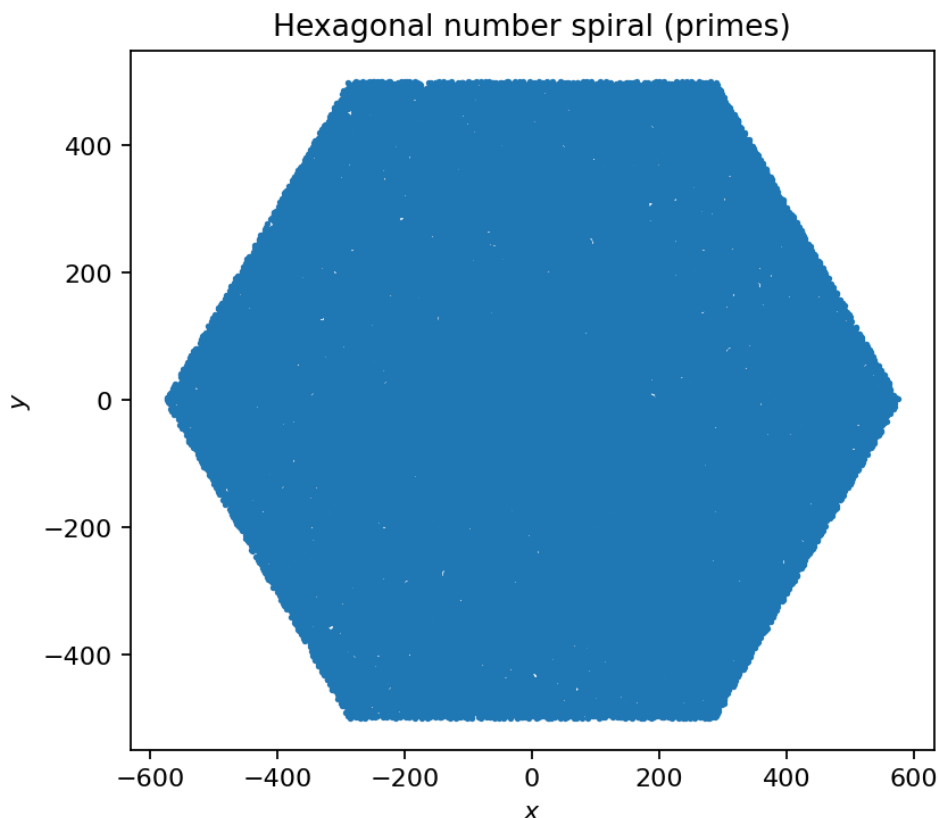
See: [Valid Tags](#).

### Highlights

- Arranges integers on a **hexagonal lattice** using a centered spiral enumeration.
- Parameterizable run: vary `--size` from the command line (default: 301).
- Writes reproducible artifacts (`params.json`, `report.md`, and figures) into `out/e126/`.







### Goal

Render a **hexagonal number spiral** and highlight primes to reveal lattice-aligned structures, then compare how that structure changes as **size** increases.

### Background (quick refresher)

#### What is a hexagonal number spiral?

A **hexagonal number spiral** places integers on a hexagonal grid, starting at the center with 1 and then spiraling outward. Compared to the square-lattice Ulam spiral, the hex lattice emphasizes different geometric alignments and may reveal different-looking prime structures. See [Kurzweg, 2020].

### Optional background pages

- *Prime numbers refresher*

### Research question

Which geometric structures appear when primes are highlighted on a hexagonal spiral, and how stable are they as **size** increases (i.e., as we include integers up to  $\text{size}^2$ )?

### Why this qualifies as a mathematical experiment

- **Finite procedure:** enumerate a finite spiral placement for integers  $1..\text{size}^2$  and run a deterministic primality test.
- **Observable(s):** lattice-aligned streaks/clusters and how visibility changes with **size**.
- **Parameter space:** **size** (and, if you extend it, coordinate convention / rendering choices).
- **Outcome:** figures and a short report capturing the key observation and caveats.
- **Reproducibility:** outputs saved to `out/e126/` with a parameter snapshot.



### Experiment design

- **Computation:** map integers  $1..size^2$  to hex-lattice coordinates using a centered spiral enumeration; mark primes.
- **Outputs:** one or more scatter plots of prime points on the hex lattice (composites omitted).
- **Artifacts written:**
  - figures/fig\_\*.png
  - params.json
  - report.md

### How to run

```
make run EXP=e126
```

Override the experiment size:

```
make run EXP=e126 ARGS="--size 501"
```

Direct invocation (always works):

```
uv run --extra dev python -m mathxlab.experiments.e126 --out out/e126 --size 501
```

### Notes / pitfalls

- Different conventions exist for hex-spiral enumeration; changing the convention can change visible structures.
- Larger `size` means more numbers ( $size^2$ ) and a longer run time.
- “Looks-true” trap: lattice-aligned streaks do not prove a statement about primes.

### Extensions

- Overlay ring boundaries (or distance-from-center contours) to relate features to the spiral geometry.
- Add simple quantitative overlays: density by hex-ring index or by lattice direction.
- Compare hex vs. square spirals at matched  $N$  (same number of integers).

### References

- Hex spiral background and construction ideas: [Kurzweg, 2020].
- Exposition connecting multiple prime visualizations: [Brockmann, 2019].
- Recreational-math context and “pattern traps”: [Gardner, 1983], [Hoffman, 1989].

See also *References*.

### Related experiments

- *E002: Even Perfect Numbers — Generator and Growth* (Even Perfect Numbers — Generator and Growth)
- *E003: Abundancy Index Landscape* (Abundancy Index Landscape)
- *E024: Ulam spiral structure* (Ulam spiral structure)
- *E124: Klauber triangle structure* (Klauber triangle structure)
- *E125: Sacks spiral structure* (Sacks spiral structure)

### Parameters (example)

- size: 301 (implied total integers: 1..90601)

### Recommended size range

The hex spiral run includes integers 1.. $\text{size}^2$ , so runtime and memory scale roughly with  $\text{size}^2$  (prime classification dominates).

A practical range that works well for most runs:

- **Minimum (still meaningful):** size = 101
- **Default / recommended:** size = 301
- **Comfortable upper range (typical laptop):** size = 801 or 1001
- **Above ~size = 1501:** expect noticeably higher runtime (and potentially memory pressure), depending on the primality implementation.

**Rule of thumb:** start with 301, then try 501, 801, 1001. Increase further only if runtime remains acceptable.

### Summary

We enumerate a hex-lattice spiral placement for the first 90,601 positive integers and highlight primes. At this scale, lattice-aligned structures can be visible: some directions show denser prime “streaks” than others.

### Key observations

- Prime points show directional structure aligned to the hex lattice.
- Some structures are subtle and become easier to see with larger **size** and careful marker sizing.

### Interpretation

The hex spiral is an alternative geometry for indexing the same integers. As with the Ulam spiral, geometric placement can visually emphasize prime-rich progressions or polynomial-like sequences. This is a strong exploratory tool — but not evidence of a global law of primes.

### Caveats

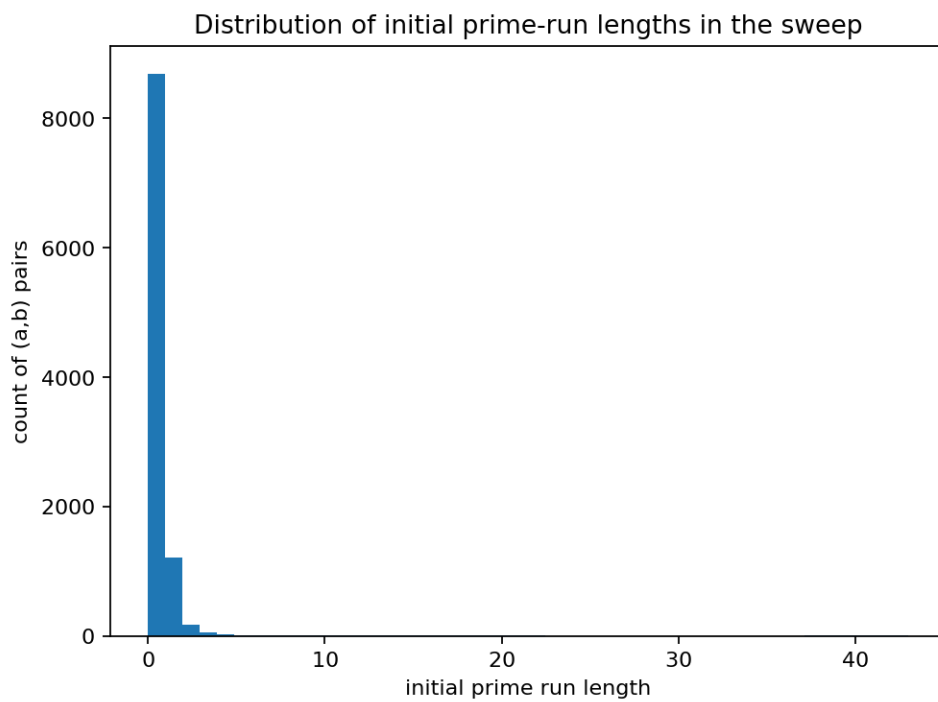
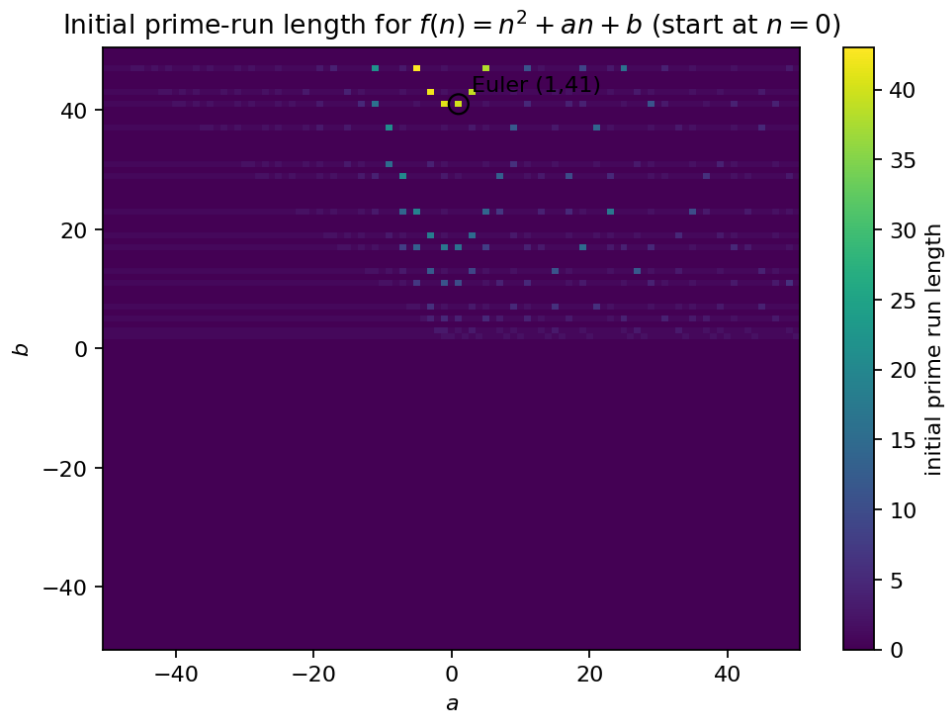
- Different enumeration conventions change the picture.
- Finite window: patterns can look stronger/weaker depending on the chosen **size** and zoom level.
- Visualization choices matter: marker size, DPI, and axis limits can hide or exaggerate structure.

## 3.3.127 E127: Quadratic prime-run atlas ( $n^2 + an + b$ )

**Tags:** number-theory, quantitative-exploration, visualization, primes, optimization See: *Valid Tags*.

### Highlights

- Heatmap of “how long primes last” for  $f(n) = n^2 + an + b$  across a small (a,b) grid.
- Euler’s  $(a, b) = (1, 41)$  appears as a visible peak, but it is not unique.
- Turns a folklore fact into a reproducible, parameterized sweep with saved artifacts.



## Goal

Explore how sensitive **initial prime streaks** are to the coefficients of a quadratic polynomial. We measure, for each  $(a,b)$ , the largest  $L$  such that  $f(0), f(1), \dots, f(L-1)$  are all prime.

## Background (quick refresher)

- *Quadratic polynomials (algebraic) refresher*
- *Euler’s prime-generating polynomial refresher*
- *Prime numbers refresher*

## Research question

Within a bounded coefficient grid, which quadratic polynomials produce the **longest initial prime runs**, and where does Euler’s polynomial sit in that landscape?

## Method

- Sweep integers  $a \in [a_{\min}, a_{\max}]$  and  $b \in [b_{\min}, b_{\max}]$ .
- For each pair, evaluate  $f(n)$  for  $n = 0..N$  and record the initial prime-run length.
- Visualize run length as a heatmap and summarize the top candidates in the report.

## How to run

```
make run EXP=e127
```

or:

```
uv run python -m mathxlab.experiments.e127
```

## Outputs

This experiment follows the standard output contract:

- `out/e127/figures/` — generated figures (PNG)
- `out/e127/report.md` — short narrative report
- `out/e127/params.json` — run parameters (stable JSON)
- `out/e127/logs/` — run logs (created by the runner/Makefile)

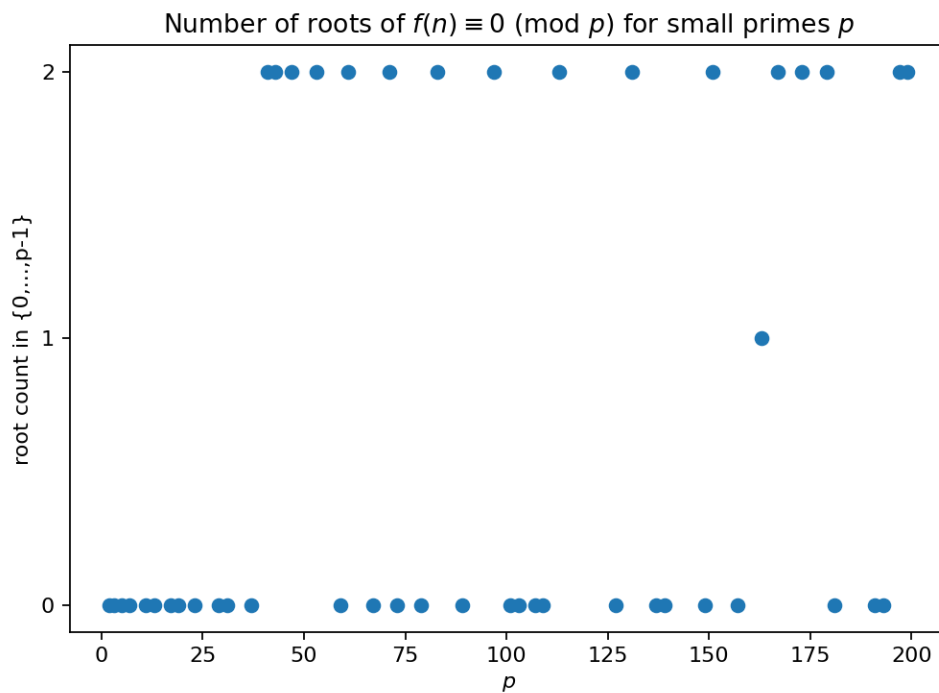
## References

- Euler’s quadratic and “lucky numbers”: contributors [2025], Weisstein [2025].
- Quadratic basics (discriminant, roots): Wikipedia contributors [2026].

## Related experiments

- *E013: Prime-polynomial counterexamples (Euler’s  $n^2 + n + 41$ )* (Euler’s polynomial: first counterexample witness)
- *E024: Ulam spiral structure* (Ulam spiral: prime-rich diagonals explained by quadratic forms)
- *E031: Admissibility and modular obstructions* (modular obstructions and admissibility)
- *E125: Sacks spiral structure* (Sacks spiral prime structure)
- *E126: Hexagonal number spiral structure* (hex-grid spiral prime structure)

### 3.3.128 E128: Quadratic modular obstructions (Euler-type)



**Tags:** number-theory, model-checking, counterexample-search, primes See: [Valid Tags](#).

#### Highlights

- Makes the “hidden reason” a prime streak must fail: modular obstructions.
- Lists explicit residue classes  $n \pmod{p}$  that force divisibility of  $f(n)$ .
- Includes the classic witness  $n = 41k$  for Euler’s  $n^2 + n + 41$ .

#### Goal

Show how congruences explain why a “prime-generating polynomial” cannot stay prime forever. For each small prime modulus  $p$ , we compute the roots of  $f(n) \equiv 0 \pmod{p}$ .

#### Background (quick refresher)

- *Euler’s prime-generating polynomial refresher*
- *Quadratic polynomials (algebraic) refresher*
- *Prime numbers refresher*

#### Research question

For Euler’s  $f(n) = n^2 + n + 41$ , which small primes  $p$  divide some values  $f(n)$ , and which residue classes  $n \pmod{p}$  generate those guaranteed composites?

#### Method

- Fix  $f(n) = n^2 + an + b$  with  $(a, b) = (1, 41)$ .
- For each prime  $p \leq p_{\max}$ , brute-force all residues  $n \in \{0, \dots, p-1\}$  and record solutions to  $f(n) \equiv 0 \pmod{p}$ .
- Plot the number of roots per prime and write a report table of residues.

## How to run

```
make run EXP=e128
```

or:

```
uv run python -m mathxlab.experiments.e128
```

## Outputs

This experiment follows the standard output contract:

- `out/e128/figures/` — generated figures (PNG)
- `out/e128/report.md` — short narrative report
- `out/e128/params.json` — run parameters (stable JSON)
- `out/e128/logs/` — run logs (created by the runner/Makefile)

## References

- Euler’s quadratic and its folklore: contributors [2025], Weisstein [2025].
- Quadratic basics: Wikipedia contributors [2026].

## Related experiments

- *E013: Prime-polynomial counterexamples (Euler’s  $n^2 + n + 41$ )* (first explicit counterexample  $f(40) = 41^2$ )
- *E031: Admissibility and modular obstructions* (admissibility / modular obstructions in prime constellations)
- *E035: Primes in arithmetic progressions mod  $q$*  (primes in arithmetic progressions)
- *E038: Bertrand’s postulate (computational verification)* (Bertrand’s postulate: primes in intervals)

### 3.3.129 E129: Euler lucky constants for $n^2 + n + b$

**Tags:** number-theory, quantitative-exploration, visualization, primes, heuristics See: [Valid Tags](#).

## Highlights

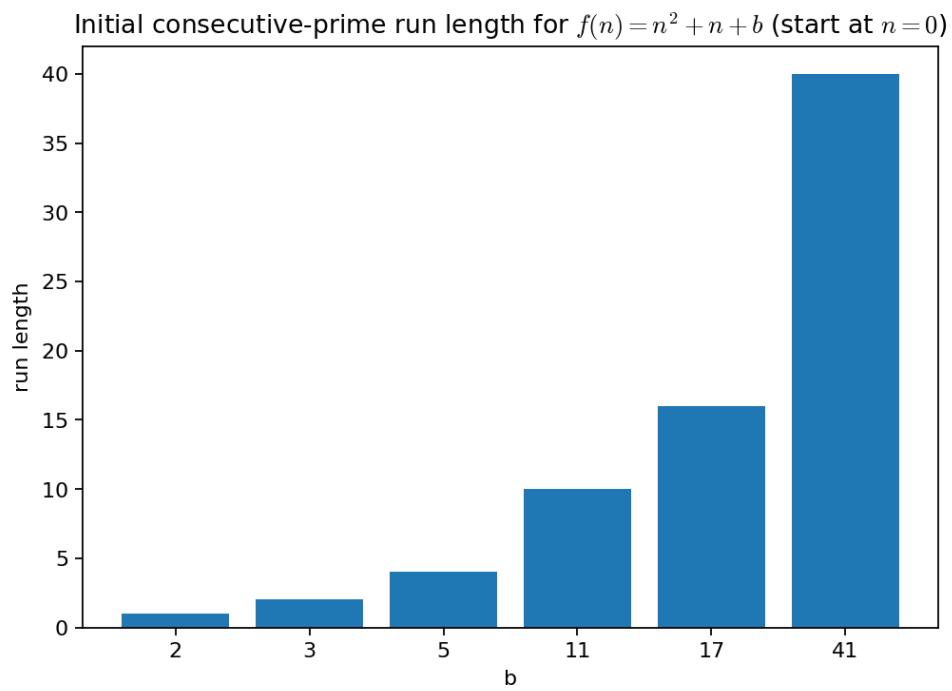
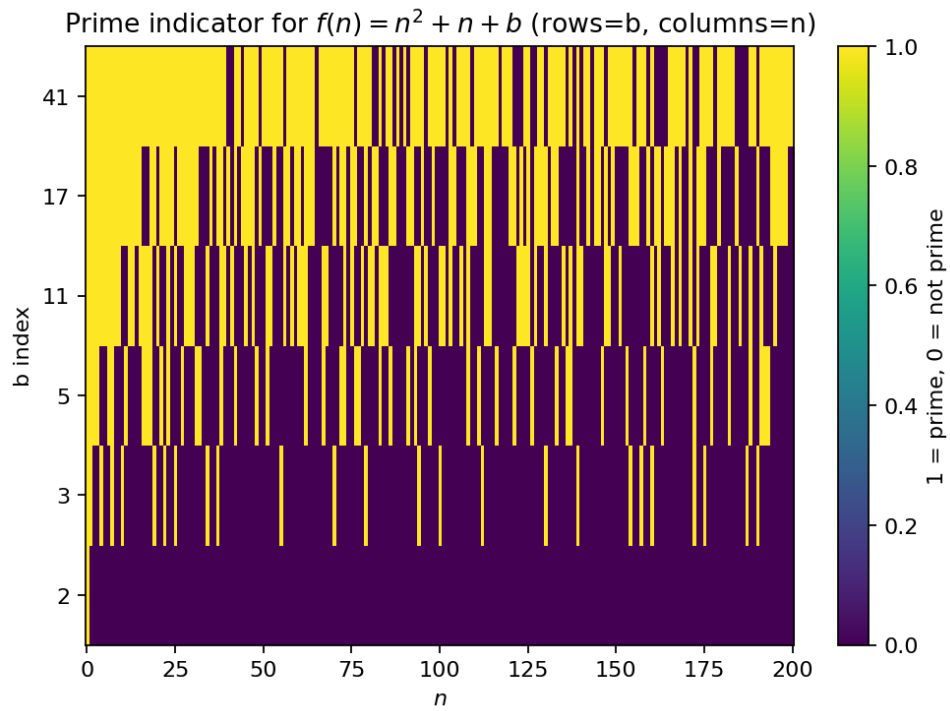
- Compares several classical “lucky” constants  $b$  side-by-side.
- Heatmap shows prime/composite patterns across  $n$  for each  $b$ .
- Report includes the first failure  $n$  and a factorization witness.

## Goal

Compare how long  $n^2 + n + b$  stays prime at the start for a small set of classical constants  $b$ . We quantify the initial prime streak length and visualize where it breaks.

## Background (quick refresher)

- *Euler’s prime-generating polynomial refresher*
- *Quadratic polynomials (algebraic) refresher*
- *Prime numbers refresher*



## Research question

Among a short list of traditional “Euler lucky numbers”  $b$ , which yields the longest initial prime streak for  $f(n) = n^2 + n + b$  and how do the failures look (first composite and its factors)?

## Method

- Choose a list of constants  $b$ .
- For each  $b$ , evaluate  $f(n)$  for  $n = 0..N$ .
- Use a sieve to classify values as prime/composite and measure the initial run length.
- Visualize a prime indicator map and a bar chart of run lengths.

## How to run

```
make run EXP=e129
```

or:

```
uv run python -m mathxlab.experiments.e129
```

## Outputs

This experiment follows the standard output contract:

- `out/e129/figures/` — generated figures (PNG)
- `out/e129/report.md` — short narrative report
- `out/e129/params.json` — run parameters (stable JSON)
- `out/e129/logs/` — run logs (created by the runner/Makefile)

## References

- Lucky numbers and Euler’s polynomial: contributors [2025], Weisstein [2025].

## Related experiments

- *E013: Prime-polynomial counterexamples (Euler’s  $n^2 + n + 41$ )* (Euler’s  $n^2 + n + 41$  counterexample witness)
- *E127: Quadratic prime-run atlas ( $n^2 + a n + b$ )* (full (a,b) atlas for  $n^2 + an + b$ )
- *E128: Quadratic modular obstructions (Euler-type)* (modular obstructions that force failures)
- *E019: Prime density and PNT visualization* (prime density and “random prime” heuristics)

### 3.3.130 Related experiments

- *E106 — Character gallery: real vs. complex* (E106 — Character gallery: real vs. complex)

## 3.4 Getting started

This project uses an **uv-only** workflow and a small Makefile wrapper to run everything consistently.

### 3.4.1 Prerequisites

You need:

- **Python 3.14**
- **uv** on your PATH



- GNU Make

#### Windows notes

- Install GNU Make (e.g., via Chocolatey).
- You can run everything from `cmd.exe`, PowerShell, or from WSL.

### 3.4.2 Verify tools

From the repository root:

```
make uv-check
make python-check
make python-info
```

### 3.4.3 First-time setup

Create a virtual environment and install dependencies:

```
make venv
make install-dev
make install-docs
```

### 3.4.4 Run the full development chain

```
make final
```

`make final` runs:

- formatting (Ruff)
- linting (Ruff)
- type checking (mypy)
- tests (pytest)
- documentation (sphinx)

Documentation can be built separately via:

```
make docs
```

### 3.4.5 Run an experiment

Example (E001):

```
make run EXP=e001 ARGS="--seed 1"
```

See the experiment overview here: *Experiments Gallery*.

### 3.4.6 Build documentation locally

```
make docs
```

Output:

- docs/\_build/html

### 3.4.7 Troubleshooting

error: Failed to spawn: sphinx-build

Sphinx is not installed in the uv environment.

Fix:

```
make install-docs
make docs
```

make python-check fails

The repo enforces Python **3.14**. Install Python 3.14, then recreate the environment:

```
make clean-venv
make venv
make install-dev
make install-docs
```

## 3.5 Development

This page describes the development workflow and the conventions used in this repository.

### 3.5.1 Workflow overview

Use the Makefile targets for everything. They wrap `uv run ...` so you do not need to activate a virtual environment manually.

Typical day-to-day:

```
make final
```

Documentation-only build:

```
make docs
```

Clean caches and build artifacts:

```
make clean
```

Remove the virtual environment (full reset):

```
make clean-venv
```

### 3.5.2 Makefile workflow

The Makefile is the **single entry point** for development tasks (env setup, quality checks, docs builds, and running experiments).

- Prefer `make final` before pushing.
- Prefer `make docs` to validate documentation changes.
- Use `make run EXP=<id>` to execute an experiment and write artifacts to `out/<id>/`.

### Dependency groups

This summary is included from the Makefile documentation:

Dependencies are organized via `pyproject.toml` extras:

- **default**: runtime dependencies needed to run the package
- **dev** (`--extra dev`): developer tooling (ruff, mypy, pytest, etc.)
- **docs** (`--extra docs`): documentation tooling (sphinx, furo, myst-parser, sphinx-design, bibtex, ...)

### What the Makefile does

- Dev / QA targets run via:

```
uv run --extra dev ...
```

- Documentation targets run via:

```
uv run --extra docs ...
```

- The docs dependency target uses:

```
uv sync --all-extras
```

(because `docs-html` also runs a small helper under the `dev` extra).

---

### Run logs

Experiments live under `mathxlab/experiments/` and can be run either directly with Python or through Make targets.

### Run an experiment module directly

```
uv run --extra dev python -m mathxlab.experiments.e001
```

### Run an experiment via Make (with logs)

```
make run EXP=e001
```

Optional arguments:

```
make run EXP=e001 ARGS="--seed 123 --n 200000"
```

### Typical output locations (convention)

Depending on the experiment runner implementation, outputs are usually placed in:

- `out/e###/` (generated artifacts, figures, manifests, logs)
- `docs/reports/`, `docs/params/`, `docs/gallery/` (published snapshots)

If you add new experiments, keep the numbering stable (`e001`, `e002`, ...) so the gallery and documentation can remain consistent.

---

## Common workflows

| Task                                | Command                                                        |
|-------------------------------------|----------------------------------------------------------------|
| Install dev dependencies            | <code>make install-dev</code>                                  |
| Complete quality check              | <code>make final</code>                                        |
| Apply auto-fixes (lint + format)    | <code>make fmt</code>                                          |
| Build HTML docs                     | <code>make docs-html</code>                                    |
| Build HTML + optional PDF docs      | <code>make docs</code>                                         |
| Run fast tests with coverage        | <code>make pytest</code>                                       |
| Run fast + slow tests with coverage | <code>make pytest-slow</code>                                  |
| Clean build artifacts               | <code>make clean</code>                                        |
| Reset environment                   | <code>make clean clean-venv &amp;&amp; make install-dev</code> |

---

## Full reference

For the complete target-by-target reference and troubleshooting, see `makefile`.

### 3.5.3 Formatting, linting, typing, tests

- Formatting: **Ruff** formatter
- Linting: **Ruff**
- Typing: **mypy**
- Tests: **pytest**

#### CI formatting behavior

In CI, formatting runs in check mode (`ruff format --check`). Locally it formats in place.

### 3.5.4 Experiment authoring guidelines

When adding a new experiment:

1. Add a new module under `mathxlab/experiments/`, e.g. `e002_...py`.
2. Prefer deterministic outputs:
  - `--seed` argument if randomness is involved
  - write results to a single `--out` directory
3. Keep the experiment runnable as a module:
  - `python -m mathxlab.experiments.e002`
4. Update the docs:
  - add a short entry to *Experiments Gallery*
  - optionally add a dedicated page under `docs/experiments/` later

### 3.5.5 Documentation

Docs are built with Sphinx + MyST.

Build locally:

```
make install-docs
make docs
```

Deployed website:

- GitHub Pages from the `docs` workflow

### 3.5.6 Contributing (high-level)

- Create a feature branch.
- Open a PR against `main`.
- CI must pass before merge.
- Keep PRs small and well-scoped.

## 3.6 Background

This section provides mathematical foundations for the experiments.

### 3.6.1 Arithmetic functions refresher

This page is a *beginner-friendly* refresher for experiments about **arithmetic functions** (i.e. functions  $f : \mathbb{N} \rightarrow \mathbb{C}$  that encode number-theoretic structure).

For deeper treatments, see [Apostol, 1976], [Niven *et al.*, 1991], and [Tenenbaum, 2015].

#### What “arithmetic function” usually means

An **arithmetic function** is any function of a positive integer  $n$ . Common themes:

- prime factorization drives the behavior of many functions
- *multiplicativity* lets you reduce to prime powers
- *averages* and *summatory functions* reveal global structure

Examples you will meet often:

- Euler’s totient  $\varphi(n)$  (coprime residues)
- Möbius  $\mu(n)$  and Mertens  $M(x) = \sum_{n \leq x} \mu(n)$
- divisor counts  $d(n)$  and sums  $\sigma_k(n)$
- prime-factor counting  $\omega(n), \Omega(n)$
- von Mangoldt  $\Lambda(n)$  (prime powers)
- Carmichael’s  $\lambda(n)$  (group exponent mod  $n$ )

#### Multiplicative vs. additive

##### Multiplicative

An arithmetic function  $f$  is **multiplicative** if

$$f(1) = 1, \quad f(ab) = f(a)f(b) \text{ whenever } \gcd(a, b) = 1.$$

If  $n = \prod p_i^{\alpha_i}$  then multiplicativity gives

$$f(n) = \prod f(p_i^{\alpha_i}).$$

Typical:  $\varphi, \mu, \sigma_k, d, J_k, \lambda$ .

## Additive

A function  $g$  is **additive** if

$$g(ab) = g(a) + g(b) \text{ whenever } \gcd(a, b) = 1,$$

and **completely additive** if the relation holds for all  $a, b$  (no gcd condition). Typical:  $\Omega(n)$  is completely additive;  $\omega(n)$  is additive.

## Why averages matter

Many experiments compare:

- pointwise behavior of  $f(n)$
- cumulative behavior  $\sum_{k \leq n} f(k)$
- distribution of  $f(n)$  over ranges

Analytic/probabilistic number theory focuses on these averages; see [Montgomery and Vaughan, 2006] and [Tenenbaum, 2015].

## Common experiment patterns

- **Value distribution:** histograms of  $f(n)$  for  $n \leq N$
- **Scatter vs. factorization features:** compare  $f(n)$  with  $\log n$ ,  $\omega(n)$ , etc.
- **Normal order:** show “typical size” vs. rare extremes (e.g. Erdős–Kac behavior)
- **Summatory oscillations:**  $M(x)$ ,  $\sum_{n \leq x} \lambda(n)$ , etc.
- **Extremal orders:** highly composite numbers maximize  $d(n)$  (see [Ramanujan, 1915])

## Experiments in this repository

- **E121** — Multiplicativity stress tests across core arithmetic functions.
- **E123** — Correlation matrix of arithmetic functions on 1..N (empirical relationships).

## 3.6.2 Average orders and the Erdős–Kac viewpoint

Many arithmetic functions are “wild” pointwise but have predictable average behavior. This page gives a short refresher on *average order*, *normal order*, and the Erdős–Kac phenomenon. See [Tenenbaum, 2015] and [Erdős and Kac, 1940].

### Average order vs. normal order

- **Average order:** a function  $A(x)$  such that

$$\sum_{n \leq x} f(n) \approx \sum_{n \leq x} A(n)$$

or  $f$  has mean value described by  $A$ .

- **Normal order:** a function  $N(n)$  such that “most” integers satisfy

$$f(n) \approx N(n).$$

A classic example: for most  $n$ ,  $\omega(n)$  is close to  $\log \log n$ .

### Erdős–Kac theorem (informal statement)

Let  $\omega(n)$  be the number of distinct prime factors. Erdős and Kac proved that the normalized variable

$$\frac{\omega(n) - \log \log n}{\sqrt{\log \log n}}$$

has an approximately standard normal distribution over  $n \leq x$  as  $x \rightarrow \infty$ . [Erdős and Kac, 1940]

### Experiment ideas

- for a chosen  $N$ , compute  $\omega(n)$  for  $n \leq N$  and histogram the normalized values
- compare the empirical mean/variance with  $\log \log N$
- explore how the fit improves as  $N$  grows

### Experiments in this repository

- **E094** — Erdős–Kac side-by-side for  $\omega$  and  $\Omega$  (normal order / scaling).

### 3.6.3 Carmichael’s $\lambda(n)$ function refresher

Carmichael’s function  $\lambda(n)$  is the exponent of the multiplicative group modulo  $n$ . It refines Euler’s theorem by giving the *smallest* universal exponent. See [Erdős *et al.*, 1991] and [Niven *et al.*, 1991].

#### Definition

Let  $(\mathbb{Z}/n\mathbb{Z})^\times$  be the multiplicative group of units modulo  $n$ . The **exponent** of a finite group  $G$  is lcm of the orders of all elements. Carmichael’s function  $\lambda(n)$  is the exponent of  $(\mathbb{Z}/n\mathbb{Z})^\times$ .

Equivalently,  $\lambda(n)$  is the smallest positive integer such that for all integers  $a$  with  $\gcd(a, n) = 1$ :

$$a^{\lambda(n)} \equiv 1 \pmod{n}.$$

Always  $\lambda(n) \mid \varphi(n)$ .

#### Prime power formula (useful in code)

For odd primes  $p$  and  $k \geq 1$ :

$$\lambda(p^k) = \varphi(p^k) = p^{k-1}(p-1).$$

For  $2^k$ :

$$\lambda(2) = 1, \quad \lambda(4) = 2, \quad \lambda(2^k) = 2^{k-2} \text{ for } k \geq 3.$$

For general  $n = \prod p_i^{\alpha_i}$ :

$$\lambda(n) = \text{lcm}(\lambda(p_i^{\alpha_i})).$$

#### Why it is interesting experimentally

- highlights differences between “group size” ( $\varphi$ ) and “group exponent” ( $\lambda$ )
- connects to orders modulo  $n$  and to cryptographic-style modular arithmetic
- has rich average-order results and rare extreme behavior (see [Erdős *et al.*, 1991])

### Experiment ideas

- compare  $\lambda(n)$  vs.  $\varphi(n)$  on ranges
- visualize distribution of  $\varphi(n)/\lambda(n)$
- record-breakers for large  $\varphi(n)/\lambda(n)$

### Experiments in this repository

- **E100** — Carmichael ( $\lambda$ ) vs Euler ( $\varphi$ ): ratios, typical size, and extremes.

### 3.6.4 Carmichael numbers

A Carmichael number is a composite  $n$  such that

$$[a^{n-1} \equiv 1 \pmod{n}]$$

for every integer  $a$  coprime to  $n$ . They are **absolute Fermat pseudoprimes**, so they defeat the naive Fermat primality test. [Prime Pages (UTM), 2025, Wikipedia contributors, 2025]

#### Key facts

- **Korselt’s criterion (1899):**  $n$  is Carmichael iff (i)  $n$  is squarefree and (ii) for every prime  $p \mid n$ , we have  $(p-1) \mid (n-1)$ . [Conrad, 2004, Wikipedia contributors, 2025]
- **At least three prime factors:** No Carmichael number is the product of only two primes. [Wikipedia contributors, 2025]
- **Infinitely many:** Alford–Granville–Pomerance proved there are infinitely many Carmichael numbers. [Alford *et al.*, 1994]
- **Historical origin:** Carmichael studied these numbers in the early 20th century. [Carmichael, 1912]

#### What to experiment with

- **Counterexample search:** Find Carmichael numbers by generating squarefree products and testing Korselt’s criterion.
- **“Fermat test is not enough”:** Show that all bases  $a$  with  $\gcd(a, n) = 1$  pass Fermat for a Carmichael  $n$ , while a Miller–Rabin test quickly finds witnesses.
- **Distribution experiments:** Count Carmichael numbers up to  $X$  and compare growth to simple heuristics.
- **Construction families:** Implement Chernick-style constructions and see when factors are prime.

#### References

See Alford *et al.* [1994], Conrad [2004], The OEIS Foundation Inc. [2025], Wikipedia contributors [2025].

### 3.6.5 Dirichlet characters refresher

This page is a *lightweight* background for experiments about primes in residue classes, Dirichlet  $L$ -functions, and prime races.

#### Core definitions

Fix an integer modulus  $q \geq 1$ . A **Dirichlet character modulo  $q$**  is a function  $\chi : \mathbb{Z} \rightarrow \mathbb{C}$  such that:

1. (Periodicity)  $\chi(n+q) = \chi(n)$  for all  $n$ .
2. (Multiplicativity)  $\chi(mn) = \chi(m)\chi(n)$  for all  $m, n$ .
3. (Support)  $\chi(n) = 0$  if  $\gcd(n, q) > 1$ .
4. For  $\gcd(n, q) = 1$ , we have  $|\chi(n)| = 1$  (so values are roots of unity).

The **principal character**  $\chi_0$  modulo  $q$  is: 
$$[\chi_0(n) = \begin{cases} 1, & \gcd(n, q) = 1, \\ 0, & \gcd(n, q) > 1. \end{cases}]$$

A character is **primitive** if it does not “factor through” a smaller modulus (its conductor equals  $q$ ).



### Orthogonality (the workhorse identity)

Let  $\chi, \psi$  be Dirichlet characters modulo  $q$ . Then (over a reduced residue system): 
$$\sum_{\substack{a \bmod q \\ \gcd(a,q)=1}} \chi(a) \overline{\psi(a)} = \begin{cases} \varphi(q), & \chi = \psi, \\ 0, & \chi \neq \psi. \end{cases}$$

A dual identity (summing over characters) is: 
$$\sum_{\chi \bmod q} \chi(a) \overline{\chi(b)} = \begin{cases} \varphi(q), & a \equiv b \pmod q, \\ 0, & \text{otherwise.} \end{cases}$$

These are the algebraic “Fourier rules” that make Dirichlet’s argument work.

#### Example: modulus $q = 4$

There are two characters modulo 4:

- $\chi_0$  (principal).
- The nontrivial character  $\chi_4$ : 
$$\chi_4(n) = \begin{cases} 0, & 2 \mid n, \\ 1, & n \equiv 1 \pmod 4, \\ -1, & n \equiv 3 \pmod 4. \end{cases}$$

This single character already explains the classic race between primes  $1 \bmod 4$  and  $3 \bmod 4$ .

### What experiments usually measure

- How  $\sum_{n \leq N} \chi(n)$  behaves (cancellation, max partial sums).
- How orthogonality isolates a residue class.
- How many primitive characters exist for a given modulus, and how they “look” as tables.

### Practical numerical caveats

- Always define  $\chi(n) = 0$  when  $\gcd(n, q) > 1$  (otherwise identities break).
- Represent  $\chi$  as a lookup table on residues  $0, \dots, q-1$  and reduce via  $n \% q$ .
- Be careful with complex floats: for most experiments, values are in  $\{0, \pm 1\}$  or small roots of unity, so you can use exact complex numbers.

### References

See *References*.

[Davenport, 2000, Ireland and Rosen, 1990, Lejeune Dirichlet, 1837, Serre, 1973]

### Experiments in this repository

- **E106** — Character gallery: real vs complex characters; conjugate pairing.
- **E107** — Primitive vs imprimitive via conductor (induced characters).
- **E108** — Orthogonality relations as a heatmap; numeric sanity checks.

### 3.6.6 Dirichlet convolution refresher

Dirichlet convolution is a key “algebra” on arithmetic functions and shows up in many proofs and experiments. See [Apostol, 1976] and [Tenenbaum, 2015].

#### Definition

For arithmetic functions  $f, g : \mathbb{N} \rightarrow \mathbb{C}$ , their **Dirichlet convolution** is

$$(f * g)(n) = \sum_{d|n} f(d) g\left(\frac{n}{d}\right).$$

The function  $1(n) \equiv 1$  acts like an identity in many formulas, and the **delta function**  $\varepsilon(n)$  (with  $\varepsilon(1) = 1$  and  $\varepsilon(n) = 0$  for  $n > 1$ ) is the convolution identity:

$$f * \varepsilon = f.$$

### Möbius inversion

If

$$F(n) = \sum_{d|n} f(d) \iff F = f * 1,$$

then **Möbius inversion** says

$$f = F * \mu.$$

This is one of the main reasons  $\mu(n)$  appears everywhere.

### Classic identities

- Sum of divisors:

$$\sigma(n) = (1 * \text{id})(n),$$

where  $\text{id}(n) = n$ .

- Totient:

$$\varphi = \mu * \text{id}.$$

- Jordan totient:

$$J_k = \mu * \text{id}^k.$$

- von Mangoldt:

$$\Lambda = \mu * \log,$$

in the sense of Dirichlet series / logarithmic derivatives of  $\zeta(s)$ .

### Dirichlet series viewpoint (why it's computationally useful)

If  $F(s) = \sum_{n \geq 1} f(n)n^{-s}$  and  $G(s) = \sum_{n \geq 1} g(n)n^{-s}$  converge absolutely, then

$$\sum_{n \geq 1} \frac{(f * g)(n)}{n^s} = F(s) G(s).$$

This “turns convolution into multiplication” and underlies many analytic estimates; see [Montgomery and Vaughan, 2006].

### Experiments in this repository

- **E102** — Dirichlet convolution identity zoo ( $1 = , 1 = \text{id}, 11 = , \text{id}1 = , \dots$ ).
- **E121** — Multiplicativity stress tests and convolution sanity checks (random coprime tests).

### 3.6.7 Dirichlet eta function (s)

The **Dirichlet eta function** is the alternating Dirichlet series  $[\eta(s) = \sum_{n \geq 1} \frac{(-1)^{n-1}}{n^s}, ]$  which converges for  $(\text{Re}(s) > 0)$  (by alternating-series arguments). It is related to the Riemann zeta function by  $[\eta(s) = \left(1 - 2^{1-s}\right) \zeta(s). ]$

### Key ideas

- **Faster convergence:** for real ( $s > 1$ ), the alternating series often converges more rapidly than the plain  $\chi$ -series.
- **Analytic continuation:** the identity above provides an analytic continuation of  $\chi(s)$  away from the factor  $(1 - 2^{1-s})$ .
- **Numerical gateway:**  $\chi(s)$  is a convenient “entry point” for simple  $\chi(s)$  numerics without complex contour methods.

### Why it matters in this project

When we want quick, small-scale experiments (e.g. comparing partial sums, smoothing, or convergence acceleration),  $\chi(s)$  provides stable numerics with minimal infrastructure.

### Experiments in this repository

- **E114** —  $(1/2 + it)$  via  $\chi(s)$ : truncation/precision stability map.

### References

See *References*.

[Edwards, 1974, Titchmarsh, 1986, Wikipedia contributors, 2025]

## 3.6.8 Dirichlet $L$ -functions refresher

Dirichlet characters package congruence information; Dirichlet  $L$ -functions turn that into analytic objects whose zeros control prime distribution in residue classes.

### Core definitions

Let  $\chi$  be a Dirichlet character modulo  $q$ . The **Dirichlet  $L$ -function** is  $[L(s, \chi) = \sum_{n=1}^{\infty} \frac{\chi(n)}{n^s}, \quad \Re(s) > 1.]$

For  $\Re(s) > 1$ , it has an **Euler product**:  $[L(s, \chi) = \prod_{p \nmid q} \left(1 - \frac{\chi(p)}{p^s}\right)^{-1}.]$

For the principal character:  $[L(s, \chi_0) = \zeta(s) \prod_{p \mid q} \left(1 - p^{-s}\right).]$

The decisive analytic fact (used in Dirichlet’s theorem) is:  $[L(1, \chi) \neq 0 \quad \text{for every nonprincipal character } \chi.]$

### What experiments usually visualize or measure

- Convergence of partial sums  $\sum_{n \leq N} \chi(n) n^{-s}$  for various  $s$ .
- Convergence of partial Euler products over primes.
- Sensitivity near  $s = 1$  (slow convergence) and how to stabilize it.

### Practical numerical caveats

- Near  $s = 1$ , both series and Euler products converge painfully slowly.
- For Euler products, compute via logs:  $[\log L(s, \chi) \approx -\sum_{p \leq P} \log \left(1 - \frac{\chi(p)}{p^s}\right)]$  to reduce catastrophic multiplication error.
- If you compare characters, always keep the same prime cutoff  $P$  to make plots comparable.

## References

See *References*.

[Davenport, 2000, Lejeune Dirichlet, 1837, Serre, 1973, Washington, 1997]

## Experiments in this repository

- **E110** — Dirichlet L-series partial sums at  $s=1$  and  $s=1/2$  (principal vs nonprincipal).
- **E111** — Euler product vs Dirichlet series truncations for  $L(s, \chi)$ : error vs cutoff.

### 3.6.9 Divisor functions $d(n)$ and $\sigma_k(n)$ refresher

Divisor functions measure how many divisors a number has, and how large they are. They are classical examples of multiplicative functions. See [Apostol, 1976].

#### Counting divisors: $d(n)$

Let  $d(n)$  (also written  $\tau(n)$ ) be the number of positive divisors of  $n$ .

If  $n = \prod p_i^{\alpha_i}$ , then

$$d(n) = \prod (\alpha_i + 1).$$

#### Sum of divisors: $\sigma_k(n)$

For  $k \geq 0$ ,

$$\sigma_k(n) = \sum_{d|n} d^k.$$

The case  $k = 1$  is the usual sum-of-divisors function  $\sigma(n)$ .

For a prime power,

$$\sigma_k(p^\alpha) = 1 + p^k + p^{2k} + \cdots + p^{\alpha k} = \frac{p^{(\alpha+1)k} - 1}{p^k - 1}.$$

Again, multiplicativity gives  $\sigma_k(n)$  from prime powers.

#### Highly composite numbers (extremal behavior)

Numbers that maximize  $d(n)$  up to a range are called **highly composite numbers**. Ramanujan's classic paper studies their structure. [Ramanujan, 1915]

#### Experiment ideas

- visualize  $d(n)$  up to  $N$  and highlight record-breakers
- compare  $\sigma(n)$  to  $n$  (abundant / perfect / deficient classification)
- log-scale plots of  $d(n)$  to make extremes visible

## Experiments in this repository

- **E096** — Record-holders for  $d(n)$  up to  $N$  (highly composite flavor).
- **E097** —  $d(n)/n$  landscape: deficient / perfect / abundant classification.
- **E098** — Extremals of  $d(n)/n^\epsilon$  across  $\epsilon$  (phase changes / superabundant intuition).

## References

See *References*.

[Alaoglu and Erdős, 1944, Apostol, 1976, Lagarias, 2002, Ramanujan, 1915, Robin, 1984]

### 3.6.10 Euler’s totient function $\varphi(n)$ refresher

Euler’s totient function  $\varphi(n)$  counts reduced residues modulo  $n$ . Standard references: [Apostol, 1976], [Niven *et al.*, 1991].

#### Definition

For  $n \geq 1$ ,

$$\varphi(n) = \#\{1 \leq k \leq n : \gcd(k, n) = 1\}.$$

Equivalently:  $\varphi(n)$  is the size of the multiplicative group  $(\mathbb{Z}/n\mathbb{Z})^\times$ .

#### Prime power formula

If  $p$  is prime and  $k \geq 1$ ,

$$\varphi(p^k) = p^k - p^{k-1} = p^k \left(1 - \frac{1}{p}\right).$$

Since  $\varphi$  is multiplicative, for  $n = \prod p_i^{\alpha_i}$ :

$$\varphi(n) = n \prod_{p|n} \left(1 - \frac{1}{p}\right).$$

#### Important properties

- **Euler’s theorem:** if  $\gcd(a, n) = 1$  then  $a^{\varphi(n)} \equiv 1 \pmod{n}$ .
- **Average order:** roughly  $\sum_{n \leq x} \varphi(n) \sim \frac{3}{\pi^2} x^2$  (analytic methods).
- **Extremes:**  $\varphi(n)$  is small when  $n$  has many small prime factors.

#### Computational notes

Given the prime factorization of  $n$ ,  $\varphi(n)$  is cheap to compute via the product formula. For experiments, you can often compute it for all  $n \leq N$  efficiently using a sieve-style method.

#### Experiments in this repository

- **E101** — Reduced residues  $(\ /q )^\times$  as an explicit set; verify size  $(q)$ .

### 3.6.11 Euler’s prime-generating polynomial refresher

One of the most famous “too good to be true” examples in experimental number theory is Euler’s quadratic

$$f(n) = n^2 + n + 41.$$

For integers  $n = 0, 1, \dots, 39$  it produces **prime numbers**. This is an unusually long initial run for such a simple polynomial, and it is a classic motivation for prime-value experiments. [contributors, 2025, Weisstein, 2025]

#### The basic facts

- For  $0 \leq n \leq 39$ ,  $f(n)$  is prime.
- At  $n = 40$ ,

$$f(40) = 40^2 + 40 + 41 = 1681 = 41^2,$$

so the prime streak must stop.

- At  $n = 41$ ,

$$f(41) = 41^2 + 41 + 41 = 41 \cdot 43,$$

again composite.

A neat way to see the “divisibility by 41” phenomenon is to write

$$f(n) = n(n + 1) + 41.$$

Working modulo 41, if  $n \equiv -1 \pmod{41}$  then  $n(n + 1) \equiv 0$  and  $f(n) \equiv 0 \pmod{41}$ . So infinitely many inputs force a factor 41.

### Why this doesn’t contradict “no prime polynomial”

A standard (and important) observation:

No non-constant polynomial with integer coefficients can take prime values for *all* integers  $n$ .

Reason: if  $f(0) = m$ , then for  $n = m$  we get  $f(m) \equiv f(0) \equiv m \pmod{m}$ , so  $m \mid f(m)$ , and  $f(m)$  is composite once it is larger than  $|m|$ .

Euler’s example is still interesting because it shows that **long prime runs can happen**, even though a “prime for all  $n$ ” polynomial is impossible.

### What to experiment with

Good experiment patterns:

- **Prime-streak length:** for a family  $n^2 + an + b$ , measure the longest initial segment  $n = 0, 1, \dots, L$  where values are prime.
- **Modular obstructions:** for each prime  $p$ , count how many residues  $n \pmod{p}$  make  $f(n) \equiv 0 \pmod{p}$ . (If there are many such residues, prime values should be rarer.)
- **Heuristic growth:** count primes among  $f(0), \dots, f(N)$  and compare against a rough  $N/\log N$  scale.
- **Visualization:** plot  $f(n)$  and mark prime/composite values; or show a heatmap over  $(a, b)$  for streak lengths.

### Practical numerical caveats

- **Big integers:**  $f(n)$  grows like  $n^2$ ; primality testing dominates runtime quickly.
- **Avoid accidental float conversion:** keep  $n$  and  $f(n)$  as Python integers throughout.
- **Sampling bias:** focusing only on “nice-looking” polynomials can be misleading; compare against random  $(a, b)$  baselines.

### References

See [References](#).

[contributors, 2025, Hardy and Wright, 2008, Weisstein, 2025]

### 3.6.12 Explicit formulas: primes zeros

“Explicit formulas” are identities that connect prime counting (typically via Chebyshev functions like  $(\psi(x))$ ) to sums over zeros of  $(s)$  (or more general L-functions). Conceptually, they are a precise form of the slogan:

**Primes are controlled by zeros.**

A prototypical form expresses  $(\psi(x))$  as a main term  $(x)$  plus oscillatory corrections coming from nontrivial zeros.

### Key ideas

- **From Euler product to primes:** differentiating  $(\log \zeta(s))$  connects  $(s)$  to the von Mangoldt function  $(\Lambda(n))$ .
- **Zeros drive oscillations:** sums over  $(\rho)$  (zeros) produce fluctuating terms that explain biases and sign changes seen in finite ranges.
- **Numerical truncation:** in experiments, one typically truncates zero sums and studies stability vs cutoff parameters.

### Why it matters in this project

This is the bridge between the “Dirichlet character / L-function” block and the “prime race / bias” phenomena: you can literally watch zero contributions modulate prime-counting curves.

### Experiments in this repository

- **E119** —  $(x) - x$  oscillations and “explicit-formula intuition” (visual).

### References

See *References*.

[Ivić, 1985, Iwaniec and Kowalski, 2004, Schoenfeld, 1976, Titchmarsh, 1986, Wikipedia contributors, 2025]

## 3.6.13 Exploratory visualizations for arithmetic functions

Several experiments in this project are intentionally “EDA-like”: they produce **heatmaps**, **atlases**, and **correlation matrices** for many functions at once. This page collects best practices so those figures remain interpretable and comparable.

### Core definitions

Given a function table  $F(n)$  for  $n = 1, \dots, N$  and a list of arithmetic functions  $f_1, \dots, f_m$ , two common derived views are:

- **Heatmap atlas:** visualize values  $f_j(n)$  as an image (rows = functions, columns =  $n$  or blocks of  $n$ ).
- **Correlation matrix:** compute an empirical correlation between columns  $f_i(n)$  and  $f_j(n)$  (Pearson or rank-based).

### What experiments usually visualize or measure

- “Texture”: squarefree bands, prime powers, smooth numbers, record-holders, etc.
- Clustering: which functions behave similarly on  $1..N$  (after normalization).
- Stability: how patterns change when  $N$  grows or when you sample on a log grid.

### Practical numerical caveats

- **Normalize first.** Raw scales differ wildly (e.g.  $\sigma(n)$  vs  $\mu(n)$ ). Typical choices: z-score, log1p, or scaling by  $n^\alpha$ .
- **Beware heavy tails.** Extremal values can dominate Pearson correlation; consider Spearman correlation or winsorization for robustness.
- **Sampling matters.** Linear  $n$  emphasizes small structure; log-spaced  $n$  emphasizes asymptotic behavior. Record the sampling rule in the figure caption.

## References

See *References*.

[Arnold, 2015, Borwein and Bailey, 2008]

## Experiments in this repository

- **E122** — Heatmap atlas of  $\zeta$ ,  $\Omega$  and related “squarefree texture” features.
- **E123** — Correlation matrix of arithmetic functions (with normalization variants).

### 3.6.14 Fermat numbers

Fermat numbers are the integers

$$F_n = 2^{2^n} + 1 \quad (n \geq 0).$$

They grow extremely quickly and are central to a classic “counterexample” story: Fermat conjectured that all  $F_n$  are prime, but  $F_5$  is composite (Euler). [Křivánek et al., 2013, Wikipedia contributors, 2025]

#### Key facts

- **Coprime property:**  $\gcd(F_m, F_n) = 1$  for  $m \neq n$ . In fact,  $F_0 F_1 \cdots F_{n-1} = F_n - 2$ . [Křivánek et al., 2013]
- **Fermat primes:** If  $F_n$  is prime, it is called a Fermat prime (known:  $n = 0..4$ ). [The OEIS Foundation Inc., 2025]
- **Pépin test (primality):** For  $n \geq 1$ ,  $F_n$  is prime iff  $3^{(F_n-1)/2} \equiv -1 \pmod{F_n}$ . [Křivánek et al., 2013]

#### What to experiment with

- **Factor hunting:** Search for small prime factors of  $F_n$  (using modular arithmetic and wheel/primorial filters).
- **Pépin vs. trial division:** Compare performance and failure modes as  $n$  increases.
- **Structure of known factors:** Many known prime factors have the form  $k \cdot 2^{n+2} + 1$ ; explore how often such candidates appear.
- **Euclid-style numbers:** Explore  $E_n = 1 + \prod_{i=0}^n p_i$  and compare “Euclid primes” vs. Fermat primes (they behave very differently).

## References

See Křivánek et al. [2013], The OEIS Foundation Inc. [2025], Wikipedia contributors [2025].

### 3.6.15 Gauss sums and the discrete Fourier viewpoint

Gauss sums are “finite Fourier transforms” of Dirichlet characters. They are a compact way to see how characters interact with additive phases, and they show up naturally in functional equations and explicit evaluations of certain  $L$ -values.

#### Core definitions

Let  $\chi$  be a Dirichlet character modulo  $q$  and write  $e_q(a) = \exp\left(\frac{2\pi i a}{q}\right)$ . The (normalized) **Gauss sum** is  $\tau(\chi) = \sum_{a=1}^q \chi(a) e_q(a)$ .

For a **primitive** character  $\chi$  modulo  $q$ , one has the fundamental magnitude law  $|\tau(\chi)| = \sqrt{q}$ . (For imprimitive characters, the behavior is more subtle and often smaller.)

A key identity connecting Gauss sums to character orthogonality is:  $\sum_{a \bmod q} \chi(a) e_q(an) = \overline{\chi}(n) \tau(\chi) \quad (\gcd(n, q) = 1)$ , and it vanishes when  $\gcd(n, q) > 1$ .



### What experiments usually visualize or measure

- The cloud of complex values  $\tau(\chi)$  for all characters modulo a fixed  $q$ .
- The “circle of radius  $\sqrt{q}$ ” phenomenon for primitive characters.
- Quadratic characters as a special case (values close to  $\pm\sqrt{q}$  or  $\pm i\sqrt{q}$  depending on  $q$ ).

### Practical numerical caveats

- Always be explicit about the representative set for residues (e.g.  $a = 1, \dots, q$ ).
- For imprimitive characters, confirm whether the implementation returns the induced character or the primitive lift; this changes  $\tau(\chi)$  drastically.
- Use complex128 (or higher) if you push  $q$  large; cancellation is extreme in the raw sum definition.

### References

See *References*.

[Berndt *et al.*, 1998, Davenport, 2000]

### Experiments in this repository

- **E109** — Gauss sums  $\tau(\chi)$ : magnitude law and geometry for characters modulo  $q$ .

### 3.6.16 Gram points and zero counting

A **Gram point** is (informally) a value  $t$  where the Riemann–Siegel theta function  $(\theta(t))$  hits an integer multiple of  $(\pi)$ . These points are useful landmarks when visualizing  $(Z(t))$  and tracking sign changes.

The **Riemann–von Mangoldt formula** gives the asymptotic count of nontrivial zeros up to height  $(T)$ :  

$$N(T) = \frac{T}{2\pi} \log \left( \frac{T}{2\pi} \right) - \frac{T}{2\pi} + O(\log T) \quad (T \rightarrow \infty),$$
with refinements that include the argument of  $\zeta$  on the critical line.

### Key ideas

- **Bookkeeping:** zero counting provides a “sanity check” for numerical root-finding: we can compare a computed zero list against  $(N(T))$ .
- **Gram blocks / Gram’s law:** empirical rules about how zeros sit between consecutive Gram points (useful, but not always true).
- **Practical numerics:** many experiments become clearer when plotted against  $(\theta(t))$  or indexed by Gram points.

### Experiments in this repository

- **E116** — Observed zero counts (via sign changes) vs Riemann–von Mangoldt estimate.

### References

See *References*.

[Odlyzko, 1992, Titchmarsh, 1986, Wikipedia contributors, 2025, Wikipedia contributors, 2025]

### 3.6.17 Hardy’s Z-function and the critical line

For real  $t$ , the **Hardy Z-function** is defined (up to standard conventions) by  $Z(t) = e^{i\theta(t)} \zeta\left(\frac{1}{2} + it\right)$ , where  $(\theta(t))$  is the Riemann–Siegel theta function. The key feature is that  **$Z(t)$  is real-valued for real  $t$** , so zeros of  $(Z(t))$  correspond to zeros of  $(\zeta(s))$  on the critical line.

### Key ideas

- **Real signal:** studying sign changes of  $(Z(t))$  is a practical way to bracket zeros on  $(\Re(s)=\tfrac{1}{2})$ .
- **Theta function:**  $(\theta(t))$  captures the “oscillatory phase” of on the critical line and is closely tied to Gram points.
- **Numerical workflows:** many computational approaches to zeta zeros are formulated in terms of  $(Z(t))$ ,  $(\theta(t))$ , and their approximations.

### Why it matters in this project

It gives a clean, visualization-friendly path from “complex -values” to “real curves” whose roots can be bracketed and counted.

### Experiments in this repository

- **E115** — Hardy Z sign-change scan and zero bracketing (bisection refinement).

### References

See *References*.

[Odlyzko, 1992, Titchmarsh, 1986, Wikipedia contributors, 2025, Wikipedia contributors, 2025]

### 3.6.18 Jordan totient $J_k(n)$ refresher

Jordan’s totient function generalizes Euler’s totient. It is multiplicative and appears naturally in group-counting problems. See [Apostol, 1976] and [Tenenbaum, 2015].

#### Definition

For a fixed integer  $k \geq 1$ , the **Jordan totient** function  $J_k(n)$  can be defined by

$$J_k(n) = n^k \prod_{p|n} \left(1 - \frac{1}{p^k}\right).$$

For  $k = 1$ , this is Euler’s totient:  $J_1(n) = \varphi(n)$ .

#### Divisor-sum identity

A useful identity is

$$\sum_{d|n} J_k(d) = n^k.$$

In Dirichlet convolution language:

$$J_k * 1 = \text{id}^k,$$

hence  $J_k = \mu * \text{id}^k$ .

#### Experiment ideas

- compare  $J_k(n)$  as  $k$  varies
- visualize  $J_2(n)$  and  $J_3(n)$  over  $n \leq N$
- compare  $J_k(n)/n^k$  as a “prime factor penalty”

### Experiments in this repository

- **E099** — Jordan totients  $J_k$ : atlas, identities ( $J_1 = \text{id}$ ), and scaling  $J_k(n)/n^k$ .

### 3.6.19 Landau’s problems refresher

At the 1912 International Congress of Mathematicians (Cambridge), Edmund Landau highlighted four “basic” open problems about prime numbers. They became known as **Landau’s problems**. [Pintz, 2009, Wikipedia contributors, 2026]

The striking aspect is that the statements are easy to explain to a beginner, but none of them has been proved so far.

#### The four problems (informal statements)

Landau’s list is usually presented as:

1. **Goldbach (binary):** Every even integer  $\geq 4$  is a sum of two primes.
2. **Twin primes:** There are infinitely many primes  $p$  such that  $p + 2$  is also prime.
3. **Legendre’s conjecture:** For every  $n \geq 1$ , there is a prime between  $n^2$  and  $(n + 1)^2$ .
4. **Primes of the form  $n^2 + 1$ :** There are infinitely many primes among  $n^2 + 1$ .

Landau’s original phrasing and historical context are discussed nicely in Pintz’s survey. [Pintz, 2009]

#### Why these problems are “simple but hard”

A recurring theme is that **primes behave like random numbers in many respects**, but not enough is known to turn probabilistic intuition into proofs.

Two typical obstacles:

- **Parity barrier:** Many sieve methods can show “almost primes” (numbers with few prime factors), but they struggle to force *exactly one* prime factor.
- **Correlation control:** Conjectures like “twin primes” require proving that primality events at two nearby integers are correlated often enough.

#### What experiments often do

Landau’s problems are perfect targets for “experimental math” because you can explore the *shape* of the evidence:

- **Goldbach:** empirical coverage, number of representations, typical smallest prime in a representation.
- **Twin primes:** counts up to  $x$ , normalized by  $x/(\log x)^2$  (heuristics).
- **Legendre:** check prime gaps near squares; visualize primes in intervals  $[n^2, (n + 1)^2]$ .
- **$n^2 + 1$  primes:** count prime values of  $n^2 + 1$  for  $n \leq N$ ; compare against heuristic  $\sim C N / \log N$  (Bateman–Horn-style intuition).

#### Practical numerical caveats

- **Cost growth:** naive primality tests become slow as ranges grow; use a fast deterministic/probabilistic test (e.g. Miller–Rabin) and cache/sieve where possible.
- **Bias in small ranges:** for small  $N$ , “constants” in asymptotics are hard to see; show uncertainty bands or multiple scales.
- **Overflow:**  $n^2$  grows quickly; use Python integers and avoid intermediate floats.

## References

See *References*.

[Guy, 2004, Pintz, 2009, Wikipedia contributors, 2026]

### 3.6.20 Liouville function $\lambda(n)$ refresher

The Liouville function is a simple, oscillatory completely multiplicative function built from  $\Omega(n)$  (the total number of prime factors with multiplicity). See [Tenenbaum, 2015].

#### Definition

Define

$$\lambda(n) = (-1)^{\Omega(n)}.$$

So  $\lambda(n) = 1$  if  $n$  has an even total number of prime factors (with multiplicity), and  $\lambda(n) = -1$  otherwise.

Examples:

- $\lambda(1) = 1$
- $\lambda(2) = -1$
- $\lambda(4) = \lambda(2^2) = 1$
- $\lambda(12) = \lambda(2^2 \cdot 3) = -1$

#### Key properties

- **Completely multiplicative:**  $\lambda(ab) = \lambda(a)\lambda(b)$  for all  $a, b$ .
- Connected to the Möbius function, but without the “squarefree = 0” behavior.

#### Summatory function experiments

A standard experiment is the summatory function

$$L(x) = \sum_{n \leq x} \lambda(n),$$

which oscillates and has deep connections to primes and zeta-function methods.

#### Experiment ideas

- compare  $L(x)$  with a random walk
- compare  $\lambda(n)$  and  $\mu(n)$  on squarefree numbers
- visualize  $\lambda(n)$  as a  $\pm 1$  texture over the integer grid

### 3.6.21 Mersenne numbers and primes refresher

This page is a *beginner-friendly* refresher for experiments about **Mersenne numbers** and **Mersenne primes**.

#### Core definitions

The **Mersenne numbers** are the integers

$$M_n = 2^n - 1 \quad (n \in \mathbb{N}).$$

A **Mersenne prime** is a Mersenne number that is prime:

$$M_p \text{ is a Mersenne prime} \iff 2^p - 1 \text{ is prime.}$$

A key (easy) fact is that if  $2^n - 1$  is prime, then  $n$  must be prime.

So, in experiments, you typically only test  $M_p$  for *prime* exponents  $p$ . [contributors, 2025]

**Key theorem / test: Lucas–Lehmer (why Mersenne primes are “computable”)**

For an odd prime exponent  $p$ , define  $M_p = 2^p - 1$  and a sequence  $\{s_k\}$  by

$$s_0 = 4, \quad s_{k+1} = s_k^2 - 2.$$

Compute this sequence *modulo*  $M_p$  at each step, and then:

$$M_p \text{ is prime} \iff s_{p-2} \equiv 0 \pmod{M_p}.$$

This is the **Lucas–Lehmer test**. It’s the workhorse behind most practical Mersenne-prime checks (and historically central to GIMPS). [contributors, 2025, Mersenne Research, 2024]

**What experiments typically visualize**

- **Growth with the exponent:** number of bits / digits of  $M_p$  as  $p$  grows.
- **Prime vs. composite behavior:** the Lucas–Lehmer residue  $s_{p-2} \bmod M_p$  across many prime exponents.
- **Factor patterns for composites:** quickly finding small factors of  $M_p$  (to avoid running LLT when a trivial factor exists).
- **Connections to perfect numbers:** if  $M_p$  is prime, then  $2^{p-1}(2^p - 1)$  is an even perfect number (Euclid–Euler). [Caldwell, n.d.]

For curated sequences (lists of exponents / known values), OEIS is a convenient “ground truth” reference. [Inc., 2025, Inc., 2025]

**Practical numerical caveats**

- **Always reduce modulo  $M_p$  in Lucas–Lehmer.**  
If you don’t, the intermediate values explode in size (each squaring roughly doubles the bit-length).
- **Test the exponent first.**  
If  $p$  is composite,  $M_p$  is automatically composite, so there’s no point running LLT.
- **Huge integers are fine, but you must be intentional.**  
Python’s big integers won’t overflow, but performance depends on bit-length and on the efficiency of modular squaring.
- **Separate “demo scale” from “real scale.”**  
For educational experiments, keep  $p$  modest (e.g.,  $p \leq 10^5$  is already huge for pure Python LLT). For large exponents, you’d rely on specialized implementations and careful FFT-based multiplication.

**References**

See *References*.

[Caldwell, 2021, contributors, 2025, contributors, 2025, Inc., 2025, Inc., 2025]

**3.6.22 Möbius function  $\mu(n)$  and Mertens function  $M(x)$  refresher**

The Möbius function  $\mu(n)$  is the main tool for inversion over divisors. The Mertens function  $M(x) = \sum_{n \leq x} \mu(n)$  is a classic “oscillatory” summatory function. See [Apostol, 1976], [Tenenbaum, 2015].

**Möbius function**

Define  $\mu(1) = 1$ . For  $n > 1$ :

- $\mu(n) = 0$  if  $n$  is divisible by the square of a prime (not squarefree)
- otherwise  $\mu(n) = (-1)^k$  where  $k$  is the number of distinct prime factors of  $n$

So for squarefree  $n = p_1 p_2 \cdots p_k$ ,

$$\mu(n) = (-1)^k.$$

### Möbius inversion (most important identity)

If

$$F(n) = \sum_{d|n} f(d),$$

then

$$f(n) = \sum_{d|n} \mu(d) F\left(\frac{n}{d}\right).$$

### Mertens function and a famous counterexample

The **Mertens function** is

$$M(x) = \sum_{n \leq x} \mu(n).$$

A historic conjecture was  $|M(x)| < \sqrt{x}$  for all  $x \geq 1$  (the **Mertens conjecture**). It was disproved by Odlyzko and te Riele. [Odlyzko and te Riele, 1985]

#### Why $M(x)$ is experiment-friendly

- $M(x)$  exhibits sign changes and irregular growth (“random walk-like” behavior).
- It is deeply connected to the Riemann zeta function and the Riemann hypothesis.

Typical experiment: plot  $M(x)$  for growing  $x$  and compare to envelopes like  $\pm\sqrt{x}$  or  $\pm x^{1/2} \log x$ .

#### Experiments in this repository

- **E105** — Mertens function  $M(x)$  at multiple scales (including rescaling views).

### 3.6.23 Prime-factor counting: $\omega(n)$ and $\Omega(n)$ refresher

Prime-factor counts are central examples of additive arithmetic functions. They are key in probabilistic number theory and in “typical behavior” experiments. See [Tenenbaum, 2015].

#### Definitions

Let  $n = \prod p_i^{\alpha_i}$ .

- $\omega(n)$ : number of **distinct** prime factors

$$\omega(n) = \#\{p : p \mid n\}.$$

- $\Omega(n)$ : number of prime factors **with multiplicity**

$$\Omega(n) = \sum_i \alpha_i.$$

Examples:

- $12 = 2^2 \cdot 3$  has  $\omega(12) = 2$  and  $\Omega(12) = 3$ .

#### Additivity

If  $\gcd(a, b) = 1$  then

$$\omega(ab) = \omega(a) + \omega(b),$$

and  $\Omega$  is even completely additive:

$$\Omega(ab) = \Omega(a) + \Omega(b) \quad \text{for all } a, b.$$

### Typical size: Erdős–Kac phenomenon

A famous theorem of Erdős and Kac says (informally) that  $\omega(n)$  behaves like a normal random variable with mean and variance  $\log \log n$  after normalization. [Erdős and Kac, 1940]

This is why histograms of  $\omega(n)$  over ranges often look Gaussian.

### Experiment ideas

- histogram of  $\omega(n)$  for  $n \leq N$  and compare to a normal curve
- compare  $\omega(n)$  vs.  $\log \log n$  (“normal order”)
- scatter  $\Omega(n)$  vs.  $\omega(n)$  to see multiplicities

### Experiments in this repository

- **E094** —  $(n)$  vs  $\Omega(n)$ : Erdős–Kac side-by-side (distribution + scaling).
- **E095** — Squarefree conditioning ( $(n) \mid 0$ ): forces  $(n) = \Omega(n)$ .
- **E122** — Heatmap atlas of  $\omega$ ,  $\Omega$  (visual textures / patterns).

## 3.6.24 Partition function $p(n)$ refresher

The partition function  $p(n)$  counts the number of ways to write  $n$  as a sum of positive integers, ignoring order (e.g.  $4 = 4 = 3 + 1 = 2 + 2 = 2 + 1 + 1 = 1 + 1 + 1 + 1$  gives  $p(4) = 5$ ). See [Andrews, 1984].

### Definition

A **partition** of  $n$  is a multiset of positive integers with sum  $n$ . The partition function  $p(n)$  is the number of such partitions.

Conventions:

- $p(0) = 1$  (empty partition)
- $p(n) = 0$  for  $n < 0$

### Generating function

The ordinary generating function is

$$\sum_{n \geq 0} p(n)q^n = \prod_{m \geq 1} \frac{1}{1 - q^m}.$$

This identity is the starting point for many algorithms and proofs.

### Growth

$p(n)$  grows very rapidly (roughly like  $\exp(C\sqrt{n})$ ), so experiments often use log-scales or focus on modular patterns.

### Experiment ideas

- compute  $p(n)$  for  $n \leq N$  via dynamic programming and plot  $\log p(n)$
- explore congruences (e.g. Ramanujan-type congruences)
- compare exact values to asymptotic approximations (later experiment)

## 3.6.25 Perfect numbers refresher

This page is a *beginner-friendly* refresher for experiments about **perfect numbers**. You only need basic number theory facts (divisors, primes) to follow it.

## Core definitions

For a positive integer  $n$ , let

- $d \mid n$  mean “ $d$  divides  $n$ ”,
- $\sigma(n) = \sum_{d \mid n} d$  be the **sum-of-divisors function**,
- $s(n) = \sum_{d \mid n, d < n} d = \sigma(n) - n$  be the sum of **proper** divisors.

A number  $n$  is **perfect** iff its proper divisors sum to itself:

$$n \text{ is perfect} \iff s(n) = n \iff \sigma(n) = 2n.$$

Examples:

- 6 is perfect because  $1 + 2 + 3 = 6$ .
- 28 is perfect because  $1 + 2 + 4 + 7 + 14 = 28$ .

## Key theorem: all even perfect numbers (Euclid–Euler)

A classic result completely characterizes **even** perfect numbers:

### Euclid–Euler theorem.

An integer  $n$  is an even perfect number **iff**

$$n = 2^{p-1}(2^p - 1)$$

where  $2^p - 1$  is prime (a **Mersenne prime**).

So every known perfect number is generated from a Mersenne prime exponent  $p$ . This is the main “generator” you’ll use in experiments. [Caldwell, n.d., Voight, 1998]

## Why $\sigma$ matters (multiplicativity)

If  $n$  factors as

$$n = \prod_{i=1}^k p_i^{a_i},$$

then

$$\sigma(n) = \prod_{i=1}^k \sigma(p_i^{a_i}) = \prod_{i=1}^k \frac{p_i^{a_i+1} - 1}{p_i - 1}.$$

This formula turns “sum all divisors” into a fast computation once you know the prime factorization.

## The big open question: odd perfect numbers

It is **unknown** whether any **odd** perfect numbers exist. A large literature proves *constraints* (congruences, size bounds, number of prime factors, etc.). Experiments can explore these constraints (and why they make brute-force search unrealistic). [Guy, 2004, Ochem and Rao, 2014, Stone, 2024]

## What experiments typically visualize

Typical “lab” questions you can turn into plots and tables:

- **Verification by computation:** compute  $\sigma(n)$  (via factorization) and check  $\sigma(n) = 2n$  for candidates.
- **Generator experiment:** produce even perfect numbers from known Mersenne exponents  $p$  and confirm perfection.
- **Growth:** number of digits / bit length of  $2^{p-1}(2^p - 1)$  as a function of  $p$ .



- **Divisor-function behavior:** compare  $\sigma(n)/n$  (abundancy index) for random  $n$  vs. perfect numbers.
- **Odd constraints (toy models):** test necessary conditions on odd  $n$  and see how restrictive they are.

For data (lists of perfect numbers and exponents), OEIS is a convenient reference. [OEIS Foundation Inc., 2025]

### Practical numerical caveats

Even with correct math, computation has a few traps:

- **Factorization dominates.** Computing  $\sigma(n)$  via divisors is slow unless you factor  $n$ . For large  $n$ , factorization becomes infeasible; prefer the Euclid–Euler generator for even perfect numbers.
- **Big integers are fine but expensive.** Python integers won’t overflow, but operations on huge numbers scale with the number of bits (so be mindful in loops and plotting).
- **Prime testing vs. proof.** Testing that  $2^p - 1$  is prime is nontrivial for large  $p$ . For experiments, use a curated list of known Mersenne prime exponents (e.g., from OEIS / GIMPS) rather than trying to discover new ones from scratch. [Mersenne Research, Inc. (GIMPS), 2024, Mersenne Research, Inc. (GIMPS), 2025]
- **Be explicit about definitions.** Some sources define “perfect” via  $\sigma(n) = 2n$ ; others via proper divisors. Use one convention consistently in code and docs.

### References

See *References*.

[Caldwell, n.d., Stone, 2024, Voight, 1998, OEIS Foundation Inc., 2025]

## 3.6.26 Pretentious number theory refresher

“Pretentious” number theory replaces some zero-driven arguments by measuring how closely a multiplicative function *pretends* to be a simple model (typically  $n^{it}$  or a Dirichlet character). It is especially effective for **comparative** experiments: *which model explains the data best?*

### Core definitions

Let  $f, g$  be multiplicative functions bounded by 1 in magnitude on primes. The **pretentious distance** up to  $x$  is  $\mathbb{D}(f, g; x)^2 = \sum_{p \leq x} \frac{1 - \operatorname{Re}(\overline{f(p)} g(p))}{p}$ .

Interpretation:

- $\mathbb{D}(f, g; x)$  small means  $f(p)$  and  $g(p)$  “agree” on most primes (in a  $1/p$ -weighted sense).
- $\mathbb{D}(f, g; x)$  large means  $f$  cannot be explained well by  $g$  on primes.

A common model family is  $g(n) = \chi(n)n^{it}$  with a Dirichlet character  $\chi$  and real  $t$ .

### What experiments usually visualize or measure

- Fit  $t$  to minimize  $\mathbb{D}(f, n^{it}; x)$  and plot the best-fit  $t(x)$ .
- Compare distances  $\mathbb{D}(f, \chi; x)$  across characters  $\chi$  to see which congruence structure matches  $f$ .
- Use the distance to explain why partial sums of  $f(n)$  may be large or small.

### Practical numerical caveats

- Always define the prime cutoff (use the same  $x$  for fair comparisons).
- Distance is dominated by small primes; if you want to study “large-prime behavior,” consider plotting contributions by prime ranges.

- When fitting  $t$ , the objective can have shallow minima; report robustness (e.g., multiple initial guesses).

## References

See *References*.

[Granville, 2009, Granville and Soundararajan, 2007]

## Experiments in this repository

- **E120** — Pretentious distance atlas for core multiplicative functions ( $\chi$ ,  $\chi$ ,  $\chi$ /n, ...).

### 3.6.27 Prime counting approximations: $\pi(x)$ , $\text{Li}(x)$ , and $R(x)$

The prime-counting function ( $\pi(x)$ ) counts primes up to  $(x)$ . Two standard smooth approximations are:

- the **logarithmic integral** ( $\mathrm{Li}(x)$ ), and
- **Riemann's R function** ( $R(x)$ ) (a Möbius-weighted combination of  $\mathrm{Li}(x^{1/n})$ ).

These approximations are central for numerical experiments around the Prime Number Theorem, error terms, and the visual “shape” of prime races.

## Key ideas

- **PNT baseline:**  $(\pi(x) \sim \frac{x}{\log x})$ , and  $(\mathrm{Li}(x))$  is often an excellent smooth baseline.
- **$R(x)$  as a refinement:**  $R(x)$  folds in prime-power information and is closely connected to explicit-formula viewpoints.
- **Error terms:** plotting  $(\pi(x) - \mathrm{Li}(x))$  or  $(\psi(x) - x)$  reveals oscillations and sign changes.

## References

See *References*.

[Titchmarsh, 1986, Wikipedia contributors, 2025]

### 3.6.28 Prime number races refresher

A **prime number race** compares prime counts in different residue classes, e.g.  $\left[\pi(x;4,3)\right]\text{ vs. }\left[\pi(x;4,1)\right]$ .

Although  $\pi(x; q, a) \sim \text{Li}(x)/\varphi(q)$  suggests “no long-term winner,” finite ranges often show persistent preferences (biases).

### The classic example: Chebyshev's bias mod 4

Empirically, for many ranges of  $x$  one observes  $[ \pi(x;4,3) > \pi(x;4,1), ]$  even though both are asymptotically equal.

Rubinstein–Sarnak analyze this phenomenon via the distribution of zeros of relevant  $L$ -functions (under standard hypotheses).

### What experiments usually visualize or measure

- The difference curve  $D(x) = \pi(x; q, a) - \pi(x; q, b)$  and its sign changes.
- The “race leaderboard” among several classes (e.g., mod 8 or mod 12).
- Histogram / empirical distribution of a normalized statistic sampled on a log-grid of  $x$  (choose a normalization and stick to it).

### Practical numerical caveats

- Bias claims depend on *how you sample  $x$*  (linear vs log-grid); be explicit.
- With small cutoffs (say  $x \leq 10^7$ ), you’ll see “apparent stability” that may later flip; that’s expected.
- Counting primes is the bottleneck; keep an eye on runtime and memory, and record parameters in your manifest.

### References

See *References*.

[Granville and Martin, 2006, Rubinstein and Sarnak, 1994]

### Experiments in this repository

- **E112** — Prime race curves  $(x;q,a) - (x;q,b)$ : sign changes and time-in-lead.

## 3.6.29 Prime numbers refresher

This page is a *beginner-friendly* refresher for experiments about **prime numbers**. It is intentionally broad: it covers the prime-related ideas most commonly explored with computation (distribution, sieves, primality testing, gaps, and prime patterns).

### Core definitions

#### Primes and composites

A positive integer  $p \geq 2$  is **prime** if its only positive divisors are 1 and  $p$ . If  $n \geq 2$  is not prime, it is **composite**.

Equivalently,

$$p \text{ is prime} \iff \forall a, b \in \mathbb{N} : p = ab \Rightarrow (a = 1 \text{ or } b = 1).$$

### Greatest common divisor and coprimality

The **greatest common divisor** of integers  $a, b$ , written  $\gcd(a, b)$ , is the largest positive integer dividing both. We say  $a$  and  $b$  are **coprime** if  $\gcd(a, b) = 1$ .

Coprimality appears constantly in modular arithmetic and in many prime-related proofs.

### Congruences (modular arithmetic)

For integers  $a, b$  and  $m \geq 2$ ,

$$a \equiv b \pmod{m}$$

means  $m$  divides  $a - b$ . Many prime experiments (residue classes, sieves, primality tests) are naturally expressed modulo  $m$ .

### Prime factorization (the “atomic” viewpoint)

The **Fundamental Theorem of Arithmetic** says every integer  $n \geq 2$  factors uniquely (up to ordering) as

$$n = \prod_{i=1}^k p_i^{\alpha_i},$$

where the  $p_i$  are distinct primes and the  $\alpha_i$  are positive integers. This makes primes the “building blocks” of the integers. [Hardy and Wright, 2008]

## Key quantitative language

### Infinitely many primes

Euclid’s classic argument shows there are infinitely many primes. This matters experimentally because it justifies questions like “how often do primes occur as numbers grow?” [Hardy and Wright, 2008]

### The prime counting function and the Prime Number Theorem

Let  $\pi(x)$  be the number of primes  $\leq x$ . The **Prime Number Theorem** (PNT) states

$$\pi(x) \sim \frac{x}{\log x} \quad (x \rightarrow \infty).$$

Informally: primes thin out, and the “typical” gap size near  $x$  is about  $\log x$ . [Apostol, 1976]

For experiments, two very common comparisons are:

- $\pi(x)$  vs.  $x / \log x$ ,
- $\pi(x)$  vs.  $\text{Li}(x)$  (the logarithmic integral), which is often a better approximation in practice. [Apostol, 1976, Rosser and Schoenfeld, 1962]

### Explicit bounds (useful sanity checks)

Many sources provide explicit inequalities for  $\pi(x)$ , the  $n$ th prime  $p_n$ , and related functions. These bounds are useful to validate computations and pick experiment ranges safely. [Axler, 2019, Dusart, 2018, Rosser and Schoenfeld, 1962]

## Computational building blocks

### Generating primes: sieves

If you need *all* primes up to a bound  $N$ , a sieve is usually best.

**Sieve of Eratosthenes (idea).** Start with a boolean array “possibly prime,” then repeatedly cross out multiples of each found prime. Time is roughly  $O(N \log \log N)$ , memory is  $O(N)$  booleans.

**Segmented sieve.** For large  $N$ , store only a window  $[L, R]$  at a time; this keeps memory bounded while preserving speed. Segmented sieves are often the right choice once  $N$  grows beyond typical RAM-friendly sizes. [Crandall and Pomerance, 2005]

### Testing primality (single numbers)

If you need to test the primality of individual (possibly large) integers, common strategies are:

- **Trial division** up to  $\sqrt{n}$  (great for small  $n$ , too slow for large),
- **Probabilistic tests** such as **Miller–Rabin**, fast and extremely reliable with good parameters, [Rabin, 1980]
- **Deterministic polynomial-time** primality testing (AKS), important theoretically but rarely used for practical large-number work. [Agrawal *et al.*, 2004]

In an experiment repo, it is common to use Miller–Rabin for speed and then either: (1) accept “probable prime” status (with a stated error bound) or (2) confirm with stronger checks for the sizes you care about. [Crandall and Pomerance, 2005]

### Factorization (often the bottleneck)

Many arithmetic functions become easy once a number is factored, but factoring large integers is hard. Even “toy” factorization methods (e.g., Pollard’s rho) make excellent experiments because performance varies wildly with input structure. [Crandall and Pomerance, 2005]

## Prime gaps and prime patterns

### Prime gaps

Let  $p_n$  be the  $n$ th prime. The **prime gap** sequence is

$$g_n = p_{n+1} - p_n.$$

Heuristically, “typical” gaps near  $p$  are size  $\log p$ , but gaps vary a lot. Two natural experimental viewpoints are:

- **local statistics**: histogram of  $g_n$  in a range,
- **record gaps**: the largest gap seen up to  $x$  (maximal gaps / first occurrences). [Caldwell, n.d., Cramér, 1936, Nicely, 1999]

A common normalization is  $g_n / \log p_n$ , which helps compare different scales.

### Twin, cousin, and sexy primes (prime pairs)

A **prime pair** is a pair of primes  $(p, p + d)$  with fixed even difference  $d$ :

- **twin primes**:  $d = 2$ ,
- **cousin primes**:  $d = 4$ ,
- **sexy primes**:  $d = 6$ .

These are all instances of “prime constellations.”

### Prime $k$ -tuples (constellations)

A **prime  $k$ -tuple** is a set of offsets  $\{h_1, \dots, h_k\}$  such that  $p + h_i$  are simultaneously prime. Hardy–Littlewood heuristics predict asymptotic counts for many such patterns (including twins). [Hardy and Littlewood, 1923]

Modern sieve breakthroughs show **bounded prime gaps** exist (there are infinitely many  $g_n$  below some constant), but we still do not know whether the twin prime conjecture is true. [Maynard, 2015, Zhang, 2014, D. H. J. Polymath, 2014]

## Primes in structure (residue classes and progressions)

### Residue classes

Primes (except 2) are odd, so they lie in residue classes modulo 2. More generally, you can study how primes distribute among residue classes modulo  $q$ . Experiments here include “prime races” and small biases in counts.

### Arithmetic progressions of primes

A classical result (Dirichlet’s theorem) says that if  $\gcd(a, q) = 1$ , then the arithmetic progression

$$a, a + q, a + 2q, \dots$$

contains infinitely many primes. A landmark modern result (Green–Tao) shows primes contain arbitrarily long arithmetic progressions. [Green and Tao, 2008]

For experiments, you can stay at an elementary level (searching for 3-term or 4-term prime APs in ranges) while still getting meaningful patterns.

### Other prime families you may include (optional)

These are often nice “side quests” that still fit under “Prime Numbers”:

- **Sophie Germain primes**:  $p$  prime and  $2p + 1$  prime,

- **safe primes:**  $q$  prime with  $(q - 1)/2$  prime,
- **Mersenne primes:**  $2^p - 1$  prime (you already have a dedicated page), [contributors, 2025]
- **palindromic / repunit primes:** “pattern-based” primes useful for search experiments.

### What experiments typically visualize

Typical “prime numbers lab” plots and tables include:

- **Prime density:**  $\pi(x)$  vs.  $x/\log x$ , and error curves  $\pi(x) - x/\log x$ . [Apostol, 1976]
- **Bounds and approximations:** compare  $\pi(x)$  with explicit inequalities (sanity checks). [Dusart, 2018, Rosser and Schoenfeld, 1962]
- **Sieve scaling:** runtime/memory vs.  $N$  for classic vs. segmented sieves. [Crandall and Pomerance, 2005]
- **Gaps:** histograms of  $g_n$ , normalized gaps  $g_n/\log p$ , record gaps vs.  $\log^2 x$ . [Cramér, 1936, Nicely, 1999]
- **Prime pairs:** counts of twin/cousin/sexy primes in  $[1, x]$ , compared to Hardy–Littlewood style heuristics. [Hardy and Littlewood, 1923]
- **Bounded gaps story:** “what bound was known when?” (Zhang  $\rightarrow$  Maynard  $\rightarrow$  Polymath) plus empirical gap data. [Maynard, 2015, Zhang, 2014, D. H. J. Polymath, 2014]
- **Primality tests:** speed vs. integer size (bits), and false-positive rates for weak tests vs. Miller–Rabin. [Agrawal *et al.*, 2004, Rabin, 1980]
- **Progressions:** frequency of 3-term prime APs in ranges; maximal AP length in a window. [Green and Tao, 2008]

For “ground truth” sequences and curated lists, OEIS and PrimePages are practical reference points. [Caldwell, n.d., OEIS Foundation Inc., 2025, OEIS Foundation Inc., 2025]

### Practical numerical caveats

- **Pick the right primitive:** if you need *many* primes up-to  $N$ , use a sieve; if you need primality of a few large numbers, use primality testing.
- **Memory matters for sieves:** a naive boolean list of length  $N$  can be huge; segmented sieves avoid this.
- **State what “prime” means in code:** if you use Miller–Rabin, your results are “probable primes” unless you add a deterministic guarantee for your size range. [Rabin, 1980]
- **Avoid floating-point traps:** use natural logs, guard against  $\log(0)$ , and remember that  $\log x$  changes slowly.
- **Be precise with pair-counting:** decide whether you count pairs by their smaller prime  $p$  (common), and avoid double-counting.
- **Separate discovery from verification:** for pattern-hunting (e.g., long APs), do fast screening first, then verify candidates carefully.

### References

See *References*.

[Agrawal *et al.*, 2004, Apostol, 1976, Cramér, 1936, Crandall and Pomerance, 2005, Dusart, 2018, Green and Tao, 2008, Hardy and Littlewood, 1923, Hardy and Wright, 2008, Maynard, 2015, Rabin, 1980, Rosser and Schoenfeld, 1962, Zhang, 2014, D. H. J. Polymath, 2014]

### 3.6.30 Primes in arithmetic progressions refresher

For integers  $q \geq 1$  and  $a$ , define  $\pi(x; q, a) := \#\{p \leq x : p \text{ prime and } p \equiv a \pmod{q}\}$ .

#### Core statements

##### Dirichlet’s theorem (existence)

If  $\gcd(a, q) = 1$ , then there are infinitely many primes  $p \equiv a \pmod{q}$ .

##### Prime number theorem in arithmetic progressions (equidistribution)

For fixed  $q$  and  $\gcd(a, q) = 1$ ,  $\pi(x; q, a) \sim \frac{\mathrm{Li}(x)}{\varphi(q)} \quad (x \rightarrow \infty)$ .

So, asymptotically, primes split evenly among the reduced residue classes.

#### What experiments usually visualize or measure

- Raw counts  $\pi(x; q, a)$  for small moduli  $q$  (2,3,4,5,8,10,12,...).
- Differences (races):  $\pi(x; q, a) - \pi(x; q, b)$ .
- “Leader changes”: values of  $x$  where the sign of the difference flips.

#### Practical numerical caveats

- Use a segmented sieve for  $\pi(x; q, a)$  if  $x$  gets large.
- For small  $x$ , visual phenomena are dominated by finite-size effects; that’s fine, but label plots honestly (e.g., “up to  $10^7$ ”).
- If you use  $\mathrm{Li}(x)$  for normalization, document the approximation you use.

#### References

See *References*.

[Davenport, 2000, Lejeune Dirichlet, 1837]

#### Experiments in this repository

- **E113** — First prime in each residue class (smallest  $p \equiv a \pmod{q}$  for all  $a \in (\mathbb{Z}/q\mathbb{Z})^\times$ ).

### 3.6.31 Primorials

The primorial of a prime  $p_k$  is

$$p_k\# = \prod_{i=1}^k p_i$$

(the product of the first  $k$  primes). Primorials are a natural “smoothing knob” in experimental number theory: they amplify small prime structure and appear in constructions like Euclid numbers  $p_k\# \pm 1$ . [Prime Pages (UTM), 2025, The OEIS Foundation Inc., 2025]

#### Key facts

- **Growth:**  $\log(p_k\#) = \sum_{i \leq k} \log p_i$ ; by the prime number theorem this is asymptotic to  $p_k$  (in a coarse sense). [Hardy and Wright, 2008]
- **Euclid numbers:**  $p_k\# + 1$  is coprime to all primes  $\leq p_k$ , but is usually composite (so: “Euclid’s proof does *not* generate primes”). [Prime Pages (UTM), 2025]

### What to experiment with

- **Euclid-number factorization:** For  $E_k = p_k \# \pm 1$ , try to find small factors and compare  $+1$  vs.  $-1$ .
- **Primorial wheels:** Use primorials to build “wheel sieves” and benchmark against a simple sieve.
- **Primorial primes / plus/minus primes:** Scan for primes of the form  $p_k \# \pm 1$  and visualize rarity.

### References

See Prime Pages (UTM) [2025], The OEIS Foundation Inc. [2025], Wikipedia contributors [2025].

### 3.6.32 Quadratic polynomials (algebraic) refresher

A **quadratic polynomial** over a field  $F$  is a degree-2 polynomial

$$f(x) = ax^2 + bx + c, \quad a \neq 0, \quad a, b, c \in F.$$

Over  $F = \mathbb{R}$  or  $\mathbb{C}$  this is the familiar object from school algebra, but many experiments treat  $f$  over other fields (like  $\mathbb{Q}$  or finite fields  $\mathbb{F}_p$ ). [Dummit and Foote, 2004, Wikipedia contributors, 2026]

#### Completing the square

A standard normal form is obtained by completing the square:

$$ax^2 + bx + c = a \left( x + \frac{b}{2a} \right)^2 - \frac{b^2 - 4ac}{4a}.$$

This shows that (over a field of characteristic  $\neq 2$ ) every quadratic is a shifted/scaled square plus a constant term.

#### Discriminant and factorization

The **discriminant** is

$$\Delta = b^2 - 4ac.$$

Over a field where square roots make sense, the roots are

$$x = \frac{-b \pm \sqrt{\Delta}}{2a}.$$

Algebraically, this means:

- $f$  **factors** over  $F$  iff  $\Delta$  is a square in  $F$  (when  $\text{char}(F) \neq 2$ ).
- If  $\Delta$  is not a square, the polynomial is **irreducible** over  $F$ .

#### Finite field case ( $\mathbb{F}_p$ , odd prime $p$ )

Over  $\mathbb{F}_p$ , “ $\Delta$  is a square” means  $\Delta$  is a quadratic residue mod  $p$ . This is the bridge between algebra and number theory: quadratic residues, Legendre symbols, and Gauss sums.

#### Why quadratics show up in prime experiments

Quadratics are the simplest non-linear polynomials, so they are a natural testbed for “prime values of polynomials”:

- Example:  $n^2 + 1$  (Landau’s 4th problem).
- Example:  $n^2 + n + 41$  (Euler’s famous prime-producing streak).

Even when a polynomial produces many primes early on, modular obstructions (e.g., always divisible by some prime for some residue class) inevitably appear.



### Practical numerical caveats

- **Stable quadratic formula:** over floating point numbers, use a stable variant when  $b^2 \gg 4ac$  to avoid catastrophic cancellation.
- **Exact arithmetic:** for modular experiments, keep everything as integers mod  $p$ ; don't mix floats.
- **Normalization:** many plots look cleaner after shifting/scaling to the monic form  $x^2 + Bx + C$  (when working over a field with  $a^{-1}$ ).

### References

See *References*.

[Dummit and Foote, 2004, Wikipedia contributors, 2026]

## 3.6.33 Riemann zeta function ( $\zeta(s)$ )

The **Riemann zeta function** ( $\zeta(s)$ ) is the Dirichlet series  $\zeta(s) = \sum_{n \geq 1} \frac{1}{n^s}$ , which converges for ( $\operatorname{Re}(s) > 1$ ) and extends (by analytic continuation) to a meromorphic function on ( $\mathbb{C}$ ) with a single simple pole at ( $s=1$ ).

### Key ideas

- **Euler product (primes):** for ( $\operatorname{Re}(s) > 1$ ),  $\zeta(s) = \prod_{p \text{ prime}} \frac{1}{1 - p^{-s}}$ , linking ( $\zeta(s)$ ) directly to prime distribution.
- **Functional equation:** ( $\zeta(s)$ ) satisfies a symmetry relating ( $\zeta(s)$ ) and ( $\zeta(1-s)$ ), which is central for studying zeros.
- **Zeros:** ( $\zeta(s)$ ) has *trivial zeros* at negative even integers and *nontrivial zeros* in the critical strip ( $0 < \operatorname{Re}(s) < 1$ ), conjecturally all on the critical line ( $\operatorname{Re}(s) = \frac{1}{2}$ ) (Riemann Hypothesis).

### Why it matters in this project

Many “analytic prime number theory” numerics (prime counting approximations, explicit formulas, prime races, etc.) are most naturally expressed in terms of ( $\zeta(s)$ ), its logarithmic derivative, and its zeros.

### Experiments in this repository

- **E117** —  $\gamma$ -factor functional-equation consistency checks on a grid of  $s$  values.
- **E118** — Partial Euler product for ( $\zeta(s)$ ): where it breaks as  $\operatorname{Re}(s)$  decreases.

### References

See *References*.

[Edwards, 1974, Ivić, 1985, Odlyzko, 1992, Titchmarsh, 1986, Wikipedia contributors, 2025]

## 3.6.34 Semiprimes

A semiprime is an integer that is the product of two primes (not necessarily distinct), i.e.

$$n = pq \quad (p, q \text{ prime}).$$

Semiprimes sit at the center of practical factorization hardness and are the basic objects behind RSA-style moduli. [Rivest *et al.*, 1978, The OEIS Foundation Inc., 2025]

### Key facts

- **Sequence:** The semiprimes form OEIS sequence A001358. [The OEIS Foundation Inc., 2025]
- **RSA context:** RSA uses  $N = pq$  and the difficulty of factoring  $N$  (when  $p, q$  are large and random-looking). [Rivest *et al.*, 1978]

- **Special cases:**  $p = q$  yields squares of primes;  $p \neq q$  yields “RSA-like” semiprimes.

#### What to experiment with

- **Distribution:** Count semiprimes up to  $X$  and compare empirical growth with simple heuristic approximations.
- **Factorization methods:** Benchmark trial division, Pollard’s rho, and (for larger sizes) ECM bindings if you later allow optional deps.
- **RSA-style generation:** Generate balanced semiprimes ( $p, q$  same bitlength) and compare factorization difficulty vs unbalanced ones.
- **Smoothness:** Explore how often  $p - 1$  or  $q - 1$  is smooth (motivates  $p - 1$  factoring methods).

#### References

See Rivest *et al.* [1978], The OEIS Foundation Inc. [2025].

### 3.6.35 Taylor series refresher

This page is a *beginner-friendly* refresher for experiments that use Taylor polynomials. You only need basic calculus (derivatives) to follow it.

#### Taylor polynomial

Assume  $f$  has enough derivatives near  $x_0$  (this is true for  $\sin(x)$ ,  $\cos(x)$ , polynomials, exponentials, etc.). The Taylor polynomial of degree  $n$  around  $x_0$  is

$$T_n(x; x_0) = \sum_{k=0}^n \frac{f^{(k)}(x_0)}{k!} (x - x_0)^k.$$

**Intuition:**  $T_n(x; x_0)$  is the polynomial that matches  $f(x_0)$  and the first  $n$  derivatives at  $x_0$ . It is usually accurate when  $x$  is close to  $x_0$ .

For  $f(x) = \sin(x)$ , the derivatives cycle, and around  $x_0 = 0$  this becomes

$$\sin(x) \approx x - \frac{x^3}{3!} + \frac{x^5}{5!} - \cdots.$$

#### Truncation error and the remainder

The approximation error is the remainder

$$R_n(x; x_0) = f(x) - T_n(x; x_0).$$

Under standard conditions, Taylor’s theorem gives a remainder representation. One common form is the (Lagrange) remainder:

$$R_n(x; x_0) = \frac{f^{(n+1)}(\xi)}{(n+1)!} (x - x_0)^{n+1} \quad \text{for some } \xi \text{ between } x \text{ and } x_0.$$

This is the key qualitative message for experiments like E001:

- the factor  $(x - x_0)^{n+1}$  makes the method **local** (good near  $x_0$ , potentially bad far away),
- increasing  $n$  helps most where  $|x - x_0|$  is small.

#### What experiments typically visualize

In a numerical experiment, you often look at

- absolute error:  $|R_n(x; x_0)|$
- relative error:  $|R_n(x; x_0)|/|f(x)|$  (careful near zeros of  $f$ )

and plot them across a domain to see where the approximation is reliable.

### Practical numerical caveats

Even when the mathematics are correct, computation can mislead:

- large  $|x - x_0|$  and high  $n$  can produce huge intermediate terms,
- subtractive cancellation can reduce accuracy,
- floating-point rounding can dominate before the theoretical truncation error does.

A common “extension” experiment is to repeat the same plots using higher precision arithmetic to separate *truncation error* from *rounding error*.

### Introductory reading

If you want a longer, *beginner-friendly* treatment (beyond this refresher), these are good starting points:

- A quick overview / definitions and examples: [Wikipedia contributors, 2025].
- A rigorous calculus textbook with a clean presentation of Taylor’s theorem and remainders: [Apostol, 1991].
- A proof-oriented classic (slower, deeper): [Spivak, 2008].
- For the numerical viewpoint (truncation vs. rounding error): [Burden *et al.*, 2015].

### References

See *References*.

[Apostol, 1991, Burden *et al.*, 2015, Spivak, 2008, Wikipedia contributors, 2025]

## 3.6.36 von Mangoldt $\Lambda(n)$ and Chebyshev functions refresher

The von Mangoldt function concentrates on prime powers and is central in analytic number theory. Chebyshev functions summarize prime distributions and are used in proofs of the Prime Number Theorem. See [Montgomery and Vaughan, 2006] and [Tenenbaum, 2015].

### von Mangoldt function

Define

$$\Lambda(n) = \begin{cases} \log p, & \text{if } n = p^k \text{ for a prime } p \text{ and } k \geq 1, \\ 0, & \text{otherwise.} \end{cases}$$

So  $\Lambda(n)$  “picks out” primes and prime powers.

### Chebyshev functions

Two common Chebyshev functions:

- $\theta(x) = \sum_{p \leq x} \log p$
- $\psi(x) = \sum_{n \leq x} \Lambda(n)$

Since  $\psi$  sums  $\log p$  over prime powers, it is smoother and often easier in analysis.

### Why these matter

- $\psi(x) \sim x$  is essentially equivalent to the Prime Number Theorem.
- $\Lambda$  is tied to the logarithmic derivative of  $\zeta(s)$  (via Dirichlet series).

### Experiment ideas

- plot  $\psi(x)$  and compare to  $x$
- plot  $\theta(x)$  and compare to  $x$
- compare contributions from primes vs. higher prime powers in  $\psi(x)$

### Experiments in this repository

- **E103** — Chebyshev  $\chi(x)$  jumps at prime powers ( $\Lambda$  support).
- **E104** — Von Mangoldt  $\Lambda$  statistics (support, value distribution, summatory behavior).
- **E119** —  $\chi(x) - x$  oscillations (prime-weighted deviation view).

### 3.6.37 Wieferich primes

A Wieferich prime (base 2) is a prime  $p$  such that

$$[2^{p-1} \equiv 1 \pmod{p^2}].$$

Only two are currently known: 1093 and 3511. [The OEIS Foundation Inc., 2025, Wikipedia contributors, 2025]

#### Key facts

- **Origin (Fermat’s Last Theorem, first case):** Wieferich proved in 1909 that if the first case of FLT fails for an odd prime exponent  $p$ , then  $p$  must be a Wieferich prime (base 2). [Wieferich, 1909]
- **Rarity:** Heuristics suggest Wieferich primes are very sparse; whether infinitely many exist is open. [Wikipedia contributors, 2025]
- **Generalization:** One can define Wieferich primes for any base  $a$  via  $a^{p-1} \equiv 1 \pmod{p^2}$ . [Prime Pages (UTM), 2025]

#### What to experiment with

- **Fast search:** Implement modular exponentiation (pow with mod) to test primes up to a bound and recover 1093, 3511.
- **Near-Wieferich primes:** Measure the  $p$ -adic valuation  $v_p(2^{p-1} - 1)$  and look for unusually large values.
- **Connections:** Explore overlaps with other “rare prime” phenomena (e.g., Wall–Sun–Sun, Wilson) as *tags* rather than hard coupling.

#### References

See Wieferich [1909], The OEIS Foundation Inc. [2025], Wikipedia contributors [2025].

## 3.7 PDF download

A PDF build of this documentation is generated by GitHub Actions and published alongside the HTML site.

- Download PDF

If the link is missing, it usually means you are viewing an older deployment or the PDF build step failed.

## 3.8 Experiment Status

This page tracks the generation and manual editing status of experiment pages.

| Experiment                           | Generated  | Last edited | Notes |
|--------------------------------------|------------|-------------|-------|
| e001_taylor_error_landscapes         | 31.12.2025 |             |       |
| e002_even_perfect_growth             | 31.12.2025 |             |       |
| e003_abundancy_index_landscape       | 31.12.2025 |             |       |
| e004_sigma_benchmark                 | 31.12.2025 |             |       |
| e005_odd_perfect_filter_pipeline     | 31.12.2025 |             |       |
| e006_near_misses                     | 31.12.2025 |             |       |
| e007_mersenne_growth                 | 31.12.2025 |             |       |
| e008_lucas_lehmer_scan               | 31.12.2025 |             |       |
| e009_small_factor_scan               | 31.12.2025 |             |       |
| e010_perfect_numbers_from_mersenne   | 31.12.2025 |             |       |
| e011_mersenne_prime_heuristic        | 31.12.2025 |             |       |
| e012_fermat_pseudoprimes             | 31.12.2025 |             |       |
| e013_prime_polynomial_counterexample | 31.12.2025 |             |       |
| e014_primorial_pm1_counterexamples   | 31.12.2025 |             |       |
| e015_wilson_test_infeasibility       | 31.12.2025 |             |       |
| e016_trial_division_vs_mr            | 31.12.2025 |             |       |
| e017_sieve_memory_segmented          | 31.12.2025 |             |       |
| e018_mr_base_choice_counterexamples  | 31.12.2025 |             |       |
| e019_prime_density_pnt               | 31.12.2025 |             |       |
| e020_pi_vs_li_numeric                | 31.12.2025 |             |       |
| e021_pi_explicit_bounds_sanity       | 31.12.2025 |             |       |
| e022_prime_race_mod4                 | 31.12.2025 |             |       |
| e023_residue_classes_modq            | 31.12.2025 |             |       |
| e024_ulam_spiral                     | 31.12.2025 | 01.01.2026  |       |
| e025_prime_gaps_nonmonotone          | 31.12.2025 |             |       |
| e026_normalized_prime_gaps           | 31.12.2025 |             |       |
| e027_record_gaps_vs_log2             | 31.12.2025 |             |       |
| e028_jumping_champions               | 31.12.2025 |             |       |
| e029_twin_primes_heuristic           | 31.12.2025 |             |       |
| e030_cousinSexy_pairs                | 31.12.2025 |             |       |
| e031_admissibility_mod_obstructions  | 31.12.2025 |             |       |
| e032_triplets_quadruplets_counts     | 31.12.2025 |             |       |
| e033_bounded_gaps_vs_twins           | 31.12.2025 |             |       |
| e034_twin_window_variance            | 31.12.2025 |             |       |
| e035_primes_in_ap                    | 31.12.2025 |             |       |
| e036_prime_ap_search                 | 31.12.2025 |             |       |
| e037_prime_free_intervals_factorial  | 31.12.2025 |             |       |
| e038_bertrand_postulate              | 31.12.2025 |             |       |
| e039_sophie_germain_safe_primes      | 31.12.2025 |             |       |
| e040_palindromic_primes_11           | 31.12.2025 |             |       |
| e041_fermat_numbers_counterexample   | 31.12.2025 |             |       |
| e042_repunit_primes_scan             | 31.12.2025 |             |       |
| e043_pollard_rho_variability         | 31.12.2025 |             |       |
| e044_ss_vs_mr                        | 31.12.2025 |             |       |
| e045_mr_deterministic_64bit          | 31.12.2025 |             |       |
| e046_prime_testing_pipeline          | 31.12.2025 |             |       |
| e047_fermat_numbers                  | 31.12.2025 |             |       |
| e048_carmichael_numbers              | 31.12.2025 |             |       |
| e049_wieferich_primes                | 31.12.2025 |             |       |
| e050_primorials_euclid_numbers       | 31.12.2025 |             |       |
| e051_semiprimes_factorization        | 31.12.2025 |             |       |

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| Experiment                                | Generated  | Last edited | Notes |
|-------------------------------------------|------------|-------------|-------|
| e052_totient_ratio_landscape              | 31.12.2025 |             |       |
| e053_inverse_totient_multiplicity         | 31.12.2025 |             |       |
| e054_mobius_squarefree_density            | 31.12.2025 |             |       |
| e055_mertens_random_walk                  | 31.12.2025 |             |       |
| e056_liouville_vs_mertens                 | 31.12.2025 |             |       |
| e057_erdos_kac_omega_distribution         | 31.12.2025 |             |       |
| e058_divisor_count_records                | 31.12.2025 |             |       |
| e059_abundancy_sigma_over_n               | 31.12.2025 |             |       |
| e060_jordan_totient_family                | 31.12.2025 |             |       |
| e061_chebyshev_psi_prime_powers           | 31.12.2025 |             |       |
| e062_carmichael_lambda_vs_phi             | 31.12.2025 |             |       |
| e063_dirichlet_convolution_identities     | 31.12.2025 |             |       |
| e064_dirichlet_character_tables           | 31.12.2025 |             |       |
| e065_dirichlet_orthogonality              | 31.12.2025 |             |       |
| e066_character_partial_sums               | 31.12.2025 |             |       |
| e067_gauss_sums                           | 31.12.2025 |             |       |
| e068_lseries_vs_euler_product             | 31.12.2025 |             |       |
| e069_l1_convergence                       | 31.12.2025 |             |       |
| e070_primes_in_progressions_counts        | 31.12.2025 |             |       |
| e071_progressions_pnt_error               | 31.12.2025 |             |       |
| e072_prime_race_mod4                      | 31.12.2025 |             |       |
| e073_prime_race_mod3                      | 31.12.2025 |             |       |
| e074_prime_race_mod8_leaderboard          | 31.12.2025 |             |       |
| e075_race_statistic_distribution          | 31.12.2025 |             |       |
| e076_theta_in_progressions                | 31.12.2025 |             |       |
| e077_indicator_via_characters             | 31.12.2025 |             |       |
| e078_max_character_sums                   | 31.12.2025 |             |       |
| e079_primitive_characters_conductors      | 31.12.2025 |             |       |
| e080_bias_fraction_curve                  | 31.12.2025 |             |       |
| e081_sign_changes_table                   | 31.12.2025 |             |       |
| e082_zeta_series_convergence              | 31.12.2025 |             |       |
| e083_series_vs_euler_product              | 31.12.2025 |             |       |
| e084_critical_line_magnitude              | 31.12.2025 |             |       |
| e085_eta_acceleration                     | 31.12.2025 |             |       |
| e086_hardy_Z_near_zeros                   | 31.12.2025 |             |       |
| e087_gram_points                          | 31.12.2025 |             |       |
| e088_zero_counting_rvm                    | 31.12.2025 |             |       |
| e089_zeta_heatmap_logabs                  | 31.12.2025 |             |       |
| e090_functional_equation_residual         | 31.12.2025 |             |       |
| e091_euler_product_critical_line          | 31.12.2025 |             |       |
| e092_mobius_series_inverse_zeta           | 31.12.2025 |             |       |
| e093_log_derivative_von_mangoldt          | 31.12.2025 |             |       |
| e094_omega_vs_bigomega                    | 31.12.2025 |             |       |
| e095_squarefree_omega_equals_bigomega     | 31.12.2025 |             |       |
| e096_tau_record_holders                   | 31.12.2025 |             |       |
| e097_sigma_over_n_abundance               | 31.12.2025 |             |       |
| e098_sigma_over_n_alpha_extremals         | 31.12.2025 |             |       |
| e099_jordan_totient_atlas                 | 31.12.2025 |             |       |
| e100_carmichael_lambda_vs_phi             | 31.12.2025 |             |       |
| e101_reduced_residues_structure           | 31.12.2025 |             |       |
| e102_convolution_identities               | 31.12.2025 |             |       |
| e103_chebyshev_psi_prime_powers           | 31.12.2025 |             |       |
| e104_von_mangoldt_statistics              | 31.12.2025 |             |       |
| e105_mertens_function_scaling             | 31.12.2025 |             |       |
| e106_dirichlet_characters_real_vs_complex | 31.12.2025 |             |       |

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| Experiment                           | Generated  | Last edited | Notes |
|--------------------------------------|------------|-------------|-------|
| e107_dirichlet_conductor_histogram   | 31.12.2025 |             |       |
| e108_dirichlet_orthogonality_heatmap | 31.12.2025 |             |       |
| e109_gauss_sums_prime_modulus        | 31.12.2025 |             |       |
| e110_L_series_partial_sums           | 31.12.2025 |             |       |
| e111_L_euler_product_vs_series       | 31.12.2025 |             |       |
| e112_prime_race_modq                 | 31.12.2025 |             |       |
| e113_first_prime_per_residue         | 31.12.2025 |             |       |
| e114_zeta_eta_truncation_error       | 31.12.2025 |             |       |
| e115_hardy_Z_sign_changes_scan       | 31.12.2025 |             |       |
| e116_zero_count_vs_rvm               | 31.12.2025 |             |       |
| e117_functional_equation_check       | 31.12.2025 |             |       |
| e118_euler_product_breakdown         | 31.12.2025 |             |       |
| e119_psi_minus_x_smoothed            | 31.12.2025 |             |       |
| e120_pretentious_distance_proxy      | 31.12.2025 |             |       |
| e121_multiplicativity_stress_tests   | 31.12.2025 |             |       |
| e122_arithmetic_function_atlas       | 31.12.2025 |             |       |
| e123_arithmetic_correlation_matrix   | 31.12.2025 |             |       |
| e124_klauber_triangle                | 01.01.2026 | 01.01.2026  |       |
| e125_sacks_spiral                    | 01.01.2026 | 01.01.2026  |       |
| e126_hexagonal_number_spiral         | 01.01.2026 | 01.01.2026  |       |
| e127_quadratic_prime_streak_atlas    | 02.01.2026 | 02.01.2026  |       |
| e128_quadratic_modular_obstructions  | 02.01.2026 | 02.01.2026  |       |
| e129_euler_lucky_constants           | 02.01.2026 | 02.01.2026  |       |

## 3.9 References





## BIBLIOGRAPHY

- [Expa] Experimental mathematics (project euclid archive). URL: <https://projecteuclid.org/journals/experimental-mathematics> (visited on 2025-12-22).
- [Expb] Experimental mathematics lab (university of luxembourg). URL: <https://math.uni.lu/eml/> (visited on 2025-12-22).
- [Expc] Experimental mathematics website. URL: <https://www.experimentalmath.info> (visited on 2025-12-22).
- [Exp92] Experimental mathematics (journal). 1992. URL: <https://www.tandfonline.com/journals/uexm20>.
- [AKS04] Manindra Agrawal, Neeraj Kayal, and Nitin Saxena. PRIMES is in P. *Annals of Mathematics*, 160(2):781–793, 2004. URL: <https://annals.math.princeton.edu/2004/160-2/p12> (visited on 2025-12-27), doi:10.4007/annals.2004.160.781.
- [AErdHos44] L. Alaoglu and Paul Erdős. On highly composite and similar numbers. *Transactions of the American Mathematical Society*, 56(3):448–469, 1944. doi:10.1090/S0002-9947-1944-0011087-2.
- [AGP94] W. R. Alford, Andrew Granville, and Carl Pomerance. There are infinitely many carmichael numbers. *Annals of Mathematics*, 139(3):703–722, 1994. URL: <https://annals.math.princeton.edu/1994/139-3/p10>.
- [And84] George E. Andrews. *The Theory of Partitions*. Volume 2 of Encyclopedia of Mathematics and its Applications. Cambridge University Press, 1984. ISBN 978-0-521-63766-4. doi:10.1017/CBO9780511608650.
- [Apo76] Tom M. Apostol. *Introduction to Analytic Number Theory*. Springer, 1976. URL: <https://link.springer.com/book/10.1007/978-1-4757-5579-4> (visited on 2025-12-27), doi:10.1007/978-1-4757-5579-4.
- [Apo91] Tom M. Apostol. *Calculus, Volume 1*. John Wiley & Sons, 2 edition, 1991. ISBN 9780471000051. URL: [https://books.google.com/books/about/Calculus\TU\textbackslashslash\}\\_\\\_Volume\TU\textbackslashslash\}\\_\\\_1.html?id=o2D4DwAAQBAJ](https://books.google.com/books/about/Calculus\TU\textbackslashslash\}_\_Volume\TU\textbackslashslash\}_\_1.html?id=o2D4DwAAQBAJ).
- [Arn15] Vladimir I. Arnold. *Experimental Mathematics*. MSRI Mathematical Circles Library. American Mathematical Society, 2015. ISBN 9780821894163. URL: <https://bookstore.ams.org/msri-13/> (visited on 2025-12-22).
- [AM14] Jeremy Avigad and Rebecca Morris. The concept of “character” in dirichlet’s theorem on primes in an arithmetic progression. *Archive for History of Exact Sciences*, 68(3):265–326, 2014. URL: <https://arxiv.org/abs/1209.3657> (visited on 2025-12-29), doi:10.1007/s00407-013-0126-0.
- [Axl19] Christian Axler. New estimates for the  $n$ th prime number. *Journal of Integer Sequences*, 22(4):Article 19.4.2, 2019. URL: <https://cs.uwaterloo.ca/journals/JIS/VOL22/Axler/axler17.pdf> (visited on 2025-12-27).

- [Axl24] Christian Axler. Effective estimates for some functions defined over primes. *Integers*, 24:A34, 2024. URL: <https://math.colgate.edu/~integers/y34/y34.pdf> (visited on 2025-12-27).
- [BB05] David H. Bailey and Jonathan M. Borwein. Experimental mathematics: examples, methods and implications. *Notices of the American Mathematical Society*, 52(5):502–514, 2005. URL: <https://www.ams.org/notices/200505/fea-borwein.pdf>.
- [BBC04] David H. Bailey, Jonathan M. Borwein, and Richard E. Crandall. Ten problems in experimental mathematics. *Experimental Mathematics*, 13(2):193–207, 2004.
- [BEW98] Bruce C. Berndt, Ronald J. Evans, and Kenneth S. Williams. *Gauss and Jacobi Sums*. Wiley-Interscience, 1998. ISBN 9780471128076.
- [BvdPSZ14] Jonathan Borwein, Alf van der Poorten, Jeffrey Shallit, and Wadim Zudilin. *Neverending Fractions: An Introduction to Continued Fractions*. Volume 23 of Australian Mathematical Society Lecture Series. Cambridge University Press, 2014. ISBN 9780521186490. URL: <https://www.cambridge.org/core/books/neverending-fractions/A3900DAB483D65CE6CB960A6B71226EE> (visited on 2025-12-22).
- [Bor05] Jonathan M. Borwein. The experimental mathematician: the pleasure of discovery and the role of proof. *International Journal of Computers for Mathematical Learning*, 10:75–108, 2005. doi:10.1007/s10758-005-6244-3.
- [Bor09] Jonathan M. Borwein. *The Crucible: An Introduction to Experimental Mathematics*. A K Peters, Wellesley, MA, USA, 2009. ISBN 9781568813438.
- [BB08] Jonathan M. Borwein and David H. Bailey. *Mathematics by Experiment: Plausible Reasoning in the 21st Century*. A K Peters, Wellesley, MA, USA, 2 edition, 2008. ISBN 9781568814421.
- [BBG04] Jonathan M. Borwein, David H. Bailey, and Roland Girgensohn. *Experimentation in Mathematics: Computational Paths to Discovery*. A K Peters, Natick, MA, USA, 2004. ISBN 9781568811369. doi:10.1201/9781439864197.
- [BBG+07] Jonathan M. Borwein, David H. Bailey, Roland Girgensohn, David Luke, and Victor H. Moll. *Experimental Mathematics in Action*. A K Peters, Wellesley, MA, USA, 2007. ISBN 9781568812717. URL: <https://carmamaths.org/resources/jon/Preprints/Books/EMA/ema.pdf> (visited on 2025-12-22).
- [BB04] Jonathan M. Borwein and Richard P. Brent. *Inquiries into Experimental Mathematics*. A K Peters, Wellesley, MA, USA, 2004. ISBN 9781568812113.
- [Bro19] Dirk Brockmann. Prime time: the distribution of primes along number spirals. Online, 2019. Explorable explanation including the Klauber triangle and Sacks spiral definitions. URL: <https://www.complexity-explorables.org/explorables/prime-time/> (visited on 2026-01-01).
- [BFB15] Richard L. Burden, J. Douglas Faires, and Annette M. Burden. *Numerical Analysis*. Cengage Learning, 10 edition, 2015. ISBN 9781305253667. URL: <https://www.cengage.com/c/numerical-analysis-10e-faires/9781305253667/>.
- [Cald.a] Chris Caldwell. Characterizing all even perfect numbers. n.d. PrimePages (The Prime Database), proof note on the Euclid–Euler characterization. URL: <https://t5k.org/notes/proofs/EvenPerfect.html> (visited on 2025-12-25).
- [Cal21] Chris K. Caldwell. Mersenne primes: history, theorems and lists. Online, 2021. PrimePages reference page collecting history, key theorems, and curated tables related to Mersenne primes and perfect numbers. URL: <https://primes.utm.edu/mersenne/> (visited on 2025-12-27).
- [Cald.b] Chris K. Caldwell. Prime gaps and related records. Online, n.d. PrimePages (The Prime Database), curated notes and links on prime gaps and record computations. URL: <https://t5k.org/notes/primegaps.html> (visited on 2025-12-27).
- [Car12] R. D. Carmichael. On composite numbers  $p$  which satisfy the fermat congruence  $a^{p-1} \equiv 1 \pmod{p}$ . *American Mathematical Monthly*, 19(2):22–27, 1912. URL: <https://www.jstor.org/stable/2974142>.
- [Con04] Keith Conrad. Korselt's criterion for carmichael numbers. <https://kconrad.math.uconn.edu/blurbs/ugradnumthy/carmichael.pdf>, 2004. Online lecture note; accessed 2025-12-28.

- [con25a] Wikipedia contributors. Lucas–lehmer primality test — wikipedia, the free encyclopedia. Online, 2025. Revision as of 01:16, 31 October 2025. Abstract: Description and correctness of the Lucas–Lehmer test used to prove primality of Mersenne numbers  $M_p = 2^p - 1$ . URL: [https://en.wikipedia.org/w/index.php?oldid=1319643028&title=Lucas%E2%80%93Lehmer\\_primality\\_test](https://en.wikipedia.org/w/index.php?oldid=1319643028&title=Lucas%E2%80%93Lehmer_primality_test) (visited on 2025-12-27).
- [con25b] Wikipedia contributors. Lucky numbers of euler. 2025. Wikipedia, revision as of 2025-01-03 (permanent link). URL: [https://en.wikipedia.org/w/index.php?oldid=1267053719&title=Lucky\\_numbers\\_of\\_Euler](https://en.wikipedia.org/w/index.php?oldid=1267053719&title=Lucky_numbers_of_Euler) (visited on 2026-01-02).
- [con25c] Wikipedia contributors. Mersenne prime — wikipedia, the free encyclopedia. Online, 2025. Revision as of 20:10, 11 December 2025. Abstract: Overview of Mersenne primes (numbers of the form  $2^p - 1$ ), their connection to perfect numbers, and known results and records. URL: [https://en.wikipedia.org/w/index.php?oldid=1326946346&title=Mersenne\\_prime](https://en.wikipedia.org/w/index.php?oldid=1326946346&title=Mersenne_prime) (visited on 2025-12-27).
- [con25d] Wikipedia contributors. Ulam spiral — wikipedia, the free encyclopedia. Online, 2025. Revision as of 16:55, 25 October 2025. Visualization of primes arranged in a spiral; discusses prime-rich lines and links to quadratic polynomials. URL: [https://en.wikipedia.org/w/index.php?oldid=1318727705&title=Ulam\\_spiral](https://en.wikipedia.org/w/index.php?oldid=1318727705&title=Ulam_spiral) (visited on 2026-01-01).
- [Cramer36] Harald Cramér. On the order of magnitude of the difference between consecutive prime numbers. *Acta Arithmetica*, 2(1):23–46, 1936. URL: <https://www.impan.pl/en/publishing-house/journals-and-series/acta-arithmetica/all/2/1/93277/on-the-order-of-magnitude-of-the-difference-between-consecutive-prime-numbers> (visited on 2025-12-27), doi:10.4064/aa-2-1-23-46.
- [CP05] Richard Crandall and Carl Pomerance. *Prime Numbers: A Computational Perspective*. Springer, 2005. URL: <https://link.springer.com/book/10.1007/978-1-4684-9316-0> (visited on 2025-12-27), doi:10.1007/978-1-4684-9316-0.
- [Dav00] Harold Davenport. *Multiplicative Number Theory*. Volume 74 of Graduate Texts in Mathematics. Springer, New York, 3 edition, 2000. ISBN 978-0-387-95097-6. URL: <https://link.springer.com/book/10.1007/978-1-4757-5927-3> (visited on 2025-12-29), doi:10.1007/978-1-4757-5927-3.
- [DF04] David S. Dummit and Richard M. Foote. *Abstract Algebra*. John Wiley & Sons, 3 edition, 2004. ISBN 9780471452348.
- [Dus18] Pierre Dusart. Explicit estimates of some functions over primes. *The Ramanujan Journal*, 45(1):227–251, 2018. URL: <https://link.springer.com/article/10.1007/s11139-016-9839-4> (visited on 2025-12-27), doi:10.1007/s11139-016-9839-4.
- [Edw74] Harold M. Edwards. *Riemann's Zeta Function*. Princeton University Press, 1974. ISBN 9780691113852.
- [ErdHosK40] Paul Erdős and Mark Kac. The gaussian law of errors in the theory of additive number theoretic functions. *American Journal of Mathematics*, 62:738–742, 1940. doi:10.2307/2371483.
- [ErdHosPS91] Paul Erdős, Carl Pomerance, and Eric Schmutz. Carmichael's lambda function. *Acta Arithmetica*, 58(4):363–385, 1991. URL: <http://eudml.org/doc/206359>.
- [FI10] John Friedlander and Henryk Iwaniec. *Opera de Cribro*. Volume 57 of Colloquium Publications. American Mathematical Society, 2010. URL: <https://bookstore.ams.org/coll-57/> (visited on 2025-12-27), doi:10.1090/coll/057.
- [Gar83] M. Gardner. *Martin Gardner's Sixth Book of Mathematical Diversions from Scientific American*. Scientific American. University of Chicago Press, 1983. ISBN 9780226282503. URL: <https://books.google.ch/books?id=fUViQgAACAAJ>.
- [Gra09] Andrew Granville. Pretentiousness in analytic number theory. *Journal de Théorie des Nombres de Bordeaux*, 21(1):159–173, 2009. doi:10.5802/jtnb.664.

- [GM06] Andrew Granville and Greg Martin. Prime number races. *The American Mathematical Monthly*, 113(1):1–33, 2006. URL: <https://arxiv.org/abs/math/0408319> (visited on 2025-12-29), doi:10.1080/00029890.2006.11920275.
- [GS07] Andrew Granville and Kannan Soundararajan. Large character sums: pretentious characters and the Pólya–Vinogradov theorem. *Journal of the American Mathematical Society*, 20(2):357–384, 2007. doi:10.1090/S0894-0347-06-00549-1.
- [GT08] Ben Green and Terence Tao. The primes contain arbitrarily long arithmetic progressions. *Annals of Mathematics*, 167(2):481–547, 2008. URL: <https://annals.math.princeton.edu/2008/167-2/p01> (visited on 2025-12-27), doi:10.4007/annals.2008.167.481.
- [Guy04] Richard K. Guy. *Unsolved Problems in Number Theory*. Springer, 3 edition, 2004. ISBN 9780387208602. URL: <https://link.springer.com/book/10.1007/978-0-387-26677-0>, doi:10.1007/978-0-387-26677-0.
- [HR74] Heini Halberstam and Hans-Egon Richert. *Sieve Methods*. Academic Press, 1974. URL: <https://www.sciencedirect.com/book/9780123182501/sieve-methods> (visited on 2025-12-27).
- [HL23] G. H. Hardy and J. E. Littlewood. Some problems of ‘partitio numerorum’; III: on the expression of a number as a sum of primes. *Acta Mathematica*, 44:1–70, 1923. URL: <https://link.springer.com/article/10.1007/BF02403921> (visited on 2025-12-27), doi:10.1007/BF02403921.
- [HW08a] G. H. Hardy and E. M. Wright. *An Introduction to the Theory of Numbers*. Oxford University Press, 6 edition, 2008. ISBN 9780199219858.
- [HW08b] G. H. Hardy and E. M. Wright. *An Introduction to the Theory of Numbers*. Oxford University Press, 6 edition, 2008. ISBN 9780199219865. URL: <https://global.oup.com/ukhe/product/an-introduction-to-the-theory-of-numbers-9780199219865> (visited on 2025-12-27).
- [Hil16] Christian Hill. The klauber triangle. Online, 2016. Blog post describing the construction and a short Python script. URL: <https://scipython.com/blog/the-klauber-triangle/> (visited on 2026-01-01).
- [Hof89] P. Hoffman. *Archimedes' Revenge: The Joys and Perils of Mathematics*. Fawcett Crest, 1989. ISBN 9780449217504. URL: [https://books.google.ch/books?id=\\_SDoAAAACAAJ](https://books.google.ch/books?id=_SDoAAAACAAJ).
- [Inc25a] OEIS Foundation Inc. A000043 — mersenne exponents. Online, 2025. OEIS entry for primes  $p$  such that  $2^p - 1$  is prime (exponents of Mersenne primes). URL: <https://oeis.org/A000043> (visited on 2025-12-27).
- [Inc25b] OEIS Foundation Inc. A001348 — mersenne numbers:  $2^p - 1$  where  $p$  is prime. Online, 2025. OEIS entry for the Mersenne numbers sequence  $2^p - 1$  with prime exponents  $p$ . URL: <https://oeis.org/A001348> (visited on 2025-12-27).
- [IR90] Kenneth Ireland and Michael Rosen. *A Classical Introduction to Modern Number Theory*. Graduate Texts in Mathematics. Springer, New York, 2 edition, 1990. ISBN 978-0-387-97329-6. URL: <https://link.springer.com/book/10.1007/978-1-4757-2103-4> (visited on 2025-12-29), doi:10.1007/978-1-4757-2103-4.
- [Ivic85] Aleksandar Ivić. *The Riemann Zeta-Function: Theory and Applications*. John Wiley & Sons, 1985.
- [IK04] Henryk Iwaniec and Emmanuel Kowalski. *Analytic Number Theory*. Volume 53 of Colloquium Publications. American Mathematical Society, 2004. URL: <https://bookstore.ams.org/coll-53/> (visited on 2025-12-27), doi:10.1090/coll/053.
- [Kur20] U. H. Kurzweg. Hexagonal-integer-spiral and primes. PDF, 11 2020. Short note proposing a hexagonal integer spiral representation and discussing primes. URL: <https://web.mae.ufl.edu/uhk/HEXAGONAL.pdf> (visited on 2026-01-01).
- [KvrvizekLvSomer13] Michal Křiváček, Florian Luca, and Ladislav Šomer. *17 Lectures on Fermat Numbers: From Number Theory to Geometry*. CMS Books in Mathematics. Springer, 2013. ISBN 9781461465250. doi:10.1007/978-1-4614-6526-7.

- [Lag02] Jeffrey C. Lagarias. An elementary problem equivalent to the Riemann hypothesis. *The American Mathematical Monthly*, 109(6):534–543, 2002. doi:10.1080/00029890.2002.11919832.
- [LD37] Peter Gustav Lejeune Dirichlet. Beweis des satzes, dass jede unbegrenzte arithmetische progression, deren erstes glied und differenz ganze zahlen ohne gemeinschaftlichen faktor sind, unendlich viele primzahlen enth"alt. *Abhandlungen der K"oniglichen Preussischen Akademie der Wissenschaften zu Berlin*, pages 45–81, 1837. URL: [https://sites.mathdoc.fr/cgi-bin/oitem?id=OE\\_DIRICHLET\\_\\_1\\_313\\_0](https://sites.mathdoc.fr/cgi-bin/oitem?id=OE_DIRICHLET__1_313_0) (visited on 2025-12-29).
- [LCD+03] Shangzhi Li, Falai Chen, Jiansong Deng, Yaohua Wu, and Yunhua Zhang. *Mathematics Experiments*. World Scientific, 2003. ISBN 9789812380500. doi:10.1142/5008.
- [May15] James Maynard. Small gaps between primes. *Annals of Mathematics*, 181(1):383–413, 2015. URL: <https://annals.math.princeton.edu/2015/181-1/p07> (visited on 2025-12-27), doi:10.4007/annals.2015.181.1.7.
- [MR24] Inc. Mersenne Research. Gimps — the math — primenet. Online, 2024. Background on the mathematics and algorithms used in the GIMPS search strategy (trial factoring, P-1, PRP testing, Lucas–Lehmer, double-checking). URL: <https://www.mersenne.org/various/math.php> (visited on 2025-12-27).
- [MR25] Inc. Mersenne Research. List of known mersenne prime numbers. Online, 2025. PrimeNet/GIMPS list of known Mersenne primes including discovery metadata and verification method. URL: <https://www.mersenne.org/primes/> (visited on 2025-12-27).
- [Mon73] H. L. Montgomery. The pair correlation of zeros of the zeta function. *Proceedings of Symposia in Pure Mathematics*, 24:181–193, 1973.
- [MV06] Hugh L. Montgomery and Robert C. Vaughan. *Multiplicative Number Theory I: Classical Theory*. Volume 97 of Cambridge Studies in Advanced Mathematics. Cambridge University Press, 2006. ISBN 978-0-521-84903-6. doi:10.1017/CBO9780511618314.
- [MM97] M. Ram Murty and V. Kumar Murty. *Non-vanishing of L-Functions and Applications*. Volume 157 of Progress in Mathematics. Birkhäuser, 1997. ISBN 9783764358013. doi:10.1007/978-3-0348-8956-8.
- [Nic99] Thomas R. Nicely. New maximal prime gaps and first occurrences. *Mathematics of Computation*, 68(227):1311–1315, 1999. URL: <https://www.ams.org/mcom/1999-68-227/S0025-5718-99-01065-0/> (visited on 2025-12-27), doi:10.1090/S0025-5718-99-01065-0.
- [NZM91] Ivan Niven, Herbert S. Zuckerman, and Hugh L. Montgomery. *An Introduction to the Theory of Numbers*. John Wiley & Sons, 5 edition, 1991. ISBN 978-0-471-62546-0.
- [OR14] Pascal Ochem and Michaël Rao. Lower bounds on odd perfect numbers. 2014. Slides (Montpellier, 2014-07-02). URL: [https://www.lirmm.fr/~ochem/opn/opn\TU\textbackslash{}\\\_slide.pdf](https://www.lirmm.fr/~ochem/opn/opn\TU\textbackslash{}\_slide.pdf) (visited on 2025-12-25).
- [Odl92] Andrew M. Odlyzko. The  $10^{20}$ -th zero of the Riemann zeta function and 175 million of its neighbors. Unpublished manuscript, 1992. Revision of a 1989 manuscript; numerical computations of high zeros of  $\zeta(s)$ . URL: <https://www-users.cse.umn.edu/~odlyzko/unpublished/zeta.10to20.1992.pdf> (visited on 2025-12-30).
- [OtR85] Andrew M. Odlyzko and Herman J. J. te Riele. Disproof of the mertens conjecture. *Journal für die reine und angewandte Mathematik*, 357:138–160, 1985. URL: <http://eudml.org/doc/152712>, doi:10.1515/crll.1985.357.138.
- [PetkovsekWZ96] Marko Petkovšek, Herbert S. Wilf, and Doron Zeilberger. *A=B*. A K Peters, Wellesley, MA, USA, 1996. ISBN 9781568810635. URL: <https://sites.math.rutgers.edu/~zeilberg/expmath/> (visited on 2025-12-22).
- [Pin09] J'anos Pintz. Landau's problems on primes. *Journal de Th'eorie des Nombres de Bordeaux*, 2009. Accessed 2026-01-02. URL: <https://www.numdam.org/item/10.5802/jtnb.676.pdf>.
- [Polya54a] George Pólya. *Mathematics and Plausible Reasoning, Volume I: Induction and Analogy in Mathematics*. Princeton University Press, Princeton, NJ, USA, 1954.



- [Polya54b] George Pólya. *Mathematics and Plausible Reasoning, Volume II: Patterns of Plausible Inference*. Princeton University Press, Princeton, NJ, USA, 1954.
- [Rab80] Michael O. Rabin. Probabilistic algorithm for testing primality. *Journal of Number Theory*, 12(1):128–138, 1980. URL: <https://www.sciencedirect.com/science/article/pii/0022314X80900840> (visited on 2025-12-27), doi:10.1016/0022-314X(80)90084-0.
- [Ram15] Srinivasa Ramanujan. Highly composite numbers. *Proceedings of the London Mathematical Society*, s2-14(1):347–409, 1915. URL: <https://ramanujan.sirinudi.org/Volumes/published/ram15.pdf>, doi:10.1112/plms/s2\_14.1.347.
- [Rib96] Paulo Ribenboim. *The New Book of Prime Number Records*. Springer, 1996. ISBN 9780387944579. URL: <https://link.springer.com/book/10.1007/978-1-4612-0759-7> (visited on 2025-12-27), doi:10.1007/978-1-4612-0759-7.
- [Rie94] Hans Riesel. *Prime Numbers and Computer Methods for Factorization*. Birkhäuser, 2 edition, 1994. URL: <https://link.springer.com/book/10.1007/978-0-8176-8298-9> (visited on 2025-12-27), doi:10.1007/978-0-8176-8298-9.
- [RSA78] Ronald L. Rivest, Adi Shamir, and Leonard Adleman. A method for obtaining digital signatures and public-key cryptosystems. *Communications of the ACM*, 21(2):120–126, 1978. doi:10.1145/359340.359342.
- [Rob84] Guy Robin. Grandes valeurs de la fonction somme des diviseurs et hypothèse de Riemann. *Journal de Mathématiques Pures et Appliquées*, 63:187–213, 1984.
- [RS62] J. Barkley Rosser and Lowell Schoenfeld. Approximate formulas for some functions of prime numbers. *Illinois Journal of Mathematics*, 6(1):64–94, 1962. URL: <https://projecteuclid.org/journals/illinois-journal-of-mathematics/volume-6/issue-1/Approximate-Formulas-for-Some-Functions-of-Prime-Numbers/ijm/1255631807.full> (visited on 2025-12-27), doi:10.1215/ijm/1255631807.
- [RS94] Michael Rubinstein and Peter Sarnak. Chebyshev's bias. *Experimental Mathematics*, 3(3):173–197, 1994. URL: <https://projecteuclid.org/journals/experimental-mathematics/volume-3/issue-3/Chebyshevs-bias/em/1048515870.full> (visited on 2025-12-29), doi:10.1080/10586458.1994.10504289.
- [Sch76] Lowell Schoenfeld. Sharper bounds for the Chebyshev functions  $\theta(x)$  and  $\psi(x)$ . II. *Mathematics of Computation*, 30(134):337–360, 1976. doi:10.1090/S0025-5718-1976-0457374-6.
- [Ser73] Jean-Pierre Serre. *A Course in Arithmetic*. Graduate Texts in Mathematics. Springer, New York, 1973. ISBN 978-0-387-90040-7. URL: <https://link.springer.com/book/10.1007/978-1-4684-9884-4> (visited on 2025-12-29), doi:10.1007/978-1-4684-9884-4.
- [SS77] Robert M. Solovay and Volker Strassen. A fast Monte-Carlo test for primality. *SIAM Journal on Computing*, 6(1):84–85, 1977. URL: <https://epubs.siam.org/doi/10.1137/0206006> (visited on 2025-12-27), doi:10.1137/0206006.
- [Spi08] Michael Spivak. *Calculus*. Publish or Perish, Inc., 4 edition, 2008. ISBN 9780914098911. URL: <https://www.amazon.com/Calculus-4th-Michael-Spivak/dp/0914098918>.
- [Sto24] Andrew Stone. Improved upper bounds for odd perfect numbers — part i. *Integers: Electronic Journal of Combinatorial Number Theory*, 2024. URL: <https://math.colgate.edu/~integers/y114/y114.pdf>.
- [Ten15] Gérald Tenenbaum. *Introduction to Analytic and Probabilistic Number Theory*. Volume 163 of Graduate Studies in Mathematics. American Mathematical Society, 3 edition, 2015. ISBN 978-1-4704-7821-6.
- [Tit86] E. C. Titchmarsh. *The Theory of the Riemann Zeta-Function*. Oxford University Press, 2 edition, 1986. ISBN 9780198533696.
- [Voi98] John Voight. Perfect numbers: an elementary introduction. 1998. Lecture notes / survey. Date: May 31, 1998; updated January 27, 2024. URL: <https://jvoight.github.io/notes/perf/elem-051015.pdf> (visited on 2025-12-25).

- [Was97] Lawrence C. Washington. *Introduction to Cyclotomic Fields*. Volume 83 of Graduate Texts in Mathematics. Springer, New York, 2 edition, 1997. ISBN 978-0-387-94762-4. URL: <https://link.springer.com/book/10.1007/978-1-4612-1934-7> (visited on 2025-12-29), doi:10.1007/978-1-4612-1934-7.
- [Wei03] Eric W. Weisstein. Perfect number. 2003. MathWorld—A Wolfram Web Resource. URL: <https://mathworld.wolfram.com/PerfectNumber.html> (visited on 2025-12-25).
- [Wei25] Eric W. Weisstein. Euler prime. 2025. MathWorld—A Wolfram Web Resource. URL: <https://mathworld.wolfram.com/EulerPrime.html> (visited on 2026-01-02).
- [Wie09] Arthur Wieferich. Zum letzten fermatschen theorem. *Journal für die reine und angewandte Mathematik*, 136:293–302, 1909.
- [WK09] George Woltman and Scott Kurowski. On the discovery of the 45th and 46th known mersenne primes. *The Fibonacci Quarterly*, 46/47(3):194–197, 2009. Abstract PDF. Describes GIMPS methods and reports the discoveries of M37156667 and M43112609. URL: [https://www.fq.math.ca/Abstracts/46\\_47-3/woltman.pdf](https://www.fq.math.ca/Abstracts/46_47-3/woltman.pdf) (visited on 2025-12-27).
- [Zha14] Yitang Zhang. Bounded gaps between primes. *Annals of Mathematics*, 179(3):1121–1174, 2014. URL: <https://annals.math.princeton.edu/2014/179-3/p07> (visited on 2025-12-27), doi:10.4007/annals.2014.179.3.7.
- [DHJPolymath14] D. H. J. Polymath. Variants of the selberg sieve, and bounded intervals containing many primes. *Research in the Mathematical Sciences*, 1:12, 2014. URL: <https://link.springer.com/article/10.1186/s40687-014-0019-3> (visited on 2025-12-27), doi:10.1186/s40687-014-0019-3.
- [MersenneResearchIncGIMPS24] Mersenne Research, Inc. (GIMPS). Mersenne prime discovery:  $2^{136279841}-1$  is prime! 2024. GIMPS press release page (52nd known Mersenne prime). URL: <https://www.mersenne.org/primes/?press=M136279841> (visited on 2025-12-25).
- [MersenneResearchIncGIMPS25] Mersenne Research, Inc. (GIMPS). Gimps milestones report. 2025. URL: [https://www.mersenne.org/report/TU\textbackslash{}\\_milestones/](https://www.mersenne.org/report/TU\textbackslash{}_milestones/) (visited on 2025-12-25).
- [OEISFInc25a] OEIS Foundation Inc. A000396: perfect numbers. 2025. The On-Line Encyclopedia of Integer Sequences (OEIS). URL: <https://oeis.org/A000396> (visited on 2025-12-25).
- [OEISFInc25b] OEIS Foundation Inc. A001097: twin primes. Online, 2025. The On-Line Encyclopedia of Integer Sequences (OEIS). URL: <https://oeis.org/A001097> (visited on 2025-12-27).
- [OEISFInc25c] OEIS Foundation Inc. A001223: prime gaps  $p_{n+1}-p_n$ . Online, 2025. The On-Line Encyclopedia of Integer Sequences (OEIS). URL: <https://oeis.org/A001223> (visited on 2025-12-27).
- [OEISFInc25d] OEIS Foundation Inc. A001359: lesser of twin primes. Online, 2025. The On-Line Encyclopedia of Integer Sequences (OEIS). URL: <https://oeis.org/A001359> (visited on 2025-12-27).
- [OEISFInc25e] OEIS Foundation Inc. A023201: primes  $p$  such that  $p+6$  is also prime (sexy primes). Online, 2025. The On-Line Encyclopedia of Integer Sequences (OEIS). URL: <https://oeis.org/A023201> (visited on 2025-12-27).
- [OEISFInc25f] OEIS Foundation Inc. A046132: primes  $p$  such that  $p-4$  is also prime (cousin prime pairs). Online, 2025. The On-Line Encyclopedia of Integer Sequences (OEIS). URL: <https://oeis.org/A046132> (visited on 2025-12-27).
- [PrimePUTM25a] Prime Pages (UTM). Carmichael number (prime glossary). <https://t5k.org/glossary/page.php?sort=CarmichaelNumber>, 2025. Online; accessed 2025-12-28.
- [PrimePUTM25b] Prime Pages (UTM). Primorial (prime glossary). <https://t5k.org/glossary/page.php?sort=Primorial>, 2025. Online; accessed 2025-12-28.
- [PrimePUTM25c] Prime Pages (UTM). Wieferich prime (prime glossary). <https://t5k.org/glossary/page.php?sort=WieferichPrime>, 2025. Online; accessed 2025-12-28.

- [TheOFInc25a] The OEIS Foundation Inc. A000215: Fermat numbers  $F(n) = 2^{(2^n)} + 1$ . <https://oeis.org/A000215>, 2025. Online; accessed 2025-12-28.
- [TheOFInc25b] The OEIS Foundation Inc. A001220: Wieferich primes (base 2). <https://oeis.org/A001220>, 2025. Online; accessed 2025-12-28.
- [TheOFInc25c] The OEIS Foundation Inc. A001358: Semiprimes. <https://oeis.org/A001358>, 2025. Online; accessed 2025-12-28.
- [TheOFInc25d] The OEIS Foundation Inc. A002110: Primorial numbers  $n\#$ . <https://oeis.org/A002110>, 2025. Online; accessed 2025-12-28.
- [TheOFInc25e] The OEIS Foundation Inc. A002997: Carmichael numbers. <https://oeis.org/A002997>, 2025. Online; accessed 2025-12-28.
- [Wikipediacontributors25a] Wikipedia contributors. Carmichael number — Wikipedia, the free encyclopedia. [https://en.wikipedia.org/w/index.php?oldid=1328580842&title=Carmichael\\_number](https://en.wikipedia.org/w/index.php?oldid=1328580842&title=Carmichael_number), 2025. Online; accessed 2025-12-28.
- [Wikipediacontributors25b] Wikipedia contributors. Fermat number — Wikipedia, the free encyclopedia. [https://en.wikipedia.org/w/index.php?oldid=1329504472&title=Fermat\\_number](https://en.wikipedia.org/w/index.php?oldid=1329504472&title=Fermat_number), 2025. Online; accessed 2025-12-28.
- [Wikipediacontributors25c] Wikipedia contributors. Harmonic number. Wikipedia, 2025. Permanent revision as of 20:03, 12 December 2025 (UTC). URL: [https://en.wikipedia.org/w/index.php?oldid=1327129204\TU\textbackslash{}&title=Harmonic\TU\textbackslash{}\\_number](https://en.wikipedia.org/w/index.php?oldid=1327129204\TU\textbackslash{}&title=Harmonic\TU\textbackslash{}_number) (visited on 2025-12-25).
- [Wikipediacontributors25d] Wikipedia contributors. Perfect number. Wikipedia, 2025. Permanent revision as of 13:46, 22 December 2025 (UTC). URL: [https://en.wikipedia.org/w/index.php?oldid=1328905011\TU\textbackslash{}&title=Perfect\TU\textbackslash{}\\_number](https://en.wikipedia.org/w/index.php?oldid=1328905011\TU\textbackslash{}&title=Perfect\TU\textbackslash{}_number) (visited on 2025-12-25).
- [Wikipediacontributors25e] Wikipedia contributors. Prime number. Wikipedia, 2025. Permanent revision as of 20:47, 2 December 2025 (UTC). URL: [https://en.wikipedia.org/w/index.php?oldid=1325385945\TU\textbackslash{}&title=Prime\TU\textbackslash{}\\_number](https://en.wikipedia.org/w/index.php?oldid=1325385945\TU\textbackslash{}&title=Prime\TU\textbackslash{}_number) (visited on 2025-12-25).
- [Wikipediacontributors25f] Wikipedia contributors. Prime-counting function — Wikipedia, the free encyclopedia. 2025. Accessed: 2025-12-29. URL: [https://en.wikipedia.org/w/index.php?title=Prime-counting\\_function&oldid=1328904428](https://en.wikipedia.org/w/index.php?title=Prime-counting_function&oldid=1328904428) (visited on 2025-12-29).
- [Wikipediacontributors25g] Wikipedia contributors. Primorial — Wikipedia, the free encyclopedia. <https://en.wikipedia.org/w/index.php?oldid=1328293471&title=Primorial>, 2025. Online; accessed 2025-12-28.
- [Wikipediacontributors25h] Wikipedia contributors. Riemann zeta function — Wikipedia, the free encyclopedia. 2025. Accessed: 2025-12-29. URL: [https://en.wikipedia.org/w/index.php?title=Riemann\\_zeta\\_function&oldid=1326434355](https://en.wikipedia.org/w/index.php?title=Riemann_zeta_function&oldid=1326434355) (visited on 2025-12-29).
- [Wikipediacontributors25i] Wikipedia contributors. Riemann–von mangoldt formula — Wikipedia, the free encyclopedia. 2025. Accessed: 2025-12-29. URL: [https://en.wikipedia.org/w/index.php?title=Riemann%E2%80%93von\\_Mangoldt\\_formula&oldid=1322838097](https://en.wikipedia.org/w/index.php?title=Riemann%E2%80%93von_Mangoldt_formula&oldid=1322838097) (visited on 2025-12-29).
- [Wikipediacontributors25j] Wikipedia contributors. Taylor series. Wikipedia, 2025. Permanent revision as of 04:10, 24 December 2025 (UTC). URL: [https://en.wikipedia.org/w/index.php?oldid=1329166943\TU\textbackslash{}&title=Taylor\TU\textbackslash{}\\_series](https://en.wikipedia.org/w/index.php?oldid=1329166943\TU\textbackslash{}&title=Taylor\TU\textbackslash{}_series) (visited on 2025-12-25).
- [Wikipediacontributors25k] Wikipedia contributors. Wieferich prime — Wikipedia, the free encyclopedia. [https://en.wikipedia.org/w/index.php?oldid=1329186350&title=Wieferich\\_prime](https://en.wikipedia.org/w/index.php?oldid=1329186350&title=Wieferich_prime), 2025. Online; accessed 2025-12-28.



- [Wikipediacontributors25l] Wikipedia contributors. Z function — Wikipedia, the free encyclopedia. 2025. Accessed: 2025-12-29. URL: [https://en.wikipedia.org/w/index.php?title=Z\\_function&oldid=1313086592](https://en.wikipedia.org/w/index.php?title=Z_function&oldid=1313086592) (visited on 2025-12-29).
- [Wikipediacontributors26a] Wikipedia contributors. Landau's problems. 2026. Accessed 2026-01-02. URL: [https://en.wikipedia.org/wiki/Landau%27s\\_problems](https://en.wikipedia.org/wiki/Landau%27s_problems).
- [Wikipediacontributors26b] Wikipedia contributors. Quadratic equation. 2026. Accessed 2026-01-02. URL: [https://en.wikipedia.org/wiki/Quadratic\\_equation](https://en.wikipedia.org/wiki/Quadratic_equation).